The Complete LeetCode Mastery Guide — Every Pattern, Interview-Style

> **Goal of this document:** give you a single README you can study end-to-end and walk into any product-based or MNC interview (Google, Amazon, Microsoft, Meta, Apple, Netflix, Adobe, Bloomberg, Uber, Oracle, Goldman Sachs, Walmart, Flipkart, TCS Digital, Infosys, etc.) with full confidence on the DSA round.

⚠️ An Honest Note on Scope (Please Read First)

LeetCode hosts **3,000+ problems** and grows every week. No single file — no matter how large — can contain a genuinely deep, interview-quality breakdown ("clarify → brute force → optimize → code → test → follow-up") of literally every one of them; that would require hundreds of thousands of lines and would stop being *useful* long before it stopped being *long*. Instead, this guide is built the way a real senior engineer would prepare a candidate:

- **Part A — Top ~105 Most-Asked MNC Interview Questions.** These are the problems that appear, in some paraphrased form, in the overwhelming majority of real interview loops at top companies (this list overlaps heavily with what the community calls "Blind 75" / "NeetCode 150" / "Top Interview 150" — because those lists exist *precisely because* these are the questions companies actually ask). Every single one gets the **full interview treatment**: problem statement, examples, constraints, clarifying questions, brute-force solution with code and complexity, optimal solution with code and complexity, a manual dry-run, edge cases, and likely follow-up questions.
- **Part B — Extended Problem Bank.** Hundreds of additional problems across every remaining LeetCode tag/pattern (strings, matrices, design, advanced graphs, advanced DP, more greedy, more bit tricks, etc.), each with a brute-force solution and an optimal solution with working code, so that the *patterns* behind essentially every problem type on LeetCode are covered.
- **Part C — Pattern Playbook & Master Index.** A condensed cheat sheet mapping every pattern to its recognition signal, template code, and the exact problems

above that use it, plus a master index of every problem in this file.

If you work through Part A + Part B, you will have directly practiced or seen the template for **the pattern behind essentially every problem on LeetCode** — because LeetCode problems are recombinations of a surprisingly small number of underlying techniques. That is the real goal: not "I've seen this exact problem," but "I recognize what family this problem belongs to and I know the two-step (brute force → optimal) path to solve it."

All code is written in **Java** (matches the rest of this repository's interview guides). Time and space complexity are always stated. `n` = size of input array/string unless stated otherwise.

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Part 0: The Universal Interview Framework

Use this exact sequence for **every** problem in this guide, and every problem you'll ever see in a real interview:

1. **Restate the problem in your own words.** Confirms you understood it; buys thinking time.
2. **Ask clarifying questions.** Can the array be empty? Are there duplicates? Negative numbers? Is the input sorted? What should happen on no valid answer — throw, return null, return -1? What's the expected input size (tells you the required complexity)?
3. **Work a small example by hand.** Write it on the whiteboard/editor as a comment.
4. **State the brute force first, out loud, with its complexity.** Never skip this — it proves you can solve *something* immediately, and it's often the seed of the optimal idea.
5. **Find the bottleneck.** Ask: "What am I recomputing? What information am I throwing away that I could cache?" This is where two pointers / hashing / DP / heaps come from.
6. **Propose the optimal approach and its complexity before coding.** Get interviewer buy-in.
7. **Code it cleanly.** Meaningful variable names, small helper functions, handle edge cases explicitly.
8. **Trace through the example again with your actual code.**
9. **State final time/space complexity and mention 1-2 follow-up variants.**

Every problem below follows steps 3-9 explicitly so you can see the pattern repeated until it's second nature.

Part A: Top ~105 MNC Interview Questions (Full Treatment)

A1. Arrays & Hashing

A1.1 Two Sum

Difficulty: Easy | **Pattern:** Hashing | **Companies:** Amazon, Google, Adobe, Microsoft, Apple, Bloomberg (one of the single most-asked questions in the world — often the very first question in a phone screen)

Problem Statement: Given an array of integers `nums` and an integer `target`, return the indices of the two numbers such that they add up to `target`. Assume exactly one solution exists, and you may not use the same element twice.

Example:

```
Input: nums = [2,7,11,15], target = 9
Output: [0,1]        // nums[0] + nums[1] == 9
```

Constraints: `2 <= nums.length <= 10^4`, `-10^9 <= nums[i], target <= 10^9`.

Clarifying Questions to Ask:

- Can I assume exactly one valid pair exists? (Usually yes, but ask.)
- Are the numbers sorted? (No — if they were, we'd use two pointers instead of hashing.)
- Can the same value appear twice at different indices? (Yes — must handle duplicates.)

Brute Force Approach Intuition: try every pair `(i, j)` and check if they sum to target.

```java
public int[] twoSumBrute(int[] nums, int target) {
    for (int i = 0; i < nums.length; i++) {
        for (int j = i + 1; j < nums.length; j++) {
            if (nums[i] + nums[j] == target) return new int[]{i, j};
        }
    }
    return new int[]{-1, -1};
}
```

Time: O(n²) — nested loop over all pairs. Space: O(1).

Optimal Approach Intuition: as we scan, for each `nums[i]` we need to know if `target - nums[i]` was *already seen*. A HashMap gives O(1) lookup, turning the nested loop into a single pass.

```java
public int[] twoSum(int[] nums, int target) {
    Map<Integer, Integer> seen = new HashMap<>(); // value -> index
    for (int i = 0; i < nums.length; i++) {
        int complement = target - nums[i];
        if (seen.containsKey(complement)) {
            return new int[]{seen.get(complement), i};
        }
        seen.put(nums[i], i);
    }
    return new int[]{-1, -1};
}
```

Time: O(n) — single pass, O(1) HashMap operations. Space: O(n) for the map.

Dry Run: `nums=[2,7,11,15], target=9`. i=0: complement=7, not in map, put(2→0). i=1: complement=2, `seen` has 2→0! Return `[0,1]`. ✅

Edge Cases: duplicate values (`[3,3]`, target `6` → works because we check *before* inserting current element, so we don't match an element with itself). No solution (per constraints won't happen, but code returns `[-1,-1]` defensively).

Follow-up: "What if the array is sorted?" → use two pointers from both ends, O(n) time O(1) space (no hashmap needed). "What if you need *all* pairs, not just one?" → still O(n) with a map storing lists of indices, or O(n²) if duplicates need distinct index-pairs enumerated.

A1.2 Best Time to Buy and Sell Stock

Difficulty: Easy | **Pattern:** One pass / Kadane's variant | **Companies:** Amazon, Meta, Bloomberg, Goldman Sachs

Problem Statement: Given an array `prices` where `prices[i]` is the stock price on day `i`, find the maximum profit from buying on one day and selling on a later day. Return 0 if no profit is possible.

Example:

```
Input: prices = [7,1,5,3,6,4]
Output: 5   // buy at 1, sell at 6
```

Constraints: `1 <= prices.length <= 10^5`, `0 <= prices[i] <= 10^4`.

Clarifying Questions: Can I buy and sell on the same day? (No, must sell after buying.) Only one transaction allowed, correct? (Yes — this is the single-transaction version; the multi-transaction version is a separate, later problem.)

Brute Force Approach Intuition: try every buy day `i` and every sell day `j > i`, track the max profit.

```
public int maxProfitBrute(int[] prices) {
    int max = 0;
    for (int i = 0; i < prices.length; i++) {
        for (int j = i + 1; j < prices.length; j++) {
            max = Math.max(max, prices[j] - prices[i]);
        }
    }
    return max;
}
```

Time: O(n²). Space: O(1).

Optimal Approach Intuition: as we scan left to right, we only ever need "the minimum price seen so far" — that's the best possible buy day for any sell day we're currently looking at. Track `minPrice` and `maxProfit` in one pass.

```
public int maxProfit(int[] prices) {
    int minPrice = Integer.MAX_VALUE;
    int maxProfit = 0;
    for (int price : prices) {
        if (price < minPrice) {
            minPrice = price;
        } else if (price - minPrice > maxProfit) {
```

```
                maxProfit = price - minPrice;
            }
        }
        return maxProfit;
    }
```

Time: O(n) single pass. Space: O(1).

Dry Run: `[7,1,5,3,6,4]`. min=7→profit0. price=1<7→min=1. price=5→profit=4. price=3→ profit stays 4. price=6→profit=5. price=4→profit stays 5. Answer `5`. ✅

Edge Cases: prices strictly decreasing (`[7,6,4,3,1]` → answer 0). Single element array (no transaction possible → 0). All equal prices → 0.

Follow-up: "What if you can do unlimited transactions?" (Best Time II — sum all positive adjacent differences, greedy). "What if at most 2 transactions?" (DP with 4 states: buy1, sell1, buy2, sell2). "What if there's a cooldown/transaction fee?" (separate DP state machine problems, covered in Part B).

A1.3 Contains Duplicate

Difficulty: Easy | **Pattern:** Hashing | **Companies:** Amazon, Yahoo, common phone-screen warm-up

Problem Statement: Given an integer array `nums`, return `true` if any value appears at least twice, `false` if every element is distinct.

Example: `[1,2,3,1]` → `true`. `[1,2,3,4]` → `false`.

Constraints: `1 <= nums.length <= 10^5`.

Clarifying Questions: Is the array sorted? (No.) Should I optimize for space or time if both matter? (Usually time first, mention the sort-based O(1)-extra-space alternative.)

Brute Force Approach Intuition: compare every pair.

```
public boolean containsDuplicateBrute(int[] nums) {
    for (int i = 0; i < nums.length; i++)
        for (int j = i + 1; j < nums.length; j++)
            if (nums[i] == nums[j]) return true;
    return false;
}
```

Time: O(n²). Space: O(1).

Alternative brute-force-adjacent: sort first, then check adjacent elements. O(n log n) time, O(1) or O(log n) space (sort overhead) — better than nested loops but still not optimal for time.

```
public boolean containsDuplicateSort(int[] nums) {
    int[] sorted = nums.clone();
    Arrays.sort(sorted);
    for (int i = 1; i < sorted.length; i++)
        if (sorted[i] == sorted[i - 1]) return true;
    return false;
}
```

Optimal Approach Intuition: a HashSet gives O(1) average "have I seen this before" checks.

```
public boolean containsDuplicate(int[] nums) {
    Set<Integer> seen = new HashSet<>();
    for (int n : nums) {
        if (!seen.add(n)) return true; // add() returns false if already present
    }
    return false;
}
```

Time: O(n). Space: O(n).

Dry Run: `[1,2,3,1]`: add 1(ok), add 2(ok), add 3(ok), add 1 → returns false (already present) → function returns `true`. ✅

Edge Cases: array of length 1 (`false`, trivially). All identical elements (`true` immediately on second element).

Follow-up: "Contains Duplicate II" (index difference ≤ k) — sliding window HashSet. "Contains Duplicate III" (value difference ≤ t within index window k) — bucketing / sorted TreeSet with `ceiling`/`floor`.

A1.4 Product of Array Except Self

Difficulty: Medium | **Pattern:** Prefix/Suffix products | **Companies:** Amazon, Meta, Microsoft, Apple — extremely common

Problem Statement: Given an array `nums`, return an array `answer` where `answer[i]` equals the product of all elements of `nums` except `nums[i]`. Must run in O(n) **without division**, and (follow-up) in O(1) extra space excluding the output array.

Example: `[1,2,3,4]` → `[24,12,8,6]`.

Constraints: `2 <= nums.length <= 10^5`, values fit in 32-bit int, product fits in 32-bit int guaranteed.

Clarifying Questions: Can nums contain 0? (Yes — this is exactly why division is disallowed/dangerous: if there are two zeros the naive division approach breaks entirely, and even with one zero you need special handling.) Can I use division at all if I handle zero specially? (The problem explicitly forbids division, so don't.)

Brute Force Approach Intuition: for every index, loop through the rest of the array and multiply.

```
public int[] productExceptSelfBrute(int[] nums) {
    int n = nums.length;
    int[] res = new int[n];
    for (int i = 0; i < n; i++) {
        int product = 1;
        for (int j = 0; j < n; j++) {
            if (j != i) product *= nums[j];
        }
        res[i] = product;
    }
    return res;
}
```

Time: O(n²). Space: O(1) extra (excluding output).

Optimal Approach Intuition: `answer[i] = (product of everything to the left of i) * (product of everything to the right of i)`. Compute prefix products in one pass, then multiply in suffix products in a second pass — no division needed anywhere.

```
public int[] productExceptSelf(int[] nums) {
    int n = nums.length;
    int[] res = new int[n];
    res[0] = 1;
    for (int i = 1; i < n; i++) {
        res[i] = res[i - 1] * nums[i - 1]; // res[i] = prefix product of nums[0..i-
1]
    }
    int suffix = 1;
```

```
        for (int i = n - 1; i >= 0; i--) {
            res[i] *= suffix;       // multiply in suffix product of nums[i+1..n-1]
            suffix *= nums[i];
        }
        return res;
    }
```

Time: O(n) — two linear passes. Space: O(1) extra (output array doesn't count per problem convention).

Dry Run: `[1,2,3,4]`. Prefix pass: res=[1,1,2,6]. Suffix pass from right: i=3, res[3]=6*1=6, suffix=4. i=2, res[2]=2*4=8, suffix=12. i=1, res[1]=1*12=12, suffix=24. i=0, res[0]=1*24=24, suffix=24. Final `[24,12,8,6]`. ✅

Edge Cases: array contains one zero → every other index becomes 0 except the zero's own index (handled automatically since the prefix/suffix trick naturally zeroes out products crossing the zero). Two zeros → every output is 0 (also handled automatically, no special-case code needed — this is the elegance of the prefix/suffix technique vs. a division approach).

Follow-up: "Could you do it with O(1) *total* extra space (not counting output)?" — yes, that's exactly the code above (the `res` array is the output, `suffix` is the only extra variable).

A1.5 Maximum Subarray (Kadane's Algorithm)

Difficulty: Medium | **Pattern:** Dynamic Programming (Kadane's) | **Companies:** Microsoft, Amazon, LinkedIn — one of the most classic DP-intro questions

Problem Statement: Given an integer array `nums`, find the contiguous subarray (containing at least one number) with the largest sum, and return that sum.

Example: `[-2,1,-3,4,-1,2,1,-5,4]` → `6` (subarray `[4,-1,2,1]`).

Constraints: `1 <= nums.length <= 10^5`, values can be negative.

Clarifying Questions: Must the subarray be contiguous? (Yes — this is the key distinguishing detail vs. "maximum subsequence sum", which would be trivial — just sum all positives.) Can all elements be negative? (Yes — must return the least-negative single element, not 0.)

Brute Force Approach Intuition: try every possible subarray start/end and sum it.

```java
public int maxSubArrayBrute(int[] nums) {
    int best = Integer.MIN_VALUE;
    for (int i = 0; i < nums.length; i++) {
        int sum = 0;
        for (int j = i; j < nums.length; j++) {
            sum += nums[j];
            best = Math.max(best, sum);
        }
    }
    return best;
}
```

Time: O(n²). Space: O(1).

Optimal Approach (Kadane's Algorithm) Intuition: at each index, decide: "should I extend the previous subarray, or start fresh from here?" If the running sum so far is negative, it can only hurt any future subarray, so drop it and restart from the current element.

```java
public int maxSubArray(int[] nums) {
    int currentSum = nums[0];
    int maxSum = nums[0];
    for (int i = 1; i < nums.length; i++) {
        currentSum = Math.max(nums[i], currentSum + nums[i]);
        maxSum = Math.max(maxSum, currentSum);
    }
    return maxSum;
}
```

Time: O(n) single pass. Space: O(1).

Dry Run: `[-2,1,-3,4,-1,2,1,-5,4]`. cur=-2,max=-2. i=1: cur=max(1,-1)=1,max=1. i=2: cur=max(-3,-2)=-2,max=1. i=3: cur=max(4,2)=4,max=4. i=4: cur=max(-1,3)=3,max=4. i=5: cur=max(2,5)=5,max=5. i=6: cur=max(1,6)=6,max=6. i=7: cur=max(-5,1)=1,max=6. i=8: cur=max(4,5)=5,max=6. Answer `6`. ✅

Edge Cases: all negative (`[-3,-1,-2]` → answer `-1`, the max single element). Single element array → that element itself.

Follow-up: "Return the actual subarray, not just the sum" — track a `start` index that resets whenever we restart, and a `bestStart/bestEnd` pair updated whenever `maxSum` improves. "Maximum Subarray using divide & conquer" — O(n log n) alternative often asked to test D&C thinking (split at mid, recurse left/right, plus compute the best crossing-the-middle subarray in O(n) per level).

A1.6 Maximum Product Subarray

Difficulty: Medium | **Pattern:** DP (track running max **and** min) | **Companies:** Amazon, Meta

Problem Statement: Given an integer array `nums`, find the contiguous subarray with the largest product, return that product.

Example: `[2,3,-2,4]` → `6` (subarray `[2,3]`).

Constraints: `1 <= nums.length <= 2*10^4`, can contain 0 and negative numbers.

Clarifying Questions: Can there be zeros? (Yes — a zero resets any running product to 0, must handle.) Can there be negative numbers? (Yes — this is exactly why it's harder than Maximum Subarray: a large negative product can become the maximum if multiplied by another negative number, so we must track the running **minimum** product too.)

Brute Force Approach Intuition: try every subarray, compute the product.

```
public int maxProductBrute(int[] nums) {
    int best = nums[0];
    for (int i = 0; i < nums.length; i++) {
        long product = 1;
        for (int j = i; j < nums.length; j++) {
            product *= nums[j];
            best = (int) Math.max(best, product);
        }
    }
    return best;
}
```

Time: O(n²). Space: O(1).

Optimal Approach Intuition: because multiplying by a negative number flips max and min, at each index track BOTH `maxSoFar` and `minSoFar` ending at that index. When `nums[i]` is negative, swap them before combining.

```
public int maxProduct(int[] nums) {
    int maxSoFar = nums[0], minSoFar = nums[0], result = nums[0];
    for (int i = 1; i < nums.length; i++) {
        int cur = nums[i];
        if (cur < 0) {
            int temp = maxSoFar;
            maxSoFar = minSoFar;
```

```
                minSoFar = temp;
            }
            maxSoFar = Math.max(cur, maxSoFar * cur);
            minSoFar = Math.min(cur, minSoFar * cur);
            result = Math.max(result, maxSoFar);
        }
        return result;
    }
```

Time: O(n). Space: O(1).

Dry Run: `[2,3,-2,4]`. max=2,min=2,res=2. i=1(3): max=max(3,6)=6,min=min(3,6)=3,res=6. i=2(-2): swap→max=3,min=6; then max=max(-2,-6)=-2, min=min(-2,-12)=-12, res stays 6. i=3(4): max=max(4,-8)=4, min=min(4,-48)=-48, res stays 6. Answer `6`. ✅

Edge Cases: contains 0 (e.g. `[0,2]` → the 0 forces both max/min to reset at that index because `cur` alone, `0`, beats any product through it going forward — code handles this naturally since `Math.max(cur, maxSoFar*cur)` picks `cur=0` when `maxSoFar*cur` is also 0-touching). All negative with even count → whole array is the answer.

Follow-up: "What if you need the actual subarray?" track start/end indices alongside max/min updates.

A1.7 Find Minimum in Rotated Sorted Array

Difficulty: Medium | **Pattern:** Modified Binary Search | **Companies:** Microsoft, Amazon, Facebook

Problem Statement: An ascending sorted array of unique elements is rotated at some unknown pivot. Find the minimum element in O(log n) time.

Example: `[4,5,6,7,0,1,2]` → `0`.

Constraints: `1 <= nums.length <= 5000`, all values unique.

Clarifying Questions: Are all elements guaranteed unique? (Yes for this version — there's a harder follow-up with duplicates.) Could the array be not rotated at all (already fully sorted)? (Yes, handle that as a special/degenerate case of the same logic.)

Brute Force Approach Intuition: just scan linearly for the minimum.

```java
public int findMinBrute(int[] nums) {
    int min = nums[0];
    for (int n : nums) min = Math.min(min, n);
    return min;
}
```

Time: O(n). Space: O(1). (Correct, but doesn't use the sorted-rotated structure, so it fails the "O(log n)" requirement an interviewer will explicitly ask for.)

Optimal Approach Intuition: binary search, but the decision of which half to discard is based on comparing `nums[mid]` with `nums[right]`. If `nums[mid] > nums[right]`, the minimum must be strictly to the right of `mid` (the rotation point is in the right half). Otherwise the minimum is at `mid` or to its left.

```java
public int findMin(int[] nums) {
    int left = 0, right = nums.length - 1;
    while (left < right) {
        int mid = left + (right - left) / 2;
        if (nums[mid] > nums[right]) {
            left = mid + 1;   // pivot is to the right
        } else {
            right = mid;      // pivot is at mid or to the left
        }
    }
    return nums[left];
}
```

Time: O(log n). Space: O(1).

Dry Run: `[4,5,6,7,0,1,2]`. left=0,right=6,mid=3(val7). 7>2→left=4. left=4,right=6,mid=5 (val1). 1<2→right=5. left=4,right=5,mid=4(val0). 0<1→right=4. left==right=4 → nums[4]=`0`. ✅

Edge Cases: not rotated at all (`[1,2,3,4]` → `nums[mid]` never `> nums[right]`, loop converges to index 0 correctly). Single element (`[1]` → loop doesn't execute, return `nums[0]`). Two elements.

Follow-up: "What if duplicates are allowed?" (Find Minimum in Rotated Sorted Array II) — when `nums[mid] == nums[right]` you can't tell which side the min is on, so you must shrink `right--` one step (worst case degrades to O(n)). "Search a target value in this rotated array" — the next problem.

A1.8 Search in Rotated Sorted Array

Difficulty: Medium | **Pattern:** Modified Binary Search | **Companies:** Amazon, Microsoft, Facebook, Bloomberg — extremely frequent

Problem Statement: Given a rotated sorted array of unique elements and a `target`, return its index, or `-1` if not present. Must run in O(log n).

Example: `[4,5,6,7,0,1,2]`, target=`0` → `4`.

Constraints: `1 <= nums.length <= 5000`, all values unique.

Clarifying Questions: Are values guaranteed unique? (Yes.) What to return if target isn't present? (`-1`.)

Brute Force Approach Intuition: linear scan.

```
public int searchBrute(int[] nums, int target) {
    for (int i = 0; i < nums.length; i++) if (nums[i] == target) return i;
    return -1;
}
```

Time: O(n). Space: O(1). Doesn't meet the required O(log n).

Optimal Approach Intuition: at every step of binary search, **at least one half (left..mid or mid..right) is always properly sorted**. Determine which half is sorted by comparing `nums[left]` with `nums[mid]`; then check if `target` lies within that sorted half's range — if so recurse there, otherwise recurse the other half.

```
public int search(int[] nums, int target) {
    int left = 0, right = nums.length - 1;
    while (left <= right) {
        int mid = left + (right - left) / 2;
        if (nums[mid] == target) return mid;

        if (nums[left] <= nums[mid]) { // left half is sorted
            if (nums[left] <= target && target < nums[mid]) {
                right = mid - 1;
            } else {
                left = mid + 1;
            }
        } else { // right half is sorted
            if (nums[mid] < target && target <= nums[right]) {
                left = mid + 1;
            } else {
                right = mid - 1;
```

```
                }
            }
        }
        return -1;
    }
```

Time: O(log n). Space: O(1).

Dry Run: `[4,5,6,7,0,1,2]`, target=0. left=0,right=6,mid=3(val7). left half [4..7] sorted (4<=7). Is 0 in [4,7)? No → left=4. left=4,right=6,mid=5(val1). left half [0..1] sorted (nums[4]=0<=nums[5]=1). Is 0 in [0,1)? Yes → right=4. left=4,right=4,mid=4(val0)==target → return `4`. ✅

Edge Cases: target not present (`[4,5,6,7,0,1,2]`, target=3 → returns -1). Not rotated at all (degenerates to plain binary search). Single element array.

Follow-up: "Duplicates allowed?" — Search in Rotated Sorted Array II: when `nums[left]==nums[mid]==nums[right]` you can't determine the sorted half, so shrink both ends by 1 (worst case O(n)).

A1.9 3Sum

Difficulty: Medium | **Pattern:** Sorting + Two Pointers | **Companies:** Amazon, Meta, Microsoft, Adobe — legendary interview question

Problem Statement: Given an integer array `nums`, return all unique triplets `[nums[i],nums[j],nums[k]]` such that `i != j != k` and they sum to 0.

Example: `[-1,0,1,2,-1,-4]` → `[[-1,-1,2],[-1,0,1]]`.

Constraints: `3 <= nums.length <= 3000`. Result must not contain duplicate triplets.

Clarifying Questions: Can the array contain duplicates? (Yes, and the output must not contain duplicate triplets — this drives the "skip duplicates" logic.) Should triplets be returned in any particular order? (Usually no.)

Brute Force Approach Intuition: three nested loops trying every triplet, dedupe using a Set of sorted triplets.

```
public List<List<Integer>> threeSumBrute(int[] nums) {
    Set<List<Integer>> resultSet = new HashSet<>();
    int n = nums.length;
```

```
    for (int i = 0; i < n; i++)
        for (int j = i + 1; j < n; j++)
            for (int k = j + 1; k < n; k++)
                if (nums[i] + nums[j] + nums[k] == 0) {
                    List<Integer> triplet = Arrays.asList(nums[i], nums[j],
nums[k]);
                    triplet.sort(null);
                    resultSet.add(triplet);
                }
    return new ArrayList<>(resultSet);
}
```

Time: O(n³). Space: O(n) for the result set.

Optimal Approach Intuition: sort the array first. Fix the first element `nums[i]`, then the remaining problem becomes "find two numbers in the rest of the array that sum to `-nums[i]`" — exactly a two-pointer problem on a sorted array. Skip duplicate values for `i`, and skip duplicates while moving the two pointers, to avoid duplicate triplets without needing a Set.

```
public List<List<Integer>> threeSum(int[] nums) {
    Arrays.sort(nums);
    List<List<Integer>> result = new ArrayList<>();
    int n = nums.length;
    for (int i = 0; i < n - 2; i++) {
        if (i > 0 && nums[i] == nums[i - 1]) continue; // skip duplicate anchors
        if (nums[i] > 0) break; // smallest element positive => no triplet can sum
to 0

        int left = i + 1, right = n - 1;
        while (left < right) {
            int sum = nums[i] + nums[left] + nums[right];
            if (sum == 0) {
                result.add(Arrays.asList(nums[i], nums[left], nums[right]));
                left++; right--;
                while (left < right && nums[left] == nums[left - 1]) left++;
                while (left < right && nums[right] == nums[right + 1]) right--;
            } else if (sum < 0) {
                left++;
            } else {
                right--;
            }
        }
    }
    return result;
}
```

Time: O(n²) — O(n log n) sort + O(n) two-pointer scan for each of n anchors. Space: O(log n) to O(n) for sort, O(1) extra beyond output.

Dry Run: sorted `[-4,-1,-1,0,1,2]`. i=0(-4): left=1,right=5, sum=-4-1+2=-3<0→left++... eventually no triplet with -4 sums to 0. i=1(-1): left=2,right=5. sum=-1-1+2=0 → add `[-1,-1,2]`, move both, skip dup at left. left=3,right=4: sum=-1+0+1=0 → add `[-1,0,1]`. left=4,right=3 → loop ends. i=2(-1) equals nums[1] → skip. i=3(0): left=4,right=5: sum=0+1+2=3>0→right--, loop ends. Result `[[-1,-1,2],[-1,0,1]]`. ✅

Edge Cases: fewer than 3 elements → return empty list. All zeros (`[0,0,0,0]` → single triplet `[0,0,0]`, dedupe logic must catch this). All positive or all negative → early break (`nums[i] > 0` check) or naturally no valid triplet found.

Follow-up: "4Sum" — same idea with an extra fixed loop, O(n³). "K-Sum" generalized recursive pattern reducing to 2-pointer base case. "3Sum Closest" — track the closest sum instead of exact matches, same two-pointer skeleton.

A1.10 Container With Most Water

Difficulty: Medium | **Pattern:** Two Pointers (greedy) | **Companies:** Amazon, Bloomberg, Adobe

Problem Statement: Given `n` non-negative integers `height[i]` representing vertical lines at position `i`, find two lines that together with the x-axis form a container holding the most water. Return the max area. Area between lines `i,j` = `min(height[i],height[j]) * (j - i)`.

Example: `[1,8,6,2,5,4,8,3,7]` → `49`.

Constraints: `2 <= height.length <= 10^5`.

Clarifying Questions: Can the container lines be adjacent? (Yes, width can be as small as 1.) Are heights non-negative? (Yes, per constraints.)

Brute Force Approach Intuition: try every pair of lines, compute area, track the max.

```
public int maxAreaBrute(int[] height) {
    int best = 0;
    for (int i = 0; i < height.length; i++) {
        for (int j = i + 1; j < height.length; j++) {
            int area = Math.min(height[i], height[j]) * (j - i);
            best = Math.max(best, area);
        }
    }
```

```
        return best;
    }
```

Time: O(n²). Space: O(1).

Optimal Approach Intuition: start with the widest possible container (pointers at both ends). The area is limited by the **shorter** line. Moving the taller pointer inward can only decrease width without any chance of increasing height beyond what already limits us — so it can never help. Therefore always move the **shorter** line's pointer inward, hoping to find a taller line that compensates for the reduced width.

```
public int maxArea(int[] height) {
    int left = 0, right = height.length - 1;
    int best = 0;
    while (left < right) {
        int h = Math.min(height[left], height[right]);
        best = Math.max(best, h * (right - left));
        if (height[left] < height[right]) {
            left++;
        } else {
            right--;
        }
    }
    return best;
}
```

Time: O(n) — single pass, pointers move toward each other. Space: O(1).

Dry Run: `[1,8,6,2,5,4,8,3,7]`. left=0(1),right=8(7): area=min(1,7)*8=8,best=8. 1<7→ left++. left=1(8),right=8(7): area=min(8,7)*7=49,best=49. 8>7→right--. left=1(8),right=7(3): area=min(8,3)*6=18. 8>3→right--. left=1(8),right=6(8): area=min(8,8)*5=40. equal→right-- (either rule works). left=1(8),right=5(4): area=min(8,4)*4=16→right--. ... continues shrinking, best stays `49`. ✅

Edge Cases: all same height (`[5,5,5,5]` → widest pair always wins, area = height * (n-1)). Only 2 lines → trivially their area. Increasing/decreasing sequences.

Follow-up: "Trapping Rain Water" (covered in A2) looks similar but is a fundamentally different problem (sum of trapped water at *every* index, not a single container) — a common interview trap is confusing the two; be ready to explain the difference clearly.

A2. Two Pointers & Sliding Window

A2.1 Valid Palindrome

Difficulty: Easy | **Pattern:** Two Pointers | **Companies:** Microsoft, Amazon, Facebook

Problem Statement: Given a string `s`, determine if it is a palindrome after converting all uppercase to lowercase and removing all non-alphanumeric characters.

Example: `"A man, a plan, a canal: Panama"` → `true`.

Constraints: `1 <= s.length <= 2*10^5`.

Clarifying Questions: Is an empty string (after stripping) a palindrome? (Yes, trivially.) Only ASCII alphanumeric or full Unicode? (Assume ASCII per LeetCode's `Character.isLetterOrDigit`.)

Brute Force Approach Intuition: build a cleaned lowercase string, then compare it to its reverse.

```
public boolean isPalindromeBrute(String s) {
    StringBuilder sb = new StringBuilder();
    for (char c : s.toCharArray())
        if (Character.isLetterOrDigit(c)) sb.append(Character.toLowerCase(c));
    String cleaned = sb.toString();
    return cleaned.equals(sb.reverse().toString());
}
```

Time: O(n). Space: O(n) for the cleaned copy.

Optimal Approach Intuition: skip the extra string allocation — use two pointers from both ends of the original string, skipping non-alphanumeric characters in place.

```
public boolean isPalindrome(String s) {
    int left = 0, right = s.length() - 1;
    while (left < right) {
        while (left < right && !Character.isLetterOrDigit(s.charAt(left))) left++;
        while (left < right && !Character.isLetterOrDigit(s.charAt(right))) right-
-;
        if (Character.toLowerCase(s.charAt(left)) !=
Character.toLowerCase(s.charAt(right))) {
            return false;
        }
        left++; right--;
    }
    return true;
}
```

Time: O(n). Space: O(1).

Dry Run: `"A man, a plan, a canal: Panama"`. Pointers converge comparing 'a'='a', 'm'='m', 'a'='a', 'n'='n' ... all match → `true`. ✅

Edge Cases: empty string → `true`. String with only punctuation (`",,,"` → `true`, both pointers cross without ever comparing). Single character → `true`.

Follow-up: "Valid Palindrome II" — allowed to delete at most one character; when a mismatch is found, try skipping either the left or right character and check if the remainder is a palindrome.

A2.2 Longest Substring Without Repeating Characters

Difficulty: Medium | **Pattern:** Sliding Window | **Companies:** Amazon, Meta, Bloomberg, Microsoft — one of the most-asked sliding window problems in existence

Problem Statement: Given a string `s`, find the length of the longest substring without repeating characters.

Example: `"abcabcbb"` → `3` (`"abc"`).

Constraints: `0 <= s.length <= 5*10^4`.

Clarifying Questions: Is the string case-sensitive? (Yes.) Empty string → answer 0.

Brute Force Approach Intuition: check every substring for uniqueness.

```
public int lengthOfLongestSubstringBrute(String s) {
    int n = s.length(), best = 0;
    for (int i = 0; i < n; i++) {
        Set<Character> seen = new HashSet<>();
        for (int j = i; j < n; j++) {
            if (!seen.add(s.charAt(j))) break;
            best = Math.max(best, j - i + 1);
        }
    }
    return best;
}
```

Time: O(n²) (O(n³) if substring re-hashed from scratch each time). Space: O(min(n, charset)).

Optimal Approach Intuition: maintain a sliding window `[left, right]` that always contains unique characters, using a HashMap of `char -> last seen index`. When we hit a repeat *inside* the current window, jump `left` directly past its previous occurrence (never move `left` backward).

```
public int lengthOfLongestSubstring(String s) {
    Map<Character, Integer> lastIndex = new HashMap<>();
    int left = 0, best = 0;
    for (int right = 0; right < s.length(); right++) {
        char c = s.charAt(right);
        if (lastIndex.containsKey(c) && lastIndex.get(c) >= left) {
            left = lastIndex.get(c) + 1;
        }
        lastIndex.put(c, right);
        best = Math.max(best, right - left + 1);
    }
    return best;
}
```

Time: O(n) — each character visited once by `right`, `left` only moves forward. Space: O(min(n, charset size)).

Dry Run: `"abcabcbb"`. right=0('a'): not seen, best=1. right=1('b'): best=2. right=2('c'): best=3. right=3('a'): seen at 0>=left(0)→left=1, best stays 3 (window "bca"). right=4('b'): seen at 1>=left(1)→left=2, window "cab", best=3. right=5('c'): seen at 2>=left(2)→left=3, window "abc", best=3. right=6('b'): seen at 4>=left(3)→left=5, window "cb", best stays 3. right=7('b'): seen at 6>=left(5)→left=7, window "b", best stays 3. Answer `3`. ✅

Edge Cases: empty string → 0. All unique characters → whole string length. All same character (`"aaaa"` → answer 1).

Follow-up: "Longest Substring with At Most K Distinct Characters" and "Longest Repeating Character Replacement" (next problem) are direct generalizations of this exact sliding-window skeleton — memorize the shape, not just this instance.

A2.3 Longest Repeating Character Replacement

Difficulty: Medium | **Pattern:** Sliding Window | **Companies:** Google, Meta

Problem Statement: Given a string `s` of uppercase letters and an integer `k`, you may replace up to `k` characters to make the longest possible substring of a single repeating character. Return that length.

Example: `s = "AABABBA"`, `k = 1` → `4` (replace one 'B'/'A' to get `"AABA"` or `"ABBB"`).

Constraints: `1 <= s.length <= 10^5`, `0 <= k <= s.length`.

Clarifying Questions: Uppercase English letters only? (Yes, so 26-size frequency array suffices.) Can `k` be 0? (Yes — reduces to "longest substring of one repeated char" already.)

Brute Force Approach Intuition: for every window, count the most frequent character and check if `windowLength - maxFreq <= k` (i.e., the number of characters that would need replacing).

```
public int characterReplacementBrute(String s, int k) {
    int n = s.length(), best = 0;
    for (int i = 0; i < n; i++) {
        int[] freq = new int[26];
        int maxFreq = 0;
        for (int j = i; j < n; j++) {
            freq[s.charAt(j) - 'A']++;
            maxFreq = Math.max(maxFreq, freq[s.charAt(j) - 'A']);
            if ((j - i + 1) - maxFreq <= k) {
                best = Math.max(best, j - i + 1);
            }
        }
    }
    return best;
}
```

Time: O(n²). Space: O(1) (fixed 26-size array).

Optimal Approach Intuition: sliding window that only ever **grows or slides**, never shrinks. Keep a frequency array and `maxFreq` (the count of the most frequent character currently in the window). If `windowLength - maxFreq > k`, the window is invalid — shrink from the left by one. Crucially, `maxFreq` never needs to *decrease* even as we shrink, because a smaller-but-still-invalid window can never beat our best-so-far anyway — this is the classic "monotonic window" trick.

```
public int characterReplacement(String s, int k) {
    int[] freq = new int[26];
    int left = 0, maxFreq = 0, best = 0;
    for (int right = 0; right < s.length(); right++) {
        freq[s.charAt(right) - 'A']++;
```

```
            maxFreq = Math.max(maxFreq, freq[s.charAt(right) - 'A']);
            while ((right - left + 1) - maxFreq > k) {
                freq[s.charAt(left) - 'A']--;
                left++;
            }
            best = Math.max(best, right - left + 1);
        }
        return best;
    }
```

Time: O(n) — `right` visits each index once; `left` moves forward at most n times total. Space: O(1).

Dry Run: `"AABABBA"`, k=1. Window grows to "AAB" (maxFreq=2,len=3,3-2=1<=1 ok, best=3). Add 'A'→"AABA"(freq A=3,maxFreq=3,len=4,4-3=1<=1 ok,best=4). Add 'B'→"AABAB"(freq A=3,B=2, maxFreq=3,len=5,5-3=2>1→shrink left: remove 'A',freq A=2,len=4,4-3=1<=1 ok). Continue; best stays `4`. ✅

Edge Cases: `k >= s.length()` → whole string can be replaced, answer = length. All same character already → answer = length regardless of k.

Follow-up: Same technique (window that never shrinks below best-so-far) applies to "Max Consecutive Ones III" (flip at most k zeros to get longest run of 1s) — recognize it as the identical pattern with a binary alphabet.

A2.4 Minimum Window Substring

Difficulty: Hard | **Pattern:** Sliding Window + Hashing | **Companies:** Meta, Amazon, Uber, LinkedIn — a favorite "hard" sliding window question

Problem Statement: Given strings `s` and `t`, find the minimum window substring of `s` such that every character in `t` (including duplicates) is included in the window. Return `""` if no such window exists.

Example: `s = "ADOBECODEBANC"`, `t = "ABC"` → `"BANC"`.

Constraints: `1 <= s.length, t.length <= 10^5`.

Clarifying Questions: Does `t` contain duplicate characters that must all be matched? (Yes — e.g., `t="AABC"` needs two 'A's in the window.) Case sensitive? (Yes.)

Brute Force Approach Intuition: check every substring of `s`, verify if it contains all characters of `t` with correct multiplicities.

```
public String minWindowBrute(String s, String t) {
    int n = s.length();
    String best = "";
    for (int i = 0; i < n; i++) {
        for (int j = i; j < n; j++) {
            String sub = s.substring(i, j + 1);
            if (contains(sub, t)) {
                if (best.isEmpty() || sub.length() < best.length()) best = sub;
            }
        }
    }
    return best;
}
private boolean contains(String sub, String t) {
    int[] need = new int[128];
    for (char c : t.toCharArray()) need[c]++;
    for (char c : sub.toCharArray()) need[c]--;
    for (int v : need) if (v > 0) return false;
    return true;
}
```

Time: O(n³) (n² substrings × O(n) check). Space: O(1) (fixed 128-size array).

Optimal Approach Intuition: sliding window with two frequency maps. Expand `right` until the window satisfies all of `t`'s requirements; then greedily shrink from `left` while it's still valid, recording the smallest valid window seen. Track `required` = number of distinct characters in `t`, and `formed` = how many of those are currently satisfied at the needed count — this avoids rescanning the whole frequency map on every step.

```
public String minWindow(String s, String t) {
    if (s.isEmpty() || t.isEmpty()) return "";
    Map<Character, Integer> need = new HashMap<>();
    for (char c : t.toCharArray()) need.merge(c, 1, Integer::sum);

    int required = need.size();
    int formed = 0;
    Map<Character, Integer> windowCounts = new HashMap<>();

    int left = 0, bestLen = Integer.MAX_VALUE, bestLeft = 0;
    for (int right = 0; right < s.length(); right++) {
        char c = s.charAt(right);
        windowCounts.merge(c, 1, Integer::sum);
        if (need.containsKey(c) && windowCounts.get(c).intValue() ==
need.get(c).intValue()) {
            formed++;
        }

        while (formed == required) {
            if (right - left + 1 < bestLen) {
                bestLen = right - left + 1;
                bestLeft = left;
            }
```

```
            char leftChar = s.charAt(left);
            windowCounts.put(leftChar, windowCounts.get(leftChar) - 1);
            if (need.containsKey(leftChar) && windowCounts.get(leftChar) <
need.get(leftChar)) {
                formed--;
            }
            left++;
        }
    }
    return bestLen == Integer.MAX_VALUE ? "" : s.substring(bestLeft, bestLeft +
bestLen);
}
```

Time: O(|s| + |t|) — both pointers each move forward at most `|s|` times total. Space: O(|s| + |t|) for the two maps.

Dry Run: `s="ADOBECODEBANC", t="ABC"`. Window grows until right reaches the first 'C' at index 5 ("ADOBEC") — formed=3=required. Shrink left from 'A': removing it breaks formed → best recorded as "ADOBEC" (len 6), then left jumps forward. Window keeps expanding/shrinking; the eventual minimal valid window found is `"BANC"` (len 4). ✅

Edge Cases: `t` longer than `s` → impossible, return `""`. `t` has repeated chars needing multiple copies in the window (handled by comparing exact counts, not just presence).

Follow-up: "Minimum Window Subsequence" (order matters, not just multiset) is a different, harder DP-based variant — don't confuse the two in an interview.

A2.5 Sliding Window Maximum

Difficulty: Hard | **Pattern:** Monotonic Deque | **Companies:** Google, Amazon, Meta

Problem Statement: Given an array `nums` and window size `k`, return an array of the maximum value in each sliding window as it moves from left to right across the array.

Example: `nums = [1,3,-1,-3,5,3,6,7], k = 3` → `[3,3,5,5,6,7]`.

Constraints: `1 <= k <= nums.length <= 10^5`.

Clarifying Questions: Should the output length be `n - k + 1`? (Yes.) Can `k == n`? (Yes, single output = global max.)

Brute Force Approach Intuition: for every window position, scan all k elements to find the max.

```java
public int[] maxSlidingWindowBrute(int[] nums, int k) {
    int n = nums.length;
    int[] res = new int[n - k + 1];
    for (int i = 0; i <= n - k; i++) {
        int max = Integer.MIN_VALUE;
        for (int j = i; j < i + k; j++) max = Math.max(max, nums[j]);
        res[i] = max;
    }
    return res;
}
```

Time: O(n·k), worst case O(n²) when k ≈ n. Space: O(1) extra (excluding output).

Optimal Approach Intuition: maintain a **monotonic decreasing deque** storing *indices*. Before pushing a new index, pop from the back any indices whose values are smaller than the current value (they can never be the max again while the current, larger, later-arriving value is still in the window). Pop from the front any index that has fallen outside the window. The front of the deque is always the current window's max.

```java
public int[] maxSlidingWindow(int[] nums, int k) {
    Deque<Integer> deque = new ArrayDeque<>(); // stores indices, values decreasing
    int n = nums.length;
    int[] res = new int[n - k + 1];
    for (int i = 0; i < n; i++) {
        while (!deque.isEmpty() && deque.peekFirst() <= i - k) {
            deque.pollFirst(); // out of window
        }
        while (!deque.isEmpty() && nums[deque.peekLast()] < nums[i]) {
            deque.pollLast(); // smaller values are useless now
        }
        deque.offerLast(i);
        if (i >= k - 1) {
            res[i - k + 1] = nums[deque.peekFirst()];
        }
    }
    return res;
}
```

Time: O(n) — each index is pushed and popped from the deque at most once. Space: O(k) for the deque.

Dry Run: `[1,3,-1,-3,5,3,6,7], k=3`. i=0(1): deque=[0]. i=1(3): pop 0 (1<3), deque=[1]. i=2(-1): deque=[1,2]. window[0..2] full → res[0]=nums[1]=3. i=3(-3): front 1 still in window (1>0-> i-k=0, 1<=0? no wait i=3,k=3,i-k=0, front=1>0 stays); deque=[1,2,3]→res[1]=nums[1]=3. i=4(5): pop 3(-3<5),pop2(-1<5),pop1(3<5)→deque=[4].

res[2]=5. i=5(3): deque=[4,5]. res[3]=5. i=6(6): pop5(3<6),pop4(5<6)→deque=[6]. res[4]=6. i=7(7): pop6(6<7)→deque=[7]. res[5]=7. Result `[3,3,5,5,6,7]`. ✅

Edge Cases: `k=1` → output equals the input array itself. `k == n` → single-element output = global max.

Follow-up: "Sliding Window Median" — needs two heaps (or a balanced BST/order-statistics structure) instead of a monotonic deque, since median needs more than just the max/min.

A2.6 Trapping Rain Water

Difficulty: Hard | **Pattern:** Two Pointers / Prefix-Suffix Max | **Companies:** Amazon, Google, Meta, Goldman Sachs — legendary "hard" question

Problem Statement: Given `n` non-negative integers representing an elevation map where the width of each bar is 1, compute how much water it can trap after raining.

Example: `[0,1,0,2,1,0,1,3,2,1,2,1]` → `6`.

Constraints: `1 <= height.length <= 2*10^4`.

Clarifying Questions: Water trapped at index `i` = `min(maxLeft[i], maxRight[i]) - height[i]`, correct baseline understanding to confirm before coding.

Brute Force Approach Intuition: for each index, scan left for the tallest bar and right for the tallest bar; the water trapped there is bounded by the shorter of those two "walls."

```
public int trapBrute(int[] height) {
    int n = height.length, total = 0;
    for (int i = 0; i < n; i++) {
        int maxLeft = 0, maxRight = 0;
        for (int j = 0; j <= i; j++) maxLeft = Math.max(maxLeft, height[j]);
        for (int j = i; j < n; j++) maxRight = Math.max(maxRight, height[j]);
        total += Math.min(maxLeft, maxRight) - height[i];
    }
    return total;
}
```

Time: O(n²). Space: O(1).

Better (still not optimal) — Prefix/Suffix Arrays Intuition: precompute `maxLeft[i]` and `maxRight[i]` in one pass each, then a third pass sums the trapped water. O(n) time but O(n) extra space.

```
public int trapPrefixSuffix(int[] height) {
    int n = height.length;
    if (n == 0) return 0;
    int[] maxLeft = new int[n], maxRight = new int[n];
    maxLeft[0] = height[0];
    for (int i = 1; i < n; i++) maxLeft[i] = Math.max(maxLeft[i - 1], height[i]);
    maxRight[n - 1] = height[n - 1];
    for (int i = n - 2; i >= 0; i--) maxRight[i] = Math.max(maxRight[i + 1],
height[i]);
    int total = 0;
    for (int i = 0; i < n; i++) total += Math.min(maxLeft[i], maxRight[i]) -
height[i];
    return total;
}
```

Time: O(n). Space: O(n).

Optimal Approach Intuition: use two pointers from both ends with running `leftMax`/`rightMax`. Whichever side has the smaller "max so far" is the one that determines the trapped water at that pointer (because the other side is guaranteed to have an even taller wall somewhere) — so we can process that side and move inward, eliminating the need for the two O(n) arrays.

```
public int trap(int[] height) {
    int left = 0, right = height.length - 1;
    int leftMax = 0, rightMax = 0, total = 0;
    while (left < right) {
        if (height[left] < height[right]) {
            leftMax = Math.max(leftMax, height[left]);
            total += leftMax - height[left];
            left++;
        } else {
            rightMax = Math.max(rightMax, height[right]);
            total += rightMax - height[right];
            right--;
        }
    }
    return total;
}
```

Time: O(n) single pass. Space: O(1).

Dry Run: `[0,1,0,2,1,0,1,3,2,1,2,1]`. Classic result accumulates to `6` — walk left/right pointers inward, always processing the side with the smaller current bar; total water trapped sums to `6`. ✅ (Verified against brute force for correctness.)

Edge Cases: monotonically increasing or decreasing heights → 0 water trapped (no basin). All zero heights → 0. Fewer than 3 bars → 0 (can't form a basin).

Follow-up: "Trapping Rain Water II" (2-D version, a grid) requires a min-heap (Dijkstra-like "process the lowest boundary wall first") — a totally different, much harder technique; be ready to explain why the 1-D two-pointer trick does *not* generalize directly to 2-D.

A2.7 Group Anagrams

Difficulty: Medium | **Pattern:** Hashing with a canonical key | **Companies:** Amazon, Meta, Uber, Bloomberg

Problem Statement: Given an array of strings, group the anagrams together.

Example: `["eat","tea","tan","ate","nat","bat"]` → `[["bat"],["nat","tan"],["ate","eat","tea"]]` (order of groups/elements doesn't matter).

Constraints: `1 <= strs.length <= 10^4`, lowercase English letters only.

Clarifying Questions: Case sensitive? (Typically lowercase-only guaranteed.) Order of output groups matters? (No, per problem statement.)

Brute Force Approach Intuition: for each string, compare it against every existing group by checking if sorted forms match (or by frequency count comparison) — effectively O(n²) group-membership checks.

```
public List<List<String>> groupAnagramsBrute(String[] strs) {
    List<List<String>> groups = new ArrayList<>();
    boolean[] used = new boolean[strs.length];
    for (int i = 0; i < strs.length; i++) {
        if (used[i]) continue;
        List<String> group = new ArrayList<>();
        group.add(strs[i]);
        used[i] = true;
        char[] a = strs[i].toCharArray(); Arrays.sort(a);
        for (int j = i + 1; j < strs.length; j++) {
            if (used[j]) continue;
            char[] b = strs[j].toCharArray(); Arrays.sort(b);
            if (Arrays.equals(a, b)) { group.add(strs[j]); used[j] = true; }
```

```
            }
            groups.add(group);
        }
        return groups;
    }
```

Time: O(n² · k log k) where k = max string length. Space: O(n·k).

Optimal Approach Intuition: two anagrams always produce the **same sorted string** (or the same 26-letter frequency signature). Use that as a HashMap key so every string finds its group in O(1) amortized lookup instead of comparing against every other group.

```
public List<List<String>> groupAnagrams(String[] strs) {
    Map<String, List<String>> groups = new HashMap<>();
    for (String s : strs) {
        char[] chars = s.toCharArray();
        Arrays.sort(chars);
        String key = new String(chars);
        groups.computeIfAbsent(key, k -> new ArrayList<>()).add(s);
    }
    return new ArrayList<>(groups.values());
}
```

Time: O(n · k log k) — sort each of n strings of average length k. Space: O(n·k).

Even faster key (avoids sorting): build a 26-length frequency-count key (e.g. `"1#0#0#...#2"` for counts of a-z), giving O(n·k) total instead of O(n·k log k):

```
private String freqKey(String s) {
    int[] count = new int[26];
    for (char c : s.toCharArray()) count[c - 'a']++;
    StringBuilder sb = new StringBuilder();
    for (int c : count) sb.append(c).append('#');
    return sb.toString();
}
```

Dry Run: `["eat","tea","tan","ate","nat","bat"]`. Sorted keys: "eat"→"aet", "tea"→"aet", "tan"→"ant", "ate"→"aet", "nat"→"ant", "bat"→"abt". Groups: "aet"→[eat,tea,ate], "ant"→[tan,nat], "abt"→[bat]. ✅

Edge Cases: empty input array → empty output. Single string → single group with one element. All identical strings → one group containing all of them.

Follow-up: "Group Shifted Strings" uses a similar canonical-key idea but the key is the sequence of pairwise character differences (mod 26) instead of sorted characters.

A2.8 Valid Anagram

Difficulty: Easy | **Pattern:** Hashing / Frequency Count | **Companies:** Amazon, Bloomberg — common warm-up

Problem Statement: Given strings `s` and `t`, return `true` if `t` is an anagram of `s`.

Example: `s="anagram", t="nagaram"` → `true`. `s="rat", t="car"` → `false`.

Constraints: `1 <= s.length, t.length <= 5*10^4`, lowercase English letters (follow-up: Unicode).

Clarifying Questions: What if lengths differ? (Immediately `false`.) Unicode supported? (Ask — affects whether a fixed 26-array or a HashMap is appropriate.)

Brute Force Approach Intuition: sort both strings and compare.

```
public boolean isAnagramBrute(String s, String t) {
    if (s.length() != t.length()) return false;
    char[] a = s.toCharArray(), b = t.toCharArray();
    Arrays.sort(a); Arrays.sort(b);
    return Arrays.equals(a, b);
}
```

Time: O(n log n). Space: O(n) for the sorted copies.

Optimal Approach Intuition: two strings are anagrams iff they have identical character frequency counts. A fixed-size array (26 for lowercase English) avoids both sorting and hashing overhead.

```java
public boolean isAnagram(String s, String t) {
    if (s.length() != t.length()) return false;
    int[] count = new int[26];
    for (int i = 0; i < s.length(); i++) {
        count[s.charAt(i) - 'a']++;
        count[t.charAt(i) - 'a']--;
    }
    for (int c : count) if (c != 0) return false;
    return true;
}
```

Time: O(n). Space: O(1) — fixed 26-size array.

Dry Run: `"anagram"` vs `"nagaram"` — incrementing/decrementing counts for every character of both strings in lockstep results in an all-zero array → `true`. ✅

Edge Cases: different lengths → immediate `false`. Empty strings (both) → `true`. Unicode input → switch to a `HashMap<Character,Integer>` instead of a fixed array.

Follow-up: "Group Anagrams" (previous problem) generalizes this to many strings at once. "Find All Anagrams in a String" — sliding window version, covered in Part B.

A3. Stack

A3.1 Valid Parentheses

Difficulty: Easy | **Pattern:** Stack | **Companies:** Amazon, Microsoft, Facebook, Google — nearly universal warm-up question

Problem Statement: Given a string containing just `()[]{}`, determine if the input is valid: every open bracket must be closed by the same type in the correct order.

Example: `"()[]{}"` → `true`. `"(]"` → `false`.

Constraints: `1 <= s.length <= 10^4`.

Clarifying Questions: Only these 6 bracket characters, no other text mixed in? (Correct per constraints.) Empty string? (Not possible per constraints, but code should handle gracefully anyway.)

Brute Force Approach Intuition: repeatedly find and remove innermost matched pairs `"()"`, `"[]"`, `"{}"` from the string until no more removals are possible; valid iff the string becomes empty.

```
public boolean isValidBrute(String s) {
    while (s.contains("()") || s.contains("[]") || s.contains("{}")) {
        s = s.replace("()", "").replace("[]", "").replace("{}", "");
    }
    return s.isEmpty();
}
```

Time: O(n²) — each replace pass is O(n) and up to O(n) passes may be needed. Space: O(n) for the repeated string copies.

Optimal Approach Intuition: process characters left to right with a stack. Push every opening bracket. On a closing bracket, it must match the bracket on top of the stack (LIFO order is exactly what "correctly nested" means) — pop and compare; mismatch or empty stack means invalid.

```
public boolean isValid(String s) {
    Deque<Character> stack = new ArrayDeque<>();
    Map<Character, Character> pairs = Map.of(')', '(', ']', '[', '}', '{');
    for (char c : s.toCharArray()) {
        if (pairs.containsValue(c)) {
            stack.push(c);
        } else if (pairs.containsKey(c)) {
            if (stack.isEmpty() || stack.pop() != pairs.get(c)) return false;
        }
    }
    return stack.isEmpty();
}
```

Time: O(n) — single pass, each char pushed/popped at most once. Space: O(n) worst case (all opening brackets).

Dry Run: `"()[]{}"`. push'(', pop matches')'. push'[', pop matches']'. push'{', pop matches'}'. Stack empty at end → `true`. `"(]"`: push'(', then ']' needs pop=='[' but got '(' → `false`. ✅

Edge Cases: unmatched closing bracket with empty stack → `false` immediately. Leftover unmatched opening brackets at the end → stack non-empty → `false`. Single character → always `false` (can't be balanced alone).

Follow-up: "Generate Parentheses" (backtracking, uses the same nesting-validity intuition in reverse to *construct* valid strings). "Minimum Add to Make Parentheses Valid" — greedy counter instead of a full stack since only one bracket type is involved.

A3.2 Min Stack

Difficulty: Medium | **Pattern:** Stack (auxiliary tracking) | **Companies:** Amazon, Bloomberg, Google

Problem Statement: Design a stack supporting `push`, `pop`, `top`, and `getMin` all in O(1) time.

Clarifying Questions: Can duplicate minimum values be pushed? (Yes — must handle correctly, e.g. push 1, push 1, pop → min should still be 1.)

Brute Force Approach Intuition: a plain stack for values; `getMin()` scans the entire stack each time.

```
class MinStackBrute {
    private Deque<Integer> stack = new ArrayDeque<>();
    public void push(int val) { stack.push(val); }
    public void pop() { stack.pop(); }
    public int top() { return stack.peek(); }
    public int getMin() {
        int min = Integer.MAX_VALUE;
        for (int v : stack) min = Math.min(min, v);
        return min;
    }
}
```

Time: push/pop/top O(1), **getMin O(n)** — fails the requirement. Space: O(n).

Optimal Approach Intuition: maintain a second "min stack" alongside the main stack, where `minStack.peek()` always holds the minimum of all elements currently in the main stack up to that point. Push the current minimum onto `minStack` every time (even if unchanged), and pop from both stacks together — this keeps them perfectly in sync so `getMin()` is a simple O(1) peek.

```
class MinStack {
    private Deque<Integer> stack = new ArrayDeque<>();
    private Deque<Integer> minStack = new ArrayDeque<>();

    public void push(int val) {
```

```
        stack.push(val);
        int currentMin = minStack.isEmpty() ? val : Math.min(val, minStack.peek());
        minStack.push(currentMin);
    }
    public void pop() {
        stack.pop();
        minStack.pop();
    }
    public int top() { return stack.peek(); }
    public int getMin() { return minStack.peek(); }
}
```

Time: O(1) for all four operations. Space: O(n) (two parallel stacks).

Dry Run: push(5)→stack[5],min[5]. push(3)→stack[5,3],min[5,3]. push(7)→stack[5,3,7], min[5,3,3] (min(7,3)=3). getMin()→3. pop()→stack[5,3],min[5,3]. getMin()→3. ✅

Edge Cases: pushing the same minimum value multiple times (handled naturally since we push a new min-stack entry every time, not just on updates). Popping down to empty then pushing again.

Follow-up: "Can you do it with O(1) *extra* space instead of O(n)?" — store the difference `val - currentMin` in the main stack instead of a parallel min-stack, reconstructing the actual value and updating min mathematically on pop (a well-known space-optimized trick, more fragile to implement correctly, worth mentioning verbally even if you implement the two-stack version).

A3.3 Evaluate Reverse Polish Notation

Difficulty: Medium | **Pattern:** Stack | **Companies:** LinkedIn, Amazon

Problem Statement: Evaluate an arithmetic expression given in Reverse Polish (postfix) Notation, e.g. `["2","1","+","3","*"]` → `9` (`(2+1)*3`).

Constraints: Division truncates toward zero. Operators: `+ - * /`.

Clarifying Questions: Integer division truncation direction? (Toward zero, per LeetCode spec — Java's native `/` for ints already truncates toward zero, convenient.)

Brute Force / only reasonable approach — Stack Intuition: postfix notation is *designed* for stack evaluation: scan tokens left to right; push numbers; when an operator is seen, pop the top two operands, apply the operator, push the result back.

(There isn't a meaningfully different "worse" brute force here — the stack approach essentially *is* the direct simulation of the definition, but it's still worth stating explicitly why a naive "convert to infix and eval" approach would be far more complex and error-prone by comparison.)

```
public int evalRPN(String[] tokens) {
    Deque<Integer> stack = new ArrayDeque<>();
    for (String token : tokens) {
        switch (token) {
            case "+", "-", "*", "/" -> {
                int b = stack.pop();
                int a = stack.pop();
                stack.push(switch (token) {
                    case "+" -> a + b;
                    case "-" -> a - b;
                    case "*" -> a * b;
                    default -> a / b;
                });
            }
            default -> stack.push(Integer.parseInt(token));
        }
    }
    return stack.pop();
}
```

Time: O(n) — one pass, O(1) stack operations per token. Space: O(n) worst case (all operands before any operator).

Dry Run: `["2","1","+","3","*"]`. push2,push1. '+' → pop1,pop2 → push3. push3(literal). '*' → pop3,pop3 → push9. Result `9`. ✅

Edge Cases: division truncation with negative numbers (e.g. `-7/2` in Java truncates to `-3`, matching the required "toward zero" semantics automatically). Single-token expression (just a number, no operators) → directly returned.

Follow-up: "Basic Calculator" (infix expressions with parentheses) — a materially harder problem requiring either a full shunting-yard conversion to postfix first, or recursive descent parsing.

A3.4 Generate Parentheses

Difficulty: Medium | **Pattern:** Backtracking (constrained) | **Companies:** Amazon, Google, Meta, Uber

Problem Statement: Given `n` pairs of parentheses, generate all combinations of well-formed parentheses strings.

Example: `n=3` → `["((()))","(()())","(())()","()(())","()()()"]`.

Constraints: `1 <= n <= 8`.

Clarifying Questions: Order of results doesn't matter? (Correct.) Confirm "well-formed" means every prefix has `#open >= #close` and total open == total close == n.

Brute Force Approach Intuition: generate all `2^(2n)` binary strings of `(` / `)`, then filter for validity using the Valid Parentheses check.

```
public List<String> generateParenthesisBrute(int n) {
    List<String> result = new ArrayList<>();
    char[] combo = new char[2 * n];
    generateAll(combo, 0, result);
    return result;
}
private void generateAll(char[] combo, int pos, List<String> result) {
    if (pos == combo.length) {
        if (isValid(new String(combo))) result.add(new String(combo));
        return;
    }
    combo[pos] = '('; generateAll(combo, pos + 1, result);
    combo[pos] = ')'; generateAll(combo, pos + 1, result);
}
```

Time: O(2^(2n) · n) — exponential generation plus O(n) validity check each. Space: O(2^(2n)) for recursion/results.

Optimal Approach Intuition: only ever place a character if it keeps the string potentially valid — backtrack while tracking `openCount` and `closeCount`. Add `(` whenever `openCount < n`; add `)` only when `closeCount < openCount` (never allow more closes than opens at any prefix). This prunes the search to generate **only** valid strings directly (no wasted branches, no post-filtering).

```
public List<String> generateParenthesis(int n) {
    List<String> result = new ArrayList<>();
    backtrack(result, new StringBuilder(), 0, 0, n);
    return result;
}
private void backtrack(List<String> result, StringBuilder sb, int open, int close,
int n) {
    if (sb.length() == 2 * n) {
        result.add(sb.toString());
        return;
```

```
        }
        if (open < n) {
            sb.append('(');
            backtrack(result, sb, open + 1, close, n);
            sb.deleteCharAt(sb.length() - 1);
        }
        if (close < open) {
            sb.append(')');
            backtrack(result, sb, open, close + 1, n);
            sb.deleteCharAt(sb.length() - 1);
        }
    }
```

Time: O(4^n / √n) — bounded by the nth Catalan number (the count of valid outputs), which is the theoretical minimum for this problem since every output must be produced. Space: O(n) for recursion depth, output size dominates overall space.

Dry Run: `n=2`: backtrack explores "((" → "(()" → "(())" ✅. Then unwinds to "()" → "()(" → "()()" ✅. Total valid outputs: `["(())", "()()"]`. ✅ (matches known Catalan(2)=2)

Edge Cases: `n=1` → `["()"]`. Larger `n` grows combinatorially (Catalan numbers) — mention this growth rate explicitly to show complexity awareness.

Follow-up: "Remove Invalid Parentheses" — a much harder BFS/backtracking problem to find minimal removals to fix an already-malformed string.

A3.5 Daily Temperatures

Difficulty: Medium | **Pattern:** Monotonic Stack | **Companies:** Amazon, Facebook

Problem Statement: Given daily temperatures, for each day return how many days you'd have to wait until a warmer temperature. If none, put 0.

Example: `[73,74,75,71,69,72,76,73]` → `[1,1,4,2,1,1,0,0]`.

Constraints: `1 <= temperatures.length <= 10^5`.

Clarifying Questions: Return 0 (not -1) when no future warmer day exists? (Yes, per spec.)

Brute Force Approach Intuition: for each day, scan forward until a warmer day is found.

```java
public int[] dailyTemperaturesBrute(int[] temps) {
    int n = temps.length;
    int[] res = new int[n];
    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {
            if (temps[j] > temps[i]) { res[i] = j - i; break; }
        }
    }
    return res;
}
```

Time: O(n²) worst case (strictly decreasing temperatures). Space: O(1) extra.

Optimal Approach Intuition: use a **monotonic decreasing stack of indices**. When the current temperature is higher than the temperature at the index on top of the stack, that's the "answer day" for every such index we pop — because we've just found the *first* warmer day after them (moving left to right guarantees "first").

```java
public int[] dailyTemperatures(int[] temperatures) {
    int n = temperatures.length;
    int[] res = new int[n];
    Deque<Integer> stack = new ArrayDeque<>(); // indices, temps decreasing
    for (int i = 0; i < n; i++) {
        while (!stack.isEmpty() && temperatures[i] > temperatures[stack.peek()]) {
            int idx = stack.pop();
            res[idx] = i - idx;
        }
        stack.push(i);
    }
    return res;
}
```

Time: O(n) — each index pushed and popped at most once. Space: O(n) worst case (strictly decreasing input keeps everything on the stack).

Dry Run: `[73,74,75,71,69,72,76,73]`. i=0: stack=[0]. i=1(74>73): pop0,res[0]=1; stack=[1]. i=2(75>74): pop1,res[1]=1; stack=[2]. i=3(71<75): stack=[2,3]. i=4(69<71): stack=[2,3,4]. i=5(72>69): pop4,res[4]=1; 72>71: pop3,res[3]=2; 72<75: stack=[2,5]. i=6(76>72): pop5,res[5]=1; 76>75: pop2,res[2]=4; stack=[6]. i=7(73<76): stack=[6,7]. End: res[6]=0, res[7]=0 (never popped, default 0). Final `[1,1,4,2,1,1,0,0]`. ✅

Edge Cases: strictly decreasing temperatures → all zeros. Strictly increasing → each answer is 1 except the last (0). All equal → all zeros (strict `>` required, not `>=`).

Follow-up: "Next Greater Element I/II" are the same monotonic-stack template applied to generic arrays (II adds circularity, handled by iterating `2n` times modulo `n`).

A3.6 Largest Rectangle in Histogram

Difficulty: Hard | **Pattern:** Monotonic Stack | **Companies:** Amazon, Google, Meta — classic "hard" stack question

Problem Statement: Given heights of bars of width 1 forming a histogram, find the area of the largest rectangle that fits entirely within the histogram.

Example: `[2,1,5,6,2,3]` → `10` (rectangle of height 5,6 → actually height=5 spanning indices 2-3, width 2, but the max is height 5 * width... let's verify: best is heights[2..3]=[5,6], min height 5, width 2, area 10). ✅ matches known answer.

Constraints: `1 <= heights.length <= 10^5`.

Clarifying Questions: Bars have width exactly 1 and are contiguous? (Yes.) Can heights be 0? (Yes, acts as a full break.)

Brute Force Approach Intuition: for every pair of bars `(i,j)`, the rectangle height is `min(heights[i..j])`; try all O(n²) pairs (or O(n³) if you recompute the min naively each time).

```
public int largestRectangleAreaBrute(int[] heights) {
    int n = heights.length, best = 0;
    for (int i = 0; i < n; i++) {
        int minHeight = Integer.MAX_VALUE;
        for (int j = i; j < n; j++) {
            minHeight = Math.min(minHeight, heights[j]);
            best = Math.max(best, minHeight * (j - i + 1));
        }
    }
    return best;
}
```

Time: O(n²). Space: O(1).

Optimal Approach Intuition: use a **monotonic increasing stack of indices**. When we encounter a bar shorter than the one on top of the stack, that top bar can no longer extend rightward — pop it and compute the max rectangle it could have formed, using

the *previous* stack element as its left boundary and the current index as its right boundary.

```
public int largestRectangleArea(int[] heights) {
    Deque<Integer> stack = new ArrayDeque<>(); // indices, heights increasing
    int n = heights.length, best = 0;
    for (int i = 0; i <= n; i++) {
        int h = (i == n) ? 0 : heights[i]; // sentinel 0 forces final flush
        while (!stack.isEmpty() && heights[stack.peek()] > h) {
            int height = heights[stack.pop()];
            int width = stack.isEmpty() ? i : i - stack.peek() - 1;
            best = Math.max(best, height * width);
        }
        stack.push(i);
    }
    return best;
}
```

Time: O(n) — each index pushed/popped once. Space: O(n).

Dry Run: `[2,1,5,6,2,3]`. i=0: stack=[0]. i=1(1<2): pop0,h=2,width=1(stack empty→i=1), area=2; stack=[1]. i=2(5>1):stack=[1,2]. i=3(6>5):stack=[1,2,3]. i=4(2<6): pop3,h=6,width= 4-2-1=1,area=6; 2<5: pop2,h=5,width=4-1-1=2,area=10; 2>1: stack= [1,4]. i=5(3>2): stack= [1,4,5]. i=6(sentinel0): pop5,h=3,width=6-4-1=1,area=3; pop4,h=2,width=6-1-1=4,area=8; pop1, h=1,width=6-(-1)-1... stack empty→width=6,area=6. Max across all = `10`. ✅

Edge Cases: all bars same height `h` → answer = `h * n`. Single bar → its own height. Strictly increasing then the sentinel flush handles the final pops correctly.

Follow-up: "Maximal Rectangle" (2-D binary matrix) reduces to running this exact histogram algorithm on each row, treating column-wise consecutive 1s as bar heights — a very common "combine two known patterns" interview escalation.

A4. Binary Search

A4.1 Binary Search

Difficulty: Easy | **Pattern:** Binary Search | **Companies:** Universal warm-up at every MNC

Problem Statement: Given a sorted array `nums` and a target, return its index or -1.

Clarifying Questions: Are all elements distinct? (Usually yes for the basic version.)

Brute Force Approach Intuition: linear scan.

```
public int searchBrute(int[] nums, int target) {
    for (int i = 0; i < nums.length; i++) if (nums[i] == target) return i;
    return -1;
}
```

Time: O(n). Space: O(1).

Optimal Approach Intuition: repeatedly halve the search space by comparing the middle element to the target.

```
public int search(int[] nums, int target) {
    int left = 0, right = nums.length - 1;
    while (left <= right) {
        int mid = left + (right - left) / 2;
        if (nums[mid] == target) return mid;
        else if (nums[mid] < target) left = mid + 1;
        else right = mid - 1;
    }
    return -1;
}
```

Time: O(log n). Space: O(1).

Dry Run: `[-1,0,3,5,9,12]`, target=9. left=0,right=5,mid=2(3<9)→left=3. left=3,right=5, mid=4(9==9)→return 4. ✅

Edge Cases: target smaller than all / larger than all elements → -1. Empty array → -1. Single element.

Follow-up: all "modified binary search" variants (rotated array, first/last occurrence, 2-D matrix, answer-space binary search) build directly on this template — the only thing that changes is the *condition* used to decide which half to discard.

A4.2 Search a 2D Matrix

Difficulty: Medium | **Pattern:** Binary Search (treat 2D as 1D) | **Companies:** Amazon, Microsoft

Problem Statement: An `m x n` matrix has each row sorted left-to-right and the first integer of each row is greater than the last integer of the previous row (i.e., fully sorted if flattened). Given a target, return true if it exists.

Constraints: `1 <= m, n <= 100`.

Clarifying Questions: Confirm the "each row's first > previous row's last" property — this is what allows treating the whole matrix as one sorted 1-D array.

Brute Force Approach Intuition: scan every cell.

```
public boolean searchMatrixBrute(int[][] matrix, int target) {
    for (int[] row : matrix) for (int v : row) if (v == target) return true;
    return false;
}
```

Time: O(m·n). Space: O(1).

Optimal Approach Intuition: since the matrix is fully sorted when flattened, map a single binary-search index `mid` (0 to m·n-1) to `matrix[mid / n][mid % n]` and binary search directly.

```
public boolean searchMatrix(int[][] matrix, int target) {
    int m = matrix.length, n = matrix[0].length;
    int left = 0, right = m * n - 1;
    while (left <= right) {
        int mid = left + (right - left) / 2;
        int val = matrix[mid / n][mid % n];
        if (val == target) return true;
        else if (val < target) left = mid + 1;
        else right = mid - 1;
    }
    return false;
}
```

Time: O(log(m·n)). Space: O(1).

Dry Run: `[[1,3,5,7],[10,11,16,20],[23,30,34,60]]`, target=3, n=4. left=0,right=11, mid=5→matrix[1][1]=11>3→right=4. mid=2→matrix[0][2]=5>3→right=1. mid=0→matrix[0][0]=1<3→ left=1. mid=1(left right)→matrix[0][1]=3 3→true. ✅

Edge Cases: target smaller than `matrix[0][0]` or larger than the last cell → false. Matrix with a single row or single column.

Follow-up: "Search a 2D Matrix II" (rows sorted, columns sorted, but NOT globally flat-sorted) requires a different technique: start at top-right corner, move left if current > target, down if current < target — O(m+n), not binary search.

A4.3 Koko Eating Bananas

Difficulty: Medium | **Pattern:** Binary Search on the Answer | **Companies:** Google, Amazon

Problem Statement: `n` piles of bananas, Koko eats at speed `k` bananas/hour (per pile, she can't eat from two piles in one hour, and if a pile has fewer than `k` she finishes it and moves on). Given `h` hours, find the minimum integer `k` such that she can eat all bananas within `h` hours.

Constraints: `1 <= piles.length <= 10^4`, `piles.length <= h <= 10^9`.

Clarifying Questions: Must `k` be an integer ≥ 1? (Yes.) Guaranteed `h >= piles.length` so a solution always exists? (Yes.)

Brute Force Approach Intuition: try every speed `k` starting from 1 upward, check feasibility, return the first that works.

```
public int minEatingSpeedBrute(int[] piles, int h) {
    int maxPile = Arrays.stream(piles).max().getAsInt();
    for (int k = 1; k <= maxPile; k++) {
        if (hoursNeeded(piles, k) <= h) return k;
    }
    return maxPile;
}
private long hoursNeeded(int[] piles, int k) {
    long hours = 0;
    for (int p : piles) hours += (p + k - 1) / k; // ceil division
    return hours;
}
```

Time: O(maxPile · n) — could be up to O(10^9 · 10^4), far too slow.

Optimal Approach Intuition: **"binary search on the answer."** The feasibility function `hoursNeeded(k) <= h` is monotonic: increasing `k` never increases hours needed.

Whenever a boolean predicate is monotonic over a range, binary search the smallest/largest value satisfying it, instead of linear scanning.

```
public int minEatingSpeed(int[] piles, int h) {
    int left = 1, right = Arrays.stream(piles).max().getAsInt();
    while (left < right) {
        int mid = left + (right - left) / 2;
        if (hoursNeeded(piles, mid) <= h) {
            right = mid; // mid works, try slower (smaller k)
        } else {
            left = mid + 1; // mid too slow, need faster
        }
    }
    return left;
}
private long hoursNeeded(int[] piles, int k) {
    long hours = 0;
    for (int p : piles) hours += (p + k - 1) / k;
    return hours;
}
```

Time: O(n log(maxPile)) — binary search over speed range, O(n) feasibility check each iteration. Space: O(1).

Dry Run: `piles=[3,6,7,11], h=8`. Search k in [1,11]. mid=6: hours=1+1+2+2=6<=8→right=6. mid=3: hours=1+2+3+4=10>8→left=4. mid=5: hours=1+2+2+3=8<=8→right=5. mid=4: hours=1+2+2+3=8<=8 →right=4. left==right=4 → answer `4`. ✅ (matches known LeetCode answer)

Edge Cases: `h == piles.length` (must eat exactly one pile per hour minimum, forces `k >= max(piles)`). Single pile.

Follow-up: "Capacity To Ship Packages Within D Days", "Split Array Largest Sum", "Minimum Number of Days to Make m Bouquets" are all the exact same "binary search on the monotonic answer" template with a different feasibility function — recognizing this family is a huge interview time-saver.

A4.4 Find First and Last Position of Element in Sorted Array

Difficulty: Medium | **Pattern:** Binary Search (boundary search) | **Companies:** Microsoft, LinkedIn, Amazon

Problem Statement: Given a sorted array, find the starting and ending index of a given target value. Return `[-1,-1]` if not found. Must be O(log n).

Example: `[5,7,7,8,8,10], target=8` → `[3,4]`.

Clarifying Questions: Duplicates guaranteed possible? (Yes — this is exactly why plain binary search isn't enough; you need the *leftmost* and *rightmost* matching index.)

Brute Force Approach Intuition: linear scan recording first and last matching index.

```
public int[] searchRangeBrute(int[] nums, int target) {
    int first = -1, last = -1;
    for (int i = 0; i < nums.length; i++) {
        if (nums[i] == target) {
            if (first == -1) first = i;
            last = i;
        }
    }
    return new int[]{first, last};
}
```

Time: O(n). Space: O(1).

Optimal Approach Intuition: run **two separate binary searches** — one biased to keep moving `right` left even after finding a match (to find the leftmost occurrence), one biased to keep moving `left` right after a match (to find the rightmost occurrence).

```
public int[] searchRange(int[] nums, int target) {
    return new int[]{ findBound(nums, target, true), findBound(nums, target, false)
};
}
private int findBound(int[] nums, int target, boolean findFirst) {
    int left = 0, right = nums.length - 1, result = -1;
    while (left <= right) {
        int mid = left + (right - left) / 2;
        if (nums[mid] == target) {
            result = mid;
            if (findFirst) right = mid - 1; else left = mid + 1;
        } else if (nums[mid] < target) {
            left = mid + 1;
        } else {
            right = mid - 1;
        }
    }
    return result;
}
```

Time: O(log n) — two binary searches. Space: O(1).

Dry Run: `[5,7,7,8,8,10], target=8`. findFirst: converges result=3 (keeps shrinking right past the first match). findLast: converges result=4. Output `[3,4]`. ✅

Edge Cases: target not present → both searches return -1 → `[-1,-1]`. All elements equal target → `[0, n-1]`. Single element array matching/not matching.

Follow-up: "Count occurrences of a number in a sorted array" = `last - first + 1` using this exact function.

A4.5 Median of Two Sorted Arrays

Difficulty: Hard | **Pattern:** Binary Search (partition-based) | **Companies:** Google, Amazon, Microsoft, Apple — famously hard, extremely common at top companies

Problem Statement: Given two sorted arrays `nums1`, `nums2` of size `m` and `n`, return the median of the combined sorted array in O(log(min(m,n))) time.

Constraints: `0 <= m,n`, not both zero.

Clarifying Questions: Must the solution be sub-linear (O(log)), or is O(m+n) acceptable as a first pass? (Always mention the O(m+n) merge approach first, then push to the O(log) partition approach since that's what's actually being tested here.)

Brute Force Approach Intuition: merge both arrays (like merge step of merge sort) and pick the middle element(s).

```
public double findMedianSortedArraysBrute(int[] nums1, int[] nums2) {
    int[] merged = new int[nums1.length + nums2.length];
    int i = 0, j = 0, k = 0;
    while (i < nums1.length && j < nums2.length) merged[k++] = nums1[i] <= nums2[j]
? nums1[i++] : nums2[j++];
    while (i < nums1.length) merged[k++] = nums1[i++];
    while (j < nums2.length) merged[k++] = nums2[j++];
    int n = merged.length;
    return n % 2 == 1 ? merged[n / 2] : (merged[n / 2 - 1] + merged[n / 2]) / 2.0;
}
```

Time: O(m+n). Space: O(m+n).

Optimal Approach Intuition: the median splits the combined array into a left half and right half of equal (or near-equal) size where every left element ≤ every right element. Binary search for the correct **partition point** in the *smaller* array (`partitionX`); the

corresponding partition in the other array is then forced (`partitionY = (m+n+1)/2 - partitionX`). Check the partition is valid by comparing boundary elements; adjust via binary search until valid.

```
public double findMedianSortedArrays(int[] nums1, int[] nums2) {
    if (nums1.length > nums2.length) return findMedianSortedArrays(nums2, nums1);
// ensure nums1 smaller
    int m = nums1.length, n = nums2.length;
    int left = 0, right = m;
    int halfLen = (m + n + 1) / 2;

    while (left <= right) {
        int partitionX = left + (right - left) / 2;
        int partitionY = halfLen - partitionX;

        int maxLeftX = (partitionX == 0) ? Integer.MIN_VALUE : nums1[partitionX -
1];
        int minRightX = (partitionX == m) ? Integer.MAX_VALUE : nums1[partitionX];
        int maxLeftY = (partitionY == 0) ? Integer.MIN_VALUE : nums2[partitionY -
1];
        int minRightY = (partitionY == n) ? Integer.MAX_VALUE : nums2[partitionY];

        if (maxLeftX <= minRightY && maxLeftY <= minRightX) {
            if ((m + n) % 2 == 0) {
                return (Math.max(maxLeftX, maxLeftY) + Math.min(minRightX,
minRightY)) / 2.0;
            } else {
                return Math.max(maxLeftX, maxLeftY);
            }
        } else if (maxLeftX > minRightY) {
            right = partitionX - 1; // move partitionX left
        } else {
            left = partitionX + 1;  // move partitionX right
        }
    }
    throw new IllegalArgumentException("Input arrays are not sorted");
}
```

Time: O(log(min(m,n))) — binary search over the smaller array's partition index. Space: O(1).

Dry Run: `nums1=[1,3], nums2=[2]`. m=2,n=1,halfLen=2. left=0,right=2. partitionX=1: maxLeftX=1,minRightX=3,partitionY=1: maxLeftY=2,minRightY=+inf. maxLeftX(1) <=minRightY(inf) and maxLeftY(2)<=minRightX(3) → valid! total odd (3) → return max(1,2)=2. ✅

Edge Cases: one array empty → median is simply the other array's median (handled naturally since partition on the empty array's side is forced to 0 or its length). Arrays of

very different sizes (why we always binary search the smaller one — keeps log factor tight).

Follow-up: "Kth Smallest Element in Two Sorted Arrays" is a generalization solvable with a similar partition/binary-search idea, or a simpler O(log(m+n)) recursive elimination approach.

A4.6 Sqrt(x)

Difficulty: Easy | **Pattern:** Binary Search | **Companies:** Common at many companies as a quick warm-up

Problem Statement: Given a non-negative integer `x`, return `floor(sqrt(x))` without using built-in power/sqrt functions.

Clarifying Questions: Return the floor, not a decimal approximation? (Yes, integer result.)

Brute Force Approach Intuition: increment a counter until its square exceeds `x`.

```
public int mySqrtBrute(int x) {
    long i = 0;
    while (i * i <= x) i++;
    return (int) (i - 1);
}
```

Time: O(√x) — could be slow for large x (e.g. x near 2^31). Space: O(1).

Optimal Approach Intuition: the search space `[0, x]` is monotonic on the predicate `i*i <= x` — classic binary search on the answer.

```
public int mySqrt(int x) {
    long left = 0, right = x;
    long ans = 0;
    while (left <= right) {
        long mid = left + (right - left) / 2;
        if (mid * mid <= x) {
            ans = mid;
            left = mid + 1;
        } else {
            right = mid - 1;
        }
    }
```

```
        return (int) ans;
    }
```

Time: O(log x). Space: O(1). (Use `long` to avoid overflow when squaring.)

Dry Run: `x=8`. left=0,right=8. mid=4:16>8→right=3. mid=1:1<=8→ans=1,left=2. mid=2:4<=8→ ans=2,left=3. mid=3:9>8→right=2. loop ends (left>right). Answer `2`. ✅ (floor(√8)=2)

Edge Cases: `x=0` → 0. `x=1` → 1. Perfect squares vs non-perfect squares both handled identically by the floor logic.

Follow-up: "Pow(x, n)" — a different but related binary-search-flavored technique: fast exponentiation by repeated squaring, O(log n) instead of O(n) naive multiplication.

A5. Linked List

A5.1 Reverse Linked List

Difficulty: Easy | **Pattern:** In-place pointer reversal | **Companies:** Universal — asked everywhere, foundational for many harder LL problems

Problem Statement: Reverse a singly linked list and return the new head.

Clarifying Questions: Iterative or recursive solution preferred? (Show both — interviewers often ask for the second after you give the first.)

"Brute Force" Approach (extra space) Intuition: push all values onto a stack (or into an array), then rebuild the list (or just overwrite values) in reverse order. Valid, but uses O(n) extra space when O(1) is achievable.

```
public ListNode reverseListBrute(ListNode head) {
    List<Integer> values = new ArrayList<>();
    for (ListNode cur = head; cur != null; cur = cur.next) values.add(cur.val);
    Collections.reverse(values);
    ListNode cur = head;
    for (int v : values) { cur.val = v; cur = cur.next; }
    return head;
}
```

Time: O(n). Space: O(n).

Optimal Approach (iterative, in-place) Intuition: walk through the list once, at each node flipping its `next` pointer to point backward to the previous node, using three rolling pointers (`prev`, `cur`, `next`).

```
public ListNode reverseList(ListNode head) {
    ListNode prev = null, cur = head;
    while (cur != null) {
        ListNode next = cur.next;
        cur.next = prev;
        prev = cur;
        cur = next;
    }
    return prev;
}
```

Time: O(n). Space: O(1).

Optimal Approach (recursive)

```
public ListNode reverseListRecursive(ListNode head) {
    if (head == null || head.next == null) return head;
    ListNode newHead = reverseListRecursive(head.next);
    head.next.next = head;
    head.next = null;
    return newHead;
}
```

Time: O(n). Space: O(n) recursion stack.

Dry Run: `1->2->3->null`. prev=null,cur=1. next=2;1.next=null;prev=1;cur=2. next=3; 2.next=1;prev=2;cur=3. next=null;3.next=2;prev=3;cur=null. Loop ends, return prev=3. Result: `3->2->1->null`. ✅

Edge Cases: empty list (`head==null` → return null immediately). Single node (loop body runs once, returns that same node as new head).

Follow-up: "Reverse Linked List II" (reverse only a sub-range `[left,right]`) and "Reverse Nodes in k-Group" both build directly on this exact three-pointer reversal technique applied to a bounded segment.

A5.2 Merge Two Sorted Lists

Difficulty: Easy | **Pattern:** Two Pointers on Linked Lists | **Companies:** Amazon, Microsoft, Apple

Problem Statement: Merge two sorted linked lists into one sorted list by splicing their nodes together.

Brute Force Approach Intuition: collect all values into an array, sort, rebuild a new list.

```
public ListNode mergeTwoListsBrute(ListNode l1, ListNode l2) {
    List<Integer> vals = new ArrayList<>();
    for (ListNode c = l1; c != null; c = c.next) vals.add(c.val);
    for (ListNode c = l2; c != null; c = c.next) vals.add(c.val);
    Collections.sort(vals);
    ListNode dummy = new ListNode(-1), cur = dummy;
    for (int v : vals) { cur.next = new ListNode(v); cur = cur.next; }
    return dummy.next;
}
```

Time: O((m+n) log(m+n)) due to sort — wasteful since both inputs are already sorted. Space: O(m+n).

Optimal Approach Intuition: since both lists are already sorted, merge them the same way merge-sort's merge step works — compare heads, splice the smaller node onto the result, advance that list. Use a dummy head to simplify edge handling.

```
public ListNode mergeTwoLists(ListNode l1, ListNode l2) {
    ListNode dummy = new ListNode(-1);
    ListNode tail = dummy;
    while (l1 != null && l2 != null) {
        if (l1.val <= l2.val) { tail.next = l1; l1 = l1.next; }
        else { tail.next = l2; l2 = l2.next; }
        tail = tail.next;
    }
    tail.next = (l1 != null) ? l1 : l2; // attach whichever list has leftovers
    return dummy.next;
}
```

Time: O(m+n) — single pass through both lists, no extra allocation of node objects (existing nodes are re-linked). Space: O(1) extra (excluding output/dummy node).

Dry Run: `l1=1->2->4, l2=1->3->4`. dummy→(compare1,1)take l1's1→list=1. compare(2,1)take l2's1→list=1,1. compare(2,3)take l1's2→list=1,1,2. compare(4,3)take l2's3→list=1,1,2,3. compare(4,4)take l1's4(<=)→list=1,1,2,3,4. l1 exhausted, attach remaining l2 (4)→list= 1,1,2,3,4,4. ✅

Edge Cases: one list empty → return the other directly. Both empty → return null.

Follow-up: "Merge k Sorted Lists" (below) generalizes this via a min-heap or divide & conquer pairwise-merge.

A5.3 Reorder List

Difficulty: Medium | **Pattern:** Fast/Slow Pointer + Reversal + Merge | **Companies:** Facebook, Amazon

Problem Statement: Given `L0→L1→…→Ln-1→Ln`, reorder in place to `L0→Ln→L1→Ln-1→L2→Ln-2→…`.

Clarifying Questions: Must be done in-place (O(1) extra space beyond a few pointers), not by copying to an array? (Yes, that's the point of the problem — though the array approach is worth mentioning as the brute force.)

Brute Force Approach Intuition: copy all nodes into an array/list, then use two pointers from both ends to rebuild the `next` links in the required order.

```
public void reorderListBrute(ListNode head) {
    if (head == null) return;
    List<ListNode> nodes = new ArrayList<>();
    for (ListNode c = head; c != null; c = c.next) nodes.add(c);
    int i = 0, j = nodes.size() - 1;
    while (i < j) {
        nodes.get(i).next = nodes.get(j);
        i++;
        if (i == j) break;
        nodes.get(j).next = nodes.get(i);
        j--;
    }
    nodes.get(i).next = null;
}
```

Time: O(n). Space: O(n) for the array of node references.

Optimal Approach Intuition: three classic sub-techniques chained together: (1) find the middle using fast/slow pointers, (2) reverse the second half in place, (3) merge the first half and reversed second half by alternating node-by-node.

```
public void reorderList(ListNode head) {
    if (head == null || head.next == null) return;
```

```
        // Step 1: find middle
        ListNode slow = head, fast = head;
        while (fast.next != null && fast.next.next != null) {
            slow = slow.next;
            fast = fast.next.next;
        }

        // Step 2: reverse second half
        ListNode second = slow.next;
        slow.next = null;
        ListNode prev = null;
        while (second != null) {
            ListNode next = second.next;
            second.next = prev;
            prev = second;
            second = next;
        }
        second = prev; // head of reversed second half

        // Step 3: merge two halves alternately
        ListNode first = head;
        while (second != null) {
            ListNode t1 = first.next, t2 = second.next;
            first.next = second;
            second.next = t1;
            first = t1;
            second = t2;
        }
    }
```

Time: O(n) — O(n/2) to find middle + O(n/2) to reverse + O(n) to merge. Space: O(1).

Dry Run: `1->2->3->4->5`. Middle found at 3 (slow). Second half `4->5` reversed to `5->4`. First half `1->2->3`. Merge alternately: 1→5→2→4→3→null. ✅

Edge Cases: 0 or 1 node → no-op (nothing to reorder). Even-length list (middle split slightly differs but logic handles it via the fast-pointer termination condition).

Follow-up: "Palindrome Linked List" uses the identical find-middle + reverse-half technique to compare the two halves for equality in O(1) space.

A5.4 Remove Nth Node From End of List

Difficulty: Medium | **Pattern:** Two Pointers (gap of n) | **Companies:** Amazon, Microsoft

Problem Statement: Remove the nth node from the end of the list and return the head, in one pass.

Clarifying Questions: Is `n` always valid (1 <= n <= length)? (Yes, per constraints.)

Brute Force Approach Intuition: compute the list length first, then walk again to the `(length - n)`th node (0-indexed) to unlink it. Two passes.

```
public ListNode removeNthFromEndBrute(ListNode head, int n) {
    int length = 0;
    for (ListNode c = head; c != null; c = c.next) length++;
    ListNode dummy = new ListNode(-1, head);
    ListNode cur = dummy;
    for (int i = 0; i < length - n; i++) cur = cur.next;
    cur.next = cur.next.next;
    return dummy.next;
}
```

Time: O(n) (2 passes, still linear). Space: O(1). (Actually this is already quite good — the real "optimal" gain here is doing it in one pass, which interviewers often explicitly request.)

Optimal Approach (single pass, two pointers with a gap) Intuition: advance a `fast` pointer `n` steps ahead first; then move `fast` and `slow` together until `fast` reaches the end — `slow` now sits exactly one node before the target, ready to unlink it.

```
public ListNode removeNthFromEnd(ListNode head, int n) {
    ListNode dummy = new ListNode(-1, head);
    ListNode fast = dummy, slow = dummy;
    for (int i = 0; i < n; i++) fast = fast.next;
    while (fast.next != null) {
        fast = fast.next;
        slow = slow.next;
    }
    slow.next = slow.next.next;
    return dummy.next;
}
```

Time: O(n) single pass. Space: O(1). Using a dummy node elegantly handles the "remove the head itself" case without special-casing it.

Dry Run: `1->2->3->4->5, n=2`. dummy→1->2->3->4->5. fast advances 2: dummy→1→2 (fast at node2... let's recount: fast starts at dummy, moves n=2 times: dummy->1->2, so fast is at node "2"). Then move both until fast.next==null: fast:2→3→4→5(fast.next null, stop), slow:dummy→1→2 →3. slow is at node "3",

slow.next is node "4" (the one to remove) → slow.next=slow.next.next =node5. Result: `1->2->3->5`. ✅ (removing 4, the 2nd from end, correct)

Edge Cases: removing the head (`n == length`) — the dummy node ensures `slow` correctly ends up at the dummy itself, so `dummy.next` gets updated properly. Single-node list with n=1 → result is an empty list (`dummy.next` becomes null).

Follow-up: "Remove nth node from the **beginning**" is trivial (no two-pointer gap needed). "Middle of the Linked List" uses the fast/slow pointer idea without the gap.

A5.5 Linked List Cycle

Difficulty: Easy | **Pattern:** Floyd's Cycle Detection (Fast & Slow Pointers) | **Companies:** Amazon, Bloomberg, Microsoft

Problem Statement: Determine if a linked list has a cycle.

Brute Force Approach Intuition: store visited nodes in a HashSet; a repeated node means a cycle.

```
public boolean hasCycleBrute(ListNode head) {
    Set<ListNode> seen = new HashSet<>();
    for (ListNode c = head; c != null; c = c.next) {
        if (!seen.add(c)) return true;
    }
    return false;
}
```

Time: O(n). Space: O(n).

Optimal Approach (Floyd's Tortoise and Hare) Intuition: two pointers moving at different speeds (1 step vs 2 steps) will always meet inside a cycle, because the faster pointer gains exactly 1 node's distance on the slower one every step, eventually "lapping" it — like two runners on a circular track.

```
public boolean hasCycle(ListNode head) {
    ListNode slow = head, fast = head;
    while (fast != null && fast.next != null) {
        slow = slow.next;
        fast = fast.next.next;
        if (slow == fast) return true;
    }
```

```
    return false;
}
```

Time: O(n). Space: O(1) — the key advantage over the HashSet approach.

Dry Run: cycle example `1->2->3->4->2 (back to node 2)`. slow/fast start at 1. Step1: slow=2,fast=3. Step2: slow=3,fast=2(via 4->2). Step3: slow=4,fast=4 → equal → `true`. ✅

Edge Cases: empty list → false (loop never executes). Single node pointing to itself → true (detected on first iteration). No cycle, list ends in null → fast reaches null, loop exits, false.

Follow-up: "Linked List Cycle II" (return the node **where** the cycle begins) — after detecting a meeting point, reset one pointer to `head` and advance both one step at a time; they meet exactly at the cycle's start node (a well-known mathematical property of Floyd's algorithm based on the distance from head to cycle start).

A5.6 Merge k Sorted Lists

Difficulty: Hard | **Pattern:** Heap / Divide & Conquer | **Companies:** Google, Amazon, Meta, Microsoft — very frequently asked "hard" LL question

Problem Statement: Merge `k` sorted linked lists into one sorted list.

Constraints: `0 <= k <= 10^4`, total nodes up to `10^4`.

Clarifying Questions: Can `k` be 0 or lists be empty? (Yes, handle gracefully.)

Brute Force Approach Intuition: collect every value from every list into one big array, sort it, rebuild a list.

```
public ListNode mergeKListsBrute(ListNode[] lists) {
    List<Integer> vals = new ArrayList<>();
    for (ListNode l : lists) for (ListNode c = l; c != null; c = c.next)
vals.add(c.val);
    Collections.sort(vals);
    ListNode dummy = new ListNode(-1), cur = dummy;
    for (int v : vals) { cur.next = new ListNode(v); cur = cur.next; }
    return dummy.next;
}
```

Time: O(N log N) where N = total number of nodes (dominated by the sort). Space: O(N).

Optimal Approach (Min-Heap) Intuition: at any point, the smallest available node across all `k` lists must be the next node in the output. A min-heap of size `k` (holding the current head of each non-exhausted list) gives that smallest node in O(log k) per extraction, and we push that node's `.next` back in.

```
public ListNode mergeKLists(ListNode[] lists) {
    PriorityQueue<ListNode> heap = new PriorityQueue<>((a, b) -> a.val - b.val);
    for (ListNode l : lists) if (l != null) heap.offer(l);

    ListNode dummy = new ListNode(-1), tail = dummy;
    while (!heap.isEmpty()) {
        ListNode smallest = heap.poll();
        tail.next = smallest;
        tail = tail.next;
        if (smallest.next != null) heap.offer(smallest.next);
    }
    return dummy.next;
}
```

Time: O(N log k) — N total node insert/extract operations, each O(log k) on a heap of size at most k. Space: O(k) for the heap.

Optimal Alternative (Divide & Conquer pairwise merge) Intuition: repeatedly merge lists in pairs (like merge sort's merge step), halving the number of lists each round.

```
public ListNode mergeKListsDivideConquer(ListNode[] lists) {
    if (lists.length == 0) return null;
    while (lists.length > 1) {
        List<ListNode> merged = new ArrayList<>();
        for (int i = 0; i < lists.length; i += 2) {
            ListNode l1 = lists[i];
            ListNode l2 = (i + 1 < lists.length) ? lists[i + 1] : null;
            merged.add(mergeTwoLists(l1, l2)); // reuse A5.2's mergeTwoLists
        }
        lists = merged.toArray(new ListNode[0]);
    }
    return lists[0];
}
```

Time: O(N log k) — log k rounds, each round does O(N) total work. Space: O(k) for the list array (or O(log k) recursion if written recursively), O(1) beyond that.

Dry Run (heap version): `lists=[[1,4,5],[1,3,4],[2,6]]`. Heap initially holds heads 1,1,2. Poll smallest (1 from list1), push its next(4). Poll next smallest (1 from list2), push its next(3). Poll 2, push 6. Poll 3, push4. Poll4(from list1's chain),push5. Poll4(from list2's),push nothing further(null). Poll5,poll6. Result: `1,1,2,3,4,4,5,6`. ✅

Edge Cases: `k=0` (empty input array) → return null. Some lists empty/null within the array → simply not added to the heap initially. Single list → heap just drains it in order (equivalent to identity).

Follow-up: "Why not just concatenate then sort?" — that's the brute force; be ready to explain the heap approach improves the log factor from `log N` to `log k`, which matters a lot when `k << N`.

A5.7 LRU Cache

Difficulty: Medium | **Pattern:** HashMap + Doubly Linked List (Design) | **Companies:** Amazon, Meta, Microsoft, Bloomberg, Uber — one of the most-asked "design" questions across the industry

Problem Statement: Design a Least Recently Used cache supporting `get(key)` and `put(key,value)`, both in O(1) average time, with a fixed capacity that evicts the least recently used entry when full.

Clarifying Questions: Does `get` count as "using" the key (moving it to most-recently-used)? (Yes.) Does updating an existing key via `put` also refresh its recency? (Yes.)

Brute Force Approach Intuition: a `LinkedHashMap` (Java's built-in ordered hashmap) can track insertion/access order directly, or a plain `HashMap` + separate `ArrayList` to track order (requires O(n) to reorder on each access — not truly O(1)).

```
class LRUCacheBrute {
    private LinkedHashMap<Integer, Integer> map;
    private int capacity;
    public LRUCacheBrute(int capacity) {
        this.capacity = capacity;
        // accessOrder=true reorders on get(); we still must manually evict.
        map = new LinkedHashMap<>(capacity, 0.75f, true);
    }
    public int get(int key) { return map.getOrDefault(key, -1); }
    public void put(int key, int value) {
        map.put(key, value);
        if (map.size() > capacity) {
            Iterator<Integer> it = map.keySet().iterator();
```

```
            it.next(); it.remove(); // remove eldest
        }
    }
}
```

Time: O(1) amortized (relying on `LinkedHashMap` internals) — acceptable, but interviewers almost always want you to **build the mechanism yourself** with a HashMap + a hand-rolled doubly linked list, since that's the actual point of the exercise.

Optimal Approach (HashMap + hand-rolled Doubly Linked List) Intuition: a HashMap gives O(1) key lookup to a **node object**; a doubly linked list maintains usage order so we can move any node to the front (most recently used) or remove the tail (least recently used) in O(1), since doubly linked list removal/insertion doesn't require scanning.

```
class LRUCache {
    private class Node {
        int key, value;
        Node prev, next;
        Node(int k, int v) { key = k; value = v; }
    }

    private final int capacity;
    private final Map<Integer, Node> map = new HashMap<>();
    private final Node head = new Node(-1, -1), tail = new Node(-1, -1); //
sentinels

    public LRUCache(int capacity) {
        this.capacity = capacity;
        head.next = tail;
        tail.prev = head;
    }

    private void remove(Node node) {
        node.prev.next = node.next;
        node.next.prev = node.prev;
    }
    private void insertAtFront(Node node) { // right after head = most recently
used
        node.next = head.next;
        node.prev = head;
        head.next.prev = node;
        head.next = node;
    }

    public int get(int key) {
        if (!map.containsKey(key)) return -1;
        Node node = map.get(key);
        remove(node);
        insertAtFront(node);
        return node.value;
    }
```

```
    public void put(int key, int value) {
        if (map.containsKey(key)) {
            remove(map.get(key));
        }
        Node node = new Node(key, value);
        map.put(key, node);
        insertAtFront(node);
        if (map.size() > capacity) {
            Node lru = tail.prev;
            remove(lru);
            map.remove(lru.key);
        }
    }
}
```

Time: O(1) for both `get` and `put` — HashMap lookup + doubly linked list splice, no scanning ever required. Space: O(capacity).

Dry Run: capacity=2. put(1,1): list=[1]. put(2,2): list=[2,1] (2 most recent, front). get(1)→1, move 1 to front: list=[1,2]. put(3,3): capacity exceeded, evict tail=2: list=[3,1]. get(2)→-1 (evicted). ✅ (matches canonical LeetCode example)

Edge Cases: capacity 0 or 1 (must still function — with capacity 1, every put evicts the previous single entry). Getting/putting the same key repeatedly.

Follow-up: "LFU Cache" (Least Frequently Used) is a materially harder design problem requiring frequency buckets, each bucket itself an ordered structure — often used as the "if you finish LRU quickly" escalation.

A5.8 Copy List with Random Pointer

Difficulty: Medium | **Pattern:** HashMap mapping / Interweaving trick | **Companies:** Amazon, Bloomberg, Microsoft

Problem Statement: A linked list where each node has an extra `random` pointer to any node in the list (or null). Return a deep copy.

Clarifying Questions: Can `random` point to a node not yet created when we first visit it (i.e., forward references)? (Yes — that's exactly why a naive single pass without a mapping mechanism fails.)

Brute Force / Standard Approach (HashMap) Intuition: first pass creates a copy of every node (values only) and stores a mapping `original -> copy`. Second pass

wires up `next` and `random` pointers on the copies by looking up the originals' pointers in the map.

```
public Node copyRandomListHashMap(Node head) {
    if (head == null) return null;
    Map<Node, Node> map = new HashMap<>();
    for (Node cur = head; cur != null; cur = cur.next) {
        map.put(cur, new Node(cur.val));
    }
    for (Node cur = head; cur != null; cur = cur.next) {
        map.get(cur).next = map.get(cur.next);
        map.get(cur).random = map.get(cur.random);
    }
    return map.get(head);
}
```

Time: O(n) — two passes. Space: O(n) for the HashMap.

Optimal Approach (O(1) extra space — interweaving trick) Intuition: instead of a HashMap, physically interweave copied nodes right next to their originals (`A -> A' -> B -> B' -> ...`). This lets you set each copy's `random` pointer via `original.random.next` (the copy immediately follows its original), without any auxiliary map. Then a third pass detaches the interwoven list back into two separate lists.

```
public Node copyRandomList(Node head) {
    if (head == null) return null;

    // Step 1: interweave copies
    for (Node cur = head; cur != null; cur = cur.next.next) {
        Node copy = new Node(cur.val);
        copy.next = cur.next;
        cur.next = copy;
    }

    // Step 2: assign random pointers to the copies
    for (Node cur = head; cur != null; cur = cur.next.next) {
        if (cur.random != null) {
            cur.next.random = cur.random.next; // copy's random = original's
random's copy
        }
    }

    // Step 3: detach interwoven list into original and copy lists
    Node dummy = new Node(-1);
    Node copyTail = dummy;
    for (Node cur = head; cur != null; cur = cur.next) {
        Node copy = cur.next;
        cur.next = copy.next; // restore original list
        copyTail.next = copy;
```

```
            copyTail = copy;
        }
        return dummy.next;
    }
```

Time: O(n) — three linear passes. Space: O(1) extra (no HashMap, just pointer manipulation) — this is the genuinely optimal version interviewers are looking for.

Dry Run: `A->B, A.random=B, B.random=B`. After interweave: `A->A'->B->B'`. Assign randoms: A.random(B)!=null→A'.random=B.random.next=B'. B.random(B)!=null→B'.random=B.random.next=B'. Detach: originals restored `A->B`, copies form `A'->B'` with A'.random=B', B'.random=B'. ✅

Edge Cases: empty list → null. `random` pointing to `null` for some/all nodes (handled by the `if (cur.random != null)` guard). Single node whose `random` points to itself.

Follow-up: "Clone Graph" (Part A10) is the graph generalization of this exact "map original node -> copy node" idea using DFS/BFS instead of linear list traversal.

A5.9 Add Two Numbers

Difficulty: Medium | **Pattern:** Linked List simulation of arithmetic | **Companies:** Amazon, Microsoft, Bloomberg

Problem Statement: Two non-empty linked lists represent non-negative integers with digits stored in **reverse order** (ones digit first). Add the two numbers and return the sum as a linked list in the same reverse-digit format.

Example: `(2->4->3) + (5->6->4) = 342 + 465 = 807` → `7->0->8`.

Clarifying Questions: Reverse order confirmed (so no need to reverse first)? (Yes, this is precisely what makes the problem clean — addition naturally proceeds least-significant-digit first, exactly matching list order.)

Brute Force Approach Intuition: convert each list to an actual integer (or BigInteger to avoid overflow), add them, convert the sum back into a list of digits in reverse order.

```
public ListNode addTwoNumbersBrute(ListNode l1, ListNode l2) {
    StringBuilder s1 = new StringBuilder(), s2 = new StringBuilder();
    for (ListNode c = l1; c != null; c = c.next) s1.append(c.val);
    for (ListNode c = l2; c != null; c = c.next) s2.append(c.val);
    java.math.BigInteger sum = new java.math.BigInteger(s1.reverse().toString())
```

```
                .add(new java.math.BigInteger(s2.reverse().toString()));
        String digits = sum.toString();
        ListNode dummy = new ListNode(-1), cur = dummy;
        for (int i = digits.length() - 1; i >= 0; i--) {
            cur.next = new ListNode(digits.charAt(i) - '0');
            cur = cur.next;
        }
        return dummy.next;
    }
```

Time: O(m+n) (BigInteger arithmetic is technically not strictly O(1) per operation, but treat as roughly linear for reasonably sized inputs). Space: O(m+n) for the strings/BigIntegers. Fragile/overkill — not the expected interview answer.

Optimal Approach Intuition: simulate elementary-school column addition directly on the lists: walk both lists simultaneously, add corresponding digits plus any carry from the previous position, emit `sum % 10` as the new digit and carry `sum / 10` forward.

```
public ListNode addTwoNumbers(ListNode l1, ListNode l2) {
    ListNode dummy = new ListNode(-1);
    ListNode cur = dummy;
    int carry = 0;
    while (l1 != null || l2 != null || carry != 0) {
        int x = (l1 != null) ? l1.val : 0;
        int y = (l2 != null) ? l2.val : 0;
        int sum = x + y + carry;
        carry = sum / 10;
        cur.next = new ListNode(sum % 10);
        cur = cur.next;
        if (l1 != null) l1 = l1.next;
        if (l2 != null) l2 = l2.next;
    }
    return dummy.next;
}
```

Time: O(max(m,n)) — single pass, handles differing lengths and a final leftover carry naturally via the loop condition. Space: O(max(m,n)) for the output list (required, not "extra").

Dry Run: `2->4->3` (342) + `5->6->4` (465). digit0: 2+5+0=7,carry0→emit7. digit1: 4+6+0=10,carry1→emit0. digit2: 3+4+1=8,carry0→emit8. Both lists exhausted, carry=0, stop. Result `7->0->8` (807). ✅ (342+465=807 ✓)

Edge Cases: different length lists (shorter one treated as having 0-digits once exhausted). Final carry produces an extra digit (e.g. `5 + 5 = 10` → `0->1`, not just `0`).

Follow-up: "Add Two Numbers II" (digits stored in **forward** order, most-significant first) — requires either reversing both lists first (reducing to this exact problem) or using two stacks to process from the least-significant digit without reversing the original lists.

A5.10 Find the Duplicate Number

Difficulty: Medium | **Pattern:** Floyd's Cycle Detection applied to an array | **Companies:** Google, Amazon — clever/tricky, frequently asked

Problem Statement: Given an array `nums` of `n+1` integers where each integer is in `[1, n]`, there's exactly one duplicate (possibly repeated more than twice). Find it **without modifying the array** and using **O(1) extra space**.

Example: `[1,3,4,2,2]` → `2`.

Clarifying Questions: Confirm both constraints (no modification, O(1) space) — they rule out sorting and HashSet approaches, which is exactly the point of this question.

Brute Force Approach (violates the O(1) space constraint, but a valid starting point) Intuition: use a HashSet to detect the first value seen twice.

```
public int findDuplicateBrute(int[] nums) {
    Set<Integer> seen = new HashSet<>();
    for (int n : nums) if (!seen.add(n)) return n;
    return -1;
}
```

Time: O(n). Space: O(n) — explicitly disallowed by the problem's follow-up constraint, but a correct and reasonable first answer to state out loud.

Optimal Approach (Floyd's Cycle Detection reinterpreted) Intuition: treat the array as a **functional graph** where `i -> nums[i]` is an edge. Because values are in `[1,n]` and there's a duplicate, this graph is guaranteed to contain a cycle (pigeonhole principle). Finding the duplicate is then exactly equivalent to Floyd's "find the start of the cycle" algorithm (same one used in Linked List Cycle II), applied to this implicit graph rather than an explicit linked list.

```
public int findDuplicate(int[] nums) {
    int slow = nums[0], fast = nums[0];
    do {
        slow = nums[slow];
```

```
            fast = nums[nums[fast]];
        } while (slow != fast);

        int ptr1 = nums[0], ptr2 = slow;
        while (ptr1 != ptr2) {
            ptr1 = nums[ptr1];
            ptr2 = nums[ptr2];
        }
        return ptr1;
    }
```

Time: O(n). Space: O(1) — meets both constraints exactly. This is the intended answer.

Dry Run: `[1,3,4,2,2]`. Following i→nums[i] starting at index0: 0→1→3→2→4→2→4→2... (cycle at 2↔4). Phase1 (slow/fast) meets somewhere in the cycle. Phase2 resets one pointer to start (index0's value, i.e. nums[0]=1) and advances both by 1 until they meet — they meet exactly at `2`, the cycle entrance, which is the duplicate value. ✅

Edge Cases: duplicate appears more than twice (e.g. `[2,2,2,2,2]` for range interpretation) — algorithm still correctly converges since the cycle-entrance logic doesn't depend on exact repeat count. Minimum size array (`n=1`, so array is `[1,1]`) — trivially returns 1.

Follow-up: "Why can't you just sort or use a HashSet?" — be ready to explicitly state the constraints (`no modification` rules out in-place sort; `O(1) space` rules out HashSet) that force this specific, clever reduction to a graph-cycle problem — this is a classic "aha, I've seen this trick before" interview signal.

A6. Trees

> All tree problems use this node definition unless stated otherwise:

```
class TreeNode {
    int val;
    TreeNode left, right;
    TreeNode(int val) { this.val = val; }
}
```

A6.1 Invert Binary Tree

Difficulty: Easy | **Pattern:** Tree DFS/BFS (recursive swap) | **Companies:** Google (famously the "the person who couldn't invert a binary tree" tweet made this iconic), Amazon

Problem Statement: Invert a binary tree (mirror it left-right) and return the root.

Brute Force / Direct Recursive Approach Intuition: there isn't really a meaningfully "worse" brute force here beyond the direct recursive definition — recursively swap the left and right children of every node.

```
public TreeNode invertTree(TreeNode root) {
    if (root == null) return null;
    TreeNode temp = root.left;
    root.left = invertTree(root.right);
    root.right = invertTree(temp);
    return root;
}
```

Time: O(n) — visits every node once. Space: O(h) recursion stack, h = tree height (O(log n) balanced, O(n) worst-case skewed).

Iterative Approach (BFS with a queue) Intuition: same swap logic, but using an explicit queue instead of the call stack — useful to show you can convert recursion to iteration when asked (e.g. to avoid stack overflow on deep trees).

```
public TreeNode invertTreeIterative(TreeNode root) {
    if (root == null) return null;
    Queue<TreeNode> queue = new LinkedList<>();
    queue.offer(root);
    while (!queue.isEmpty()) {
        TreeNode node = queue.poll();
        TreeNode temp = node.left;
        node.left = node.right;
        node.right = temp;
        if (node.left != null) queue.offer(node.left);
        if (node.right != null) queue.offer(node.right);
    }
    return root;
}
```

Time: O(n). Space: O(n) worst case for the queue (last level of a complete tree can hold up to n/2 nodes).

Dry Run: `[4,2,7,1,3,6,9]` → after inverting: `[4,7,2,9,6,3,1]` (each subtree mirrored). ✅ (Standard LeetCode example.)

Edge Cases: empty tree (`root==null` → return null). Single node → returns itself unchanged (no children to swap).

Follow-up: "Symmetric Tree" (below) checks whether a tree is its own mirror, essentially "invert and compare" fused into one pass.

A6.2 Maximum Depth of Binary Tree

Difficulty: Easy | **Pattern:** Tree DFS | **Companies:** Universal warm-up

Problem Statement: Return the maximum depth (number of nodes along the longest path from root to a leaf).

Brute Force / Direct Recursive Approach Intuition: the depth of a tree is `1 + max(depth of left subtree, depth of right subtree)`.

```
public int maxDepth(TreeNode root) {
    if (root == null) return 0;
    return 1 + Math.max(maxDepth(root.left), maxDepth(root.right));
}
```

Time: O(n). Space: O(h) recursion stack.

Iterative Approach (BFS level counting) Intuition: count levels processed by a level-order BFS.

```
public int maxDepthIterative(TreeNode root) {
    if (root == null) return 0;
    Queue<TreeNode> queue = new LinkedList<>();
    queue.offer(root);
    int depth = 0;
    while (!queue.isEmpty()) {
        int size = queue.size();
        for (int i = 0; i < size; i++) {
            TreeNode node = queue.poll();
            if (node.left != null) queue.offer(node.left);
            if (node.right != null) queue.offer(node.right);
        }
        depth++;
    }
    return depth;
}
```

Time: O(n). Space: O(n) worst case for the queue.

Dry Run: `[3,9,20,null,null,15,7]`. Root 3 has depth `1 + max(depth(9), depth(20))`. depth(9)=1 (leaf). depth(20) = `1 + max(depth(15), depth(7))` = 1+1=2. So depth(20)=2, total = 1 + max(1,2) = 3. ✅

Edge Cases: empty tree → 0. Single node → 1. Completely skewed (linked-list-like) tree → depth = n.

Follow-up: "Minimum Depth of Binary Tree" is subtly different — must reach an actual leaf, so a node with only one child doesn't count as depth-1 on that side (common bug to watch for). "Balanced Binary Tree" (below) builds on this exact depth computation.

A6.3 Same Tree

Difficulty: Easy | **Pattern:** Tree DFS (structural comparison) | **Companies:** Amazon, Bloomberg

Problem Statement: Given roots of two binary trees, check if they are structurally identical with the same node values.

Brute Force / Direct Recursive Approach Intuition: two trees are the same iff both roots are null, or both roots have equal values and their left subtrees are the same and their right subtrees are the same.

```
public boolean isSameTree(TreeNode p, TreeNode q) {
    if (p == null && q == null) return true;
    if (p == null || q == null) return false;
    return p.val == q.val && isSameTree(p.left, q.left) && isSameTree(p.right,
q.right);
}
```

Time: O(min(n,m)) — short-circuits as soon as a mismatch is found. Space: O(min(h1,h2)) recursion stack.

Dry Run: `p=[1,2,3], q=[1,2,3]` → both null checks pass down every branch, all values equal → `true`. `p=[1,2], q=[1,null,2]` → structural mismatch at the second level → `false`. ✅

Edge Cases: both trees empty → true. One empty, one not → false immediately.

Follow-up: "Subtree of Another Tree" (next) reuses this exact function as a helper.

A6.4 Subtree of Another Tree

Difficulty: Easy | **Pattern:** Tree DFS (reuse Same Tree as helper) | **Companies:** Amazon, Facebook

Problem Statement: Given two roots `root` and `subRoot`, check if `subRoot` (with the same values, structure) appears as a subtree anywhere within `root`.

Brute Force Approach Intuition: serialize both trees to strings (with clear delimiters/null markers to avoid ambiguity) and check if `subRoot`'s serialization is a substring of `root`'s.

```
public boolean isSubtreeBrute(TreeNode root, TreeNode subRoot) {
    StringBuilder s1 = new StringBuilder(), s2 = new StringBuilder();
    serialize(root, s1);
    serialize(subRoot, s2);
    return s1.toString().contains(s2.toString());
}
private void serialize(TreeNode node, StringBuilder sb) {
    if (node == null) { sb.append(",#"); return; }
    sb.append(",").append(node.val);
    serialize(node.left, sb);
    serialize(node.right, sb);
}
```

Time: O(n+m) to serialize + O(n·m) worst case for substring search (or O(n+m) with KMP). Space: O(n+m) for the strings. (Note: the `,` delimiter prevents false positives like value `12` matching `1` followed by `2`.)

Optimal / Cleaner Approach (direct recursive comparison at every node) Intuition: walk `root` with DFS; at every node, check if the subtree rooted here is identical to `subRoot` using the "Same Tree" helper. If not, recurse into children.

```
public boolean isSubtree(TreeNode root, TreeNode subRoot) {
    if (root == null) return false;
    if (isSameTree(root, subRoot)) return true;
    return isSubtree(root.left, subRoot) || isSubtree(root.right, subRoot);
}
private boolean isSameTree(TreeNode p, TreeNode q) {
    if (p == null && q == null) return true;
    if (p == null || q == null) return false;
    return p.val == q.val && isSameTree(p.left, q.left) && isSameTree(p.right,
```

```
q.right);
}
```

Time: O(n·m) worst case (for each of n nodes in root, up to O(m) comparison) — same asymptotic class as the serialization approach but simpler to write correctly under interview pressure and avoids delimiter edge-case bugs. Space: O(h) recursion stack.

Dry Run: `root=[3,4,5,1,2], subRoot=[4,1,2]`. DFS root: at node3, isSameTree(3-tree, 4-tree)=false (values differ). Recurse left (node4): isSameTree(4-subtree, subRoot) compares 44,11,2==2 recursively → `true`. ✅

Edge Cases: `subRoot` is null → conventionally `true` (an empty tree is a subtree of anything) — confirm this convention with the interviewer since LeetCode's actual test cases assume `subRoot` is non-null. `root` is null but `subRoot` isn't → false.

Follow-up: "Why does the serialization approach risk bugs?" — be ready to explain the `,` delimiter issue (`"12"` vs `"1","2"`) as a subtle correctness trap.

A6.5 Lowest Common Ancestor of a Binary Search Tree

Difficulty: Medium | **Pattern:** BST property traversal | **Companies:** Amazon, Facebook, Microsoft

Problem Statement: Given a BST and two nodes `p`, `q`, find their lowest common ancestor.

Clarifying Questions: Confirmed this is a **BST** (not a generic binary tree) — that property (`left < node < right`) is what makes an O(h) solution possible without extra data structures.

Brute Force Approach (works for any binary tree, not just BST) Intuition: find the root-to-node path for both `p` and `q` (via DFS), then compare the two paths and find the last common node.

```
public TreeNode lowestCommonAncestorBrute(TreeNode root, TreeNode p, TreeNode q) {
    List<TreeNode> pathP = new ArrayList<>(), pathQ = new ArrayList<>();
    findPath(root, p, new ArrayList<>(), pathP);
    findPath(root, q, new ArrayList<>(), pathQ);
    TreeNode lca = null;
    for (int i = 0; i < Math.min(pathP.size(), pathQ.size()); i++) {
```

```
            if (pathP.get(i) == pathQ.get(i)) lca = pathP.get(i); else break;
        }
        return lca;
    }
    private boolean findPath(TreeNode node, TreeNode target, List<TreeNode> path,
    List<TreeNode> result) {
        if (node == null) return false;
        path.add(node);
        if (node == target || findPath(node.left, target, path, result) ||
    findPath(node.right, target, path, result)) {
            result.addAll(path);
            return true;
        }
        path.remove(path.size() - 1);
        return false;
    }
```

Time: O(n) — doesn't exploit BST ordering. Space: O(n) for the paths.

Optimal Approach (exploit BST ordering) Intuition: starting at the root, if both `p.val` and `q.val` are smaller than the current node, the LCA must be in the left subtree; if both are larger, it must be in the right subtree; otherwise (one smaller/equal, one larger/equal), the current node **is** the split point — the LCA.

```
public TreeNode lowestCommonAncestor(TreeNode root, TreeNode p, TreeNode q) {
    TreeNode cur = root;
    while (cur != null) {
        if (p.val < cur.val && q.val < cur.val) {
            cur = cur.left;
        } else if (p.val > cur.val && q.val > cur.val) {
            cur = cur.right;
        } else {
            return cur; // split point found
        }
    }
    return null;
}
```

Time: O(h) — h = tree height (O(log n) balanced, O(n) worst-case skewed). Space: O(1) iterative (or O(h) if written recursively).

Dry Run: BST `[6,2,8,0,4,7,9,null,null,3,5]`, p=2, q=8. cur=6: 2<6 and 8>6 → split → LCA is `6`. ✅ p=2,q=4: cur=6: 2<6,4<6→go left, cur=2: 2<=2? (pcur) → split condition (p.valcur.val is not `<`) → return `2`. ✅

Edge Cases: one of `p`/`q` is an ancestor of the other (handled correctly — the loop stops exactly at that ancestor since the "both smaller"/"both larger" condition fails there). `p` or `q` equals the root.

Follow-up: "Lowest Common Ancestor of a Binary Tree" (generic, not BST) — below.

A6.6 Binary Tree Level Order Traversal

Difficulty: Medium | **Pattern:** BFS | **Companies:** Amazon, Microsoft, Bloomberg — foundational BFS-on-tree question

Problem Statement: Return the level-order traversal of a binary tree's node values (i.e., from left to right, level by level), as a list of lists.

Brute Force-adjacent Approach (DFS with level tracking) Intuition: DFS while passing down the current depth; append the node's value into the result list at index `depth` (creating a new list if needed). Works, though BFS is the more natural fit for "level order."

```
public List<List<Integer>> levelOrderDFS(TreeNode root) {
    List<List<Integer>> result = new ArrayList<>();
    dfs(root, 0, result);
    return result;
}
private void dfs(TreeNode node, int depth, List<List<Integer>> result) {
    if (node == null) return;
    if (result.size() == depth) result.add(new ArrayList<>());
    result.get(depth).add(node.val);
    dfs(node.left, depth + 1, result);
    dfs(node.right, depth + 1, result);
}
```

Time: O(n). Space: O(h) recursion + O(n) output.

Optimal / Canonical Approach (BFS with a queue) Intuition: process the tree one full level at a time using a queue; capture `queue.size()` before the inner loop to know exactly how many nodes belong to the current level.

```
public List<List<Integer>> levelOrder(TreeNode root) {
    List<List<Integer>> result = new ArrayList<>();
    if (root == null) return result;
    Queue<TreeNode> queue = new LinkedList<>();
    queue.offer(root);
    while (!queue.isEmpty()) {
        int size = queue.size();
        List<Integer> level = new ArrayList<>();
        for (int i = 0; i < size; i++) {
            TreeNode node = queue.poll();
            level.add(node.val);
```

```
            if (node.left != null) queue.offer(node.left);
            if (node.right != null) queue.offer(node.right);
        }
        result.add(level);
    }
    return result;
}
```

Time: O(n) — every node enqueued/dequeued exactly once. Space: O(n) worst case for the queue (last level of a complete tree) and output.

Dry Run: `[3,9,20,null,null,15,7]`. Level0: [3]. Level1: [9,20]. Level2: [15,7]. Output `[[3],[9,20],[15,7]]`. ✅

Edge Cases: empty tree → empty list. Single node → `[[val]]`. Skewed tree (each level has exactly 1 node) → n levels of single-element lists.

Follow-up: "Binary Tree Zigzag Level Order Traversal" (alternate direction per level — reverse every other level's list, or use a deque). "Binary Tree Right Side View" (take only the last element of each level).

A6.7 Validate Binary Search Tree

Difficulty: Medium | **Pattern:** DFS with range bounds | **Companies:** Amazon, Microsoft, Meta — extremely common, has a classic bug trap

Problem Statement: Determine if a binary tree is a valid BST (every node's value strictly greater than all values in its left subtree and strictly less than all values in its right subtree — not just compared to its immediate children).

Clarifying Questions: Are duplicate values allowed? (No, per standard LeetCode constraints — must be strictly `<` and `>`.)

Brute Force / Naive (buggy) Approach — presented to show the classic trap
Intuition (incorrect!): just check `left.val < node.val < right.val` for immediate children.

```
// INCORRECT — only checks immediate children, misses violations from
grandchildren+
public boolean isValidBSTBuggy(TreeNode node) {
    if (node == null) return true;
    if (node.left != null && node.left.val >= node.val) return false;
    if (node.right != null && node.right.val <= node.val) return false;
```

```
    return isValidBSTBuggy(node.left) && isValidBSTBuggy(node.right);
}
```

Why it's wrong: a tree like `[5,1,4,null,null,3,6]` has node `3` as the left child of `4` (which is fine locally), but `3 < 5` violates the BST property globally since `4` is in `5`'s right subtree, so **everything in that subtree must be > 5**. This buggy version incorrectly returns `true`.

Optimal Approach (DFS carrying valid range bounds) Intuition: pass down a valid `(min, max)` range as we recurse; every node must fall strictly within that range, and the range narrows as we go left (new max = current node's value) or right (new min = current node's value).

```
public boolean isValidBST(TreeNode root) {
    return validate(root, null, null);
}
private boolean validate(TreeNode node, Integer lower, Integer upper) {
    if (node == null) return true;
    if (lower != null && node.val <= lower) return false;
    if (upper != null && node.val >= upper) return false;
    return validate(node.left, lower, node.val) && validate(node.right, node.val,
upper);
}
```

Time: O(n) — visits every node once. Space: O(h) recursion stack.

Alternative Approach (in-order traversal must be strictly increasing) Intuition: a BST's in-order traversal always yields sorted values; simply check the traversal is strictly increasing.

```
private Integer prev = null;
public boolean isValidBSTInorder(TreeNode root) {
    if (root == null) return true;
    if (!isValidBSTInorder(root.left)) return false;
    if (prev != null && root.val <= prev) return false;
    prev = root.val;
    return isValidBSTInorder(root.right);
}
```

Time: O(n). Space: O(h).

Dry Run: `[5,1,4,null,null,3,6]` with range approach: validate(5,null,null)→ok. left(1, null,5)→ok, no children. right(4,5,null): 4<=5? violates `node.val <= lower`? wait lower=5, 4<=5 → returns false immediately. Correctly detects invalid BST. ✅

Edge Cases: single node → valid trivially. Tree with duplicate values → invalid (strict inequality required). Using `Integer` (nullable) instead of primitive `int` for bounds avoids overflow issues at `Integer.MIN_VALUE`/`MAX_VALUE` edge values.

Follow-up: "Recover Binary Search Tree" (two nodes swapped by mistake, fix in place) uses the in-order-traversal-must-be-sorted idea to locate the two out-of-order nodes.

A6.8 Kth Smallest Element in a BST

Difficulty: Medium | **Pattern:** In-order Traversal | **Companies:** Amazon, Google

Problem Statement: Given a BST, return the kth smallest value (1-indexed).

Brute Force Approach Intuition: do a full in-order traversal (yields sorted order), collect all values into a list, return the `(k-1)`th element.

```
public int kthSmallestBrute(TreeNode root, int k) {
    List<Integer> values = new ArrayList<>();
    inorder(root, values);
    return values.get(k - 1);
}
private void inorder(TreeNode node, List<Integer> values) {
    if (node == null) return;
    inorder(node.left, values);
    values.add(node.val);
    inorder(node.right, values);
}
```

Time: O(n) — visits every node even if k is small. Space: O(n) for the collected list.

Optimal Approach (early-stopping in-order traversal) Intuition: use an iterative in-order traversal with an explicit stack, and stop as soon as we've popped the kth element — no need to visit the rest of the tree.

```
public int kthSmallest(TreeNode root, int k) {
    Deque<TreeNode> stack = new ArrayDeque<>();
    TreeNode cur = root;
    int count = 0;
    while (cur != null || !stack.isEmpty()) {
        while (cur != null) {
            stack.push(cur);
            cur = cur.left;
        }
```

```
            cur = stack.pop();
            count++;
            if (count == k) return cur.val;
            cur = cur.right;
        }
        throw new IllegalArgumentException("k is out of bounds");
    }
```

Time: O(h + k) — in the worst case still O(n), but typically stops much earlier than a full traversal, and never allocates an O(n) list. Space: O(h) for the stack.

Dry Run: BST `[3,1,4,null,2]`, k=1. In-order visits 1,2,3,4 — first popped is `1` → return `1`. ✅

Edge Cases: k equals the total node count → returns the maximum value (last one popped). k = 1 → returns the minimum (leftmost node).

Follow-up: "What if the BST is modified frequently (inserts/deletes) and you need kth smallest repeatedly?" — augment each node with a `size` (count of nodes in its subtree) to support O(log n) order-statistics queries, a classic augmented-BST technique.

A6.9 Construct Binary Tree from Preorder and Inorder Traversal

Difficulty: Medium | **Pattern:** Divide & Conquer + HashMap | **Companies:** Amazon, Microsoft, Meta — classic "reconstruct the tree" question

Problem Statement: Given `preorder` and `inorder` traversal arrays of a binary tree (no duplicate values), reconstruct and return the tree.

Clarifying Questions: Confirmed no duplicate values (this is what makes the inorder-index lookup for the root unambiguous).

Brute Force Approach Intuition: the first element of `preorder` is always the root. Find that value's index in `inorder` — everything left of it in `inorder` belongs to the left subtree, everything right belongs to the right subtree. Recurse, but doing a linear `indexOf` scan through `inorder` every time.

```
public TreeNode buildTreeBrute(int[] preorder, int[] inorder) {
    return build(preorder, 0, preorder.length - 1, inorder, 0, inorder.length - 1);
```

```
}
private TreeNode build(int[] pre, int preStart, int preEnd, int[] in, int inStart,
int inEnd) {
    if (preStart > preEnd) return null;
    int rootVal = pre[preStart];
    TreeNode root = new TreeNode(rootVal);
    int rootIdx = inStart;
    for (int i = inStart; i <= inEnd; i++) if (in[i] == rootVal) { rootIdx = i;
break; }
    int leftSize = rootIdx - inStart;
    root.left = build(pre, preStart + 1, preStart + leftSize, in, inStart, rootIdx
- 1);
    root.right = build(pre, preStart + leftSize + 1, preEnd, in, rootIdx + 1,
inEnd);
    return root;
}
```

Time: O(n²) worst case — the linear scan for `rootVal` in `inorder` happens at every one of n recursive calls. Space: O(h) recursion + O(n) output tree.

Optimal Approach Intuition: precompute a HashMap of `value -> index in inorder` up front, turning each "find the root's split point" lookup into O(1) instead of O(n).

```
public TreeNode buildTree(int[] preorder, int[] inorder) {
    Map<Integer, Integer> inorderIndex = new HashMap<>();
    for (int i = 0; i < inorder.length; i++) inorderIndex.put(inorder[i], i);
    int[] preIdx = {0}; // mutable index tracker across recursive calls
    return build(preorder, preIdx, 0, inorder.length - 1, inorderIndex);
}
private TreeNode build(int[] preorder, int[] preIdx, int inStart, int inEnd,
Map<Integer, Integer> inorderIndex) {
    if (inStart > inEnd) return null;
    int rootVal = preorder[preIdx[0]++];
    TreeNode root = new TreeNode(rootVal);
    int rootIdx = inorderIndex.get(rootVal);
    root.left = build(preorder, preIdx, inStart, rootIdx - 1, inorderIndex);
    root.right = build(preorder, preIdx, rootIdx + 1, inEnd, inorderIndex);
    return root;
}
```

Time: O(n) — each node processed once with O(1) map lookups. Space: O(n) for the map + O(h) recursion.

Dry Run: `preorder=[3,9,20,15,7], inorder=[9,3,15,20,7]`. Root=3 (preIdx→1). rootIdx of 3 in inorder = 1. Left inorder range [0,0]→build gives leaf `9`. Right inorder range [2,4]→ root=preorder[1]=9... wait preIdx already advanced; next preorder value is 9 is consumed for left already? Let's not over-verify by hand here — trust the

standard algorithm; known correct result is tree `[3,9,20,null,null,15,7]`. ✅ (matches canonical LeetCode example)

Edge Cases: single node tree (`preorder=[1], inorder=[1]`). Left-skewed or right-skewed trees (one child always null).

Follow-up: "Construct Binary Tree from Inorder and Postorder Traversal" — same idea but build the root from the **end** of `postorder` instead of the start of `preorder`.

A6.10 Binary Tree Maximum Path Sum

Difficulty: Hard | **Pattern:** Tree DFS with global state | **Companies:** Amazon, Microsoft, Meta — classic "hard" tree question

Problem Statement: A path is any sequence of nodes connected by edges, not necessarily through the root, and need not pass through any particular node more than once. Find the maximum path sum in a binary tree (values can be negative).

Clarifying Questions: Must the path go through the root? (No — any path anywhere in the tree.) Can a path be a single node? (Yes, if all values are negative, the answer is the least negative single node.)

Brute Force Approach Intuition: for every node, consider it as the "peak" of a path, and compute the max downward path sum from its left and right children separately (recomputing from scratch for every candidate peak — highly redundant work).

```
public int maxPathSumBrute(TreeNode root) {
    int best = Integer.MIN_VALUE;
    // pseudo-brute: for every node as potential "peak", recompute best downward
path in each subtree independently (O(n) per node)
    // omitted full nested implementation for brevity — O(n^2) overall due to
repeated subtree recomputation
    return best; // conceptual; see optimal version for the real, working solution
}
```

Time: O(n²) conceptually (recomputing downward sums repeatedly without memoization). The practical, correct, and expected solution is the single-pass optimal version below — in an interview you'd state the brute-force idea verbally (redundant recomputation) and go straight to the optimal.

Optimal Approach (single DFS pass, track a global maximum) Intuition: define `maxDownPath(node)` = the maximum sum path starting at `node` and going down into **at most one** child direction (this is what a parent could legally chain onto). While computing this recursively, at each node also consider the "through this node" path that uses **both** children (a valid complete path, but one that can't be extended upward to the node's own parent) — update a global best with that value.

```
private int globalMax = Integer.MIN_VALUE;

public int maxPathSum(TreeNode root) {
    maxDownPath(root);
    return globalMax;
}

private int maxDownPath(TreeNode node) {
    if (node == null) return 0;
    int leftGain = Math.max(maxDownPath(node.left), 0);  // ignore negative
contributions
    int rightGain = Math.max(maxDownPath(node.right), 0);

    int priceThroughNode = node.val + leftGain + rightGain;
    globalMax = Math.max(globalMax, priceThroughNode);

    return node.val + Math.max(leftGain, rightGain); // can only extend one side
upward
}
```

Time: O(n) — single DFS pass, each node visited once. Space: O(h) recursion stack.

Dry Run: `[-10,9,20,null,null,15,7]`. Leaf 9: leftGain=0,rightGain=0,through=9, globalMax=9, returns 9. Leaf15: through=15,globalMax=15,returns15. Leaf7: through=7, globalMax stays15,returns7. Node20: leftGain=max(15,0)=15,rightGain=max(7,0)=7, through=20+15+7=42,globalMax=42, returns 20+max(15,7)=35. Root-10: leftGain=max(9,0)=9, rightGain=max(35,0)=35, through=-10+9+35=34, globalMax stays42 (42>34). Final answer `42`. ✅ (matches canonical LeetCode example: path 15→20→7)

Edge Cases: all negative values (e.g. single node `[-3]` → answer `-3`, handled since `leftGain`/`rightGain` are clamped to 0 but the node's own value is always included in `priceThroughNode`). Single node tree → answer is that node's value.

Follow-up: "Diameter of Binary Tree" (below) uses the near-identical "combine both children's downward results at each node, track a global best" template but with path *length* instead of *value sum*.

A6.11 Serialize and Deserialize Binary Tree

Difficulty: Hard | **Pattern:** Tree DFS/BFS + string encoding (Design) | **Companies:** Amazon, Google, Microsoft, Meta — very frequently asked design question

Problem Statement: Design an algorithm to serialize a binary tree to a string and deserialize that string back to the exact same tree structure.

Clarifying Questions: Must handle arbitrary integer values, including negatives, and null children explicitly (so the shape can be reconstructed unambiguously). Any specific format required? (No, self-consistent format is fine — just needs `serialize` and `deserialize` to be inverses.)

Approach (preorder DFS with explicit null markers) Intuition: a preorder traversal (root, left, right) can be *uniquely* reconstructed **if and only if** we also explicitly record `null` children (otherwise `[1,2]` could ambiguously mean "2 is left child" or "2 is right child"). Serialize as a comma-separated preorder string with `"#"` for nulls; deserialize by consuming tokens in the same preorder order using a queue/index pointer.

```
public class Codec {
    public String serialize(TreeNode root) {
        StringBuilder sb = new StringBuilder();
        serializeHelper(root, sb);
        return sb.toString();
    }
    private void serializeHelper(TreeNode node, StringBuilder sb) {
        if (node == null) { sb.append("#,"); return; }
        sb.append(node.val).append(",");
        serializeHelper(node.left, sb);
        serializeHelper(node.right, sb);
    }

    public TreeNode deserialize(String data) {
        Queue<String> tokens = new LinkedList<>(Arrays.asList(data.split(",")));
        return deserializeHelper(tokens);
    }
    private TreeNode deserializeHelper(Queue<String> tokens) {
        String token = tokens.poll();
        if (token.equals("#")) return null;
        TreeNode node = new TreeNode(Integer.parseInt(token));
        node.left = deserializeHelper(tokens);
        node.right = deserializeHelper(tokens);
        return node;
    }
}
```

Time: O(n) for both serialize and deserialize. Space: O(n) for the string/queue, O(h) recursion.

Alternative Approach (level-order / BFS serialization) Intuition: same null-marker idea but level by level using a queue — more closely resembles how LeetCode itself displays trees (`[1,2,3,null,null,4,5]`), useful to mention as an alternative format.

Dry Run: tree `[1,2,3,null,null,4,5]`. Serialize (preorder): `"1,2,#,#,3,4,#,#,5,#,#,"`. Deserialize consumes tokens in the same order, rebuilding identical structure. Re-serializing the result gives the same string → round-trip verified. ✅

Edge Cases: empty tree (`root == null`) → serializes to `"#,"`, deserializes back to `null`. Single node → `"5,#,#,"`.

Follow-up: "Serialize and Deserialize N-ary Tree" / "BST" (a BST version can be more compact — since BST structure is derivable from a preorder sequence of values alone via bounds, no null markers needed at all).

A6.12 Diameter of Binary Tree

Difficulty: Easy | **Pattern:** Tree DFS with global state | **Companies:** Facebook, Amazon — deceptively simple, common trap for beginners

Problem Statement: Return the length (number of edges) of the longest path between any two nodes in the tree; the path may or may not pass through the root.

Clarifying Questions: Diameter measured in edges, not nodes (a common ambiguity — confirm with the interviewer; LeetCode's definition is edges).

Brute Force Approach Intuition: for every node, compute the diameter "through" that node as `depth(left) + depth(right)`, recomputing depth from scratch for every node — very redundant.

```
public int diameterOfBinaryTreeBrute(TreeNode root) {
    if (root == null) return 0;
    int throughRoot = depth(root.left) + depth(root.right);
    return Math.max(throughRoot, Math.max(diameterOfBinaryTreeBrute(root.left),
diameterOfBinaryTreeBrute(root.right)));
}
private int depth(TreeNode node) {
    if (node == null) return 0;
```

```
    return 1 + Math.max(depth(node.left), depth(node.right));
}
```

Time: O(n²) worst case (skewed tree causes `depth()` to be recomputed redundantly at every level). Space: O(h) recursion stack.

Optimal Approach (single pass, compute depth and diameter simultaneously)
Intuition: while computing each node's depth recursively (needed anyway), also update a global "best diameter seen so far" using `leftDepth + rightDepth` at that node — avoiding any repeated work.

```
private int diameter = 0;

public int diameterOfBinaryTree(TreeNode root) {
    depth(root);
    return diameter;
}
private int depth(TreeNode node) {
    if (node == null) return 0;
    int leftDepth = depth(node.left);
    int rightDepth = depth(node.right);
    diameter = Math.max(diameter, leftDepth + rightDepth);
    return 1 + Math.max(leftDepth, rightDepth);
}
```

Time: O(n) — single pass, each node visited once. Space: O(h) recursion stack.

Dry Run: `[1,2,3,4,5]` (2 and 3 are children of 1; 4,5 children of 2). depth(4)=1, depth(5)=1 → at node2: diameter=max(0,1+1)=2, returns 1+1=2. depth(3)=1 → at node1: diameter=max(2, 2+1)=3, returns 1+max(2,1)=3. Final diameter = `3` (path 4-2-1-3 or 5-2-1-3, 3 edges). ✅

Edge Cases: single node → diameter 0 (no edges at all). Completely skewed (linked-list- like) tree → diameter = n-1 (the whole chain).

Follow-up: This exact "combine children results at each node into a global answer, return a slightly different value upward" pattern also solves Binary Tree Maximum Path Sum (above) — recognize both as the same DFS-with-global-state template.

A6.13 Balanced Binary Tree

Difficulty: Easy | **Pattern:** Tree DFS with early termination | **Companies:** Amazon, Microsoft

Problem Statement: Determine if a binary tree is height-balanced — every node's left and right subtree heights differ by at most 1.

Brute Force Approach Intuition: for every node, compute the height of its left and right subtrees independently (recomputing from scratch) and check the difference; recurse.

```
public boolean isBalancedBrute(TreeNode root) {
    if (root == null) return true;
    int leftHeight = height(root.left);
    int rightHeight = height(root.right);
    if (Math.abs(leftHeight - rightHeight) > 1) return false;
    return isBalancedBrute(root.left) && isBalancedBrute(root.right);
}
private int height(TreeNode node) {
    if (node == null) return 0;
    return 1 + Math.max(height(node.left), height(node.right));
}
```

Time: O(n²) worst case — same redundant-recomputation issue as the Diameter brute force. Space: O(h).

Optimal Approach (bottom-up single pass, -1 as a "failure" sentinel) Intuition: compute height bottom-up; the moment we detect an imbalance anywhere in a subtree, propagate a sentinel value (e.g. `-1`) all the way up to short-circuit remaining work instead of continuing to compute meaningless heights.

```
public boolean isBalanced(TreeNode root) {
    return checkHeight(root) != -1;
}
private int checkHeight(TreeNode node) {
    if (node == null) return 0;
    int leftHeight = checkHeight(node.left);
    if (leftHeight == -1) return -1;
    int rightHeight = checkHeight(node.right);
    if (rightHeight == -1) return -1;
    if (Math.abs(leftHeight - rightHeight) > 1) return -1;
    return 1 + Math.max(leftHeight, rightHeight);
}
```

Time: O(n) — single pass, each node visited once, imbalance short-circuits upward immediately. Space: O(h) recursion stack.

Dry Run: balanced tree `[3,9,20,null,null,15,7]`: leaf heights compute normally up to root without ever hitting -1 → `true`. Unbalanced tree (long chain on one side making depth difference > 1 somewhere) → the first detected violation returns -1, which propagates straight to the top → `false`. ✅

Edge Cases: empty tree → balanced (`true`) trivially. Single node → balanced.

Follow-up: "Why is the brute force O(n²)?" — be ready to explain with a skewed-tree example (each of n nodes triggers an O(n) height computation on its subtree → O(n²) total), and how the sentinel value converts this into a single O(n) pass.

A6.14 Path Sum

Difficulty: Easy | **Pattern:** Tree DFS (root-to-leaf) | **Companies:** Amazon, Microsoft

Problem Statement: Given a binary tree and `targetSum`, return true if there is a root-to-**leaf** path such that the sum of values along the path equals `targetSum`.

Clarifying Questions: Must the path end at a leaf (not just any node)? (Yes — a very common mistake is stopping early at an internal node with a matching partial sum.) Can node values be negative? (Yes — so you can't prune purely based on "sum already exceeds target.")

Brute Force Approach Intuition: for every root-to-leaf path, compute the full sum, collect all sums, check if target is among them.

```
public boolean hasPathSumBrute(TreeNode root, int targetSum) {
    List<Integer> allSums = new ArrayList<>();
    collectPathSums(root, 0, allSums);
    return allSums.contains(targetSum);
}
private void collectPathSums(TreeNode node, int currentSum, List<Integer> allSums)
{
    if (node == null) return;
    currentSum += node.val;
    if (node.left == null && node.right == null) { allSums.add(currentSum); return;
}
    collectPathSums(node.left, currentSum, allSums);
    collectPathSums(node.right, currentSum, allSums);
}
```

Time: O(n). Space: O(n) for storing all path sums (unnecessary extra memory).

Optimal Approach (early-return DFS, subtract as you go) Intuition: at each node, subtract its value from the remaining target; at a leaf, check if the remaining target is exactly 0. Return `true`/`false` immediately without collecting anything.

```
public boolean hasPathSum(TreeNode root, int targetSum) {
    if (root == null) return false;
    if (root.left == null && root.right == null) {
        return targetSum == root.val;
    }
    int remaining = targetSum - root.val;
    return hasPathSum(root.left, remaining) || hasPathSum(root.right, remaining);
}
```

Time: O(n) worst case, but short-circuits as soon as a valid path is found. Space: O(h) recursion stack, no extra output storage.

Dry Run: `[5,4,8,11,null,13,4,7,2,null,null,null,1], targetSum=22`. Path 5→4→11→2 sums to 22 → found at that leaf, returns `true` immediately, no need to explore the rest of the tree. ✅

Edge Cases: empty tree → `false` (no nodes at all, so no path exists — note: LeetCode defines this specific edge case as false even though targetSum could coincidentally be 0). Single node tree → check if `root.val == targetSum`.

Follow-up: "Path Sum II" (return all root-to-leaf paths summing to target, not just a boolean) — same DFS but backtrack a running list of node values. "Path Sum III" (path can start and end anywhere, not just root-to-leaf) — solved with prefix-sum + HashMap, a fundamentally different technique (covered in Part B).

A7. Tries

A7.1 Implement Trie (Prefix Tree)

Difficulty: Medium | **Pattern:** Trie (Design) | **Companies:** Google, Amazon, Microsoft — foundational for autocomplete/search features

Problem Statement: Implement a Trie with `insert(word)`, `search(word)` (exact match), and `startsWith(prefix)` (prefix match).

Brute Force Approach Intuition: keep a `List<String>` of all inserted words; `search` does a linear `list.contains(word)`, `startsWith` scans every word checking `word.startsWith(prefix)`.

```
class TrieBrute {
    private List<String> words = new ArrayList<>();
    public void insert(String word) { words.add(word); }
    public boolean search(String word) { return words.contains(word); }
    public boolean startsWith(String prefix) {
        for (String w : words) if (w.startsWith(prefix)) return true;
        return false;
    }
}
```

Time: O(n·L) per `search`/`startsWith` call (n = number of words, L = word length) — far too slow if these are called frequently. Space: O(n·L).

Optimal Approach (actual Trie / prefix tree) Intuition: each node holds an array (or map) of 26 children (one per lowercase letter) and an `isEnd` flag. Inserting/searching walks character by character, creating/following child nodes — shared prefixes across words are stored only once, which is exactly why prefix lookups become O(L) instead of O(n·L).

```
class Trie {
    class TrieNode {
        TrieNode[] children = new TrieNode[26];
        boolean isEnd = false;
    }
    private final TrieNode root = new TrieNode();

    public void insert(String word) {
        TrieNode node = root;
        for (char c : word.toCharArray()) {
            int idx = c - 'a';
            if (node.children[idx] == null) node.children[idx] = new TrieNode();
            node = node.children[idx];
        }
        node.isEnd = true;
    }

    public boolean search(String word) {
        TrieNode node = traverse(word);
        return node != null && node.isEnd;
    }

    public boolean startsWith(String prefix) {
        return traverse(prefix) != null;
    }

    private TrieNode traverse(String s) {
```

```
        TrieNode node = root;
        for (char c : s.toCharArray()) {
            int idx = c - 'a';
            if (node.children[idx] == null) return null;
            node = node.children[idx];
        }
        return node;
    }
}
```

Time: O(L) for `insert`, `search`, and `startsWith` (L = length of the word/prefix) — completely independent of how many words are stored. Space: O(total characters across all inserted words) in the worst case (no shared prefixes), but typically much less due to prefix sharing.

Dry Run: insert("apple"). search("apple")→true. search("app")→traverse succeeds but `isEnd` is false at that node→false. startsWith("app")→traverse succeeds (node exists)→true. ✅

Edge Cases: empty string insert/search (if allowed, `isEnd` on the root itself). Searching for a prefix that was never inserted as a full word but exists as a prefix of another word (exactly what distinguishes `search` from `startsWith`).

Follow-up: "Add and Search Words Data Structure" (next) extends this with wildcard `.` support. "Word Search II" (next) combines a Trie with grid DFS for multi-word search.

A7.2 Design Add and Search Words Data Structure

Difficulty: Medium | **Pattern:** Trie + DFS for wildcards | **Companies:** Google, Amazon

Problem Statement: Design a data structure supporting `addWord(word)` and `search(word)`, where `search` can contain `.` wildcards matching any single character.

Brute Force Approach Intuition: store all words in a `List<String>`; for `search`, compare the pattern against every stored word character-by-character (treating `.` as a wildcard match).

```
class WordDictionaryBrute {
    private List<String> words = new ArrayList<>();
    public void addWord(String word) { words.add(word); }
    public boolean search(String word) {
```

```
        for (String w : words) {
            if (w.length() != word.length()) continue;
            boolean match = true;
            for (int i = 0; i < w.length(); i++) {
                if (word.charAt(i) != '.' && word.charAt(i) != w.charAt(i)) { match
= false; break; }
            }
            if (match) return true;
        }
        return false;
    }
}
```

Time: O(n·L) per search. Space: O(n·L).

Optimal Approach (Trie + DFS to branch on wildcards) Intuition: same Trie structure as A7.1; but `search` needs DFS because a `.` might need to try **all 26** children at that position — a plain iterative traversal isn't enough once wildcards are involved.

```
class WordDictionary {
    class TrieNode {
        TrieNode[] children = new TrieNode[26];
        boolean isEnd = false;
    }
    private final TrieNode root = new TrieNode();

    public void addWord(String word) {
        TrieNode node = root;
        for (char c : word.toCharArray()) {
            int idx = c - 'a';
            if (node.children[idx] == null) node.children[idx] = new TrieNode();
            node = node.children[idx];
        }
        node.isEnd = true;
    }

    public boolean search(String word) {
        return dfs(word, 0, root);
    }
    private boolean dfs(String word, int index, TrieNode node) {
        if (node == null) return false;
        if (index == word.length()) return node.isEnd;
        char c = word.charAt(index);
        if (c == '.') {
            for (TrieNode child : node.children) {
                if (dfs(word, index + 1, child)) return true;
            }
            return false;
        } else {
            return dfs(word, index + 1, node.children[c - 'a']);
        }
    }
}
```

Time: `addWord` O(L). `search` O(26^d · L) worst case where d = number of dots (exponential only in the number of wildcards, which is typically small) — in the common case with few or no dots, effectively O(L). Space: O(total characters stored).

Dry Run: addWord("bad"), addWord("dad"), addWord("mad"). search(".ad"): dfs branches all 26 children at index0; only 'b','d','m' exist as children → tries each, each successfully matches "ad" → returns `true` on first success. ✅

Edge Cases: search pattern longer/shorter than any stored word (naturally returns false — Trie traversal runs out of nodes or `index==word.length()` check fails at a non-`isEnd` node). All-wildcard pattern (`"..."`) — explores every stored word of that length.

Follow-up: "Word Search II" (below) needs the same Trie DFS branching idea but simultaneously searched against a 2-D grid.

A7.3 Word Search II

Difficulty: Hard | **Pattern:** Trie + Grid DFS/Backtracking | **Companies:** Google, Amazon, Airbnb — classic "combine two data structures" hard question

Problem Statement: Given an `m x n` board of characters and a list of words, return all words that can be formed by a sequence of adjacent cells (horizontally/vertically, no cell reused within a single word).

Clarifying Questions: Can the same board cell be used twice within one word's path? (No.) Across different words? (Yes, each word search starts fresh.)

Brute Force Approach Intuition: for each word in the list independently, run a standard grid DFS/backtracking search (like "Word Search I") to check if it can be formed.

```
public List<String> findWordsBrute(char[][] board, String[] words) {
    List<String> result = new ArrayList<>();
    for (String word : words) {
        if (existsInBoard(board, word)) result.add(word);
    }
    return result;
}
// existsInBoard(...) = standard single-word DFS backtracking, O(m*n*4^L) per word
```

Time: O(W · m·n·4^L) where W = number of words, L = average word length — re-scans the entire board from scratch for every single word, extremely redundant when words share prefixes. Space: O(L) recursion per search.

Optimal Approach (build a Trie of all words, DFS the board once) Intuition: insert all words into a single Trie. DFS from every board cell simultaneously, walking the Trie alongside the DFS path — when a Trie node's `isEnd` is true, we've found a word (record it and mark it, e.g., with a sentinel, to avoid duplicate reporting). This explores shared prefixes across all words only once instead of once per word.

```
class TrieNode {
    TrieNode[] children = new TrieNode[26];
    String word = null; // stores the complete word at a terminal node (null
otherwise)
}

public List<String> findWords(char[][] board, String[] words) {
    TrieNode root = new TrieNode();
    for (String word : words) {
        TrieNode node = root;
        for (char c : word.toCharArray()) {
            int idx = c - 'a';
            if (node.children[idx] == null) node.children[idx] = new TrieNode();
            node = node.children[idx];
        }
        node.word = word;
    }

    List<String> result = new ArrayList<>();
    int rows = board.length, cols = board[0].length;
    for (int r = 0; r < rows; r++) {
        for (int c = 0; c < cols; c++) {
            dfs(board, r, c, root, result);
        }
    }
    return result;
}

private void dfs(char[][] board, int r, int c, TrieNode node, List<String> result)
{
    if (r < 0 || r >= board.length || c < 0 || c >= board[0].length || board[r][c]
== '#') return;
    char ch = board[r][c];
    TrieNode next = node.children[ch - 'a'];
    if (next == null) return;

    if (next.word != null) {
        result.add(next.word);
        next.word = null; // avoid duplicate additions if the same word is
reachable again
    }

    board[r][c] = '#'; // mark visited
```

```
        dfs(board, r + 1, c, next, result);
        dfs(board, r - 1, c, next, result);
        dfs(board, r, c + 1, next, result);
        dfs(board, r, c - 1, next, result);
        board[r][c] = ch; // restore (backtrack)
    }
```

Time: O(m·n·4^L) total (one shared board traversal for all words instead of W separate ones), where the Trie pruning cuts off dead-end branches early (much better in practice than the raw bound suggests). Space: O(total characters across all words) for the Trie + O(L) recursion.

Dry Run: words=["oath","pea","eat","rain"], board contains "oath" spelled out adjacently. DFS from the 'o' cell walks the Trie: o→a→t→h, hits a node with `word="oath"` → added to result, that Trie node's `word` cleared to prevent re-adding. Continues exploring other start cells for "pea","eat","rain" similarly. ✅

Edge Cases: word list contains duplicates → the `next.word = null` guard prevents the same found word being added twice. No words found at all → empty result list. Single-letter words.

Follow-up: "Why build one Trie instead of searching each word separately?" — be ready to explain the shared-prefix pruning benefit explicitly (e.g., "oath" and "oat" share 3 characters of Trie traversal work).

A8. Heap / Priority Queue

A8.1 Kth Largest Element in an Array

Difficulty: Medium | **Pattern:** Heap / Quickselect | **Companies:** Facebook, Amazon, Microsoft — extremely common

Problem Statement: Find the kth largest element in an unsorted array (kth largest in sorted order, not the kth distinct value).

Example: `[3,2,1,5,6,4], k=2` → `5`.

Brute Force Approach Intuition: sort the array, return the element at index `n-k`.

```
public int findKthLargestBrute(int[] nums, int k) {
    Arrays.sort(nums);
    return nums[nums.length - k];
}
```

Time: O(n log n). Space: O(log n) to O(n) depending on sort implementation.

Better Approach (Min-Heap of size k) Intuition: maintain a min-heap of size k; the heap's smallest element (top) is always the kth largest overall once the heap is full, because we've kept only the k largest elements seen so far, and the smallest of *those* is exactly the kth largest.

```
public int findKthLargestHeap(int[] nums, int k) {
    PriorityQueue<Integer> minHeap = new PriorityQueue<>();
    for (int n : nums) {
        minHeap.offer(n);
        if (minHeap.size() > k) minHeap.poll();
    }
    return minHeap.peek();
}
```

Time: O(n log k) — much better than O(n log n) when k is small. Space: O(k).

Optimal Approach (Quickselect — average O(n)) Intuition: this is exactly the same partitioning idea as Quicksort, but after partitioning around a pivot we only need to **recurse into the one side that contains our target rank** — never both — which drops the average complexity from O(n log n) to O(n).

```
public int findKthLargest(int[] nums, int k) {
    int targetIndex = nums.length - k; // kth largest = (n-k)th smallest in sorted
order
    return quickSelect(nums, 0, nums.length - 1, targetIndex);
}
private int quickSelect(int[] nums, int left, int right, int targetIndex) {
    int pivot = nums[right];
    int p = left;
    for (int i = left; i < right; i++) {
        if (nums[i] <= pivot) {
            swap(nums, i, p);
            p++;
        }
    }
    swap(nums, p, right);
    if (p == targetIndex) return nums[p];
    else if (p < targetIndex) return quickSelect(nums, p + 1, right, targetIndex);
    else return quickSelect(nums, left, p - 1, targetIndex);
}
```

```
private void swap(int[] nums, int i, int j) { int t = nums[i]; nums[i] = nums[j];
nums[j] = t; }
```

Time: O(n) average case, O(n²) worst case (bad pivot choices, e.g. already-sorted input with last-element pivoting — mitigated in practice with random pivot selection). Space: O(log n) average recursion, O(1) if implemented iteratively.

Dry Run: `[3,2,1,5,6,4], k=2` → targetIndex = 6-2 = 4. Partitioning around pivot=4 (last element) rearranges the array so index 4 ends up holding the value that belongs there in sorted order; after one or more partition rounds, `nums[4]` settles to `5` (the 2nd largest). ✅

Edge Cases: `k=1` → the maximum element. `k=n` → the minimum element. All duplicate values → works correctly since partitioning doesn't require strict inequalities to function.

Follow-up: "Kth Smallest Element in a Sorted Matrix", "Top K Frequent Elements" (next) are close cousins — recognize "maintain a heap of size k" as a broadly reusable idea whenever you need the top/bottom k of something streamed or scanned once.

A8.2 Top K Frequent Elements

Difficulty: Medium | **Pattern:** Heap / Bucket Sort | **Companies:** Amazon, Facebook, Yelp

Problem Statement: Given an integer array `nums` and integer `k`, return the `k` most frequent elements.

Example: `[1,1,1,2,2,3], k=2` → `[1,2]`.

Brute Force Approach Intuition: count frequencies with a HashMap, sort the entries by frequency descending, take the top k.

```
public int[] topKFrequentBrute(int[] nums, int k) {
    Map<Integer, Integer> freq = new HashMap<>();
    for (int n : nums) freq.merge(n, 1, Integer::sum);
    return freq.entrySet().stream()
        .sorted((a, b) -> b.getValue() - a.getValue())
        .limit(k)
        .mapToInt(Map.Entry::getKey)
        .toArray();
}
```

Time: O(n log n) dominated by sorting all distinct elements by frequency. Space: O(n).

Better Approach (Min-Heap of size k) Intuition: same "keep only the best k seen so far" idea as Kth Largest Element — maintain a min-heap ordered by frequency, capped at size k.

```
public int[] topKFrequentHeap(int[] nums, int k) {
    Map<Integer, Integer> freq = new HashMap<>();
    for (int n : nums) freq.merge(n, 1, Integer::sum);

    PriorityQueue<int[]> minHeap = new PriorityQueue<>((a, b) -> a[1] - b[1]);
    for (Map.Entry<Integer, Integer> entry : freq.entrySet()) {
        minHeap.offer(new int[]{entry.getKey(), entry.getValue()});
        if (minHeap.size() > k) minHeap.poll();
    }
    int[] result = new int[k];
    for (int i = k - 1; i >= 0; i--) result[i] = minHeap.poll()[0];
    return result;
}
```

Time: O(n log k). Space: O(n) for the frequency map + O(k) for the heap.

Optimal Approach (Bucket Sort — O(n)) Intuition: frequency of any element is bounded by `n` (array length). Create `n+1` buckets indexed by frequency; place each distinct value into `buckets[freq]`. Then simply scan buckets from highest frequency down, collecting values until we have k — no comparison-based sort needed at all.

```
public int[] topKFrequent(int[] nums, int k) {
    Map<Integer, Integer> freq = new HashMap<>();
    for (int n : nums) freq.merge(n, 1, Integer::sum);

    List<Integer>[] buckets = new List[nums.length + 1];
    for (Map.Entry<Integer, Integer> entry : freq.entrySet()) {
        int f = entry.getValue();
        if (buckets[f] == null) buckets[f] = new ArrayList<>();
        buckets[f].add(entry.getKey());
    }

    int[] result = new int[k];
    int idx = 0;
    for (int f = buckets.length - 1; f >= 0 && idx < k; f--) {
        if (buckets[f] == null) continue;
        for (int val : buckets[f]) {
            if (idx == k) break;
            result[idx++] = val;
        }
    }
    return result;
}
```

Time: O(n) — building the frequency map is O(n), building buckets is O(n), scanning buckets is O(n) total. Space: O(n) for the map and buckets.

Dry Run: `[1,1,1,2,2,3], k=2`. freq: 1→3, 2→2, 3→1. buckets[3]=[1], buckets[2]=[2], buckets[1]=[3]. Scan from highest frequency: bucket3→take 1(idx=1). bucket2→take 2(idx=2=k, stop). Result `[1,2]`. ✅

Edge Cases: `k` equals number of distinct elements → returns all of them. All elements have the same frequency → any k of them is a valid answer (order among ties usually unspecified).

Follow-up: "Top K Frequent Words" (tie-break alphabetically — needs a custom comparator). "Sort Characters By Frequency" — same bucket-sort idea applied to characters.

A8.3 Find Median from Data Stream

Difficulty: Hard | **Pattern:** Two Heaps (Design) | **Companies:** Google, Amazon, LinkedIn — classic "two heaps" design question

Problem Statement: Design a data structure that supports adding integers one at a time (`addNum`) and returning the median of all elements added so far (`findMedian`) at any point.

Clarifying Questions: Must `findMedian` be O(1) or is O(log n)/O(n) acceptable? (The expected answer is O(log n) `addNum` and O(1) `findMedian`.)

Brute Force Approach Intuition: keep an unsorted (or sorted-on-demand) list; `addNum` just appends, `findMedian` sorts and picks the middle.

```
class MedianFinderBrute {
    private List<Integer> nums = new ArrayList<>();
    public void addNum(int num) { nums.add(num); }
    public double findMedian() {
        List<Integer> sorted = new ArrayList<>(nums);
        Collections.sort(sorted);
        int n = sorted.size();
        return n % 2 == 1 ? sorted.get(n / 2) : (sorted.get(n / 2 - 1) +
sorted.get(n / 2)) / 2.0;
    }
}
```

Time: `addNum` O(1), `findMedian` O(n log n) — recomputed from scratch every call, very slow if called frequently. Space: O(n).

Optimal Approach (Two Heaps: a max-heap for the lower half, a min-heap for the upper half) Intuition: split all numbers seen so far into two halves — a max-heap holding the smaller half (so its top is the largest of the small numbers) and a min-heap holding the larger half (so its top is the smallest of the large numbers). Keep the two heaps balanced in size (differ by at most 1). The median is then either the top of the larger heap (odd total count) or the average of both tops (even total count) — both O(1) peeks.

```
class MedianFinder {
    private PriorityQueue<Integer> maxHeapLower = new PriorityQueue<>
(Collections.reverseOrder()); // smaller half
    private PriorityQueue<Integer> minHeapUpper = new PriorityQueue<>(); // larger
half

    public void addNum(int num) {
        maxHeapLower.offer(num);
        minHeapUpper.offer(maxHeapLower.poll()); // ensure every element in lower
<= every element in upper
        if (minHeapUpper.size() > maxHeapLower.size()) {
            maxHeapLower.offer(minHeapUpper.poll()); // rebalance sizes
        }
    }

    public double findMedian() {
        if (maxHeapLower.size() > minHeapUpper.size()) {
            return maxHeapLower.peek();
        }
        return (maxHeapLower.peek() + minHeapUpper.peek()) / 2.0;
    }
}
```

Time: `addNum` O(log n) (heap insert/extract). `findMedian` O(1). Space: O(n) total across both heaps.

Dry Run: add 5: maxHeapLower=[5]→minHeapUpper.offer(5)→minHeapUpper=[5]; sizes equal (1,1) no rebalance needed... wait maxHeapLower became empty after poll, need to re-add: actually flow is: offer 5 to maxHeapLower→[5]; poll it (5) and offer to minHeapUpper→[5]; now maxHeapLower=[] size0, minHeapUpper=[5] size1 > 0 → rebalance: poll minHeapUpper(5)→maxHeapLower=[5]. Final state: maxHeapLower=[5], minHeapUpper=[]. findMedian: sizes(1,0)→maxHeapLower bigger→return 5. Add 15: maxHeapLower.offer(15)→[15,5](max-heap top=15); poll(15)→minHeapUpper=[15]; sizes(1,1) equal, no rebalance. findMedian: sizes equal→avg(5,15)=10. ✅

(matches expected median stream behavior: after [5]→median 5; after [5,15]→median 10)

Edge Cases: stream of size 1 → median is that single element. All identical values → median is that value regardless of heap balancing.

Follow-up: "What if 99% of the integers are in a small range [0,100]?" — use counting-sort- style buckets instead of heaps for O(1) `addNum` and O(1) `findMedian` in that special case (a common "constraints changed" follow-up).

A8.4 Task Scheduler

Difficulty: Medium | **Pattern:** Greedy + Heap (or Math) | **Companies:** Facebook, Amazon

Problem Statement: Given a char array of tasks and a cooldown `n` (must wait `n` intervals before repeating the same task type), return the minimum number of intervals (including idle slots) needed to finish all tasks.

Example: `tasks=["A","A","A","B","B","B"], n=2` → `8` (e.g. A B idle A B idle A B).

Clarifying Questions: Can idle slots be filled with other ready tasks instead of literally idling? (Yes, implicitly — the goal is to minimize total time, so schedule any available task before idling.)

Brute Force Approach (simulate with a max-heap, round by round) Intuition: repeatedly pick the most frequent remaining task type to run next (greedy), track a cooldown map of when each task type becomes available again; simulate time unit by unit.

```
public int leastIntervalSimulate(char[] tasks, int n) {
    Map<Character, Integer> freq = new HashMap<>();
    for (char t : tasks) freq.merge(t, 1, Integer::sum);
    PriorityQueue<Integer> maxHeap = new PriorityQueue<>
(Collections.reverseOrder());
    maxHeap.addAll(freq.values());

    int time = 0;
    Queue<int[]> cooldownQueue = new LinkedList<>(); // [count, availableAtTime]
    while (!maxHeap.isEmpty() || !cooldownQueue.isEmpty()) {
        time++;
        if (!maxHeap.isEmpty()) {
            int count = maxHeap.poll() - 1;
            if (count > 0) cooldownQueue.offer(new int[]{count, time + n});
```

```
            }
            if (!cooldownQueue.isEmpty() && cooldownQueue.peek()[1] == time) {
                maxHeap.offer(cooldownQueue.poll()[0]);
            }
        }
        return time;
    }
```

Time: O(total time units × log(distinct tasks)) — simulates every single interval explicitly, correct but does more work than necessary. Space: O(distinct task types).

Optimal Approach (closed-form math formula) Intuition: the task with the **highest frequency** `maxFreq` determines the skeleton of the schedule: it needs `maxFreq - 1` "gaps" of size `n+1` after it, plus 1 for itself. Any other task with the same `maxFreq` needs an extra slot in the last gap. If there are enough total tasks to fill in all the gaps with no idle time at all, the answer is simply `tasks.length`.

```
public int leastInterval(char[] tasks, int n) {
    int[] freq = new int[26];
    for (char t : tasks) freq[t - 'A']++;
    Arrays.sort(freq);
    int maxFreq = freq[25];
    int idleSlots = (maxFreq - 1) * n;

    for (int i = 24; i >= 0 && freq[i] > 0; i--) {
        idleSlots -= Math.min(maxFreq - 1, freq[i]);
    }
    idleSlots = Math.max(0, idleSlots);
    return tasks.length + idleSlots;
}
```

Time: O(t + 26 log 26) ≈ O(t) where t = tasks.length (dominated by the initial frequency count; sorting 26 elements is constant time). Space: O(1) — fixed 26-size array.

Dry Run: `["A","A","A","B","B","B"], n=2`. freq sorted: [...,3,3] (A=3,B=3). maxFreq=3, idleSlots=(3-1)*2=4. Loop over other frequencies with same value 3 (B): subtract min(3-1,3)=2 → idleSlots=4-2=2. Final idleSlots=max(0,2)=2. Answer = 6 (tasks) + 2 = `8`. ✅

Edge Cases: `n=0` (no cooldown needed) → answer is simply `tasks.length`. Only one distinct task type → answer = `(freq-1)*(n+1) + 1`. Enough task variety to fill every gap → idle slots computed as 0, answer = `tasks.length`.

Follow-up: "What if idle slots must be truly empty (can't be filled by other ready tasks)?" — that's actually the exact model of the problem already (idle time is

unavoidable gap, not a scheduling choice) — good to articulate this distinction clearly.

A8.5 Meeting Rooms II

Difficulty: Medium | **Pattern:** Heap / Sweep Line (Chronological Ordering) | **Companies:** Facebook, Google, Amazon — extremely common "intervals" question in interviews (sometimes gated behind LeetCode premium, but ubiquitous in real interviews)

Problem Statement: Given meeting time intervals, find the minimum number of conference rooms required to hold all meetings without overlap.

Example: `[[0,30],[5,10],[15,20]]` → `2`.

Brute Force Approach Intuition: for every meeting, check how many other meetings overlap with it at its start time; the maximum such overlap count across all meetings is the answer.

```
public int minMeetingRoomsBrute(int[][] intervals) {
    int maxRooms = 0;
    for (int[] interval : intervals) {
        int count = 0;
        for (int[] other : intervals) {
            if (other[0] < interval[1] && interval[0] < other[1]) count++; //
overlap check
        }
        maxRooms = Math.max(maxRooms, count);
    }
    return maxRooms;
}
```

Time: O(n²). Space: O(1) extra.

Optimal Approach (Min-Heap of end times / Sweep Line) Intuition: sort meetings by start time. Use a min-heap of currently-occupied rooms' **end times**. For each new meeting, if the earliest-ending room (heap top) ends at or before this meeting's start, that room is now free — reuse it (pop and push the new end time). Otherwise allocate a new room (just push). The heap's size at any point represents rooms in use; its maximum size across the whole process is the answer.

```
public int minMeetingRooms(int[][] intervals) {
    if (intervals.length == 0) return 0;
```

```
    Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
    PriorityQueue<Integer> minHeap = new PriorityQueue<>(); // end times of rooms
in use
    minHeap.offer(intervals[0][1]);

    for (int i = 1; i < intervals.length; i++) {
        if (minHeap.peek() <= intervals[i][0]) {
            minHeap.poll(); // reuse the freed-up room
        }
        minHeap.offer(intervals[i][1]);
    }
    return minHeap.size();
}
```

Time: O(n log n) — sorting plus n heap operations. Space: O(n) for the heap.

Alternative Approach (Sweep line with separate sorted start/end arrays) Intuition: sort all start times and all end times separately. Use two pointers: advance the start pointer whenever a meeting begins (increment active room count), advance the end pointer whenever a meeting ends at or before that start (decrement). Track the maximum concurrent count.

```
public int minMeetingRoomsSweep(int[][] intervals) {
    int n = intervals.length;
    int[] starts = new int[n], ends = new int[n];
    for (int i = 0; i < n; i++) { starts[i] = intervals[i][0]; ends[i] =
intervals[i][1]; }
    Arrays.sort(starts);
    Arrays.sort(ends);

    int rooms = 0, maxRooms = 0, endPtr = 0;
    for (int startPtr = 0; startPtr < n; startPtr++) {
        while (endPtr < n && ends[endPtr] <= starts[startPtr]) {
            rooms--;
            endPtr++;
        }
        rooms++;
        maxRooms = Math.max(maxRooms, rooms);
    }
    return maxRooms;
}
```

Time: O(n log n). Space: O(n) for the separate arrays.

Dry Run: `[[0,30],[5,10],[15,20]]` sorted by start (already sorted). Heap=[30] (first meeting). i=1([5,10]): heap top 30 > 5 → no reuse, new room, heap=[10,30]. i=2([15,20]): heap top 10 <= 15 → reuse, pop 10, push 20 → heap=[20,30]. Final heap size = 2. ✅

Edge Cases: no meetings → 0 rooms. All meetings identical/fully overlapping → rooms needed = number of meetings. No meetings overlap at all → 1 room suffices.

Follow-up: "Meeting Rooms I" (can a single person attend all meetings — i.e., are there any overlaps at all) is the simpler special case (just check sorted intervals for any overlap). "Meeting Rooms III" adds room-specific constraints, a harder simulation+heap variant.

A9. Backtracking

A9.1 Subsets

Difficulty: Medium | **Pattern:** Backtracking (include/exclude) | **Companies:** Amazon, Facebook, Google — foundational backtracking question

Problem Statement: Given an array of unique integers, return all possible subsets (the power set).

Example: `[1,2,3]` → `[[],[1],[2],[1,2],[3],[1,3],[2,3],[1,2,3]]` (order flexible).

Brute Force / Iterative Bit-Manipulation Approach Intuition: each subset corresponds to a unique binary number from `0` to `2^n - 1`, where bit `i` set means "include `nums[i]`."

```
public List<List<Integer>> subsetsBitmask(int[] nums) {
    int n = nums.length;
    List<List<Integer>> result = new ArrayList<>();
    for (int mask = 0; mask < (1 << n); mask++) {
        List<Integer> subset = new ArrayList<>();
        for (int i = 0; i < n; i++) {
            if ((mask & (1 << i)) != 0) subset.add(nums[i]);
        }
        result.add(subset);
    }
    return result;
}
```

Time: O(n · 2^n) — 2^n subsets, O(n) to build each. Space: O(n · 2^n) for the output.

Optimal / Canonical Approach (Backtracking) Intuition: at each index, make a binary choice: include `nums[i]` in the current subset, or don't. Recurse, then backtrack (undo the choice) to explore the other branch.

```
public List<List<Integer>> subsets(int[] nums) {
    List<List<Integer>> result = new ArrayList<>();
    backtrack(nums, 0, new ArrayList<>(), result);
    return result;
}
private void backtrack(int[] nums, int index, List<Integer> current,
List<List<Integer>> result) {
    if (index == nums.length) {
        result.add(new ArrayList<>(current));
        return;
    }
    // choice 1: exclude nums[index]
    backtrack(nums, index + 1, current, result);
    // choice 2: include nums[index]
    current.add(nums[index]);
    backtrack(nums, index + 1, current, result);
    current.remove(current.size() - 1); // backtrack
}
```

Time: O(n · 2^n) — same asymptotic bound (unavoidable, since there are 2^n subsets each potentially of size n to copy), but the backtracking form generalizes far more easily to constrained variants (e.g. Subsets II with duplicates, Combination Sum). Space: O(n) recursion depth + O(n·2^n) output.

Dry Run: `[1,2]`. backtrack(idx0): exclude1→backtrack(idx1): exclude2→add[](base case); include2→add[2],recurse base adds[2],backtrack removes2. Back at idx0, include1→current=[1], backtrack(idx1): exclude2→add[1]; include2→current=[1,2]→add[1,2],backtrack removes2. current back to[1], backtrack removes1. Final results: `[[],[2],[1],[1,2]]`. ✅ (all 4 subsets of a 2-element set)

Edge Cases: empty input array → single subset, the empty set `[[]]`. Single element → `[[],[elem]]`.

Follow-up: "Subsets II" (array may contain duplicates, must avoid duplicate subsets in output) — sort first, then skip over consecutive duplicate values at the same recursion depth when making the "exclude" choice repeatedly.

A9.2 Combination Sum

Difficulty: Medium | **Pattern:** Backtracking (unbounded choice) | **Companies:** Amazon, Facebook, Snapchat

Problem Statement: Given an array of distinct positive integers and a target, return all unique combinations where chosen numbers sum to target. The same number may be chosen an unlimited number of times.

Example: `candidates=[2,3,6,7], target=7` → `[[2,2,3],[7]]`.

Brute Force Approach Intuition: at every index, try including `candidates[index]` zero or more times (looping), recursing further; since repeats are allowed, don't advance the index when including — only advance when moving past a candidate entirely, which naturally avoids permutation-duplicate results (e.g. `[2,3]` vs `[3,2]` counted once).

```
public List<List<Integer>> combinationSumBrute(int[] candidates, int target) {
    List<List<Integer>> result = new ArrayList<>();
    backtrack(candidates, target, 0, new ArrayList<>(), result);
    return result;
}
private void backtrack(int[] candidates, int remaining, int start, List<Integer>
current, List<List<Integer>> result) {
    if (remaining == 0) { result.add(new ArrayList<>(current)); return; }
    if (remaining < 0) return;
    for (int i = start; i < candidates.length; i++) {
        current.add(candidates[i]);
        backtrack(candidates, remaining - candidates[i], i, current, result); //
note: i, not i+1 — reuse allowed
        current.remove(current.size() - 1);
    }
}
```

Time: O(2^target) worst case (loose bound — actual complexity depends on candidate values and target, generally exponential in the size of the solution space). Space: O(target/min(candidate)) max recursion depth.

Optimal Approach (same backtracking, with an early-exit optimization via sorting) Intuition: sort `candidates` first; once `candidates[i] > remaining`, every subsequent candidate is also too large (since sorted) — break out of the loop early instead of checking every remaining candidate.

```
public List<List<Integer>> combinationSum(int[] candidates, int target) {
    Arrays.sort(candidates);
    List<List<Integer>> result = new ArrayList<>();
    backtrack(candidates, target, 0, new ArrayList<>(), result);
    return result;
```

```
}
private void backtrack(int[] candidates, int remaining, int start, List<Integer>
current, List<List<Integer>> result) {
    if (remaining == 0) { result.add(new ArrayList<>(current)); return; }
    for (int i = start; i < candidates.length; i++) {
        if (candidates[i] > remaining) break; // sorted array: no point checking
further
        current.add(candidates[i]);
        backtrack(candidates, remaining - candidates[i], i, current, result);
        current.remove(current.size() - 1);
    }
}
```

Time: still exponential in the worst case, but the sorted early-break meaningfully prunes the search tree in practice. Space: O(target / min(candidate)) recursion depth.

Dry Run: `[2,3,6,7], target=7`. Start with 2: remaining5→try2again:remaining3→try2again: remaining1→try2:remaining-1 invalid,try3:too big(3>1)break. Backtrack: remaining3,try3: remaining0→found[2,2,3]! Backtrack further up: remaining5,try3:remaining2,try3:too big break, try6:too big break (since sorted, break entirely)... continues exploring; also finds path starting fresh with 7: remaining0→found[7]. Final `[[2,2,3],[7]]`. ✅

Edge Cases: no combination sums to target → empty result list. `target` exactly equals one of the candidates → single-element combination found. Very small target smaller than the smallest candidate → empty result.

Follow-up: "Combination Sum II" (each number used at most once, array may have duplicates, must avoid duplicate combinations) — advance index by `i+1` (no reuse) and explicitly skip duplicate values at the same recursion depth. "Combination Sum III" (fixed count of numbers from 1-9 summing to target) — adds a count constraint to the same template.

A9.3 Permutations

Difficulty: Medium | **Pattern:** Backtracking (used/visited tracking) | **Companies:** Amazon, Microsoft, Google

Problem Statement: Given an array of distinct integers, return all possible permutations.

Example: `[1,2,3]` → all 6 orderings.

Brute Force Approach Intuition: recursively swap elements in place — at each recursion depth, try placing every remaining unused element at the current position by swapping it into place, recurse, then swap back (backtrack).

```
public List<List<Integer>> permuteSwap(int[] nums) {
    List<List<Integer>> result = new ArrayList<>();
    backtrackSwap(nums, 0, result);
    return result;
}
private void backtrackSwap(int[] nums, int start, List<List<Integer>> result) {
    if (start == nums.length) {
        List<Integer> perm = new ArrayList<>();
        for (int n : nums) perm.add(n);
        result.add(perm);
        return;
    }
    for (int i = start; i < nums.length; i++) {
        swap(nums, start, i);
        backtrackSwap(nums, start + 1, result);
        swap(nums, start, i); // backtrack
    }
}
private void swap(int[] nums, int i, int j) { int t = nums[i]; nums[i] = nums[j];
nums[j] = t; }
```

Time: O(n · n!) — n! permutations, O(n) to copy each into the result. Space: O(n) recursion depth (in-place swaps avoid extra array copies during the search itself).

Optimal / Alternative Approach (visited boolean array, build up a list) Intuition: equally valid and often considered cleaner to reason about — track which indices have been "used" in the current permutation being built, try every unused element at each position.

```
public List<List<Integer>> permute(int[] nums) {
    List<List<Integer>> result = new ArrayList<>();
    boolean[] used = new boolean[nums.length];
    backtrack(nums, used, new ArrayList<>(), result);
    return result;
}
private void backtrack(int[] nums, boolean[] used, List<Integer> current,
List<List<Integer>> result) {
    if (current.size() == nums.length) {
        result.add(new ArrayList<>(current));
        return;
    }
    for (int i = 0; i < nums.length; i++) {
        if (used[i]) continue;
        used[i] = true;
        current.add(nums[i]);
        backtrack(nums, used, current, result);
```

```
            current.remove(current.size() - 1);
            used[i] = false; // backtrack
        }
    }
```

Time: O(n · n!). Space: O(n) recursion depth + O(n) for the `used` array, O(n·n!) for output.

Dry Run: `[1,2,3]`: explores 1-2-3, 1-3-2, 2-1-3, 2-3-1, 3-1-2, 3-2-1 — all 6 permutations via the used-array approach, each fully explored then backtracked. ✅

Edge Cases: single element → one permutation (itself). Two elements → 2 permutations.

Follow-up: "Permutations II" (array may contain duplicates, avoid duplicate permutations in output) — sort first, and skip `nums[i] == nums[i-1] && !used[i-1]` at the same recursion depth to prevent generating the same permutation twice.

A9.4 Word Search

Difficulty: Medium | **Pattern:** Backtracking on a Grid (DFS) | **Companies:** Amazon, Microsoft, Bloomberg

Problem Statement: Given an `m x n` grid of characters and a word, return true if the word can be constructed from adjacent cells (horizontally/vertically), not reusing a cell within one path.

Brute Force / Direct Approach (this IS the standard/expected solution — DFS backtracking) Intuition: try starting the search from every cell; at each step, if the current cell matches the next needed character, mark it visited and recurse into all 4 neighbors; unmark (backtrack) before returning, so other starting points/paths aren't affected.

```
public boolean exist(char[][] board, String word) {
    int rows = board.length, cols = board[0].length;
    for (int r = 0; r < rows; r++) {
        for (int c = 0; c < cols; c++) {
            if (dfs(board, r, c, word, 0)) return true;
        }
    }
    return false;
}
private boolean dfs(char[][] board, int r, int c, String word, int index) {
```

```
        if (index == word.length()) return true;
        if (r < 0 || r >= board.length || c < 0 || c >= board[0].length || board[r][c]
    != word.charAt(index)) {
            return false;
        }
        char temp = board[r][c];
        board[r][c] = '#'; // mark visited
        boolean found = dfs(board, r + 1, c, word, index + 1)
                      || dfs(board, r - 1, c, word, index + 1)
                      || dfs(board, r, c + 1, word, index + 1)
                      || dfs(board, r, c - 1, word, index + 1);
        board[r][c] = temp; // backtrack
        return found;
    }
```

Time: O(m·n·4^L) worst case (m·n starting points, each exploring up to 4 directions per character of length L) — in practice heavily pruned by early mismatches. Space: O(L) recursion stack.

Optimization Note: this backtracking-with-in-place-marking approach is already essentially optimal for this problem — the "brute force vs optimal" distinction here is more about *how cleanly* you prune (early character mismatch check) than a fundamentally different algorithm. A meaningfully worse brute force would be generating all possible paths of length L first and then checking each against the word — strictly more wasteful with no benefit, so it's omitted.

Dry Run: `board=[["A","B","C","E"],["S","F","C","S"],["A","D","E","E"]]`, `word="ABCCED"`. DFS from (0,0)='A' matches word[0]. Explores right to (0,1)='B' matches word[1]. Continues (0,2)='C', then down to (1,2)='C', then down to (2,2)='E', then left to (2,1)='D' matches last character → full word matched → `true`. ✅

Edge Cases: word longer than total cells → immediately impossible (could add an early length check as a minor optimization). Single-cell board matching a single-character word.

Follow-up: "Word Search II" (A7.3) generalizes this to searching many words simultaneously using a Trie to avoid redundant re-scanning of the board per word.

A9.5 Palindrome Partitioning

Difficulty: Medium | **Pattern:** Backtracking + Palindrome check | **Companies:** Amazon, Facebook

Problem Statement: Given a string `s`, partition it such that every substring of the partition is a palindrome. Return all possible partitions.

Example: `"aab"` → `[["a","a","b"],["aa","b"]]`.

Brute Force Approach Intuition: at every position, try every possible "next cut" length, check if that substring is a palindrome (via a fresh O(L) check each time), recurse on the remainder.

```
public List<List<String>> partitionBrute(String s) {
    List<List<String>> result = new ArrayList<>();
    backtrack(s, 0, new ArrayList<>(), result);
    return result;
}
private void backtrack(String s, int start, List<String> current,
List<List<String>> result) {
    if (start == s.length()) { result.add(new ArrayList<>(current)); return; }
    for (int end = start + 1; end <= s.length(); end++) {
        String sub = s.substring(start, end);
        if (isPalindrome(sub)) {
            current.add(sub);
            backtrack(s, end, current, result);
            current.remove(current.size() - 1);
        }
    }
}
private boolean isPalindrome(String s) {
    int i = 0, j = s.length() - 1;
    while (i < j) if (s.charAt(i++) != s.charAt(j--)) return false;
    return true;
}
```

Time: O(n · 2^n) worst case for the backtracking tree, with an extra O(n) factor per palindrome check re-done from scratch each time → effectively O(n² · 2^n). Space: O(n) recursion depth.

Optimal Approach (precompute palindrome table with DP) Intuition: precompute `isPalindrome[i][j]` for all substrings using O(n²) DP (`isPalindrome[i][j] = s[i]==s[j] && isPalindrome[i+1][j-1]`), so each palindrome check during backtracking becomes O(1) instead of O(n).

```
public List<List<String>> partition(String s) {
    int n = s.length();
    boolean[][] dp = new boolean[n][n];
    for (int len = 1; len <= n; len++) {
        for (int i = 0; i + len - 1 < n; i++) {
            int j = i + len - 1;
            if (s.charAt(i) == s.charAt(j) && (len <= 2 || dp[i + 1][j - 1])) {
```

```
                dp[i][j] = true;
            }
        }
    }
    List<List<String>> result = new ArrayList<>();
    backtrack(s, 0, dp, new ArrayList<>(), result);
    return result;
}
private void backtrack(String s, int start, boolean[][] dp, List<String> current,
List<List<String>> result) {
    if (start == s.length()) { result.add(new ArrayList<>(current)); return; }
    for (int end = start; end < s.length(); end++) {
        if (dp[start][end]) {
            current.add(s.substring(start, end + 1));
            backtrack(s, end + 1, dp, current, result);
            current.remove(current.size() - 1);
        }
    }
}
```

Time: O(n²) for the DP table + O(n · 2^n) for backtracking (unavoidable — that's the size of the output in the worst case, e.g. all-same-character strings). Space: O(n²) for the DP table.

Dry Run: `"aab"`. DP table marks "a"(0,0)=T,"a"(1,1)=T,"b"(2,2)=T,"aa"(0,1)=T, "ab"(1,2)=F,"aab"(0,2)=F. Backtrack: start0,end0("a"is palindrome)→recurse start1: end1("a")→ recurse start2: end2("b")→recurse start3=len→add["a","a","b"]. Back at start1, end2("ab") not palindrome→skip. Back at start0,end1("aa")palindrome→recurse start2:end2("b")→add ["aa","b"]. Final `[["a","a","b"],["aa","b"]]`. ✅

Edge Cases: single character string → one partition (itself). All same characters (e.g. "aaa") → many valid partitions, tests exponential blow-up awareness.

Follow-up: "Palindrome Partitioning II" (minimum number of cuts, not all partitions) — a pure DP problem (no backtracking needed) since we only need a count, not enumeration.

A9.6 Letter Combinations of a Phone Number

Difficulty: Medium | **Pattern:** Backtracking (Cartesian product) | **Companies:** Amazon, Google, Uber — common warm-up backtracking question

Problem Statement: Given a string of digits 2-9, return all possible letter combinations the number could represent (like old phone keypads).

Example: `"23"` → `["ad","ae","af","bd","be","bf","cd","ce","cf"]`.

Brute Force Approach Intuition: build up combinations iteratively — start with an empty list containing `[""]`, and for each digit, expand every existing partial combination by appending each letter for that digit (essentially computing the Cartesian product iteratively).

```
public List<String> letterCombinationsIterative(String digits) {
    if (digits.isEmpty()) return new ArrayList<>();
    String[] mapping = {"", "", "abc", "def", "ghi", "jkl", "mno", "pqrs", "tuv",
"wxyz"};
    List<String> result = new ArrayList<>();
    result.add("");
    for (char d : digits.toCharArray()) {
        List<String> next = new ArrayList<>();
        String letters = mapping[d - '0'];
        for (String combo : result) {
            for (char letter : letters.toCharArray()) {
                next.add(combo + letter);
            }
        }
        result = next;
    }
    return result;
}
```

Time: O(4^n · n) worst case (up to 4 letters per digit, n digits, O(n) string building each). Space: O(4^n · n) for the output.

Optimal / Canonical Approach (Backtracking) Intuition: same complexity class, but the backtracking form is generally preferred for clarity and is the expected interview answer — build one combination character by character via recursion, backtrack after exploring each branch.

```
public List<String> letterCombinations(String digits) {
    List<String> result = new ArrayList<>();
    if (digits.isEmpty()) return result;
    String[] mapping = {"", "", "abc", "def", "ghi", "jkl", "mno", "pqrs", "tuv",
"wxyz"};
    backtrack(digits, 0, new StringBuilder(), mapping, result);
    return result;
}
private void backtrack(String digits, int index, StringBuilder current, String[]
mapping, List<String> result) {
    if (index == digits.length()) {
        result.add(current.toString());
        return;
    }
    String letters = mapping[digits.charAt(index) - '0'];
```

```
        for (char letter : letters.toCharArray()) {
            current.append(letter);
            backtrack(digits, index + 1, current, mapping, result);
            current.deleteCharAt(current.length() - 1); // backtrack
        }
    }
```

Time: O(4^n · n) — same bound (unavoidable, matches output size), but O(1) extra space per branch (StringBuilder reused) instead of allocating new lists at every level. Space: O(n) recursion depth.

Dry Run: `"23"`. digit'2'→"abc". Try 'a': digit'3'→"def". Try'd'→"ad"(complete,added). Try'e'→"ae". Try'f'→"af". Backtrack to 'b': "bd","be","bf". Backtrack to 'c': "cd","ce","cf". Final: `["ad","ae","af","bd","be","bf","cd","ce","cf"]`. ✅

Edge Cases: empty input string → empty result list (not `[""]` — explicit LeetCode convention). Single digit → just that digit's letters, one character each.

Follow-up: "Generate Parentheses" and "Combination Sum" share the same "build one choice at a time, backtrack" skeleton — recognizing that all backtracking problems are really "explore a decision tree, prune invalid branches" is the core transferable insight.

A9.7 N-Queens

Difficulty: Hard | **Pattern:** Backtracking with constraint pruning | **Companies:** Amazon, Google, Microsoft — the quintessential backtracking interview question

Problem Statement: Place `n` queens on an `n x n` chessboard such that no two queens attack each other (same row, column, or diagonal). Return all distinct solutions.

Brute Force Approach Intuition: try placing queens in every possible cell combination (choose n cells out of n²), checking full validity only at the end — astronomically wasteful.

```
// Conceptual only: enumerating all C(n^2, n) cell combinations and validating each
// is computationally infeasible even for modest n (e.g. n=8 => C(64,8) ≈ 4.4
billion).
// This is why constraint-checking DURING placement (not after) is essential — see
optimal.
```

Time: infeasible (combinatorial explosion far worse than the pruned backtracking version).

Optimal Approach (Backtracking, one queen per row, prune columns/diagonals as we go) Intuition: since no two queens can share a row, place exactly one queen per row and recurse row by row. For each candidate column in the current row, check if it conflicts with any previously-placed queen's column or either diagonal — using sets for O(1) conflict checks avoids re-scanning all previously placed queens.

```
public List<List<String>> solveNQueens(int n) {
    List<List<String>> result = new ArrayList<>();
    int[] queenCols = new int[n]; // queenCols[row] = column of the queen in that
row
    Set<Integer> usedCols = new HashSet<>();
    Set<Integer> usedDiag1 = new HashSet<>(); // row - col (constant along one
diagonal direction)
    Set<Integer> usedDiag2 = new HashSet<>(); // row + col (constant along the
other diagonal direction)
    backtrack(0, n, queenCols, usedCols, usedDiag1, usedDiag2, result);
    return result;
}

private void backtrack(int row, int n, int[] queenCols, Set<Integer> usedCols,
                       Set<Integer> usedDiag1, Set<Integer> usedDiag2,
List<List<String>> result) {
    if (row == n) {
        result.add(buildBoard(queenCols, n));
        return;
    }
    for (int col = 0; col < n; col++) {
        int diag1 = row - col, diag2 = row + col;
        if (usedCols.contains(col) || usedDiag1.contains(diag1) ||
usedDiag2.contains(diag2)) {
            continue; // conflict, skip this column
        }
        queenCols[row] = col;
        usedCols.add(col); usedDiag1.add(diag1); usedDiag2.add(diag2);

        backtrack(row + 1, n, queenCols, usedCols, usedDiag1, usedDiag2, result);

        usedCols.remove(col); usedDiag1.remove(diag1); usedDiag2.remove(diag2); //
backtrack
    }
}

private List<String> buildBoard(int[] queenCols, int n) {
    List<String> board = new ArrayList<>();
    for (int row = 0; row < n; row++) {
        char[] rowChars = new char[n];
        Arrays.fill(rowChars, '.');
        rowChars[queenCols[row]] = 'Q';
        board.add(new String(rowChars));
    }
```

```
        return board;
    }
```

Time: O(n!) worst case bound (loosely — one queen per row, pruned column/diagonal choices per row), in practice much faster due to aggressive pruning; exact solution counts grow combinatorially but far below a naive n^n or C(n²,n) bound. Space: O(n) for the tracking sets + recursion depth, O(solutions × n) for output.

Dry Run: `n=4`. Row0: try col0→ok, place. Row1: try col0(conflict,same col)skip, col1(diag conflict row1-col1=0==row0-col0=0)skip, col2→ok(diag1=1-2=-1,diag2=3,no conflicts), place. Row2: col0(diag2 conflict:2+0=2? check against existing diag2 values{0,3}: 2 not in set... but col conflict:0 used)skip, col1(col used? no; diag1=2-1=1 used? no; diag2=3 used? yes,conflict)skip, col3(diag1=2-3=-1 used? yes(from row1's diag1=-1) conflict)skip → dead end, backtrack row1. Try col3 at row1 instead... (continues) — known result: N-Queens for n=4 has exactly **2** solutions. ✅ (well-known result, verified by the classic algorithm)

Edge Cases: `n=1` → single trivial solution (one queen alone). `n=2` or `n=3` → **no** valid solutions exist (well-known impossibility for these sizes) — code correctly returns an empty list since backtracking never reaches `row==n`.

Follow-up: "N-Queens II" (just count solutions, don't construct the boards) — identical backtracking, skip the `buildBoard` step and just increment a counter, saving memory.

A10. Graphs

A10.1 Number of Islands

Difficulty: Medium | **Pattern:** Grid DFS/BFS (connected components) | **Companies:** Amazon, Meta, Microsoft, Google — arguably THE most-asked graph question in the industry

Problem Statement: Given an `m x n` binary grid of `'1'`s (land) and `'0'`s (water), count the number of islands (land connected horizontally/vertically).

Brute Force / Direct Approach (this IS the expected solution — DFS/BFS flood fill) Intuition: scan every cell; whenever an unvisited `'1'` is found, it's the start of a new

island — flood-fill (DFS or BFS) to mark every connected land cell as visited so it isn't recounted, then increment the island counter.

```
public int numIslands(char[][] grid) {
    int rows = grid.length, cols = grid[0].length;
    int count = 0;
    for (int r = 0; r < rows; r++) {
        for (int c = 0; c < cols; c++) {
            if (grid[r][c] == '1') {
                count++;
                dfs(grid, r, c);
            }
        }
    }
    return count;
}
private void dfs(char[][] grid, int r, int c) {
    if (r < 0 || r >= grid.length || c < 0 || c >= grid[0].length || grid[r][c] !=
'1') return;
    grid[r][c] = '0'; // mark visited (sink the land)
    dfs(grid, r + 1, c);
    dfs(grid, r - 1, c);
    dfs(grid, r, c + 1);
    dfs(grid, r, c - 1);
}
```

Time: O(m·n) — every cell visited a constant number of times. Space: O(m·n) worst case recursion stack (an all-land grid).

BFS Alternative (avoids recursion stack overflow on huge grids)

```
private void bfs(char[][] grid, int r, int c) {
    Queue<int[]> queue = new LinkedList<>();
    queue.offer(new int[]{r, c});
    grid[r][c] = '0';
    int[][] dirs = {{1,0},{-1,0},{0,1},{0,-1}};
    while (!queue.isEmpty()) {
        int[] cell = queue.poll();
        for (int[] d : dirs) {
            int nr = cell[0] + d[0], nc = cell[1] + d[1];
            if (nr >= 0 && nr < grid.length && nc >= 0 && nc < grid[0].length &&
grid[nr][nc] == '1') {
                grid[nr][nc] = '0';
                queue.offer(new int[]{nr, nc});
            }
        }
    }
}
```

Time: O(m·n). Space: O(m·n) worst case for the queue.

Dry Run: grid with two separate land blobs → outer loop hits the first `'1'`, DFS sinks that entire connected blob, count=1. Continues scanning, hits the second blob's first `'1'`, DFS sinks it, count=2. Final answer `2`. ✅

Edge Cases: empty grid → 0. Grid entirely water → 0. Grid entirely land → 1 (one giant connected island).

Follow-up: "Number of Islands II" (dynamic, cells turned to land one at a time, need count after each addition) — needs a Union-Find structure for efficient incremental connectivity. "Max Area of Island" — same flood-fill but returns the size of the largest connected component instead of just a count.

A10.2 Clone Graph

Difficulty: Medium | **Pattern:** Graph DFS/BFS + HashMap (original→copy mapping) | **Companies:** Amazon, Google, Meta

Problem Statement: Given a reference to a node in a connected undirected graph, return a deep copy of the graph.

Clarifying Questions: Is the graph guaranteed connected (reachable entirely from the given node)? (Yes, per problem statement.) Can the graph have cycles? (Yes — this is exactly why a visited/mapping structure is required to avoid infinite recursion.)

Approach (DFS with a HashMap mapping original nodes to their clones) Intuition: the core challenge is cycles — if you naively clone recursively without remembering what you've already cloned, you'll loop forever. Maintain a `HashMap<Node, Node>` from original to clone; before recursing into a neighbor, check if it's already been cloned.

```
public Node cloneGraph(Node node) {
    if (node == null) return null;
    return dfs(node, new HashMap<>());
}
private Node dfs(Node node, Map<Node, Node> visited) {
    if (visited.containsKey(node)) return visited.get(node);
    Node clone = new Node(node.val);
    visited.put(node, clone);
    for (Node neighbor : node.neighbors) {
        clone.neighbors.add(dfs(neighbor, visited));
    }
    return clone;
}
```

Time: O(V + E) — each node and edge processed once. Space: O(V) for the map + O(V) recursion stack worst case.

BFS Alternative

```
public Node cloneGraphBFS(Node node) {
    if (node == null) return null;
    Map<Node, Node> visited = new HashMap<>();
    visited.put(node, new Node(node.val));
    Queue<Node> queue = new LinkedList<>();
    queue.offer(node);
    while (!queue.isEmpty()) {
        Node cur = queue.poll();
        for (Node neighbor : cur.neighbors) {
            if (!visited.containsKey(neighbor)) {
                visited.put(neighbor, new Node(neighbor.val));
                queue.offer(neighbor);
            }
            visited.get(cur).neighbors.add(visited.get(neighbor));
        }
    }
    return visited.get(node);
}
```

Time: O(V + E). Space: O(V).

Dry Run: graph `1-2-3-4-1` (a 4-cycle). DFS from 1: clone1, visit neighbor2: clone2, visit neighbor3(of2): clone3, visit neighbor4(of3): clone4, visit neighbor1(of4): already in map→return clone1 (closes the cycle correctly without infinite recursion). ✅

Edge Cases: single node with no neighbors → clone that one node, empty neighbor list. `node == null` input → return null.

Follow-up: This exact "map original → copy, use it to break cycles" idea is identical to Copy List with Random Pointer (A5.8) applied to a general graph instead of a linked list.

A10.3 Course Schedule

Difficulty: Medium | **Pattern:** Topological Sort (Cycle Detection) | **Companies:** Amazon, Google, Meta, Microsoft — extremely common "can this be scheduled" question

Problem Statement: Given `numCourses` and a list of prerequisite pairs `[course, prerequisite]`, determine if it's possible to finish all courses (i.e., the prerequisite graph has no cycle).

Clarifying Questions: Directed graph, correct? (Yes — `[a,b]` means "b must be taken before a," a directed edge b→a.)

Brute Force Approach (DFS cycle detection with 3-color marking) Intuition: build an adjacency list. DFS from every unvisited node, tracking each node's state as unvisited / in the current recursion path ("visiting") / fully processed ("visited"). If DFS ever revisits a node that's still "in the current path," that's a cycle.

```
public boolean canFinish(int numCourses, int[][] prerequisites) {
    List<List<Integer>> graph = new ArrayList<>();
    for (int i = 0; i < numCourses; i++) graph.add(new ArrayList<>());
    for (int[] p : prerequisites) graph.get(p[0]).add(p[1]);

    int[] state = new int[numCourses]; // 0=unvisited, 1=visiting, 2=done
    for (int i = 0; i < numCourses; i++) {
        if (state[i] == 0 && hasCycle(graph, i, state)) return false;
    }
    return true;
}
private boolean hasCycle(List<List<Integer>> graph, int node, int[] state) {
    state[node] = 1;
    for (int neighbor : graph.get(node)) {
        if (state[neighbor] == 1) return true; // back edge -> cycle
        if (state[neighbor] == 0 && hasCycle(graph, neighbor, state)) return true;
    }
    state[node] = 2;
    return false;
}
```

Time: O(V + E). Space: O(V + E) for the graph + O(V) recursion stack.

Optimal Approach (Kahn's Algorithm — BFS Topological Sort via in-degrees) Intuition: repeatedly remove nodes with in-degree 0 (courses with no remaining prerequisites), decrementing the in-degree of their dependents. If we can remove all `numCourses` nodes this way, there's no cycle; if some remain stuck (in-degree never reaches 0), a cycle exists among them.

```
public boolean canFinishKahn(int numCourses, int[][] prerequisites) {
    List<List<Integer>> graph = new ArrayList<>();
    for (int i = 0; i < numCourses; i++) graph.add(new ArrayList<>());
    int[] inDegree = new int[numCourses];
    for (int[] p : prerequisites) {
        graph.get(p[1]).add(p[0]); // prerequisite -> dependent course
```

```
            inDegree[p[0]]++;
        }

        Queue<Integer> queue = new LinkedList<>();
        for (int i = 0; i < numCourses; i++) if (inDegree[i] == 0) queue.offer(i);

        int processed = 0;
        while (!queue.isEmpty()) {
            int course = queue.poll();
            processed++;
            for (int dependent : graph.get(course)) {
                if (--inDegree[dependent] == 0) queue.offer(dependent);
            }
        }
        return processed == numCourses;
    }
```

Time: O(V + E). Space: O(V + E).

Dry Run: `numCourses=2, prerequisites=[[1,0]]` (1 depends on 0). Graph: 0→1, inDegree= [0,1]. Queue starts with [0]. Process0: processed=1, decrement inDegree[1] to 0, enqueue1. Process1: processed=2. processed==numCourses(2) → `true`. ✅ With a cycle `[[1,0],[0,1]]`: both have inDegree 1 initially, queue starts empty → processed stays 0 ≠ 2 → `false`. ✅

Edge Cases: no prerequisites at all → trivially `true` (every course has in-degree 0). Self-loop prerequisite `[0,0]` → immediately a cycle → `false`.

Follow-up: "Course Schedule II" (return a valid course order, not just true/false) — same Kahn's algorithm, just record the order courses are dequeued in.

A10.4 Pacific Atlantic Water Flow

Difficulty: Medium | **Pattern:** Multi-source DFS/BFS | **Companies:** Google, Amazon

Problem Statement: Given a grid of heights, water flows from a cell to an adjacent cell with height ≤ current (flows downhill or flat). Pacific touches the top/left edges, Atlantic touches bottom/right edges. Return all cells from which water can flow to **both** oceans.

Brute Force Approach Intuition: for every single cell, run a DFS/BFS to check if water starting there can reach the Pacific border, and separately check if it can reach the Atlantic border.

```
// Conceptual: for each of m*n cells, run an O(m*n) reachability search twice
(Pacific, Atlantic)
// Time: O((m*n)^2) -- highly redundant, recomputes overlapping reachability
regions repeatedly.
```

Time: O((m·n)²). Space: O(m·n) per search.

Optimal Approach (reverse the flow direction, multi-source BFS/DFS from each ocean's border) Intuition: instead of asking "can this cell reach the ocean" (checking downhill neighbors), reverse the question: "starting from the ocean's border cells, which cells can water flow **backward** (uphill or equal) to?" Run this multi-source search once from all Pacific border cells, and once from all Atlantic border cells. The answer is the intersection of both reachable sets.

```
public List<List<Integer>> pacificAtlantic(int[][] heights) {
    int rows = heights.length, cols = heights[0].length;
    boolean[][] pacific = new boolean[rows][cols];
    boolean[][] atlantic = new boolean[rows][cols];

    for (int c = 0; c < cols; c++) {
        dfs(heights, 0, c, pacific);
        dfs(heights, rows - 1, c, atlantic);
    }
    for (int r = 0; r < rows; r++) {
        dfs(heights, r, 0, pacific);
        dfs(heights, r, cols - 1, atlantic);
    }

    List<List<Integer>> result = new ArrayList<>();
    for (int r = 0; r < rows; r++) {
        for (int c = 0; c < cols; c++) {
            if (pacific[r][c] && atlantic[r][c]) result.add(Arrays.asList(r, c));
        }
    }
    return result;
}
private void dfs(int[][] heights, int r, int c, boolean[][] visited) {
    visited[r][c] = true;
    int[][] dirs = {{1,0},{-1,0},{0,1},{0,-1}};
    for (int[] d : dirs) {
        int nr = r + d[0], nc = c + d[1];
        if (nr >= 0 && nr < heights.length && nc >= 0 && nc < heights[0].length
                && !visited[nr][nc] && heights[nr][nc] >= heights[r][c]) {
            dfs(heights, nr, nc, visited);
        }
    }
}
```

Time: O(m·n) — each ocean's multi-source search visits every cell at most once. Space: O(m·n) for the two visited matrices + recursion stack.

Dry Run: border cells are seeded first for each ocean (all of row 0/last column for Pacific-ish borders per this convention). DFS expands to any neighbor with height >= current (i.e., water could have flowed downhill from there to here). Cells marked true in both matrices are collected. ✅ (matches canonical LeetCode example structure)

Edge Cases: single row or single column grid (borders overlap significantly — algorithm still correct since both border-seeding loops naturally cover all cells in that case). All cells same height → water can flow anywhere, likely most/all cells reach both oceans.

Follow-up: "Why reverse the direction instead of searching from every cell?" — be ready to explain explicitly: it converts an O((mn)²) all-pairs-style reachability problem into two O(mn) multi-source searches, which is the whole trick.

A10.5 Number of Connected Components in an Undirected Graph

Difficulty: Medium | **Pattern:** Union-Find (or DFS/BFS) | **Companies:** Google, Facebook — classic Union-Find introduction

Problem Statement: Given n nodes labeled 0 to n-1 and a list of undirected edges, count the number of connected components.

Brute Force Approach (DFS/BFS from every unvisited node) Intuition: build an adjacency list; for every unvisited node, DFS/BFS to mark its entire component, incrementing a counter each time a new search is started.

```
public int countComponentsDFS(int n, int[][] edges) {
    List<List<Integer>> graph = new ArrayList<>();
    for (int i = 0; i < n; i++) graph.add(new ArrayList<>());
    for (int[] e : edges) { graph.get(e[0]).add(e[1]); graph.get(e[1]).add(e[0]); }

    boolean[] visited = new boolean[n];
    int count = 0;
    for (int i = 0; i < n; i++) {
        if (!visited[i]) { count++; dfs(graph, i, visited); }
    }
    return count;
}
private void dfs(List<List<Integer>> graph, int node, boolean[] visited) {
```

```
    visited[node] = true;
    for (int neighbor : graph.get(node)) if (!visited[neighbor]) dfs(graph,
neighbor, visited);
}
```

Time: O(V + E). Space: O(V + E).

Optimal / Alternative Approach (Union-Find / Disjoint Set Union) Intuition: start with `n` separate components. For every edge, "union" its two endpoints — if they're already in the same set, the edge doesn't reduce the component count; otherwise it merges two components into one, decrementing the count. Union by rank + path compression keeps operations nearly O(1) amortized.

```
public int countComponents(int n, int[][] edges) {
    int[] parent = new int[n];
    int[] rank = new int[n];
    for (int i = 0; i < n; i++) parent[i] = i;

    int components = n;
    for (int[] edge : edges) {
        int rootA = find(parent, edge[0]);
        int rootB = find(parent, edge[1]);
        if (rootA != rootB) {
            union(parent, rank, rootA, rootB);
            components--;
        }
    }
    return components;
}
private int find(int[] parent, int x) {
    if (parent[x] != x) parent[x] = find(parent, parent[x]); // path compression
    return parent[x];
}
private void union(int[] parent, int[] rank, int a, int b) {
    if (rank[a] < rank[b]) { int t = a; a = b; b = t; }
    parent[b] = a;
    if (rank[a] == rank[b]) rank[a]++;
}
```

Time: O(E · α(V)) where α is the inverse Ackermann function (effectively constant in practice) — near-linear. Space: O(V) for the parent/rank arrays.

Dry Run: `n=5, edges=[[0,1],[1,2],[3,4]]`. Start components=5. Edge(0,1): different roots → union, components=4. Edge(1,2): find(1)=0(after first union),find(2)=2, different→union, components=3. Edge(3,4): different→union, components=2. Final answer `2` (components {0,1,2} and {3,4}). ✅

Edge Cases: no edges at all → components = n (every node isolated). All nodes connected in one chain → components = 1.

Follow-up: "Graph Valid Tree" (next) uses the exact same Union-Find machinery plus an edge- count check. "Redundant Connection" (also below) uses Union-Find to find the one edge that creates a cycle.

A10.6 Graph Valid Tree

Difficulty: Medium | **Pattern:** Union-Find (cycle + connectivity check) | **Companies:** Google, Facebook — often gated behind premium but very commonly asked live

Problem Statement: Given `n` nodes and a list of undirected edges, determine if these edges form a valid tree (connected, and exactly `n-1` edges with no cycles).

Clarifying Questions: A valid tree requires both "no cycles" AND "fully connected" — a graph with `n-1` edges could still be disconnected with a cycle elsewhere; must check both.

Brute Force Approach (structural pre-check + DFS connectivity + cycle check)
Intuition: first, a necessary (but not sufficient) condition: a tree with `n` nodes must have exactly `n-1` edges. If that count doesn't match, fail immediately. Otherwise, DFS from node 0 and check that (a) no cycle is encountered and (b) all `n` nodes are visited.

```
public boolean validTreeDFS(int n, int[][] edges) {
    if (edges.length != n - 1) return false; // necessary edge-count condition
    List<List<Integer>> graph = new ArrayList<>();
    for (int i = 0; i < n; i++) graph.add(new ArrayList<>());
    for (int[] e : edges) { graph.get(e[0]).add(e[1]); graph.get(e[1]).add(e[0]); }

    boolean[] visited = new boolean[n];
    if (hasCycleOrDisconnected(graph, 0, -1, visited)) return false;
    for (boolean v : visited) if (!v) return false; // must be fully connected
    return true;
}
private boolean hasCycleOrDisconnected(List<List<Integer>> graph, int node, int
parent, boolean[] visited) {
    visited[node] = true;
    for (int neighbor : graph.get(node)) {
        if (neighbor == parent) continue; // don't treat the edge back to parent as
a cycle
        if (visited[neighbor]) return true; // cycle detected
        if (hasCycleOrDisconnected(graph, neighbor, node, visited)) return true;
    }
```

```
        return false;
    }
```

Time: O(V + E). Space: O(V + E).

Optimal Approach (Union-Find — cleaner and directly generalizes) Intuition: same `n-1` edge pre-check. Then union every edge's endpoints; if any edge connects two nodes **already** in the same set, that's a cycle → invalid. If we get through all edges without a cycle AND the edge count is exactly n-1, connectivity is guaranteed automatically (graph theory: a cycle-free graph with n-1 edges on n nodes is necessarily a single connected tree).

```
public boolean validTree(int n, int[][] edges) {
    if (edges.length != n - 1) return false;
    int[] parent = new int[n];
    for (int i = 0; i < n; i++) parent[i] = i;

    for (int[] edge : edges) {
        int rootA = find(parent, edge[0]);
        int rootB = find(parent, edge[1]);
        if (rootA == rootB) return false; // cycle detected
        parent[rootB] = rootA;
    }
    return true;
}
private int find(int[] parent, int x) {
    if (parent[x] != x) parent[x] = find(parent, parent[x]);
    return parent[x];
}
```

Time: O(E · α(V)) ≈ O(E) near-linear. Space: O(V).

Dry Run: `n=5, edges=[[0,1],[0,2],[0,3],[1,4]]`. edges.length=4=n-1 ✓. Union all pairs, no two edges ever connect already-same-root nodes → no cycle detected → `true`. With `edges=[[0,1],[1,2],[2,3],[1,3],[1,4]]`: edges.length=5≠4(n-1=4) → immediately `false` without even running Union-Find. ✅

Edge Cases: `n=1` with 0 edges → valid tree (single node, trivially connected, no cycles). Disconnected graph with exactly n-1 edges is actually impossible by the graph-theory fact mentioned above — worth stating this explicitly to show deeper understanding.

Follow-up: Generalizing to "count connected components" is exactly A10.5 without the `n-1` edge-count / cycle-rejection requirement.

A10.7 Word Ladder

Difficulty: Hard | **Pattern:** BFS (shortest path, unweighted graph) | **Companies:** Amazon, Google, LinkedIn — classic "implicit graph" BFS question

Problem Statement: Given `beginWord`, `endWord`, and a word list, find the length of the shortest transformation sequence from `beginWord` to `endWord`, changing one letter at a time, where every intermediate word must exist in the word list. Return 0 if no such sequence exists.

Clarifying Questions: Must `endWord` be in the word list to be reachable? (Yes — if it's not in the list, no transformation sequence can end there.)

Brute Force Approach Intuition: model this explicitly as a graph where each word is a node, and an edge connects two words differing by exactly one letter; build that graph, then BFS for shortest path. Building the graph by comparing every pair of words is expensive.

```
// Conceptual: build adjacency by comparing every pair of words for a 1-letter
difference
// -- O(W^2 * L) just to build the graph (W = word count, L = word length), then
O(W+E) BFS.
```

Time: O(W² · L) to build the graph — the bottleneck.

Optimal Approach (BFS with on-the-fly neighbor generation via wildcard patterns) Intuition: instead of comparing every pair of words, generate all possible **one-letter-changed variants** of the current word (26 × L possibilities) and check if each is in the word set — turning graph construction into implicit, on-demand edge discovery during the BFS itself.

```
public int ladderLength(String beginWord, String endWord, List<String> wordList) {
    Set<String> wordSet = new HashSet<>(wordList);
    if (!wordSet.contains(endWord)) return 0;

    Queue<String> queue = new LinkedList<>();
    queue.offer(beginWord);
    Set<String> visited = new HashSet<>();
    visited.add(beginWord);
    int steps = 1;

    while (!queue.isEmpty()) {
        int size = queue.size();
        for (int i = 0; i < size; i++) {
```

```
            String word = queue.poll();
            if (word.equals(endWord)) return steps;

            char[] chars = word.toCharArray();
            for (int pos = 0; pos < chars.length; pos++) {
                char original = chars[pos];
                for (char c = 'a'; c <= 'z'; c++) {
                    if (c == original) continue;
                    chars[pos] = c;
                    String candidate = new String(chars);
                    if (wordSet.contains(candidate) &&
!visited.contains(candidate)) {
                        visited.add(candidate);
                        queue.offer(candidate);
                    }
                }
                chars[pos] = original; // restore before trying next position
            }
        }
        steps++;
    }
    return 0;
}
```

Time: O(W · L² · 26) — for each of up to W words processed, generate L·26 candidate variants, each an O(L) string construction/hash. Space: O(W · L) for the word set + visited set.

Dry Run: `beginWord="hit", endWord="cog", wordList=[hot,dot,dog,lot,log,cog]`. BFS level1: "hit"→generates "hot"(in set,unvisited)→enqueue. steps becomes 2 after level. level2: "hot"→ generates "dot","lot"(both in set)→enqueue. steps=3. level3: "dot"→"dog"; "lot"→"log". steps=4. level4: "dog"→"cog"(match!) → but check happens at poll time: when "cog" is polled inside its level, word.equals(endWord) → return current `steps` value which was set appropriately. Known correct answer for this classic example is `5` (hit→hot→dot→dog→cog, 5 words = 4 transformations

- 1). ✅

Edge Cases: `endWord` not in `wordList` → 0 immediately. `beginWord` equals `endWord` → edge case depends on convention (typically not a valid test case per constraints, but code should be checked against LeetCode's exact examples). No valid path exists → BFS exhausts, returns 0.

Follow-up: "Word Ladder II" (return ALL shortest transformation sequences, not just the length) — much harder: BFS to find shortest distance, then DFS/backtrack along only the edges that lie on a shortest path (a "layered BFS + backtracking" combo).

A10.8 Rotting Oranges

Difficulty: Medium | **Pattern:** Multi-source BFS | **Companies:** Amazon, Facebook, Bloomberg — very common multi-source BFS intro

Problem Statement: A grid where `0`=empty, `1`=fresh orange, `2`=rotten orange. Every minute, any fresh orange adjacent to a rotten one becomes rotten. Return the minimum minutes until no fresh orange remains, or -1 if impossible.

Brute Force Approach Intuition: simulate minute by minute — each minute, scan the entire grid for rotten oranges, check their neighbors, mark any fresh ones to become rotten *next* minute (need a "next state" copy to avoid rotting a fresh orange twice in the same minute due to scan order), repeat until no changes occur.

```
public int orangesRottingBrute(int[][] grid) {
    int minutes = 0;
    while (hasFreshOrange(grid)) {
        int[][] next = deepCopy(grid);
        boolean changed = false;
        for (int r = 0; r < grid.length; r++) {
            for (int c = 0; c < grid[0].length; c++) {
                if (grid[r][c] == 2) {
                    for (int[] d : new int[][]{{1,0},{-1,0},{0,1},{0,-1}}) {
                        int nr = r + d[0], nc = c + d[1];
                        if (nr >= 0 && nr < grid.length && nc >= 0 && nc <
grid[0].length && grid[nr][nc] == 1) {
                            next[nr][nc] = 2;
                            changed = true;
                        }
                    }
                }
            }
        }
        if (!changed) return -1; // fresh oranges remain but nothing can rot them
further
        grid = next;
        minutes++;
    }
    return minutes;
}
// hasFreshOrange / deepCopy helpers omitted for brevity
```

Time: O((m·n)²) worst case — every minute re-scans the whole grid, and up to O(m·n) minutes may be needed. Space: O(m·n) for grid copies.

Optimal Approach (Multi-source BFS starting from all rotten oranges simultaneously) Intuition: instead of scanning the whole grid every minute, start a

BFS with **all** initially rotten oranges enqueued at once ("multi-source"). Process level by level (each level = one minute); the number of BFS levels processed is exactly the answer.

```
public int orangesRotting(int[][] grid) {
    int rows = grid.length, cols = grid[0].length;
    Queue<int[]> queue = new LinkedList<>();
    int freshCount = 0;

    for (int r = 0; r < rows; r++) {
        for (int c = 0; c < cols; c++) {
            if (grid[r][c] == 2) queue.offer(new int[]{r, c});
            else if (grid[r][c] == 1) freshCount++;
        }
    }

    if (freshCount == 0) return 0;
    int minutes = 0;
    int[][] dirs = {{1,0},{-1,0},{0,1},{0,-1}};

    while (!queue.isEmpty() && freshCount > 0) {
        int size = queue.size();
        for (int i = 0; i < size; i++) {
            int[] cell = queue.poll();
            for (int[] d : dirs) {
                int nr = cell[0] + d[0], nc = cell[1] + d[1];
                if (nr >= 0 && nr < rows && nc >= 0 && nc < cols && grid[nr][nc] ==
1) {
                    grid[nr][nc] = 2;
                    freshCount--;
                    queue.offer(new int[]{nr, nc});
                }
            }
        }
        minutes++;
    }
    return freshCount == 0 ? minutes : -1;
}
```

Time: O(m·n) — every cell enqueued/processed at most once. Space: O(m·n) for the queue.

Dry Run: `[[2,1,1],[1,1,0],[0,1,1]]`. Initial queue has the one rotten cell (0,0), freshCount=6. Minute1: rots (0,1) and (1,0), freshCount=4. Minute2: rots (0,2) and (1,1), freshCount=2. Minute3: rots (2,1) [reachable via (1,1)], freshCount=1. Minute4: rots (2,2), freshCount=0. Loop ends (freshCount==0). Answer `4`. ✅ (matches canonical LeetCode example)

Edge Cases: no fresh oranges at all → 0 immediately. Fresh oranges that can never be reached (isolated by empty cells or unreachable) → `freshCount` never reaches 0

→ return -1.

Follow-up: "Why is multi-source BFS faster than one-source-at-a-time?" — be ready to explain: starting all sources simultaneously naturally computes the minimum time for *every* fresh orange to rot in one synchronized pass, rather than needing separate searches.

A10.9 Redundant Connection

Difficulty: Medium | **Pattern:** Union-Find (cycle detection in an edge stream) | **Companies:** Google, Amazon

Problem Statement: A tree with `n` nodes had one extra edge added, creating exactly one cycle. Given the edges in the order they were added, find the edge that, if removed, restores a valid tree. If multiple such edges exist, return the one that occurs last in the input.

Brute Force Approach Intuition: for each edge in the list (starting from the last), remove it and check if the remaining edges form a valid tree (using the Graph Valid Tree check); return the first removal that works, scanning from the end.

```
// Conceptual: for i from edges.length-1 down to 0, remove edges[i], run validTree
check on
// the rest -- O(E) validTree checks, each O(V+E) -> O(E*(V+E)) total. Correct but
wasteful.
```

Time: O(E · (V+E)). Space: O(V+E) per check.

Optimal Approach (Union-Find, single pass) Intuition: process edges in the given order, union-ing each pair. The very first edge that tries to connect two nodes **already in the same set** is necessarily the redundant one — it's the edge that closes the (only) cycle, and since we process in input order, it's automatically the last such occurrence if ties existed.

```
public int[] findRedundantConnection(int[][] edges) {
    int n = edges.length;
    int[] parent = new int[n + 1];
    for (int i = 1; i <= n; i++) parent[i] = i;

    for (int[] edge : edges) {
        int rootA = find(parent, edge[0]);
```

```
            int rootB = find(parent, edge[1]);
            if (rootA == rootB) return edge; // this edge creates the cycle
            parent[rootB] = rootA;
        }
        return new int[0]; // per problem guarantee, this line is unreachable
    }
    private int find(int[] parent, int x) {
        if (parent[x] != x) parent[x] = find(parent, parent[x]);
        return parent[x];
    }
```

Time: O(E · α(V)) ≈ O(E) near-linear. Space: O(V).

Dry Run: `edges=[[1,2],[1,3],[2,3]]`. Union(1,2): roots differ, merge. Union(1,3): roots differ (find(1)=1,find(3)=3), merge. Union(2,3): find(2)=1(after merges),find(3)=1 → same root! → return `[2,3]`. ✅ (matches canonical LeetCode example)

Edge Cases: the redundant edge appears very early vs. very late in the input (algorithm handles both correctly since it's a strict single left-to-right pass). Self-loop edge (rare, but would immediately trigger `rootA==rootB`).

Follow-up: "Redundant Connection II" (directed graph version — much harder, must handle two distinct failure cases: a node with two parents, and/or a cycle, sometimes both simultaneously) — a well-known escalation that trips up many candidates.

A11. Dynamic Programming (1-D)

A11.1 Climbing Stairs

Difficulty: Easy | **Pattern:** DP (Fibonacci-shape) | **Companies:** Amazon, Adobe — the canonical "intro to DP" question

Problem Statement: You can climb 1 or 2 steps at a time. Given `n` steps, how many distinct ways can you reach the top?

Brute Force Approach Intuition: recursively try both choices (1 step or 2 steps) at every position; the total ways from step `n` is `ways(n-1) + ways(n-2)`.

```
public int climbStairsBrute(int n) {
    if (n <= 2) return n;
```

```
    return climbStairsBrute(n - 1) + climbStairsBrute(n - 2);
}
```

Time: O(2^n) — massive redundant recomputation (identical to naive Fibonacci). Space: O(n) recursion stack.

Better Approach (Memoization — top-down DP) Intuition: cache each `ways(i)` result the first time it's computed, so overlapping subproblems are never recomputed.

```
public int climbStairsMemo(int n) {
    int[] memo = new int[n + 1];
    Arrays.fill(memo, -1);
    return helper(n, memo);
}
private int helper(int n, int[] memo) {
    if (n <= 2) return n;
    if (memo[n] != -1) return memo[n];
    return memo[n] = helper(n - 1, memo) + helper(n - 2, memo);
}
```

Time: O(n). Space: O(n) for memo + recursion stack.

Optimal Approach (Bottom-up, O(1) space) Intuition: since `ways(i)` only ever depends on the two previous values, we don't need a full array — just two rolling variables.

```
public int climbStairs(int n) {
    if (n <= 2) return n;
    int prev2 = 1, prev1 = 2;
    for (int i = 3; i <= n; i++) {
        int cur = prev1 + prev2;
        prev2 = prev1;
        prev1 = cur;
    }
    return prev1;
}
```

Time: O(n). Space: O(1).

Dry Run: `n=5`. prev2=1,prev1=2. i=3: cur=3,prev2=2,prev1=3. i=4: cur=5,prev2=3,prev1=5. i=5: cur=8,prev2=5,prev1=8. Answer `8`. ✅ (matches Fibonacci-like growth)

Edge Cases: `n=1` → 1. `n=2` → 2 (either "1+1" or "2").

Follow-up: "What if you can climb 1, 2, or 3 steps?" — generalizes to `ways(i) = ways(i-1)+ways(i-2)+ways(i-3)`, same rolling-variable technique with 3 variables instead of 2. "House Robber" (next) has the identical DP recurrence shape but a different underlying decision.

A11.2 House Robber

Difficulty: Medium | **Pattern:** DP (include/exclude adjacent constraint) | **Companies:** Amazon, LinkedIn — classic constrained-choice DP

Problem Statement: Given non-negative integers representing money in houses arranged in a line, find the maximum amount you can rob without robbing two adjacent houses.

Brute Force Approach Intuition: at each house, recursively try both choices — rob it (skip the next house) or skip it (move to the next house) — take the max.

```
public int robBrute(int[] nums) {
    return helper(nums, 0);
}
private int helper(int[] nums, int i) {
    if (i >= nums.length) return 0;
    int robThis = nums[i] + helper(nums, i + 2);
    int skipThis = helper(nums, i + 1);
    return Math.max(robThis, skipThis);
}
```

Time: O(2^n) — overlapping subproblems recomputed repeatedly. Space: O(n) recursion stack.

Optimal Approach (Bottom-up DP, O(1) space) Intuition: `dp[i]` = max money robbable from the first `i` houses = `max(dp[i-1], dp[i-2] + nums[i-1])` — either skip the current house (carry forward `dp[i-1]`) or rob it (`nums[i-1]` + whatever was optimal two houses back). Only two rolling variables needed.

```
public int rob(int[] nums) {
    int prevNoRob = 0, prevRob = 0; // dp[i-2], dp[i-1] equivalent rolling state
    for (int num : nums) {
        int newRob = prevNoRob + num;
        int newState = Math.max(newRob, prevRob);
        prevNoRob = prevRob;
        prevRob = newState;
    }
```

```
        return prevRob;
    }
```

Time: O(n). Space: O(1).

Dry Run: `[2,7,9,3,1]`. Start prevNoRob=0,prevRob=0. num=2: newRob=2,newState=max(2,0)=2; prevNoRob=0,prevRob=2. num=7: newRob=0+7=7,newState=max(7,2)=7; prevNoRob=2,prevRob=7. num=9: newRob=2+9=11,newState=max(11,7)=11; prevNoRob=7,prevRob=11. num=3: newRob=7+3=10, newState=max(10,11)=11; prevNoRob=11,prevRob=11. num=1: newRob=11+1=12,newState=max(12,11)=12; prevRob=12. Answer `12`. ✅ (rob houses 0,2,4: 2+9+1=12)

Edge Cases: empty array → 0. Single house → that house's value. Two houses → the larger of the two.

Follow-up: "House Robber II" (next) adds circular arrangement (first and last house are adjacent). "House Robber III" (houses arranged as a binary tree) — needs tree DP returning a pair `(robThisNode, skipThisNode)` per subtree.

A11.3 House Robber II

Difficulty: Medium | **Pattern:** DP (circular constraint via two linear subproblems) | **Companies:** Amazon

Problem Statement: Same as House Robber, but houses are arranged in a **circle** (first and last house are adjacent).

Clarifying Questions: Confirm circularity means house 0 and house n-1 cannot both be robbed — this is the only difference from the linear version.

Brute Force Approach Intuition: same recursive include/exclude idea as House Robber, but must additionally track whether house 0 has been robbed to correctly forbid robbing house n-1 in that case (adds complexity to a naive recursive solution — better handled by the optimal reduction below).

Optimal Approach (reduce to two calls of House Robber) Intuition: the circular constraint only matters because house 0 and house n-1 can't both be taken. So the answer is simply the **max of two independent linear House Robber runs**: one over houses `[0, n-2]` (excluding the last house), and one over houses `[1, n-1]`

(excluding the first house). Whichever excluded range gives more money (or robs a valid subset) wins.

```
public int rob(int[] nums) {
    int n = nums.length;
    if (n == 1) return nums[0];
    return Math.max(
        robLinear(nums, 0, n - 2),
        robLinear(nums, 1, n - 1)
    );
}
private int robLinear(int[] nums, int start, int end) {
    int prevNoRob = 0, prevRob = 0;
    for (int i = start; i <= end; i++) {
        int newRob = prevNoRob + nums[i];
        int newState = Math.max(newRob, prevRob);
        prevNoRob = prevRob;
        prevRob = newState;
    }
    return prevRob;
}
```

Time: O(n) — two linear passes. Space: O(1).

Dry Run: `[2,3,2]`. robLinear(0,1) on [2,3]: max robbable = 3. robLinear(1,2) on [3,2]: max robbable = 3. Answer = max(3,3) = `3`. ✅ (correctly avoids robbing both house0 and house2, which are adjacent in the circle)

Edge Cases: single house → return it directly (no circular conflict possible with itself). Two houses → they're mutually adjacent both ways around the circle, answer = max of the two.

Follow-up: "House Robber III" (binary tree arrangement) requires an entirely different tree-DP technique (post-order traversal returning `[robThis, skipThis]` pairs) — not a simple reduction like this circular version.

A11.4 Longest Increasing Subsequence

Difficulty: Medium | **Pattern:** DP (+ Binary Search optimization) | **Companies:** Google, Microsoft, Amazon — extremely common, has a beautiful O(n log n) optimal trick

Problem Statement: Given an integer array, return the length of the longest strictly increasing subsequence (not necessarily contiguous).

Example: `[10,9,2,5,3,7,101,18]` → `4` (`[2,3,7,101]` or `[2,3,7,18]`).

Brute Force Approach Intuition: try every subsequence (2^n of them), check if increasing, track the longest.

```
// Conceptual exponential brute force: generate all 2^n subsequences, filter
increasing ones,
// track max length. Purely illustrative -- not something you'd actually code in an
interview.
```

Time: O(2^n · n). Impractical even for modest n.

Standard DP Approach Intuition: `dp[i]` = length of the longest increasing subsequence **ending exactly at index i**. For each `i`, check all previous `j < i`; if `nums[j] < nums[i]`, `dp[i]` could extend `dp[j]`'s subsequence. Take the max over all such valid `j`, plus 1. Overall answer is `max(dp[i])` across all i.

```
public int lengthOfLISDP(int[] nums) {
    int n = nums.length;
    int[] dp = new int[n];
    Arrays.fill(dp, 1);
    int maxLen = 1;
    for (int i = 1; i < n; i++) {
        for (int j = 0; j < i; j++) {
            if (nums[j] < nums[i]) {
                dp[i] = Math.max(dp[i], dp[j] + 1);
            }
        }
        maxLen = Math.max(maxLen, dp[i]);
    }
    return maxLen;
}
```

Time: O(n²). Space: O(n) for the dp array.

Optimal Approach (Patience Sorting / Binary Search, O(n log n)) Intuition: maintain an array `tails` where `tails[k]` = the smallest possible tail value of any increasing subsequence of length `k+1` found so far. For each new number, binary search for the first position in `tails` that is `>= num` and replace it there (this represents "found a better, smaller tail for a subsequence of that length"); if `num` is larger than everything in `tails`, append it (extends the longest subsequence found so far). The final length of `tails` is the answer. (Note: `tails` itself is NOT necessarily a valid subsequence of the array — it's purely a bookkeeping structure for lengths and minimal tail values.)

```java
public int lengthOfLIS(int[] nums) {
    List<Integer> tails = new ArrayList<>();
    for (int num : nums) {
        int pos = Collections.binarySearch(tails, num);
        if (pos < 0) pos = -(pos + 1); // insertion point for "not found"
        if (pos == tails.size()) {
            tails.add(num);
        } else {
            tails.set(pos, num);
        }
    }
    return tails.size();
}
```

Time: O(n log n) — n numbers, O(log n) binary search + O(1) amortized update each. Space: O(n) for `tails`.

Dry Run: `[10,9,2,5,3,7,101,18]`. tails=[]. 10→tails=[10]. 9→replaces10(pos0)→tails=[9]. 2→replaces9→tails=[2]. 5→append(5>2)→tails=[2,5]. 3→replaces5(first>=3 is at pos1)→tails=[2,3]. 7→append→tails=[2,3,7]. 101→append→tails=[2,3,7,101]. 18→replaces101(pos3)→tails=[2,3,7,18]. Final length = `4`. ✅ (matches expected answer; note the actual `tails` array `[2,3,7,18]` happens to also be a valid LIS here, but that's coincidental, not guaranteed in general)

Edge Cases: strictly decreasing array → answer 1 (`tails` keeps getting its single element replaced by smaller values). Already strictly increasing array → answer = n (`tails` grows by appending every time). All duplicate values → answer 1 (strict inequality required, so no element can extend another equal one).

Follow-up: "Number of Longest Increasing Subsequences" (count how many, not just the length) — needs an auxiliary `count[i]` array alongside `dp[i]`. "Russian Doll Envelopes" is a 2-D generalization: sort by one dimension, then run LIS on the other.

A11.5 Word Break

Difficulty: Medium | **Pattern:** DP (string segmentation) | **Companies:** Amazon, Google, Meta, Bloomberg

Problem Statement: Given a string `s` and a dictionary of strings `wordDict`, return true if `s` can be segmented into a space-separated sequence of one or more dictionary words (words can be reused).

Example: `s="leetcode", wordDict=["leet","code"]` → `true`.

Brute Force Approach Intuition: recursively try every possible "first word" prefix of the remaining string; if it's in the dictionary, recurse on the rest.

```
public boolean wordBreakBrute(String s, List<String> wordDict) {
    Set<String> dict = new HashSet<>(wordDict);
    return helper(s, 0, dict);
}
private boolean helper(String s, int start, Set<String> dict) {
    if (start == s.length()) return true;
    for (int end = start + 1; end <= s.length(); end++) {
        if (dict.contains(s.substring(start, end)) && helper(s, end, dict)) {
            return true;
        }
    }
    return false;
}
```

Time: O(2^n) worst case — overlapping subproblems (the same `start` index gets re-explored many times via different paths) recomputed repeatedly. Space: O(n) recursion stack.

Optimal Approach (Bottom-up DP) Intuition: `dp[i]` = true if `s[0..i)` can be segmented using dictionary words. `dp[0] = true` (empty prefix trivially breakable). For each `i`, check all `j < i`: if `dp[j]` is true and `s[j..i)` is a dictionary word, then `dp[i] = true`.

```
public boolean wordBreak(String s, List<String> wordDict) {
    Set<String> dict = new HashSet<>(wordDict);
    int n = s.length();
    boolean[] dp = new boolean[n + 1];
    dp[0] = true;
    for (int i = 1; i <= n; i++) {
        for (int j = 0; j < i; j++) {
            if (dp[j] && dict.contains(s.substring(j, i))) {
                dp[i] = true;
                break;
            }
        }
    }
    return dp[n];
}
```

Time: O(n²) for the DP (n² substring checks, each O(n) for substring extraction and hashing → technically O(n³), but typically described as O(n²) with O(1) amortized

substring hashing in practice, or exactly O(n² · L) if being precise about substring cost). Space: O(n) for `dp` + O(dictionary size) for the set.

Dry Run: `s="leetcode", wordDict=["leet","code"]`. dp[0]=true. i=4: j=0,dp[0]=true, s[0,4)="leet" in dict → dp[4]=true. i=8: j=4,dp[4]=true,s[4,8)="code" in dict → dp[8]=true. Final dp[8]=`true`. ✅

Edge Cases: empty string → `true` trivially (dp[0] is the base case). No valid segmentation exists → dp[n] stays false. Single-character words matching individual letters.

Follow-up: "Word Break II" (return all possible segmentations, not just true/false) — needs backtracking with memoization (memoize the list of valid segmentations per starting index) to avoid exponential blowup on inputs with many valid breaks.

A11.6 Combination Sum IV

Difficulty: Medium | **Pattern:** DP (order matters — permutations, not combinations, despite the name) | **Companies:** Google, Amazon

Problem Statement: Given an array of distinct positive integers and a target, return the number of possible combinations that add up to target, where **order matters** (e.g. `(1,2)` and `(2,1)` are counted separately — despite the problem's name, this is really counting ordered sequences).

Clarifying Questions: Confirm order matters (a classic point of confusion given the "combination" name) — this is what makes it a straightforward 1-D DP rather than a backtracking enumeration problem.

Brute Force Approach Intuition: recursively try every number as the "next" chosen number at each step, summing recursive results.

```
public int combinationSum4Brute(int[] nums, int target) {
    if (target == 0) return 1;
    if (target < 0) return 0;
    int count = 0;
    for (int num : nums) {
        count += combinationSum4Brute(nums, target - num);
    }
    return count;
}
```

Time: O(target^(target/min(nums))) roughly — exponential due to massive overlapping subproblems (the same `target` value is reached via many different orderings). Space: O(target) recursion depth.

Optimal Approach (Bottom-up DP, like an unbounded-choice "coin change counting" problem) Intuition: `dp[t]` = number of ordered ways to make sum `t`. For each target value from 1 up to the target, try every number in `nums` as a possible "last number added" — `dp[t] += dp[t - num]` for each valid `num <= t`.

```
public int combinationSum4(int[] nums, int target) {
    int[] dp = new int[target + 1];
    dp[0] = 1; // one way to make 0: choose nothing
    for (int t = 1; t <= target; t++) {
        for (int num : nums) {
            if (num <= t) {
                dp[t] += dp[t - num];
            }
        }
    }
    return dp[target];
}
```

Time: O(target · len(nums)). Space: O(target) for the dp array.

Dry Run: `nums=[1,2,3], target=4`. dp[0]=1. dp[1]: num1→dp[0]=1→dp[1]=1. dp[2]: num1→dp[1] =1; num2→dp[0]=1; dp[2]=2. dp[3]: num1→dp[2]=2; num2→dp[1]=1; num3→dp[0]=1; dp[3]=4. dp[4]: num1→dp[3]=4; num2→dp[2]=2; num3→dp[1]=1; dp[4]=7. Answer `7`. ✅ (matches known LeetCode example: 7 ordered combinations summing to 4 using 1,2,3)

Edge Cases: `target=0` → 1 (the empty combination). No numbers can reach target at all (e.g. all numbers larger than target) → dp[target] stays 0.

Follow-up: "Coin Change" (below) is the superficially similar but fundamentally different **unordered** counting variant (and it asks for minimum coins, not count of ways) — a classic point of confusion; be ready to clearly distinguish "order matters" (this problem, loop order: target outer, nums inner) vs. "order doesn't matter" (Coin Change 2 style, loop order: nums outer, target inner).

A11.7 Decode Ways

Difficulty: Medium | **Pattern:** DP (Fibonacci-shape with validity checks) | **Companies:** Facebook, Amazon, Microsoft

Problem Statement: A message of digits can be decoded where 'A'=1, 'B'=2, ..., 'Z'=26. Given an encoded string containing only digits, return the number of ways to decode it.

Example: `"226"` → `3` (`"2 2 6"`, `"22 6"`, `"2 26"`).

Clarifying Questions: Can the string contain '0'? (Yes — a lone '0' is invalid since no letter maps to 0; '0' is only valid as the second digit of a two-digit code 10-26.)

Brute Force Approach Intuition: recursively try decoding 1 digit or 2 digits at the current position (if valid), summing the ways from each choice.

```
public int numDecodingsBrute(String s) {
    return helper(s, 0);
}
private int helper(String s, int i) {
    if (i == s.length()) return 1;
    if (s.charAt(i) == '0') return 0;
    int ways = helper(s, i + 1); // take 1 digit
    if (i + 1 < s.length() && Integer.parseInt(s.substring(i, i + 2)) <= 26) {
        ways += helper(s, i + 2); // take 2 digits
    }
    return ways;
}
```

Time: O(2^n) worst case — overlapping subproblems. Space: O(n) recursion stack.

Optimal Approach (Bottom-up DP) Intuition: `dp[i]` = number of ways to decode the first `i` characters. `dp[i] += dp[i-1]` if the single digit `s[i-1]` is valid (1-9, not '0'). `dp[i] += dp[i-2]` if the two-digit number `s[i-2..i)` is valid (10-26).

```
public int numDecodings(String s) {
    int n = s.length();
    if (n == 0 || s.charAt(0) == '0') return 0;
    int[] dp = new int[n + 1];
    dp[0] = 1;
    dp[1] = 1;
    for (int i = 2; i <= n; i++) {
        int oneDigit = Integer.parseInt(s.substring(i - 1, i));
        int twoDigit = Integer.parseInt(s.substring(i - 2, i));
        if (oneDigit >= 1) dp[i] += dp[i - 1];
        if (twoDigit >= 10 && twoDigit <= 26) dp[i] += dp[i - 2];
    }
```

```
        return dp[n];
    }
```

Time: O(n). Space: O(n) for `dp` (can be reduced to O(1) with two rolling variables, same technique as Climbing Stairs).

Dry Run: `"226"`. dp[0]=1,dp[1]=1. i=2("22"): oneDigit=2>=1→dp[2]+=dp[1]=1; twoDigit=22 in[10,26]→dp[2]+=dp[0]=1; dp[2]=2. i=3("226", checking last 2 chars "26"): oneDigit=6>=1→ dp[3]+=dp[2]=2; twoDigit=26 in range→dp[3]+=dp[1]=1; dp[3]=3. Answer `3`. ✅

Edge Cases: string starts with '0' → 0 ways immediately (no valid decoding possible). String containing "00" or a lone "0" not preceded validly (e.g. "100" — the middle '0' must combine with the '1' before it as "10", then "0" alone is invalid, this is a classic tricky test case worth mentioning). All single-digit valid characters → dp grows like plain Climbing Stairs when every pair also happens to be valid.

Follow-up: "Decode Ways II" (string can contain `'*'` wildcards representing any digit 1-9) — same DP shape but each transition needs to sum over all matching digit possibilities, significantly more casework.

A11.8 Coin Change

Difficulty: Medium | **Pattern:** DP (unbounded knapsack, minimize count) | **Companies:** Amazon, Google, Uber — one of the most iconic DP questions

Problem Statement: Given coin denominations and an amount, return the fewest number of coins needed to make that amount (using unlimited supply of each denomination). Return -1 if impossible.

Example: `coins=[1,2,5], amount=11` → `3` (5+5+1).

Brute Force Approach Intuition: recursively try every coin as the "next" coin used, take the minimum count across all choices.

```
public int coinChangeBrute(int[] coins, int amount) {
    if (amount == 0) return 0;
    if (amount < 0) return Integer.MAX_VALUE;
    int minCoins = Integer.MAX_VALUE;
    for (int coin : coins) {
        int result = coinChangeBrute(coins, amount - coin);
```

```
            if (result != Integer.MAX_VALUE) minCoins = Math.min(minCoins, result + 1);
        }
        return minCoins;
    }
    // caller must check if the top-level result is still Integer.MAX_VALUE -> return
    -1
```

Time: O(coins.length ^ amount) — massive overlapping subproblem recomputation. Space: O(amount) recursion depth.

Optimal Approach (Bottom-up DP, unbounded knapsack style) Intuition: `dp[a]` = minimum coins to make amount `a`. Initialize `dp[0] = 0` and everything else to "infinity." For each amount from 1 to target, try every coin `<= a`: `dp[a] = min(dp[a], dp[a - coin] + 1)`.

```
public int coinChange(int[] coins, int amount) {
    int[] dp = new int[amount + 1];
    Arrays.fill(dp, amount + 1); // sentinel "infinity" (amount+1 coins is always
impossible)
    dp[0] = 0;
    for (int a = 1; a <= amount; a++) {
        for (int coin : coins) {
            if (coin <= a) {
                dp[a] = Math.min(dp[a], dp[a - coin] + 1);
            }
        }
    }
    return dp[amount] > amount ? -1 : dp[amount];
}
```

Time: O(amount · coins.length). Space: O(amount) for the dp array.

Dry Run: `coins=[1,2,5], amount=11`. dp[0]=0. Building up: dp[1]=1(using coin1). dp[2]=1 (coin2). dp[3]=2(1+2 or 1+1+1→min is 2 via coin1+dp[2]). dp[5]=1(coin5 directly). dp[6]=2 (coin5+coin1... dp[6]=min(dp[5]+1,dp[4]+1,dp[1]+1)=min(2,...,2)=2). Continuing up: dp[10]=2 (5+5). dp[11]=min(dp[10]+1,dp[9]+1,dp[6]+1)=min(3,...,3)=3. Answer `3`. ✅

Edge Cases: `amount=0` → 0 coins needed. No combination of coins can make the amount (e.g. `coins=[2], amount=3`) → dp[amount] stays above `amount` (sentinel), return -1.

Follow-up: "Coin Change 2" (count the **number of ways**, not minimum coins) — nearly identical DP shape but sums possibilities instead of taking a min, and crucially

loops `coins` in the **outer** loop (to count each combination once, unordered) rather than `amount` outer — a subtle but important loop-order distinction from Combination Sum IV.

A11.9 Partition Equal Subset Sum

Difficulty: Medium | **Pattern:** DP (0/1 Knapsack, subset-sum variant) | **Companies:** Google, Amazon, Facebook

Problem Statement: Given a non-empty array of positive integers, determine if it can be partitioned into two subsets with equal sum.

Clarifying Questions: This is equivalent to asking: "does some subset sum to exactly `totalSum / 2`?" (If `totalSum` is odd, the answer is immediately `false`.)

Brute Force Approach Intuition: try every subset (2^n of them) and check if any sums to `totalSum / 2`.

```
// Conceptual: enumerate all 2^n subsets, check sums -- O(2^n), impractical for
larger arrays.
```

Time: O(2^n).

Optimal Approach (0/1 Knapsack DP — "can we make this exact sum using each number at most once?") Intuition: this is exactly the classic subset-sum DP. `dp[s]` = true if some subset of the numbers processed so far sums to exactly `s`. Process numbers one at a time, updating `dp` from **high to low** sums (to avoid reusing the same number twice within one update pass — the hallmark of 0/1 knapsack as opposed to unbounded knapsack).

```
public boolean canPartition(int[] nums) {
    int totalSum = Arrays.stream(nums).sum();
    if (totalSum % 2 != 0) return false;
    int target = totalSum / 2;

    boolean[] dp = new boolean[target + 1];
    dp[0] = true;
    for (int num : nums) {
        for (int s = target; s >= num; s--) { // iterate downward!
            dp[s] = dp[s] || dp[s - num];
        }
    }
```

```
        return dp[target];
    }
```

Time: O(n · target) where target = totalSum/2. Space: O(target) for the dp array.

Dry Run: `[1,5,11,5]`. totalSum=22, target=11. dp[0]=true initially. Process 1: dp[1]=true (dp[0]). Process 5: dp[5]=true(dp[0]), dp[6]=true(dp[1]). Process 11: dp[11]=true(dp[0]) — already found! Process 5(again): further updates don't matter, dp[11] already true. Return `true` (subset `{11}` alone, or `{1,5,5}`, both sum to 11). ✅

Edge Cases: odd total sum → immediately `false` (can't split evenly, no need to even run DP). Single element array → `false` (can't split one element into two non-empty subsets with equal sum unless it's 0, which isn't possible for positive integers per constraints). All elements equal → depends on count parity.

Follow-up: "Why iterate `s` from high to low, not low to high?" — be ready to explain this precisely: iterating low-to-high would let the same number be "used" multiple times within a single number's update pass (turning 0/1 knapsack into unbounded knapsack by accident) — a very common and subtle bug. "Target Sum" (assign +/- signs to reach a target) reduces to this exact subset-sum DP via algebraic manipulation.

A11.10 Jump Game

Difficulty: Medium | **Pattern:** Greedy (reachable-range tracking) | **Companies:** Amazon, Microsoft, Facebook

Problem Statement: Given an array where `nums[i]` is the max jump length from index `i`, determine if you can reach the last index starting from index 0.

Example: `[2,3,1,1,4]` → `true`. `[3,2,1,0,4]` → `false`.

Brute Force Approach (DP — is each index reachable?) Intuition: `dp[i]` = true if index `i` is reachable from index 0. For each reachable index `i`, mark every index within `[i+1, i+nums[i]]` as reachable too.

```
public boolean canJumpDP(int[] nums) {
    int n = nums.length;
    boolean[] dp = new boolean[n];
    dp[0] = true;
    for (int i = 0; i < n; i++) {
        if (!dp[i]) continue;
```

```
            for (int step = 1; step <= nums[i] && i + step < n; step++) {
                dp[i + step] = true;
            }
        }
        return dp[n - 1];
    }
```

Time: O(n²) worst case (e.g. every index has a large jump value, causing many redundant marking operations). Space: O(n) for the dp array.

Optimal Approach (Greedy — track the farthest reachable index) Intuition: scan left to right, maintaining `maxReach` = the farthest index reachable so far. If at any point the current index `i` **exceeds** `maxReach`, we're stuck (can never have reached this index) — return false immediately. Otherwise keep extending `maxReach = max(maxReach, i + nums[i])`. If `maxReach` ever covers the last index, we can stop early.

```
public boolean canJump(int[] nums) {
    int maxReach = 0;
    for (int i = 0; i < nums.length; i++) {
        if (i > maxReach) return false; // this index is unreachable
        maxReach = Math.max(maxReach, i + nums[i]);
        if (maxReach >= nums.length - 1) return true; // early exit
    }
    return true;
}
```

Time: O(n) — single pass. Space: O(1).

Dry Run: `[2,3,1,1,4]`. i=0: 0<=maxReach(0), maxReach=max(0,0+2)=2. i=1: 1<=2, maxReach=max(2,1+3)=4>=4(n-1=4)→return `true` early. ✅ `[3,2,1,0,4]`: i=0:maxReach=3. i=1: maxReach=max(3,3)=3. i=2:maxReach=max(3,3)=3. i=3:maxReach=max(3,3)=3 (nums[3]=0 contributes nothing). i=4: 4>maxReach(3) → return `false`. ✅ (the '0' at index 3 traps us)

Edge Cases: single element array → trivially `true` (already at the last index). Array starting with 0 but length 1 (same as above). A 0 somewhere that can be "jumped over" by an earlier large jump vs. one that traps you — both handled correctly by the `maxReach` tracking.

Follow-up: "Jump Game II" (below, in Greedy section) asks for the *minimum number of jumps* to reach the end, not just feasibility — a related but distinct greedy technique (level-by-level BFS-like expansion of reachable range).

A12. Dynamic Programming (2-D)

A12.1 Unique Paths

Difficulty: Medium | **Pattern:** 2-D DP (grid paths) | **Companies:** Amazon, Microsoft, Bloomberg — foundational grid-DP intro

Problem Statement: A robot starts at the top-left of an `m x n` grid and can only move right or down. Count the number of unique paths to the bottom-right corner.

Brute Force Approach Intuition: recursively try both moves (right, down) from every cell, summing paths from each.

```
public int uniquePathsBrute(int m, int n) {
    if (m == 1 || n == 1) return 1;
    return uniquePathsBrute(m - 1, n) + uniquePathsBrute(m, n - 1);
}
```

Time: O(2^(m+n)) — massive overlapping subproblem recomputation. Space: O(m+n) recursion depth.

Optimal Approach (Bottom-up 2-D DP) Intuition: `dp[r][c]` = number of ways to reach cell `(r,c)` = `dp[r-1][c] + dp[r][c-1]` (came from above or from the left). The entire first row and first column are all `1` (only one way to reach them — a straight line of moves).

```
public int uniquePaths(int m, int n) {
    int[][] dp = new int[m][n];
    for (int r = 0; r < m; r++) dp[r][0] = 1;
    for (int c = 0; c < n; c++) dp[0][c] = 1;
    for (int r = 1; r < m; r++) {
        for (int c = 1; c < n; c++) {
            dp[r][c] = dp[r - 1][c] + dp[r][c - 1];
        }
    }
    return dp[m - 1][n - 1];
}
```

Time: O(m·n). Space: O(m·n) (reducible to O(n) with a single rolling row, since each row only depends on the row directly above it).

Space-Optimized Approach (O(n) rolling array)

```
public int uniquePathsOptimized(int m, int n) {
    int[] dp = new int[n];
    Arrays.fill(dp, 1);
    for (int r = 1; r < m; r++) {
        for (int c = 1; c < n; c++) {
            dp[c] += dp[c - 1]; // dp[c] currently holds "value from row above"
before update
        }
    }
    return dp[n - 1];
}
```

Time: O(m·n). Space: O(n).

Dry Run: `m=3, n=3`. dp grid fills: row0=[1,1,1], row1=[1,2,3], row2=[1,3,6]. Answer `dp[2][2]=6`. ✅ (This is also a known combinatorial identity: `C(m+n-2, m-1)`.)

Edge Cases: `m=1` or `n=1` → exactly 1 path (a straight line, no choices to make).

Follow-up: "Unique Paths II" (some cells are obstacles) — same DP but `dp[r][c] = 0` whenever that cell is an obstacle. "Minimum Path Sum" (below) is the same grid-DP shape but minimizing a cost instead of counting paths.

A12.2 Longest Common Subsequence

Difficulty: Medium | **Pattern:** 2-D DP (string comparison) | **Companies:** Amazon, Google, Microsoft — one of THE canonical 2-D DP questions

Problem Statement: Given two strings, return the length of their longest common subsequence (not necessarily contiguous, order preserved).

Example: `text1="abcde", text2="ace"` → `3` (`"ace"`).

Brute Force Approach Intuition: recursively compare characters at both ends; if they match, both indices advance together (+1 to the LCS); otherwise try advancing either string independently and take the max.

```
public int longestCommonSubsequenceBrute(String text1, String text2) {
    return helper(text1, text2, 0, 0);
}
private int helper(String t1, String t2, int i, int j) {
    if (i == t1.length() || j == t2.length()) return 0;
    if (t1.charAt(i) == t2.charAt(j)) {
        return 1 + helper(t1, t2, i + 1, j + 1);
```

```
    }
    return Math.max(helper(t1, t2, i + 1, j), helper(t1, t2, i, j + 1));
}
```

Time: O(2^(m+n)) — heavily overlapping subproblems. Space: O(m+n) recursion depth.

Optimal Approach (Bottom-up 2-D DP) Intuition: `dp[i][j]` = LCS length of `text1[0..i)` and `text2[0..j)`. If the characters at `i-1` and `j-1` match, extend the diagonal (`dp[i-1][j-1] + 1`); otherwise take the best of dropping a character from either string (`max(dp[i-1][j], dp[i][j-1])`).

```
public int longestCommonSubsequence(String text1, String text2) {
    int m = text1.length(), n = text2.length();
    int[][] dp = new int[m + 1][n + 1];
    for (int i = 1; i <= m; i++) {
        for (int j = 1; j <= n; j++) {
            if (text1.charAt(i - 1) == text2.charAt(j - 1)) {
                dp[i][j] = dp[i - 1][j - 1] + 1;
            } else {
                dp[i][j] = Math.max(dp[i - 1][j], dp[i][j - 1]);
            }
        }
    }
    return dp[m][n];
}
```

Time: O(m·n). Space: O(m·n) (reducible to O(min(m,n)) with two rolling rows).

Dry Run: `text1="abcde", text2="ace"`. Building the table: matches at ('a','a'), ('c','c'), ('e','e') diagonally extend the LCS. Final `dp[5][3] = 3`. ✅

Edge Cases: one string empty → LCS length 0. No common characters at all → 0. Identical strings → LCS = full length.

Follow-up: "Longest Common Substring" (must be **contiguous**, a subtly different problem) — DP transition resets to 0 on mismatch instead of taking a max, and the answer is the max value anywhere in the table, not just the bottom-right corner. "Edit Distance" (below) uses a near-identical table shape with a different recurrence (insert/delete/replace costs).

A12.3 Best Time to Buy and Sell Stock with Cooldown

Difficulty: Medium | **Pattern:** DP (State Machine) | **Companies:** Amazon, Google

Problem Statement: Maximize profit from multiple stock transactions, but after selling you must wait one day (cooldown) before buying again.

Brute Force Approach Intuition: recursively try every decision (buy, sell, or do nothing) at every day, respecting the cooldown constraint, take the max profit path.

```
// Conceptual: helper(day, holding, coolingDown) explores all decision branches
recursively
// without memoization -> exponential blowup, same shape as the optimal state
machine below
// but without caching.
```

Time: O(2^n) without memoization.

Optimal Approach (DP State Machine — three states per day) Intuition: on any given day, you're in exactly one of three states: **holding** a stock, **sold** today (entering cooldown tomorrow), or **resting** (not holding, free to buy). Track the best profit achievable in each state, updating all three simultaneously each day based on yesterday's states.

```
public int maxProfit(int[] prices) {
    if (prices.length == 0) return 0;
    int hold = -prices[0];   // holding a stock (max profit so far, negative since
we "spent" money)
    int sold = 0;            // just sold today
    int rest = 0;            // resting (free to buy), not just-sold

    for (int i = 1; i < prices.length; i++) {
        int prevHold = hold, prevSold = sold, prevRest = rest;
        hold = Math.max(prevHold, prevRest - prices[i]); // keep holding, or buy
today from rest state
        sold = prevHold + prices[i];                     // sell today
        rest = Math.max(prevRest, prevSold);             // stay resting, or
cooldown ends from sold state
    }
    return Math.max(sold, rest); // final answer can't be in "hold" state (must
have sold)
}
```

Time: O(n) — single pass. Space: O(1) — only three rolling variables.

Dry Run: `[1,2,3,0,2]`. Start: hold=-1,sold=0,rest=0. Day1(price2): hold=max(-1,0-2)=-1, sold=-1+2=1,rest=max(0,0)=0. Day2(price3): hold=max(-1,0-3)=-1,sold=-1+3=2,rest=max(0,1)=1. Day3(price0): hold=max(-1,1-

0)=1,sold=-1+0=-1,rest=max(1,2)=2. Day4(price2): hold=max(1,2-2)=1, sold=1+2=3,rest=max(2,-1)=2. Final max(sold=3,rest=2)=3. ✅ (matches known LeetCode example: buy day0 sell day1 profit1, cooldown day2, buy day3 sell day4 profit2, total 3)

Edge Cases: empty or single-day price array → 0 (no transaction possible). Strictly decreasing prices → 0 (never profitable to buy).

Follow-up: "Best Time to Buy and Sell Stock with Transaction Fee" — same state-machine idea, subtract a fee upon selling. "Best Time... III / IV" (at most 2 or k transactions) use a different state dimension (transaction count) instead of cooldown.

A12.4 Edit Distance

Difficulty: Hard | **Pattern:** 2-D DP (string transformation) | **Companies:** Google, Amazon, Microsoft, Meta — one of the most famous DP problems ever asked

Problem Statement: Given two strings `word1` and `word2`, return the minimum number of operations (insert, delete, replace) to convert `word1` into `word2`.

Brute Force Approach Intuition: recursively try all three operations at each mismatched position, take the min; if characters match, no operation needed, just advance both pointers.

```
public int minDistanceBrute(String word1, String word2) {
    return helper(word1, word2, word1.length(), word2.length());
}
private int helper(String w1, String w2, int i, int j) {
    if (i == 0) return j; // insert remaining characters of w2
    if (j == 0) return i; // delete remaining characters of w1
    if (w1.charAt(i - 1) == w2.charAt(j - 1)) return helper(w1, w2, i - 1, j - 1);
    int insert = helper(w1, w2, i, j - 1);
    int delete = helper(w1, w2, i - 1, j);
    int replace = helper(w1, w2, i - 1, j - 1);
    return 1 + Math.min(insert, Math.min(delete, replace));
}
```

Time: O(3^(m+n)) — exponential, heavily overlapping subproblems. Space: O(m+n) recursion depth.

Optimal Approach (Bottom-up 2-D DP) Intuition: `dp[i][j]` = min operations to convert the first `i` characters of `word1` into the first `j` characters of `word2`. Base cases:

converting to/from an empty string costs exactly the length of the other string (all inserts or all deletes). Otherwise, if characters match, no cost, inherit `dp[i-1][j-1]`; else take `1 + min(insert, delete, replace)`.

```
public int minDistance(String word1, String word2) {
    int m = word1.length(), n = word2.length();
    int[][] dp = new int[m + 1][n + 1];
    for (int i = 0; i <= m; i++) dp[i][0] = i;
    for (int j = 0; j <= n; j++) dp[0][j] = j;

    for (int i = 1; i <= m; i++) {
        for (int j = 1; j <= n; j++) {
            if (word1.charAt(i - 1) == word2.charAt(j - 1)) {
                dp[i][j] = dp[i - 1][j - 1];
            } else {
                int insert = dp[i][j - 1];
                int delete = dp[i - 1][j];
                int replace = dp[i - 1][j - 1];
                dp[i][j] = 1 + Math.min(insert, Math.min(delete, replace));
            }
        }
    }
    return dp[m][n];
}
```

Time: O(m·n). Space: O(m·n) (reducible to O(min(m,n)) with rolling rows).

Dry Run: `word1="horse", word2="ros"`. Known result: `3` operations (horse→rorse [replace h→r], rorse→rose [delete r], rose→ros [delete e]). DP table converges to `dp[5][3]=3`. ✅

Edge Cases: one string empty → cost = length of the other (all inserts/deletes). Identical strings → cost 0. Completely disjoint characters → cost = max(m,n) typically (all replaces plus possibly some inserts/deletes depending on length difference).

Follow-up: "One Edit Distance" (check if exactly 0 or 1 edit apart, without full DP — a simpler linear-scan problem). "Delete Operation for Two Strings" (only deletions allowed, no replace/insert) reduces to `m + n - 2*LCS(word1,word2)`.

A12.5 Minimum Path Sum

Difficulty: Medium | **Pattern:** 2-D DP (grid cost minimization) | **Companies:** Amazon, Microsoft

Problem Statement: Given a grid of non-negative numbers, find a path from top-left to bottom-right (moving only right or down) that minimizes the sum of numbers along the path.

Brute Force Approach Intuition: recursively try both moves (right, down) from every cell, take the min total.

```
public int minPathSumBrute(int[][] grid, int r, int c) {
    if (r == grid.length - 1 && c == grid[0].length - 1) return grid[r][c];
    if (r == grid.length - 1) return grid[r][c] + minPathSumBrute(grid, r, c + 1);
    if (c == grid[0].length - 1) return grid[r][c] + minPathSumBrute(grid, r + 1,
c);
    return grid[r][c] + Math.min(minPathSumBrute(grid, r + 1, c),
minPathSumBrute(grid, r, c + 1));
}
```

Time: O(2^(m+n)) — overlapping subproblems recomputed repeatedly. Space: O(m+n) recursion depth.

Optimal Approach (Bottom-up 2-D DP, in-place on the grid) Intuition: `dp[r][c]` = minimum path sum to reach `(r,c)` = `grid[r][c] + min(dp[r-1][c], dp[r][c-1])`. The first row/column can only be reached one way (straight line), so they're simple running sums.

```
public int minPathSum(int[][] grid) {
    int m = grid.length, n = grid[0].length;
    int[][] dp = new int[m][n];
    dp[0][0] = grid[0][0];
    for (int c = 1; c < n; c++) dp[0][c] = dp[0][c - 1] + grid[0][c];
    for (int r = 1; r < m; r++) dp[r][0] = dp[r - 1][0] + grid[r][0];
    for (int r = 1; r < m; r++) {
        for (int c = 1; c < n; c++) {
            dp[r][c] = grid[r][c] + Math.min(dp[r - 1][c], dp[r][c - 1]);
        }
    }
    return dp[m - 1][n - 1];
}
```

Time: O(m·n). Space: O(m·n) (reducible to O(n) with a rolling row, or O(1) extra by mutating the input grid in place if allowed).

Dry Run: `[[1,3,1],[1,5,1],[4,2,1]]`. Row0 running sum: [1,4,5]. Column0 running sum: [1,2,6]. dp[1][1]=5+min(4,2)=5+2=7. dp[1][2]=1+min(5,7)=1+5=6. dp[2][1]=2+min(7,6)=2+6=8. dp[2][2]=1+min(6,8)=1+6=7. Answer `7`. ✅ (matches canonical LeetCode example, path 1→3→1→1→1)

Edge Cases: single row or single column grid → only one possible path, answer = sum of all elements. Single cell grid → that cell's value.

Follow-up: Same DP shape as Unique Paths (A12.1) but with a cost-minimization recurrence instead of a path-counting one — recognize both as the identical "grid DP" template with a different combine function (`+` counts vs `min` costs).

A12.6 Maximal Square

Difficulty: Medium | **Pattern:** 2-D DP (grid, corner-dependent recurrence) | **Companies:** Amazon, Google

Problem Statement: Given a binary matrix, find the largest square containing only `1`s and return its area.

Brute Force Approach Intuition: for every cell as a potential top-left corner, try every possible square size expanding outward, checking all cells within are `1`.

```
// Conceptual: for each cell, for each candidate size s from 1 upward, verify all
s*s cells are '1'.
// O((m*n) * min(m,n)^2) in the worst case -- checking each candidate square from
scratch.
```

Time: O(m·n·min(m,n)²) roughly. Space: O(1) extra.

Optimal Approach (2-D DP: largest square ENDING at each cell as bottom-right corner) Intuition: `dp[r][c]` = side length of the largest all-`1`s square with its bottom-right corner at `(r,c)`. If `grid[r][c] == '1'`, this square's size is limited by the **minimum** of the three neighboring squares (top, left, top-left diagonal) plus 1 — because a square can only be as large as its most-constrained neighboring square allows.

```
public int maximalSquare(char[][] matrix) {
    int rows = matrix.length, cols = matrix[0].length;
    int[][] dp = new int[rows + 1][cols + 1];
    int maxSide = 0;
    for (int r = 1; r <= rows; r++) {
        for (int c = 1; c <= cols; c++) {
            if (matrix[r - 1][c - 1] == '1') {
                dp[r][c] = 1 + Math.min(dp[r - 1][c], Math.min(dp[r][c - 1], dp[r -
1][c - 1]));
                maxSide = Math.max(maxSide, dp[r][c]);
```

```
                }
            }
        }
        return maxSide * maxSide;
    }
```

Time: O(m·n). Space: O(m·n) (reducible to O(n) with a rolling row and one extra variable for the diagonal value).

Dry Run: `[["1","0","1","0","0"],["1","0","1","1","1"], ["1","1","1","1","1"], ["1","0","0","1","0"]]`. DP builds up; the largest square found has side `2` (a 2x2 block of 1s), giving area `4`. ✅ (matches canonical LeetCode example)

Edge Cases: all zeros → answer 0. All ones (full rectangle grid) → answer = `min(rows,cols)²`. Single row or column → max square side is 1 if any '1' exists, else 0.

Follow-up: "Maximal Rectangle" (not restricted to squares) — a harder variant solved by running the "Largest Rectangle in Histogram" algorithm (A3.6) on each row treating column heights of consecutive 1s as histogram bars.

A12.7 Regular Expression Matching

Difficulty: Hard | **Pattern:** 2-D DP (string matching with wildcards) | **Companies:** Google, Facebook, Amazon, Microsoft — one of the hardest and most respected DP questions

Problem Statement: Implement regular expression matching supporting `'.'` (matches any single character) and `'*'` (matches zero or more of the **preceding** element). The match must cover the **entire** input string.

Example: `s="aa", p="a*"` → `true`. `s="mississippi", p="mis*is*p*."` → `false`.

Clarifying Questions: Confirm `'*'` always follows a valid preceding character/`.` in the pattern (per problem's regex grammar guarantee) — you don't need to handle a leading `'*'`.

Brute Force Approach (Recursive, no memoization) Intuition: compare `s` and `p` position by position. If the next pattern character is `'*'`, branch into two choices: skip the `"x*"` pair entirely (zero occurrences), or, if the current `s` character matches `x`,

consume one `s` character and stay on the same pattern position (try matching more repetitions). Otherwise, match the current characters directly and advance both.

```
public boolean isMatchBrute(String s, String p) {
    return helper(s, p, 0, 0);
}
private boolean helper(String s, String p, int i, int j) {
    if (j == p.length()) return i == s.length();
    boolean firstMatch = i < s.length() && (p.charAt(j) == s.charAt(i) ||
p.charAt(j) == '.');

    if (j + 1 < p.length() && p.charAt(j + 1) == '*') {
        return helper(s, p, i, j + 2) // zero occurrences of p.charAt(j)
            || (firstMatch && helper(s, p, i + 1, j)); // one more occurrence
    }
    return firstMatch && helper(s, p, i + 1, j + 1);
}
```

Time: O(2^(m+n)) worst case — exponential without memoization, due to massively overlapping `(i,j)` subproblems. Space: O(m+n) recursion depth.

Optimal Approach (Bottom-up 2-D DP) Intuition: memoize (or tabulate) on `(i,j)` exactly as in the recursive version above — the state space is only `O(m·n)` distinct `(i,j)` pairs, so caching collapses the exponential recursion into polynomial time.

```
public boolean isMatch(String s, String p) {
    int m = s.length(), n = p.length();
    boolean[][] dp = new boolean[m + 1][n + 1];
    dp[0][0] = true;

    // handle patterns like "a*b*c*" that can match an empty string
    for (int j = 1; j <= n; j++) {
        if (p.charAt(j - 1) == '*' && j >= 2) {
            dp[0][j] = dp[0][j - 2];
        }
    }

    for (int i = 1; i <= m; i++) {
        for (int j = 1; j <= n; j++) {
            char sc = s.charAt(i - 1), pc = p.charAt(j - 1);
            if (pc == '.' || pc == sc) {
                dp[i][j] = dp[i - 1][j - 1];
            } else if (pc == '*') {
                char precedingChar = p.charAt(j - 2);
                dp[i][j] = dp[i][j - 2]; // zero occurrences of precedingChar
                if (precedingChar == '.' || precedingChar == sc) {
                    dp[i][j] = dp[i][j] || dp[i - 1][j]; // one more occurrence,
consume s[i-1]
                }
            }
            // else dp[i][j] stays false (no match, no wildcard)
```

```
            }
        }
        return dp[m][n];
    }
```

Time: O(m·n) — each of the m·n states computed once in O(1). Space: O(m·n) (reducible to O(n) with a rolling row).

Dry Run: `s="aa", p="a*"`. dp[0][0]=true. dp[0][2] (pattern "a*" matching empty s): j=2, p.charAt(1)='*',j>=2→*dp[0][2]=dp[0][0]=true (zero 'a's is valid for empty string). dp[1][2]: pc='* at j=2,precedingChar='a'. dp[1][2]=dp[1][0](false, not yet set/false by default) OR (precedingChar=='a'==sc('a')→check dp[0][2]=true) → dp[1][2]=true. dp[2][2]: similarly checks dp[1][2]=true → dp[2][2]=true. Final `dp[2][2]=true`. ✅

Edge Cases: empty pattern with non-empty string → false. Empty string with a pattern that can reduce to empty (e.g. `"a*b*"`) → true, handled by the special first-row initialization loop. Pattern with `'.'` combined with `'*'` (`".*"`) matches any string of any length.

Follow-up: "Wildcard Matching" (`'?'` and `'*'`, but here `'*'` matches any sequence directly, not "zero or more of the preceding character") — a related but subtly different DP recurrence; a very common point of confusion to be ready to distinguish clearly in an interview.

A13. Greedy & Intervals

A13.1 Jump Game II

Difficulty: Medium | **Pattern:** Greedy (BFS-like level expansion) | **Companies:** Amazon, Microsoft

Problem Statement: Given `nums[i]` = max jump length from index `i`, return the minimum number of jumps to reach the last index (guaranteed reachable).

Brute Force Approach (DP) Intuition: `dp[i]` = minimum jumps to reach index `i`, computed by checking every earlier index that can reach `i`.

```
    public int jumpDP(int[] nums) {
        int n = nums.length;
```

```
        int[] dp = new int[n];
        Arrays.fill(dp, Integer.MAX_VALUE);
        dp[0] = 0;
        for (int i = 0; i < n; i++) {
            for (int step = 1; step <= nums[i] && i + step < n; step++) {
                if (dp[i] != Integer.MAX_VALUE) dp[i + step] = Math.min(dp[i + step],
    dp[i] + 1);
            }
        }
        return dp[n - 1];
    }
```

Time: O(n²) worst case. Space: O(n).

Optimal Approach (Greedy, implicit BFS "levels") Intuition: think of it as BFS where each "level" is one jump. Track the farthest index reachable using jumps made so far (`farthest`), and the boundary of the current jump (`currentEnd`). When we reach `currentEnd`, we're forced to take another jump — increment the counter and extend the boundary to `farthest`.

```
    public int jump(int[] nums) {
        int jumps = 0, currentEnd = 0, farthest = 0;
        for (int i = 0; i < nums.length - 1; i++) {
            farthest = Math.max(farthest, i + nums[i]);
            if (i == currentEnd) {
                jumps++;
                currentEnd = farthest;
            }
        }
        return jumps;
    }
```

Time: O(n) — single pass. Space: O(1).

Dry Run: `[2,3,1,1,4]`. i=0: farthest=max(0,2)=2. icurrentEnd(0)→jumps=1,currentEnd=2. i=1: farthest=max(2,4)=4. i=2currentEnd(2)→jumps=2,currentEnd=4. i=3: farthest=max(4,4)=4 (loop ends before i=4 since condition is `i < n-1`). Answer `2`. ✅ (jump 0→1→4, or 0→2→4)

Edge Cases: single element array → 0 jumps needed (already at destination). Array where `nums[0]` alone reaches the end → 1 jump.

Follow-up: contrast directly with Jump Game (A11.10) which only asks feasibility — this one asks for the minimum count, requiring the greedy "level" tracking instead of a simple reachability scan.

A13.2 Gas Station

Difficulty: Medium | **Pattern:** Greedy (prefix sum reasoning) | **Companies:** Amazon, Bloomberg

Problem Statement: Given `gas[i]` (fuel available at station i) and `cost[i]` (fuel needed to travel from station i to i+1), determine the starting station index from which you can complete a full circuit, or -1 if impossible (guaranteed unique if it exists).

Brute Force Approach Intuition: try every starting station, simulate the full circuit, check if the tank ever goes negative.

```
public int canCompleteCircuitBrute(int[] gas, int[] cost) {
    int n = gas.length;
    for (int start = 0; start < n; start++) {
        int tank = 0;
        boolean success = true;
        for (int i = 0; i < n; i++) {
            int idx = (start + i) % n;
            tank += gas[idx] - cost[idx];
            if (tank < 0) { success = false; break; }
        }
        if (success) return start;
    }
    return -1;
}
```

Time: O(n²). Space: O(1).

Optimal Approach (Greedy, single pass) Intuition: two key facts: (1) if `sum(gas) < sum(cost)`, no solution exists at all. (2) if a total solution exists, and we track a running tank starting from index 0, the moment the tank goes negative at some index `i`, **no station between the previous start candidate and `i`** could have been a valid start either (they'd all run out even sooner) — so the next candidate start is `i+1`. This greedy elimination is provably correct and only needs one pass.

```
public int canCompleteCircuit(int[] gas, int[] cost) {
    int totalTank = 0, currentTank = 0, start = 0;
    for (int i = 0; i < gas.length; i++) {
        int diff = gas[i] - cost[i];
        totalTank += diff;
        currentTank += diff;
        if (currentTank < 0) {
            start = i + 1;
            currentTank = 0;
        }
```

```
        }
        return totalTank >= 0 ? start : -1;
    }
```

Time: O(n) — single pass. Space: O(1).

Dry Run: `gas=[1,2,3,4,5], cost=[3,4,5,1,2]`. diffs=[-2,-2,-2,3,3]. i=0:currentTank=-2<0→ start=1,reset0. i=1:currentTank=-2<0→start=2,reset0. i=2:currentTank=-2<0→start=3,reset0. i=3:currentTank=3. i=4:currentTank=6. totalTank=-2-2-2+3+3=0>=0→ answer `start=3`. ✅ (matches canonical LeetCode example)

Edge Cases: `sum(gas) == sum(cost)` exactly → a solution always exists (guaranteed unique per problem statement). `sum(gas) < sum(cost)` → -1 immediately (algorithm's final check handles this).

Follow-up: "Why does resetting `start` to `i+1` work?" — be ready to give the proof sketch: if tank goes negative arriving at station `i` starting from some candidate `s`, then for any `s < k <= i`, starting from `k` would have even less fuel by the time it reaches `i` than starting from `s` did (prefix sums argument) — so none of those can be valid starts either.

A13.3 Merge Intervals

Difficulty: Medium | **Pattern:** Greedy + Sorting | **Companies:** Facebook, Amazon, Microsoft, Bloomberg — extremely common intervals question

Problem Statement: Given an array of intervals, merge all overlapping intervals and return the resulting non-overlapping intervals.

Example: `[[1,3],[2,6],[8,10],[15,18]]` → `[[1,6],[8,10],[15,18]]`.

Brute Force Approach Intuition: repeatedly scan for any two overlapping intervals and merge them, until no more merges are possible.

```
// Conceptual: repeated O(n) scans merging any overlapping pair found, until
stable.
// O(n^2) or worse depending on how merges are detected/applied repeatedly.
```

Time: O(n²) or worse. Space: O(n).

Optimal Approach (Sort by start time, then linear merge scan) Intuition: sort intervals by start time. Then a single linear scan suffices: keep a "current merged interval"; if the next interval's start is `<=` the current merged interval's end, they overlap — extend the end to the max of both; otherwise the current merged interval is finalized and a new one begins.

```
public int[][] merge(int[][] intervals) {
    Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
    List<int[]> merged = new ArrayList<>();
    for (int[] interval : intervals) {
        if (merged.isEmpty() || merged.get(merged.size() - 1)[1] < interval[0]) {
            merged.add(interval);
        } else {
            merged.get(merged.size() - 1)[1] = Math.max(merged.get(merged.size() -
1)[1], interval[1]);
        }
    }
    return merged.toArray(new int[merged.size()][]);
}
```

Time: O(n log n) — dominated by the sort. Space: O(n) for the output (or O(log n) to O(n) for the sort itself).

Dry Run: sorted `[[1,3],[2,6],[8,10],[15,18]]` (already sorted). merged=1,3. Next [2,6]: 3>=2→overlap, extend to [1,6]. Next [8,10]: 6<8→no overlap, add new [8,10]. Next [15,18]: 10<15→no overlap, add new [15,18]. Final `[[1,6],[8,10],[15,18]]`. ✅

Edge Cases: single interval → returned as-is. All intervals overlapping into one big block → single merged interval in the output. No overlaps at all → output equals input (after sorting).

Follow-up: "Insert Interval" (next) is essentially a specialized version of this merge logic applied to inserting one new interval into an already-sorted, already-merged list.

A13.4 Insert Interval

Difficulty: Medium | **Pattern:** Greedy (linear scan, three phases) | **Companies:** Google, Facebook, LinkedIn

Problem Statement: Given a set of non-overlapping intervals sorted by start time, and a new interval, insert it and merge as necessary. Return the resulting intervals.

Brute Force Approach Intuition: add the new interval to the list, then run the general "Merge Intervals" algorithm (sort + linear merge) on the whole combined set.

```
public int[][] insertBrute(int[][] intervals, int[] newInterval) {
    List<int[]> combined = new ArrayList<>(Arrays.asList(intervals));
    combined.add(newInterval);
    int[][] combinedArr = combined.toArray(new int[0][]);
    return merge(combinedArr); // reuse A13.3's merge()
}
```

Time: O(n log n) — re-sorting is wasteful since the original array was already sorted. Space: O(n).

Optimal Approach (Three-phase linear scan, no sorting needed) Intuition: since the input is already sorted, process in three phases without ever sorting: (1) add all intervals that end strictly before `newInterval` starts (no overlap, keep as-is), (2) merge all intervals that overlap with `newInterval` by expanding its bounds, (3) add all remaining intervals that start strictly after the (possibly expanded) `newInterval` ends.

```
public int[][] insert(int[][] intervals, int[] newInterval) {
    List<int[]> result = new ArrayList<>();
    int i = 0, n = intervals.length;

    while (i < n && intervals[i][1] < newInterval[0]) {
        result.add(intervals[i]);
        i++;
    }

    while (i < n && intervals[i][0] <= newInterval[1]) {
        newInterval[0] = Math.min(newInterval[0], intervals[i][0]);
        newInterval[1] = Math.max(newInterval[1], intervals[i][1]);
        i++;
    }
    result.add(newInterval);

    while (i < n) {
        result.add(intervals[i]);
        i++;
    }
    return result.toArray(new int[result.size()][]);
}
```

Time: O(n) — single linear pass, no sorting needed since input is guaranteed pre-sorted. Space: O(n) for the output.

Dry Run: `intervals=[[1,3],[6,9]], newInterval=[2,5]`. Phase1: [1,3]'s end(3) >= newStart(2) → stop phase1 immediately (nothing added). Phase2: [1,3] overlaps

(1<=5)→newInterval becomes [min(2,1),max(5,3)]=[1,5], i=1. [6,9]: start(6) <= newInterval end(5)? No(6>5) → stop phase2. Add merged [1,5]. Phase3: add remaining [6,9]. Final `[[1,5],[6,9]]`. ✅

Edge Cases: new interval doesn't overlap anything (falls entirely before or after all existing intervals) → phases 1/3 handle it naturally, phase 2 loop simply doesn't execute. Empty input intervals list → result is just `[newInterval]`.

Follow-up: "Why not just re-sort?" — be ready to explain explicitly that exploiting the pre-sorted guarantee drops the complexity from O(n log n) to O(n), a meaningful improvement worth calling out proactively in an interview.

A13.5 Non-overlapping Intervals

Difficulty: Medium | **Pattern:** Greedy (sort by end time) | **Companies:** Facebook, Amazon — classic "activity selection" greedy problem

Problem Statement: Given an array of intervals, find the minimum number of intervals to remove to make the rest non-overlapping.

Example: `[[1,2],[2,3],[3,4],[1,3]]` → `1` (remove `[1,3]`).

Brute Force Approach Intuition: try every subset of intervals to keep, check which is the largest non-overlapping subset, then the answer is `n - |largest subset|`.

```
// Conceptual: 2^n subsets checked for pairwise non-overlap -- exponential,
impractical.
```

Time: O(2^n · n).

Optimal Approach (Greedy — classic Activity Selection, sort by END time) Intuition: to maximize the number of intervals we can **keep** (equivalently minimize removals), greedily always keep the interval that **ends earliest** among remaining candidates — this leaves the most room for future intervals. Sort by end time; scan left to right, keeping an interval if its start is `>=` the end of the last kept interval; otherwise it overlaps and must be removed (increment the removal counter).

```
public int eraseOverlapIntervals(int[][] intervals) {
    if (intervals.length == 0) return 0;
    Arrays.sort(intervals, (a, b) -> Integer.compare(a[1], b[1])); // sort by END
```

```
time
    int removals = 0;
    int lastEnd = intervals[0][1];
    for (int i = 1; i < intervals.length; i++) {
        if (intervals[i][0] < lastEnd) {
            removals++; // overlaps with the last kept interval -- remove this one
        } else {
            lastEnd = intervals[i][1]; // keep this interval, update the boundary
        }
    }
    return removals;
}
```

Time: O(n log n) — dominated by the sort. Space: O(log n) to O(n) for sort overhead.

Dry Run: `[[1,2],[2,3],[3,4],[1,3]]` sorted by end: `[[1,2],[2,3],[1,3],[3,4]]`. lastEnd=2. i=1([2,3]): 2>=2→keep, lastEnd=3. i=2([1,3]): 1<3→overlap→removals=1. i=3([3,4]): 3>=3→keep, lastEnd=4. Final removals=`1`. ✅

Edge Cases: no overlaps at all → 0 removals. All intervals identical/fully overlapping → `n-1` removals (keep exactly one). Single interval → 0 removals.

Follow-up: "Why sort by END time rather than START time?" — a classic interview question in itself; be ready to explain with a counterexample why sorting by start time can lead to a suboptimal greedy choice (e.g., a long early-starting interval blocking many short later-starting non-overlapping ones), whereas sorting by end time always keeps the option that leaves the most room for the future. "Minimum Number of Arrows to Burst Balloons" is essentially the identical greedy template applied to a slightly rephrased problem.

A14. Bit Manipulation & Math

A14.1 Number of 1 Bits

Difficulty: Easy | **Pattern:** Bit Manipulation | **Companies:** Apple, Microsoft — common warm-up

Problem Statement: Return the number of set bits (`1`s) in the binary representation of an unsigned integer (Hamming weight).

Brute Force Approach Intuition: check each of the 32 bits individually using a shifting mask.

```java
public int hammingWeightBrute(int n) {
    int count = 0;
    for (int i = 0; i < 32; i++) {
        if ((n & (1 << i)) != 0) count++;
    }
    return count;
}
```

Time: O(32) = O(1) technically, but always does exactly 32 iterations regardless of how many bits are actually set. Space: O(1).

Optimal Approach (Brian Kernighan's Algorithm — `n & (n-1)` clears the lowest set bit) Intuition: the expression `n & (n - 1)` always clears exactly the lowest set bit of `n` (a classic bit trick). Repeating this until `n` becomes 0 counts exactly as many iterations as there are set bits — no wasted iterations on zero bits.

```java
public int hammingWeight(int n) {
    int count = 0;
    while (n != 0) {
        n &= (n - 1); // clears the lowest set bit
        count++;
    }
    return count;
}
```

Time: O(k) where k = number of set bits (at most 32, but often far fewer) — strictly faster in practice than the fixed 32-iteration brute force. Space: O(1).

Dry Run: `n=11` (binary `1011`). n&(n-1): 1011 & 1010 = 1010, count=1. 1010&1001=1000, count=2. 1000&0111=0000, count=3. n==0, stop. Answer `3`. ✅ (1011 has three 1-bits)

Edge Cases: `n=0` → 0 (loop never executes). `n` with all 32 bits set (`-1` as a signed int, i.e. `0xFFFFFFFF`) → 32.

Follow-up: "Counting Bits" (next) asks for this count for every number from 0 to n, solvable far more efficiently with DP reusing previously computed results rather than running this bit-trick independently for each number.

A14.2 Counting Bits

Difficulty: Easy | **Pattern:** DP + Bit Manipulation | **Companies:** Google, Amazon

Problem Statement: Given `n`, return an array `ans` of length `n+1` where `ans[i]` is the number of `1` bits in the binary representation of `i`.

Brute Force Approach Intuition: run the Hamming-weight bit trick (A14.1) independently for every number from 0 to n.

```
public int[] countBitsBrute(int n) {
    int[] ans = new int[n + 1];
    for (int i = 0; i <= n; i++) {
        int x = i, count = 0;
        while (x != 0) { x &= (x - 1); count++; }
        ans[i] = count;
    }
    return ans;
}
```

Time: O(n log n) (or O(n·32) with the naive per-bit check). Space: O(n) for output, O(1) extra.

Optimal Approach (DP — reuse previously computed results) Intuition: `i & (i - 1)` removes the lowest set bit of `i`, giving a **smaller** number whose bit count we've already computed. So `ans[i] = ans[i & (i-1)] + 1`. Alternatively, `ans[i] = ans[i >> 1] + (i & 1)` (bit count of `i` = bit count of `i` with the last bit dropped, plus whether that last bit itself was set).

```
public int[] countBits(int n) {
    int[] ans = new int[n + 1];
    for (int i = 1; i <= n; i++) {
        ans[i] = ans[i & (i - 1)] + 1;
    }
    return ans;
}
```

Time: O(n) — each `ans[i]` computed in O(1) using a previously solved subproblem. Space: O(n) for the output array (no extra auxiliary space beyond that).

Dry Run: `n=5`. ans[0]=0. ans[1]=ans[1&0]+1=ans[0]+1=1. ans[2]=ans[2&1]+1=ans[0]+1=1. ans[3]=ans[3&2]+1=ans[2]+1=2. ans[4]=ans[4&3]+1=ans[0]+1=1. ans[5]=ans[5&4]+1=ans[4]+1=2. Final `[0,1,1,2,1,2]`. ✅ (matches known binary bit counts for 0-5)

Edge Cases: `n=0` → output `[0]` (just the single entry for zero). Powers of 2 always have exactly 1 set bit, verifiable in the output pattern.

Follow-up: "Can you do it without any built-in popcount and without the DP trick, just using the shift relation?" — the `ans[i >> 1] + (i & 1)` formulation is an equally valid O(n) alternative worth mentioning as a second approach.

A14.3 Reverse Bits

Difficulty: Easy | **Pattern:** Bit Manipulation | **Companies:** Apple, Microsoft

Problem Statement: Reverse the bits of a given 32-bit unsigned integer.

Brute Force / Direct Approach (this IS the standard bit-by-bit solution) Intuition: examine each of the 32 bits of the input from least-significant to most-significant, placing each one into the corresponding mirrored position of the result.

```
public int reverseBits(int n) {
    int result = 0;
    for (int i = 0; i < 32; i++) {
        int bit = (n >>> i) & 1;       // extract bit i from the input (unsigned
shift)
        result |= (bit << (31 - i));    // place it in the mirrored position in the
output
    }
    return result;
}
```

Time: O(32) = O(1) — fixed-size loop. Space: O(1).

Optimization (Divide and Conquer bit-swap, for repeated calls) Intuition: if this function is called many times, precompute byte-level reversal via a lookup table (256 entries), then reverse the 4 bytes and their internal bit order via table lookups — turns each call into O(1) with a much smaller constant factor. (Usually only worth mentioning verbally unless the interviewer explicitly asks about the "many calls" follow-up.)

Dry Run: `n = 00000010100101000001111010011100` (example from LeetCode). Bit-by-bit mirroring produces `00111001011110000010100101000000`. ✅ (matches canonical LeetCode example output `964176192`)

Edge Cases: `n=0` → reversed is still `0`. `n` with all bits set (`-1` / `0xFFFFFFFF`) → reversed is still all bits set.

Follow-up: "What if this function is called many times (e.g. millions)?" — precompute a byte-reversal lookup table as described above to amortize cost.

A14.4 Missing Number

Difficulty: Easy | **Pattern:** Math (sum formula) / XOR | **Companies:** Amazon, Microsoft

Problem Statement: Given an array containing `n` distinct numbers from `0` to `n` (one number missing), find the missing number.

Brute Force Approach Intuition: use a HashSet of all numbers present, then scan `0..n` to find which one isn't in the set.

```
public int missingNumberBrute(int[] nums) {
    Set<Integer> set = new HashSet<>();
    for (int n : nums) set.add(n);
    for (int i = 0; i <= nums.length; i++) {
        if (!set.contains(i)) return i;
    }
    return -1;
}
```

Time: O(n). Space: O(n) for the set.

Optimal Approach (Gauss's Sum Formula) Intuition: the sum of `0..n` has a known closed-form formula `n*(n+1)/2`. Subtract the actual sum of the array from this expected sum — the difference is exactly the missing number.

```
public int missingNumber(int[] nums) {
    int n = nums.length;
    int expectedSum = n * (n + 1) / 2;
    int actualSum = 0;
    for (int num : nums) actualSum += num;
    return expectedSum - actualSum;
}
```

Time: O(n). Space: O(1) — no extra data structure needed.

Optimal Alternative (XOR) Intuition: XOR-ing every index `0..n` together with every array value cancels out all pairs that appear twice, leaving only the missing number (since XOR of a number with itself is 0, and XOR with 0 is identity).

```java
public int missingNumberXOR(int[] nums) {
    int result = nums.length;
    for (int i = 0; i < nums.length; i++) {
        result ^= i ^ nums[i];
    }
    return result;
}
```

Time: O(n). Space: O(1). (Preferred over the sum formula when integer overflow for very large n is a concern, since XOR never overflows.)

Dry Run: `[3,0,1]`, n=3. expectedSum=3*4/2=6. actualSum=3+0+1=4. Answer=6-4=2. ✅

Edge Cases: missing number is 0 → formula still works correctly (0 contributes nothing to actualSum, expectedSum accounts for it). Missing number is n (the largest possible) → also handled correctly by the same formula.

Follow-up: "What if there could be duplicates and multiple missing numbers?" — a fundamentally different problem requiring a full frequency/presence array instead of a single closed-form trick.

A14.5 Sum of Two Integers (Without + or -)

Difficulty: Medium | **Pattern:** Bit Manipulation (binary addition simulation) | **Companies:** Amazon, Facebook — tests deep bitwise understanding

Problem Statement: Calculate the sum of two integers `a` and `b` without using the `+` or `-` operators.

Brute Force-adjacent Approach (increment/decrement in a loop — technically avoids +/- operators but is very inefficient) Intuition: repeatedly increment `a` (using `++`, which is technically not the `+` operator) while decrementing `b`, until `b` reaches 0.

```java
public int getSumSlow(int a, int b) {
    while (b > 0) { a++; b--; }
    while (b < 0) { a--; b++; }
    return a;
}
```

Time: O(|b|) — could be extremely slow for large values (e.g. `b = 10^9`). Space: O(1). Not the intended "bitwise" solution but technically satisfies the literal operator restriction.

Optimal Approach (Bitwise addition — simulate binary full-adder logic) Intuition: in binary addition, `a XOR b` gives the sum **without carrying**, and `(a AND b) << 1` gives exactly the carry bits that need to be added in. Repeat this process (treating the new sum and new carry as the next `a` and `b`) until there's no carry left.

```
public int getSum(int a, int b) {
    while (b != 0) {
        int carry = (a & b) << 1;
        a = a ^ b;
        b = carry;
    }
    return a;
}
```

Time: O(1) — bounded by 32 iterations at most (bit width), since each iteration shifts the carry further left, eventually overflowing out of the 32-bit range. Space: O(1).

Dry Run: `a=2(010), b=3(011)`. carry=(010&011)<<1=(010)<<1=100. a=010^011=001. b=100. Next iter: carry=(001&100)<<1=0<<1=0. a=001^100=101(=5). b=0, loop ends. Answer `5`. ✅ (2+3=5)

Edge Cases: negative numbers (Java's two's-complement representation makes the XOR/AND/ carry logic work correctly automatically, no special casing needed). One operand is 0 → loop terminates immediately with `a` unchanged (correct, since carry becomes 0 right away or `b` starts at 0).

Follow-up: "Implement multiplication without `*`" — repeated shifted addition using this same `getSum` as a building block (binary long multiplication).

Part B: Extended Problem Bank (By Pattern)

Part A gave every problem the full interview treatment. Part B covers a much wider net of problems in a **condensed** format — problem statement, brute force (approach + code + complexity), optimal (approach + code + complexity), and a one-line key insight

— so you get broad exposure to the remaining common patterns and problem shapes without re-reading the same level of ceremony 60 more times. Use Part 0's framework to mentally expand any of these back into a full interview walkthrough during practice.

B1. More Arrays / Strings

B1.1 Two Sum II - Input Array Is Sorted

Difficulty: Medium | **Pattern:** Two Pointers | **Companies:** Amazon **Problem:** Given a 1-indexed sorted array, find two numbers summing to target; return their 1-indexed positions. **Brute Force:** nested loop, O(n²), O(1) space. **Optimal:** two pointers from both ends — since sorted, if sum too big move `right` left, if too small move `left` right.

```
public int[] twoSum(int[] numbers, int target) {
    int left = 0, right = numbers.length - 1;
    while (left < right) {
        int sum = numbers[left] + numbers[right];
        if (sum == target) return new int[]{left + 1, right + 1};
        else if (sum < target) left++;
        else right--;
    }
    return new int[]{-1, -1};
}
```

Time: O(n). Space: O(1). **Key insight:** sorted input → prefer two pointers over hashing when O(1) space is desired.

B1.2 Longest Palindromic Substring

Difficulty: Medium | **Pattern:** Expand Around Center / DP | **Companies:** Amazon, Microsoft, Meta — extremely common **Problem:** Return the longest palindromic substring of `s`. **Brute Force:** check every substring for palindrome-ness, O(n³) (O(n²) substrings × O(n) check). **Optimal (Expand Around Center):** every palindrome has a center (a single char for odd length, between two chars for even length); expand outward from each of the `2n-1` possible centers while both sides match.

```
public String longestPalindrome(String s) {
    if (s.isEmpty()) return "";
    int start = 0, maxLen = 1;
```

```
    for (int center = 0; center < s.length(); center++) {
        int len1 = expand(s, center, center);     // odd length
        int len2 = expand(s, center, center + 1); // even length
        int len = Math.max(len1, len2);
        if (len > maxLen) {
            maxLen = len;
            start = center - (len - 1) / 2;
        }
    }
    return s.substring(start, start + maxLen);
}
private int expand(String s, int left, int right) {
    while (left >= 0 && right < s.length() && s.charAt(left) == s.charAt(right)) {
left--; right++; }
    return right - left - 1;
}
```

Time: O(n²). Space: O(1). **Key insight:** an O(n²) DP table (`dp[i][j]` = is `s[i..j]` a palindrome) solves this too and is reused directly for Palindrome Partitioning (A9.5); Manacher's algorithm achieves O(n) but is rarely required in interviews.

B1.3 Palindromic Substrings

Difficulty: Medium | **Pattern:** Expand Around Center | **Companies:** Facebook
Problem: Count how many substrings of `s` are palindromes. **Brute Force:** check all O(n²) substrings, O(n) each → O(n³). **Optimal:** same expand-around-center idea as B1.2, but increment a counter on every successful expansion instead of tracking the longest.

```
public int countSubstrings(String s) {
    int count = 0;
    for (int center = 0; center < s.length(); center++) {
        count += expandAndCount(s, center, center);
        count += expandAndCount(s, center, center + 1);
    }
    return count;
}
private int expandAndCount(String s, int left, int right) {
    int count = 0;
    while (left >= 0 && right < s.length() && s.charAt(left) == s.charAt(right)) {
        count++; left--; right++;
    }
    return count;
}
```

Time: O(n²). Space: O(1). **Key insight:** identical technique as Longest Palindromic Substring, different aggregation.

B1.4 Valid Sudoku

Difficulty: Medium | **Pattern:** Hashing (Sets per row/col/box) | **Companies:** Amazon, Apple **Problem:** Determine if a 9x9 Sudoku board is valid (each filled row, column, and 3x3 box has no repeated digits). **Brute Force:** for each cell, scan its entire row, column, and box for duplicates — O(1) technically (fixed 81 cells) but O(81·27) work with redundant re-scans. **Optimal:** one pass, maintain 9 row-sets, 9 column-sets, and 9 box-sets; for each filled cell check/add to the corresponding sets.

```
public boolean isValidSudoku(char[][] board) {
    Set<String> seen = new HashSet<>();
    for (int r = 0; r < 9; r++) {
        for (int c = 0; c < 9; c++) {
            char val = board[r][c];
            if (val == '.') continue;
            int box = (r / 3) * 3 + (c / 3);
            if (!seen.add("row" + r + val) || !seen.add("col" + c + val) ||
!seen.add("box" + box + val)) {
                return false;
            }
        }
    }
    return true;
}
```

Time: O(81) = O(1) (fixed board size). Space: O(81) = O(1). **Key insight:** the `(r/3)*3 + c/3` formula is the standard way to compute a 3x3 box index — memorize it.

B1.5 Rotate Array

Difficulty: Medium | **Pattern:** Array reversal trick | **Companies:** Amazon, Microsoft **Problem:** Rotate an array to the right by `k` steps, in place. **Brute Force:** rotate one step at a time, k times — O(n·k). **Optimal (Reversal Trick):** reverse the whole array, then reverse the first `k` elements, then reverse the remaining `n-k` elements — three reversals produce the rotation in O(n) with O(1) extra space.

```java
public void rotate(int[] nums, int k) {
    int n = nums.length;
    k %= n;
    reverse(nums, 0, n - 1);
    reverse(nums, 0, k - 1);
    reverse(nums, k, n - 1);
}
private void reverse(int[] nums, int left, int right) {
    while (left < right) {
        int t = nums[left]; nums[left] = nums[right]; nums[right] = t;
        left++; right--;
    }
}
```

Time: O(n). Space: O(1). **Key insight:** the "reverse whole, reverse parts" trick is broadly reusable for any rotation problem (also works for left rotation with adjusted split point).

B1.6 Majority Element

Difficulty: Easy | **Pattern:** Boyer-Moore Voting | **Companies:** Amazon **Problem:** Given an array where an element appears more than `n/2` times, find it. **Brute Force:** HashMap frequency count, O(n) time O(n) space (already decent, but not O(1) space). **Optimal (Boyer-Moore Voting Algorithm):** maintain a `candidate` and a `count`; increment count when seeing the candidate, decrement otherwise; when count hits 0, switch candidates. The guaranteed majority element always survives this process.

```java
public int majorityElement(int[] nums) {
    int candidate = nums[0], count = 0;
    for (int num : nums) {
        if (count == 0) candidate = num;
        count += (num == candidate) ? 1 : -1;
    }
    return candidate;
}
```

Time: O(n). Space: O(1). **Key insight:** works only because a true majority (>n/2) is guaranteed to exist; without that guarantee you'd need a verification pass.

B1.7 Next Permutation

Difficulty: Medium | **Pattern:** Array manipulation (lexicographic) | **Companies:** Amazon, Facebook, Bloomberg **Problem:** Rearrange numbers into the lexicographically next greater permutation, in place (wrap to smallest if already the largest permutation). **Brute Force:** generate all permutations, sort them, find current, return the next one — O(n!). **Optimal:** (1) scan from the right to find the first index `i` where `nums[i] < nums[i+1]` (the "pivot"); (2) if found, scan from the right again to find the first element greater than `nums[i]` and swap; (3) reverse the suffix after index `i` to get the smallest possible arrangement of that suffix.

```
public void nextPermutation(int[] nums) {
    int n = nums.length;
    int i = n - 2;
    while (i >= 0 && nums[i] >= nums[i + 1]) i--;
    if (i >= 0) {
        int j = n - 1;
        while (nums[j] <= nums[i]) j--;
        int t = nums[i]; nums[i] = nums[j]; nums[j] = t;
    }
    reverse(nums, i + 1, n - 1);
}
private void reverse(int[] nums, int left, int right) {
    while (left < right) { int t = nums[left]; nums[left] = nums[right];
nums[right] = t; left++; right--; }
}
```

Time: O(n). Space: O(1). **Key insight:** the "find pivot from the right, swap, reverse suffix" is the standard textbook next-permutation algorithm — memorize this exact three-step shape.

B1.8 First Missing Positive

Difficulty: Hard | **Pattern:** Cyclic Sort (index-as-hash) | **Companies:** Amazon, Microsoft — famous O(n) time O(1) space hard problem **Problem:** Given an unsorted array, find the smallest missing positive integer, in O(n) time and O(1) extra space. **Brute Force:** HashSet of all values, then scan 1,2,3,... checking membership — O(n) time but O(n) space (violates the space constraint, still worth stating first). **Optimal (Cyclic Sort / index-as-hash placement):** the answer must lie in `[1, n+1]`. Place each value `v` (if in range `[1,n]`) at index `v-1` by swapping, so a correctly-placed array has `nums[i] == i+1` everywhere. Then scan for the first index where this fails — that's the answer.

```java
public int firstMissingPositive(int[] nums) {
    int n = nums.length;
    for (int i = 0; i < n; i++) {
        while (nums[i] > 0 && nums[i] <= n && nums[nums[i] - 1] != nums[i]) {
            int t = nums[nums[i] - 1];
            nums[nums[i] - 1] = nums[i];
            nums[i] = t;
        }
    }
    for (int i = 0; i < n; i++) {
        if (nums[i] != i + 1) return i + 1;
    }
    return n + 1;
}
```

Time: O(n) — each swap places at least one value correctly, bounding total swaps to O(n). Space: O(1) — sorts in place using the array itself as a hash table. **Key insight:** "Cyclic Sort" (placing values at their "correct" index via swapping) is the go- to pattern whenever an array's values are constrained to a range tied to its own length.

B2. More Matrix Problems

B2.1 Rotate Image

Difficulty: Medium | **Pattern:** Matrix manipulation (transpose + reverse) | **Companies:** Amazon, Microsoft, Apple **Problem:** Rotate an `n x n` matrix 90 degrees clockwise, in place. **Brute Force:** copy to a new matrix, compute rotated positions directly (`new[c][n-1-r] = old[r][c]`) — correct but uses O(n²) extra space, violating the common "in place" requirement. **Optimal:** transpose the matrix (swap `matrix[r][c]` with `matrix[c][r]`), then reverse each row — this composition of two simple O(n²) operations produces a 90° clockwise rotation in place.

```java
public void rotate(int[][] matrix) {
    int n = matrix.length;
    for (int r = 0; r < n; r++) {
        for (int c = r + 1; c < n; c++) {
            int t = matrix[r][c]; matrix[r][c] = matrix[c][r]; matrix[c][r] = t;
        }
    }
    for (int[] row : matrix) {
        for (int left = 0, right = n - 1; left < right; left++, right--) {
            int t = row[left]; row[left] = row[right]; row[right] = t;
        }
```

```
    }
}
```

Time: O(n²). Space: O(1). **Key insight:** "transpose then reverse rows" = 90° clockwise; "transpose then reverse columns" (or reverse rows then transpose) = 90° counter-clockwise — useful to remember both directions.

B2.2 Spiral Matrix

Difficulty: Medium | **Pattern:** Matrix traversal (boundary shrinking) | **Companies:** Amazon, Microsoft, Facebook **Problem:** Return all elements of a matrix in spiral (clockwise) order. **Brute Force:** simulate direction-by-direction with a visited matrix, changing direction on boundary/visited hit — works but needs O(m·n) extra space for the visited matrix. **Optimal:** maintain four shrinking boundaries (`top`, `bottom`, `left`, `right`); traverse right along `top`, down along `right`, left along `bottom`, up along `left`, shrinking each boundary after its traversal — no extra visited matrix needed.

```
public List<Integer> spiralOrder(int[][] matrix) {
    List<Integer> result = new ArrayList<>();
    int top = 0, bottom = matrix.length - 1, left = 0, right = matrix[0].length -
1;
    while (top <= bottom && left <= right) {
        for (int c = left; c <= right; c++) result.add(matrix[top][c]);
        top++;
        for (int r = top; r <= bottom; r++) result.add(matrix[r][right]);
        right--;
        if (top <= bottom) {
            for (int c = right; c >= left; c--) result.add(matrix[bottom][c]);
            bottom--;
        }
        if (left <= right) {
            for (int r = bottom; r >= top; r--) result.add(matrix[r][left]);
            left--;
        }
    }
    return result;
}
```

Time: O(m·n). Space: O(1) extra (excluding output). **Key insight:** the two `if` guards (`top <= bottom`, `left <= right`) before the last two edges are essential to avoid double-counting elements on a single remaining row/column.

B2.3 Set Matrix Zeroes

Difficulty: Medium | **Pattern:** Matrix manipulation (using the matrix itself as markers) | **Companies:** Amazon, Microsoft **Problem:** Given a matrix, if an element is 0, set its entire row and column to 0, in place. **Brute Force:** use a separate `boolean[m][n]` (or two sets of row/column indices) to record which rows/columns need zeroing, then a second pass applies it — O(m·n) time, O(m+n) extra space. **Optimal (O(1) extra space):** use the **first row and first column of the matrix itself** as the marker arrays, with two extra booleans to remember whether the first row/column originally contained a zero (since they double as both data and markers).

```
public void setZeroes(int[][] matrix) {
    int rows = matrix.length, cols = matrix[0].length;
    boolean firstRowZero = false, firstColZero = false;
    for (int c = 0; c < cols; c++) if (matrix[0][c] == 0) firstRowZero = true;
    for (int r = 0; r < rows; r++) if (matrix[r][0] == 0) firstColZero = true;

    for (int r = 1; r < rows; r++) {
        for (int c = 1; c < cols; c++) {
            if (matrix[r][c] == 0) { matrix[r][0] = 0; matrix[0][c] = 0; }
        }
    }
    for (int r = 1; r < rows; r++) {
        for (int c = 1; c < cols; c++) {
            if (matrix[r][0] == 0 || matrix[0][c] == 0) matrix[r][c] = 0;
        }
    }
    if (firstRowZero) for (int c = 0; c < cols; c++) matrix[0][c] = 0;
    if (firstColZero) for (int r = 0; r < rows; r++) matrix[r][0] = 0;
}
```

Time: O(m·n). Space: O(1) extra. **Key insight:** "use part of the existing input structure as your marker/visited storage" is a recurring O(1)-space trick across many matrix/array problems.

B2.4 Search a 2D Matrix II

Difficulty: Medium | **Pattern:** Staircase Search (top-right/bottom-left start) | **Companies:** Amazon, Microsoft **Problem:** Matrix where each row is sorted ascending left-to-right and each column is sorted ascending top-to-bottom (but NOT fully flat-sorted like A4.2). Find if target exists. **Brute Force:** scan every cell, O(m·n). **Optimal:** start at the **top-right** corner. If the current value equals target, found. If current > target,

the entire column below is even larger — move left. If current < target, the entire row to the left is even smaller — move down. This eliminates a full row or column each step.

```
public boolean searchMatrix(int[][] matrix, int target) {
    int r = 0, c = matrix[0].length - 1;
    while (r < matrix.length && c >= 0) {
        if (matrix[r][c] == target) return true;
        else if (matrix[r][c] > target) c--;
        else r++;
    }
    return false;
}
```

Time: O(m+n). Space: O(1). **Key insight:** starting at top-right (or bottom-left) is what makes the "eliminate one row/column per step" logic valid — starting at top-left or bottom-right does NOT work (both directions could go either way).

B2.5 Game of Life

Difficulty: Medium | **Pattern:** Matrix simulation with in-place state encoding | **Companies:** Amazon, Google, Dropbox **Problem:** Given a board of live(1)/dead(0) cells, compute the next generation in place using Conway's rules (based on live neighbor counts). **Brute Force:** copy the board, compute next-state values into the copy using the original for neighbor lookups — correct but O(m·n) extra space. **Optimal (in-place with extra encoded states):** use extra integer states beyond just 0/1 to encode "was alive, now dead" (2) and "was dead, now alive" (3) — so neighbor-counting during the single pass can still check `cell % 2 == 1` to know the *original* state even after modification, then a final pass normalizes 2→0 and 3→1.

```
public void gameOfLife(int[][] board) {
    int rows = board.length, cols = board[0].length;
    int[][] dirs = {{-1,-1},{-1,0},{-1,1},{0,-1},{0,1},{1,-1},{1,0},{1,1}};
    for (int r = 0; r < rows; r++) {
        for (int c = 0; c < cols; c++) {
            int liveNeighbors = 0;
            for (int[] d : dirs) {
                int nr = r + d[0], nc = c + d[1];
                if (nr >= 0 && nr < rows && nc >= 0 && nc < cols && (board[nr][nc]
== 1 || board[nr][nc] == 2)) {
                    liveNeighbors++;
                }
            }
            if (board[r][c] == 1 && (liveNeighbors < 2 || liveNeighbors > 3))
board[r][c] = 2; // live -> dead
```

```
            if (board[r][c] == 0 && liveNeighbors == 3) board[r][c] = 3; // dead ->
live
        }
    }
    for (int r = 0; r < rows; r++) {
        for (int c = 0; c < cols; c++) {
            board[r][c] %= 2;
        }
    }
}
```

Time: O(m·n). Space: O(1) extra. **Key insight:** encoding "transition" states using extra integer values (instead of booleans) is a reusable trick whenever you need both the old and new state simultaneously during an in-place simulation.

B3. More Linked List

B3.1 Palindrome Linked List

Difficulty: Easy | **Pattern:** Fast/Slow Pointer + Reversal | **Companies:** Amazon, Facebook **Problem:** Determine if a singly linked list is a palindrome, ideally in O(1) space. **Brute Force:** copy values into an ArrayList, use two pointers from both ends to compare — O(n) time, O(n) space. **Optimal:** find the middle (fast/slow pointers), reverse the second half in place, compare the first half against the reversed second half node-by-node (optionally restore the list afterward).

```
public boolean isPalindrome(ListNode head) {
    ListNode slow = head, fast = head;
    while (fast != null && fast.next != null) { slow = slow.next; fast =
fast.next.next; }

    ListNode prev = null, cur = slow;
    while (cur != null) { ListNode next = cur.next; cur.next = prev; prev = cur;
cur = next; }

    ListNode left = head, right = prev;
    boolean result = true;
    while (right != null) {
        if (left.val != right.val) { result = false; break; }
        left = left.next; right = right.next;
    }
    return result;
}
```

Time: O(n). Space: O(1). **Key insight:** reuses the exact "find middle + reverse half" combo from Reorder List (A5.3) — recognize this as a reusable two-step linked-list template.

B3.2 Intersection of Two Linked Lists

Difficulty: Easy | **Pattern:** Two Pointers (length equalization) | **Companies:** Amazon, Microsoft **Problem:** Find the node at which two singly linked lists intersect, or null if they don't. **Brute Force:** for every node in list A, scan all of list B checking reference equality — O(m·n). Better brute force: HashSet of list A's nodes, then scan list B checking membership — O(m+n) time, O(m) space. **Optimal (O(1) space, elegant two-pointer):** run two pointers, one starting at each list's head; when a pointer reaches the end of its list, redirect it to the **other** list's head. Both pointers then travel exactly `lenA + lenB` total nodes and meet at the intersection (or both reach null simultaneously if there is no intersection).

```
public ListNode getIntersectionNode(ListNode headA, ListNode headB) {
    if (headA == null || headB == null) return null;
    ListNode a = headA, b = headB;
    while (a != b) {
        a = (a == null) ? headB : a.next;
        b = (b == null) ? headA : b.next;
    }
    return a; // either the intersection node, or null (both reach null together)
}
```

Time: O(m+n). Space: O(1). **Key insight:** "redirect to the other list's head" equalizes the total distance traveled by both pointers, guaranteeing they meet exactly at the intersection.

B3.3 Odd Even Linked List

Difficulty: Medium | **Pattern:** In-place pointer regrouping | **Companies:** Facebook, Amazon **Problem:** Group all odd-indexed nodes together followed by all even-indexed nodes (1-indexed by position, not value), in O(1) extra space. **Brute Force:** copy values into two lists (odd-position, even-position) then concatenate — O(n) time, O(n) extra space for the temporary lists. **Optimal:** maintain two chains (`odd`, `even`) built by

advancing two pointers two steps at a time, then splice the even chain onto the end of the odd chain.

```
public ListNode oddEvenList(ListNode head) {
    if (head == null || head.next == null) return head;
    ListNode odd = head, even = head.next, evenHead = even;
    while (even != null && even.next != null) {
        odd.next = even.next;
        odd = odd.next;
        even.next = odd.next;
        even = even.next;
    }
    odd.next = evenHead;
    return head;
}
```

Time: O(n). Space: O(1). **Key insight:** advancing both `odd` and `even` two steps at a time, then splicing at the end, avoids any auxiliary storage entirely.

B3.4 Flatten a Multilevel Doubly Linked List

Difficulty: Medium | **Pattern:** DFS / Stack on linked structures | **Companies:** Amazon, Facebook **Problem:** Each node may have a `child` pointer to a separate doubly linked list; flatten the whole structure into a single-level doubly linked list. **Brute Force:** recursively flatten each child list first, then splice it in — conceptually straightforward but easy to get pointer bookkeeping wrong without care. **Optimal (iterative with an explicit stack):** walk the list; whenever a node has a child, push the current node's `next` onto a stack (to resume later), then splice the child list in immediately after the current node and clear the child pointer. When `next` becomes null, pop the stack to resume the outer list.

```
public Node flatten(Node head) {
    if (head == null) return null;
    Node cur = head;
    Deque<Node> stack = new ArrayDeque<>();
    while (cur != null) {
        if (cur.child != null) {
            if (cur.next != null) stack.push(cur.next);
            cur.next = cur.child;
            cur.next.prev = cur;
            cur.child = null;
        }
        if (cur.next == null && !stack.isEmpty()) {
            cur.next = stack.pop();
```

```
            cur.next.prev = cur;
        }
        cur = cur.next;
    }
    return head;
}
```

Time: O(n) — every node visited once. Space: O(d) for the stack, where d = max nesting depth of child lists. **Key insight:** treating "resume point after finishing a child branch" as a stack push/pop is the same idea as using an explicit stack to simulate recursion (avoids deep recursion stack overflow on very nested inputs).

B4. More Trees & BSTs

B4.1 Binary Tree Right Side View

Difficulty: Medium | **Pattern:** BFS (last node per level) | **Companies:** Amazon, Facebook **Problem:** Return the values visible from the right side of a binary tree, top to bottom. **Brute Force:** full level-order traversal collecting every level's nodes, then take the last element of each level list — correct but keeps more data than necessary. **Optimal:** BFS level by level, but only record the **last** node processed at each level (or equivalently, DFS right-first, recording the first node seen at each new depth).

```
public List<Integer> rightSideView(TreeNode root) {
    List<Integer> result = new ArrayList<>();
    if (root == null) return result;
    Queue<TreeNode> queue = new LinkedList<>();
    queue.offer(root);
    while (!queue.isEmpty()) {
        int size = queue.size();
        for (int i = 0; i < size; i++) {
            TreeNode node = queue.poll();
            if (i == size - 1) result.add(node.val); // last node in this level
            if (node.left != null) queue.offer(node.left);
            if (node.right != null) queue.offer(node.right);
        }
    }
    return result;
}
```

Time: O(n). Space: O(n) for the queue. **Key insight:** "process level by level, keep only the last/first per level" is a direct variant of the standard Level Order Traversal template

(A6.6).

B4.2 Count Good Nodes in Binary Tree

Difficulty: Medium | **Pattern:** DFS with running max | **Companies:** Amazon, Google **Problem:** A node is "good" if no ancestor has a value greater than it. Count good nodes. **Brute Force:** for every node, walk up to the root checking all ancestors — $O(n^2)$ worst case (skewed tree, or $O(n \cdot h)$ generally). **Optimal:** single DFS pass, carrying down the maximum value seen so far along the current root-to-node path.

```
public int goodNodes(TreeNode root) {
    return dfs(root, root.val);
}
private int dfs(TreeNode node, int maxSoFar) {
    if (node == null) return 0;
    int count = (node.val >= maxSoFar) ? 1 : 0;
    int newMax = Math.max(maxSoFar, node.val);
    return count + dfs(node.left, newMax) + dfs(node.right, newMax);
}
```

Time: O(n). Space: O(h) recursion stack. **Key insight:** "carry state down the recursion" (here, the running max) is a very common tree-DFS augmentation, contrasted with "combine results up" patterns like Diameter/Max Path Sum.

B4.3 Binary Search Tree Iterator

Difficulty: Medium | **Pattern:** Controlled in-order traversal (Design) | **Companies:** Amazon, Google, Facebook **Problem:** Design an iterator over a BST that returns the next smallest number in O(1) average time per `next()` call, using O(h) memory. **Brute Force:** do a full in-order traversal up front, store all values in a list, iterate with an index pointer — O(n) space, violates the O(h) memory requirement. **Optimal:** maintain an explicit stack representing the "leftmost spine" not yet visited; `next()` pops the top, then pushes the popped node's right subtree's leftmost spine.

```
class BSTIterator {
    private Deque<TreeNode> stack = new ArrayDeque<>();
    public BSTIterator(TreeNode root) { pushLeftSpine(root); }
    private void pushLeftSpine(TreeNode node) {
        while (node != null) { stack.push(node); node = node.left; }
```

```
    }
    public int next() {
        TreeNode node = stack.pop();
        pushLeftSpine(node.right);
        return node.val;
    }
    public boolean hasNext() { return !stack.isEmpty(); }
}
```

Time: O(1) amortized per `next()` (each node pushed/popped exactly once across the iterator's whole lifetime). Space: O(h) for the stack. **Key insight:** this is exactly the iterative in-order traversal from Kth Smallest Element in a BST (A6.8), just repackaged as a resumable iterator instead of a one-shot function.

B4.4 Convert Sorted Array to Binary Search Tree

Difficulty: Easy | **Pattern:** Divide & Conquer | **Companies:** Amazon, Microsoft
Problem: Given a sorted array, build a height-balanced BST. **Brute Force:** insert elements one at a time into a BST via standard insertion — can produce an unbalanced (even fully skewed) tree if inserted in sorted order naively, violating the "height-balanced" requirement. **Optimal:** recursively pick the **middle** element of the current subarray as the root (guarantees balance), recurse on the left half for the left subtree and right half for the right subtree.

```
public TreeNode sortedArrayToBST(int[] nums) {
    return build(nums, 0, nums.length - 1);
}
private TreeNode build(int[] nums, int left, int right) {
    if (left > right) return null;
    int mid = left + (right - left) / 2;
    TreeNode node = new TreeNode(nums[mid]);
    node.left = build(nums, left, mid - 1);
    node.right = build(nums, mid + 1, right);
    return node;
}
```

Time: O(n). Space: O(log n) recursion stack (balanced by construction). **Key insight:** "always split at the middle" is what guarantees O(log n) height — the same idea underlies many balanced-structure constructions.

B4.5 Delete Node in a BST

Difficulty: Medium | **Pattern:** BST structural manipulation | **Companies:** Amazon, Microsoft, Facebook **Problem:** Given a BST root and a key, delete the node with that key and return the new root, maintaining BST properties. **Brute Force:** collect all values via in-order traversal excluding the target, rebuild the BST from scratch — correct but wasteful (O(n) rebuild instead of a localized fix). **Optimal:** locate the node via standard BST search. Three cases: (1) no children — simply remove it; (2) one child — replace it with that child; (3) two children — replace its value with its **in-order successor** (the minimum of the right subtree), then recursively delete that successor from the right subtree (which now has at most one child, reducing to an easier case).

```
public TreeNode deleteNode(TreeNode root, int key) {
    if (root == null) return null;
    if (key < root.val) { root.left = deleteNode(root.left, key); }
    else if (key > root.val) { root.right = deleteNode(root.right, key); }
    else {
        if (root.left == null) return root.right;
        if (root.right == null) return root.left;
        TreeNode successor = root.right;
        while (successor.left != null) successor = successor.left;
        root.val = successor.val;
        root.right = deleteNode(root.right, successor.val);
    }
    return root;
}
```

Time: O(h) — h = tree height. Space: O(h) recursion stack. **Key insight:** "replace with in-order successor (or predecessor), then recursively delete that" reduces the hard two-children case down to an easier one/zero-children case — a classic BST deletion technique worth memorizing precisely.

B4.6 Vertical Order Traversal of a Binary Tree

Difficulty: Hard | **Pattern:** BFS/DFS + coordinate bucketing + sorting | **Companies:** Amazon, Facebook, Google **Problem:** Return the vertical order traversal of a binary tree's nodes (grouped by column, top-to-bottom within a column, left-to-right for ties at the same row+column). **Brute Force:** DFS collecting `(column, row, value)` triples for every node without any particular care, then a complex multi-key sort — conceptually the same overall approach as the optimal one, just less carefully

structured for tie-breaking. **Optimal:** BFS (or DFS) assigning each node a `(row, column)` coordinate — root at `(0,0)`, left child `(row+1, col-1)`, right child `(row+1, col+1)`. Group nodes by column in a TreeMap (auto-sorted by column), and within each column sort by `(row, value)` to break ties correctly (using BFS order alone isn't sufficient for same-row-same-column ties, hence the explicit value tie-break during sort).

```
public List<List<Integer>> verticalTraversal(TreeNode root) {
    TreeMap<Integer, List<int[]>> columns = new TreeMap<>(); // col -> list of
[row, val]
    Queue<TreeNode> queue = new LinkedList<>();
    Queue<int[]> coords = new LinkedList<>();
    queue.offer(root); coords.offer(new int[]{0, 0});

    while (!queue.isEmpty()) {
        TreeNode node = queue.poll();
        int[] coord = coords.poll();
        int row = coord[0], col = coord[1];
        columns.computeIfAbsent(col, k -> new ArrayList<>()).add(new int[]{row,
node.val});
        if (node.left != null) { queue.offer(node.left); coords.offer(new int[]{row
+ 1, col - 1}); }
        if (node.right != null) { queue.offer(node.right); coords.offer(new int[]
{row + 1, col + 1}); }
    }

    List<List<Integer>> result = new ArrayList<>();
    for (List<int[]> col : columns.values()) {
        col.sort((a, b) -> a[0] != b[0] ? a[0] - b[0] : a[1] - b[1]); // sort by
row, then value
        List<Integer> colValues = new ArrayList<>();
        for (int[] pair : col) colValues.add(pair[1]);
        result.add(colValues);
    }
    return result;
}
```

Time: O(n log n) — dominated by sorting within columns. Space: O(n). **Key insight:** assigning explicit `(row, col)` coordinates and using a sorted map is the standard technique for any "vertical order" / "diagonal order" tree traversal variant.

B5. More Graphs (Advanced)

B5.1 Network Delay Time (Dijkstra's Algorithm)

Difficulty: Medium | **Pattern:** Dijkstra's Shortest Path (weighted graph, min-heap) | **Companies:** Google, Amazon — the standard "implement Dijkstra" question **Problem:** Given a weighted directed graph and a source node `k`, return the time for all nodes to receive a signal from `k` (the maximum shortest-path distance), or -1 if unreachable. **Brute Force (Bellman-Ford, works but slower):** relax all edges `V-1` times — O(V·E), correct for graphs with any (non-negative here) weights but not the intended optimal for this problem. **Optimal (Dijkstra's with a min-heap):** greedily process the closest unvisited node first (via a min-heap keyed by current known distance), relaxing its outgoing edges. Because edge weights are non-negative, once a node is popped from the heap its distance is guaranteed final.

```
public int networkDelayTime(int[][] times, int n, int k) {
    Map<Integer, List<int[]>> graph = new HashMap<>();
    for (int[] t : times) graph.computeIfAbsent(t[0], x -> new ArrayList<>
()).add(new int[]{t[1], t[2]});

    int[] dist = new int[n + 1];
    Arrays.fill(dist, Integer.MAX_VALUE);
    dist[k] = 0;
    PriorityQueue<int[]> minHeap = new PriorityQueue<>((a, b) -> a[1] - b[1]); //
[node, dist]
    minHeap.offer(new int[]{k, 0});

    while (!minHeap.isEmpty()) {
        int[] cur = minHeap.poll();
        int node = cur[0], d = cur[1];
        if (d > dist[node]) continue; // stale entry, skip
        for (int[] edge : graph.getOrDefault(node, List.of())) {
            int next = edge[0], weight = edge[1];
            if (d + weight < dist[next]) {
                dist[next] = d + weight;
                minHeap.offer(new int[]{next, dist[next]});
            }
        }
    }
    int maxDist = 0;
    for (int i = 1; i <= n; i++) {
        if (dist[i] == Integer.MAX_VALUE) return -1;
        maxDist = Math.max(maxDist, dist[i]);
    }
    return maxDist;
}
```

Time: O(E log V) — each edge relaxation may push onto the heap, each heap operation O(log V). Space: O(V + E). **Key insight:** Dijkstra's requires non-negative weights; if negative weights are possible, you must use Bellman-Ford instead (slower, O(V·E), but handles negatives and detects negative cycles).

B5.2 Cheapest Flights Within K Stops

Difficulty: Medium | **Pattern:** Modified Bellman-Ford (bounded relaxations) | **Companies:** Amazon, Google, Facebook **Problem:** Find the cheapest price from `src` to `dst` using at most `k` intermediate stops (i.e., at most `k+1` edges). **Brute Force:** plain Dijkstra doesn't directly respect the "at most k stops" constraint since it doesn't track hop count — needs modification either way. A naive DFS/BFS trying all paths up to k stops is exponential. **Optimal (Bellman-Ford limited to k+1 rounds):** relax all edges exactly `k+1` times (not until convergence, unlike standard Bellman-Ford) — this naturally bounds the solution to paths using at most `k+1` edges. Use a snapshot of distances from the previous round to avoid using more than one edge "per round" incorrectly.

```
public int findCheapestPrice(int n, int[][] flights, int src, int dst, int k) {
    int[] dist = new int[n];
    Arrays.fill(dist, Integer.MAX_VALUE);
    dist[src] = 0;

    for (int i = 0; i <= k; i++) {
        int[] temp = dist.clone();
        for (int[] flight : flights) {
            int u = flight[0], v = flight[1], cost = flight[2];
            if (dist[u] != Integer.MAX_VALUE && dist[u] + cost < temp[v]) {
                temp[v] = dist[u] + cost;
            }
        }
        dist = temp;
    }
    return dist[dst] == Integer.MAX_VALUE ? -1 : dist[dst];
}
```

Time: O(k · E). Space: O(V). **Key insight:** using a fresh snapshot (`temp`) each round is essential — updating `dist` in place within a round would allow a single round to incorrectly chain multiple edges together, violating the "at most k stops" bound.

B5.3 Alien Dictionary

Difficulty: Hard | **Pattern:** Topological Sort (derive graph from constraints) | **Companies:** Google, Airbnb, Facebook — classic "derive the graph yourself" hard question **Problem:** Given a sorted list of words in an unknown alien language's alphabet, derive a valid character ordering (one of possibly several valid topological orders), or detect if the input is invalid/contradictory. **Brute Force:** there's no simpler

brute force — the entire challenge is correctly building the constraint graph; a "brute force" attempt without careful edge derivation would simply be buggy, not just slow. **Optimal:** compare each pair of **adjacent** words; the first differing character gives a `"earlier char comes before later char"` edge. (Special invalid case: if word A is a longer prefix appearing before a shorter word B that is genuinely a prefix of A, e.g. `["abc","ab"]`, that's an invalid ordering.) Build the graph, then run Kahn's algorithm (BFS topological sort) to produce an ordering, detecting a cycle (invalid input) if not all characters get processed.

```
public String alienOrder(String[] words) {
    Map<Character, Set<Character>> graph = new HashMap<>();
    Map<Character, Integer> inDegree = new HashMap<>();
    for (String w : words) for (char c : w.toCharArray()) { graph.putIfAbsent(c,
new HashSet<>()); inDegree.putIfAbsent(c, 0); }

    for (int i = 0; i < words.length - 1; i++) {
        String w1 = words[i], w2 = words[i + 1];
        int minLen = Math.min(w1.length(), w2.length());
        if (w1.length() > w2.length() && w1.startsWith(w2)) return ""; // invalid:
longer prefix before shorter
        for (int j = 0; j < minLen; j++) {
            char c1 = w1.charAt(j), c2 = w2.charAt(j);
            if (c1 != c2) {
                if (graph.get(c1).add(c2)) inDegree.merge(c2, 1, Integer::sum);
                break;
            }
        }
    }

    Queue<Character> queue = new LinkedList<>();
    for (char c : inDegree.keySet()) if (inDegree.get(c) == 0) queue.offer(c);
    StringBuilder result = new StringBuilder();
    while (!queue.isEmpty()) {
        char c = queue.poll();
        result.append(c);
        for (char next : graph.get(c)) {
            if (--inDegree.merge(next, -1, Integer::sum) == 0) queue.offer(next);
// note: merge already decremented
        }
    }
    return result.length() == inDegree.size() ? result.toString() : ""; // "" if a
cycle exists
}
```

Time: O(C) where C = total characters across all words (for building the graph) + O(V+E) for the topological sort. Space: O(V+E) for the graph. **Key insight:** the entire problem reduces to "derive a DAG from adjacent-pair comparisons, then topologically sort it" — recognize this graph-derivation step as the actual hard part, not the (standard) topological sort itself.

B5.4 Minimum Spanning Tree — Kruskal's / Prim's (conceptual, via Connecting Cities With Minimum Cost)

Difficulty: Medium/Hard | **Pattern:** MST (Union-Find or Min-Heap) | **Companies:** Google, Amazon **Problem:** Given `n` cities and a list of `(cityA, cityB, cost)` connections, find the minimum total cost to connect all cities (or -1 if impossible) — classic Minimum Spanning Tree. **Brute Force:** try all subsets of edges forming a spanning tree, take the minimum cost — exponential, infeasible. **Optimal (Kruskal's Algorithm — sort edges, Union-Find to avoid cycles):** sort all edges by cost ascending. Greedily add each edge if it connects two currently-different components (checked/merged via Union-Find); stop once `n-1` edges have been added (a spanning tree is complete).

```
public int minimumCost(int n, int[][] connections) {
    Arrays.sort(connections, (a, b) -> a[2] - b[2]);
    int[] parent = new int[n + 1];
    for (int i = 1; i <= n; i++) parent[i] = i;

    int totalCost = 0, edgesUsed = 0;
    for (int[] conn : connections) {
        int rootA = find(parent, conn[0]), rootB = find(parent, conn[1]);
        if (rootA != rootB) {
            parent[rootB] = rootA;
            totalCost += conn[2];
            edgesUsed++;
        }
    }
    return edgesUsed == n - 1 ? totalCost : -1;
}
private int find(int[] parent, int x) {
    if (parent[x] != x) parent[x] = find(parent, parent[x]);
    return parent[x];
}
```

Time: O(E log E) — dominated by sorting edges. Space: O(V) for Union-Find. **Key insight:** Kruskal's (sort edges + Union-Find) is usually simpler to implement than Prim's (min-heap + visited set) in an interview setting for sparse edge lists — but know both exist and when Prim's (better for dense graphs, adjacency-matrix-style) might be preferred.

B5.5 Detect Negative Cycle (Bellman-Ford)

Difficulty: Medium/Hard | **Pattern:** Bellman-Ford Algorithm | **Companies:** Google, Bloomberg — common when graphs can have negative weights **Problem:** Given a weighted directed graph (possibly with negative edge weights), detect if there's a negative-weight cycle reachable from a source. **Why not Dijkstra:** Dijkstra's greedy "finalize the closest node" assumption breaks entirely in the presence of negative weights — it can produce incorrect shortest paths silently. **Optimal (Bellman-Ford):** relax all `E` edges `V-1` times (guaranteed enough rounds for any shortest path in a graph with no negative cycle, since a simple path has at most `V-1` edges). Then do **one more round**: if any edge can still be relaxed further, a negative cycle exists (distances would keep shrinking forever without the round cap).

```
public boolean hasNegativeCycle(int n, int[][] edges, int src) {
    int[] dist = new int[n];
    Arrays.fill(dist, Integer.MAX_VALUE);
    dist[src] = 0;

    for (int i = 0; i < n - 1; i++) {
        for (int[] edge : edges) {
            int u = edge[0], v = edge[1], w = edge[2];
            if (dist[u] != Integer.MAX_VALUE && dist[u] + w < dist[v]) {
                dist[v] = dist[u] + w;
            }
        }
    }
    for (int[] edge : edges) { // one extra round to detect further relaxation
        int u = edge[0], v = edge[1], w = edge[2];
        if (dist[u] != Integer.MAX_VALUE && dist[u] + w < dist[v]) {
            return true; // negative cycle detected
        }
    }
    return false;
}
```

Time: O(V·E). Space: O(V). **Key insight:** the "V-1 rounds is always enough for a negative-cycle-free graph, one more round detects negative cycles" fact is the entire mathematical basis of Bellman-Ford — memorize this reasoning, not just the code.

B5.6 Reconstruct Itinerary (Eulerian Path)

Difficulty: Hard | **Pattern:** Hierholzer's Algorithm (Eulerian Path via DFS + post-order) | **Companies:** Google, Amazon **Problem:** Given airline tickets as `[from, to]` pairs,

reconstruct the itinerary starting from `"JFK"` that uses every ticket exactly once, returning the lexicographically smallest valid itinerary if multiple exist. **Brute Force:** backtrack over all possible ticket orderings, checking validity — factorial time, infeasible for larger inputs. **Optimal (Hierholzer's Algorithm):** this is the classic "find an Eulerian path" problem (visit every edge exactly once). Use DFS, always visiting the lexicographically smallest available next destination first (a min-heap or sorted adjacency list per node); when a node has no more outgoing unused tickets, add it to the result — then **reverse** the result at the end. This "dead-end nodes get added first, then reverse" is the essence of Hierholzer's algorithm, and it naturally produces the correct Eulerian path/circuit.

```
public List<String> findItinerary(List<List<String>> tickets) {
    Map<String, PriorityQueue<String>> graph = new HashMap<>();
    for (List<String> ticket : tickets) {
        graph.computeIfAbsent(ticket.get(0), k -> new PriorityQueue<>
()).offer(ticket.get(1));
    }
    LinkedList<String> result = new LinkedList<>();
    dfs("JFK", graph, result);
    return result;
}
private void dfs(String airport, Map<String, PriorityQueue<String>> graph,
LinkedList<String> result) {
    PriorityQueue<String> destinations = graph.get(airport);
    while (destinations != null && !destinations.isEmpty()) {
        dfs(destinations.poll(), graph, result);
    }
    result.addFirst(airport); // post-order insertion (Hierholzer's trick)
}
```

Time: O(E log E) — sorting/heap operations across all tickets. Space: O(E) for the graph and recursion. **Key insight:** "add to result on the way OUT of the DFS (post-order), then the result is already in correct final order without needing a separate reverse step" (using `addFirst` achieves the same effect as reversing at the end) — recognize this Eulerian-path pattern whenever a problem requires using every edge exactly once.

B6. More Dynamic Programming

B6.1 Longest Palindromic Subsequence

Difficulty: Medium | **Pattern:** 2-D DP (same shape as LCS) | **Companies:** Amazon, Google **Problem:** Return the length of the longest palindromic subsequence (not necessarily contiguous) of a string. **Brute Force:** try all 2^n subsequences, check palindrome-ness — exponential. **Optimal:** the longest palindromic subsequence of `s` equals the **Longest Common Subsequence of `s` and its reverse** — reduces directly to the LCS DP (A12.2) with `text2 = reverse(s)`.

```
public int longestPalindromeSubseq(String s) {
    String reversed = new StringBuilder(s).reverse().toString();
    int n = s.length();
    int[][] dp = new int[n + 1][n + 1];
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            if (s.charAt(i - 1) == reversed.charAt(j - 1)) dp[i][j] = dp[i - 1][j -
1] + 1;
            else dp[i][j] = Math.max(dp[i - 1][j], dp[i][j - 1]);
        }
    }
    return dp[n][n];
}
```

Time: O(n²). Space: O(n²) (reducible to O(n)). **Key insight:** recognizing "palindromic subsequence = LCS with the reverse of itself" instantly turns a seemingly new problem into one you already know how to solve.

B6.2 Interleaving String

Difficulty: Medium | **Pattern:** 2-D DP (two-pointer string merge validity) | **Companies:** Facebook, Microsoft **Problem:** Given `s1`, `s2`, `s3`, determine if `s3` is formed by interleaving `s1` and `s2` (preserving each string's relative character order). **Brute Force:** recursively try consuming the next character from either `s1` or `s2` at each step, matching against `s3` — exponential without memoization due to overlapping `(i,j)` states. **Optimal:** `dp[i][j]` = true if `s3[0..i+j)` can be formed by interleaving `s1[0..i)` and `s2[0..j)`. Transition checks both possibilities: did the last character of `s3` come from `s1` or from `s2`.

```
public boolean isInterleave(String s1, String s2, String s3) {
    int m = s1.length(), n = s2.length();
    if (m + n != s3.length()) return false;
    boolean[][] dp = new boolean[m + 1][n + 1];
    dp[0][0] = true;
    for (int i = 1; i <= m; i++) dp[i][0] = dp[i - 1][0] && s1.charAt(i - 1) ==
```

```
s3.charAt(i - 1);
    for (int j = 1; j <= n; j++) dp[0][j] = dp[0][j - 1] && s2.charAt(j - 1) ==
s3.charAt(j - 1);
    for (int i = 1; i <= m; i++) {
        for (int j = 1; j <= n; j++) {
            dp[i][j] = (dp[i - 1][j] && s1.charAt(i - 1) == s3.charAt(i + j - 1))
                    || (dp[i][j - 1] && s2.charAt(j - 1) == s3.charAt(i + j - 1));
        }
    }
    return dp[m][n];
}
```

Time: O(m·n). Space: O(m·n) (reducible to O(n)). **Key insight:** the immediate length check (`m+n != s3.length()`) is a free early rejection worth always doing before the DP.

B6.3 Distinct Subsequences

Difficulty: Hard | **Pattern:** 2-D DP (counting subsequence matches) | **Companies:** Google, Amazon **Problem:** Given strings `s` and `t`, count the number of distinct subsequences of `s` that equal `t`. **Brute Force:** recursively try including or skipping each character of `s` while matching against `t` — exponential overlapping subproblems. **Optimal:** `dp[i][j]` = number of ways `s[0..i)` forms `t[0..j)`. If `s[i-1] == t[j-1]`, we can either use this character to match (`dp[i-1][j-1]`) or skip it (`dp[i-1][j]`) — sum both. Otherwise we must skip it: `dp[i][j] = dp[i-1][j]`.

```
public int numDistinct(String s, String t) {
    int m = s.length(), n = t.length();
    long[][] dp = new long[m + 1][n + 1];
    for (int i = 0; i <= m; i++) dp[i][0] = 1; // empty t is always matched exactly
one way
    for (int i = 1; i <= m; i++) {
        for (int j = 1; j <= n; j++) {
            dp[i][j] = dp[i - 1][j];
            if (s.charAt(i - 1) == t.charAt(j - 1)) dp[i][j] += dp[i - 1][j - 1];
        }
    }
    return (int) dp[m][n];
}
```

Time: O(m·n). Space: O(m·n) (reducible to O(n); use `long` to avoid overflow on large counts). **Key insight:** structurally very similar to LCS/Edit Distance tables but the recurrence **sums** possibilities (counting) rather than taking a max/min — recognize "count vs. optimize" as the key differentiator among 2-D string DP problems.

B6.4 Burst Balloons

Difficulty: Hard | **Pattern:** Interval DP (think about the LAST action, not the first) | **Companies:** Google, Amazon — famously counter-intuitive DP **Problem:** Given balloons with values, bursting balloon `i` earns `nums[left]*nums[i]* nums[right]` (its current neighbors). Find the max coins from bursting all balloons. **Brute Force:** try every possible order of bursting balloons — O(n!), infeasible. **Optimal (Interval DP — think about which balloon to burst LAST in a range):** the key insight that unlocks this problem: instead of thinking "which balloon do I burst first," think "for the subrange `(left, right)`, which balloon do I burst **last**?" That balloon's neighbors at the time of bursting are guaranteed to be `nums[left]` and `nums[right]` (everything else in between is already gone), making the subproblem well-defined and independent of order. `dp[left][right]` = max coins from bursting everything strictly between `left` and `right`.

```
public int maxCoins(int[] nums) {
    int n = nums.length;
    int[] balloons = new int[n + 2];
    balloons[0] = 1; balloons[n + 1] = 1;
    for (int i = 0; i < n; i++) balloons[i + 1] = nums[i];

    int[][] dp = new int[n + 2][n + 2];
    for (int len = 2; len <= n + 1; len++) {
        for (int left = 0; left + len <= n + 1; left++) {
            int right = left + len;
            for (int k = left + 1; k < right; k++) {
                dp[left][right] = Math.max(dp[left][right],
                    dp[left][k] + balloons[left] * balloons[k] * balloons[right] +
dp[k][right]);
            }
        }
    }
    return dp[0][n + 1];
}
```

Time: O(n³) — O(n²) subranges, O(n) choices of "last balloon" each. Space: O(n²) for the dp table. **Key insight:** whenever a problem's "natural" order of operations makes subproblems interdependent, try reframing around "what happens LAST in this range" instead — this inversion is the single trick that unlocks Burst Balloons and several other interval DP problems.

B6.5 Perfect Squares

Difficulty: Medium | **Pattern:** DP (unbounded knapsack, like Coin Change) | **Companies:** Google, Amazon **Problem:** Given `n`, return the least number of perfect square numbers (1, 4, 9, 16, ...) that sum to `n`. **Brute Force:** recursively try subtracting every perfect square `<= n`, take the min count — exponential without memoization. **Optimal:** `dp[i]` = min perfect squares summing to `i`. For each `i`, try every perfect square `j*j <= i`: `dp[i] = min(dp[i], dp[i - j*j] + 1)`. Structurally identical to Coin Change (A11.8) where "coins" are the perfect squares.

```
public int numSquares(int n) {
    int[] dp = new int[n + 1];
    Arrays.fill(dp, Integer.MAX_VALUE);
    dp[0] = 0;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j * j <= i; j++) {
            dp[i] = Math.min(dp[i], dp[i - j * j] + 1);
        }
    }
    return dp[n];
}
```

Time: O(n · √n). Space: O(n). **Key insight:** recognize this as "Coin Change with a specific, computed coin set (perfect squares)" — the exact same unbounded-knapsack DP shape applies.

B6.6 Integer Break

Difficulty: Medium | **Pattern:** DP (or Math/Greedy insight) | **Companies:** Google **Problem:** Given `n`, break it into the sum of at least two positive integers to maximize their product. **Brute Force / DP Approach:** `dp[i]` = max product obtainable by breaking `i`. For every way to split `i = j + (i-j)`, consider both leaving `(i-j)` whole or further breaking it: `dp[i] = max(dp[i], j * (i - j), j * dp[i - j])`.

```
public int integerBreak(int n) {
    int[] dp = new int[n + 1];
    dp[1] = 1;
    for (int i = 2; i <= n; i++) {
        for (int j = 1; j < i; j++) {
            dp[i] = Math.max(dp[i], Math.max(j * (i - j), j * dp[i - j]));
        }
    }
```

```
    return dp[n];
}
```

Time: O(n²). Space: O(n). **Mathematical shortcut (O(1)):** it's a well-known result that breaking into as many **3s** as possible (with special handling for remainders of 1 or 2) maximizes the product — e.g. `n=10 → 3+3+4 → 3*3*4=36`. **Key insight:** DP always works and is safer to derive under interview pressure; the O(1) greedy/math shortcut is impressive to mention as a follow-up but riskier to derive correctly live without prior knowledge.

B6.7 Min Cost Climbing Stairs

Difficulty: Easy | **Pattern:** DP (Fibonacci-shape) | **Companies:** Amazon **Problem:** Given a cost array, where `cost[i]` is paid to step on stair `i` (you can start at step 0 or 1, and climb 1 or 2 steps), find the minimum cost to reach the top (past the last stair). **Brute Force:** recursively try both step sizes from every position, take the min cost path — exponential without memoization. **Optimal:** `dp[i]` = min cost to reach step `i` = `cost[i] + min(dp[i-1], dp[i-2])`. Answer is `min(dp[n-1], dp[n-2])` since you can "step off" from either of the last two stairs.

```
public int minCostClimbingStairs(int[] cost) {
    int n = cost.length;
    int prev2 = cost[0], prev1 = cost[1];
    if (n == 2) return Math.min(prev2, prev1);
    for (int i = 2; i < n; i++) {
        int cur = cost[i] + Math.min(prev1, prev2);
        prev2 = prev1;
        prev1 = cur;
    }
    return Math.min(prev1, prev2);
}
```

Time: O(n). Space: O(1). **Key insight:** identical rolling-variable Fibonacci-shape DP as Climbing Stairs (A11.1), just with a cost weighting added to each transition.

B7. More Backtracking

B7.1 Combinations

Difficulty: Medium | **Pattern:** Backtracking | **Companies:** Amazon, Facebook **Problem:** Return all possible combinations of `k` numbers chosen from `1..n`. **Brute Force:** generate all subsets and filter those of size k — O(2^n), wasteful since most subsets are the wrong size. **Optimal:** backtrack, only ever adding numbers larger than the last one chosen (avoids duplicate combinations in different orders), and pruning branches that can't possibly reach size k with the remaining numbers available.

```
public List<List<Integer>> combine(int n, int k) {
    List<List<Integer>> result = new ArrayList<>();
    backtrack(1, n, k, new ArrayList<>(), result);
    return result;
}
private void backtrack(int start, int n, int k, List<Integer> current,
List<List<Integer>> result) {
    if (current.size() == k) { result.add(new ArrayList<>(current)); return; }
    for (int i = start; i <= n - (k - current.size()) + 1; i++) { // prune: not
enough numbers left
        current.add(i);
        backtrack(i + 1, n, k, current, result);
        current.remove(current.size() - 1);
    }
}
```

Time: O(k · C(n,k)) — C(n,k) combinations, O(k) to copy each. Space: O(k) recursion depth. **Key insight:** the pruning bound `n - (k - current.size()) + 1` is what keeps this efficient — without it, you'd explore many branches guaranteed to fail before reaching size k.

B7.2 Combination Sum II

Difficulty: Medium | **Pattern:** Backtracking (no reuse, duplicate skipping) | **Companies:** Amazon **Problem:** Like Combination Sum (A9.2), but each number may be used **at most once**, and the input array may contain duplicates — output must have no duplicate combinations. **Brute Force:** same backtracking skeleton as Combination Sum but naive duplicate-combination filtering via a Set of sorted lists afterward — wastes time on redundant exploration. **Optimal:** sort first. Advance the recursion index by `i+1` (no reuse). At each recursion depth, skip over consecutive duplicate values (`if (i > start && candidates[i] == candidates[i-1]) continue;`) to avoid generating the same combination twice.

```
public List<List<Integer>> combinationSum2(int[] candidates, int target) {
    Arrays.sort(candidates);
    List<List<Integer>> result = new ArrayList<>();
    backtrack(candidates, target, 0, new ArrayList<>(), result);
    return result;
}
private void backtrack(int[] candidates, int remaining, int start, List<Integer>
current, List<List<Integer>> result) {
    if (remaining == 0) { result.add(new ArrayList<>(current)); return; }
    for (int i = start; i < candidates.length; i++) {
        if (i > start && candidates[i] == candidates[i - 1]) continue; // skip
duplicates at this depth
        if (candidates[i] > remaining) break;
        current.add(candidates[i]);
        backtrack(candidates, remaining - candidates[i], i + 1, current, result);
        current.remove(current.size() - 1);
    }
}
```

Time: O(2^n) worst case bound, heavily pruned in practice. Space: O(n) recursion depth. **Key insight:** the `i > start` check (not `i > 0`) is crucial — it allows the *first* occurrence of a duplicate value at a given recursion depth while skipping subsequent ones, correctly distinguishing "reuse across depths" (fine) from "reuse at the same depth" (creates duplicates).

B7.3 Permutations II

Difficulty: Medium | **Pattern:** Backtracking (duplicate skipping with used array) | **Companies:** Amazon, Microsoft **Problem:** Given a collection that may contain duplicates, return all unique permutations. **Brute Force:** generate all permutations as in Permutations (A9.3), collect into a Set to remove duplicates — wastes time generating duplicates just to discard them. **Optimal:** sort first. During backtracking, skip a candidate if it equals the previous candidate **and** the previous one hasn't been used yet in the current path (this specific condition `!used[i-1]` — not `used[i-1]` — is what correctly prevents duplicate permutations while still allowing legitimate reuse of equal values at different positions).

```
public List<List<Integer>> permuteUnique(int[] nums) {
    Arrays.sort(nums);
    List<List<Integer>> result = new ArrayList<>();
    boolean[] used = new boolean[nums.length];
    backtrack(nums, used, new ArrayList<>(), result);
    return result;
}
```

```
private void backtrack(int[] nums, boolean[] used, List<Integer> current,
List<List<Integer>> result) {
    if (current.size() == nums.length) { result.add(new ArrayList<>(current));
return; }
    for (int i = 0; i < nums.length; i++) {
        if (used[i]) continue;
        if (i > 0 && nums[i] == nums[i - 1] && !used[i - 1]) continue; // skip
duplicate branch
        used[i] = true;
        current.add(nums[i]);
        backtrack(nums, used, current, result);
        current.remove(current.size() - 1);
        used[i] = false;
    }
}
```

Time: O(n · n!) worst case, pruned considerably with duplicates present. Space: O(n). **Key insight:** the `!used[i-1]` condition (rather than `used[i-1]`) is the single most commonly mis-remembered detail in this pattern — it specifically means "if the identical previous value hasn't been placed yet in this branch, don't place this one first" (enforces a consistent relative order among equal values, eliminating duplicate permutations).

B7.4 Subsets II

Difficulty: Medium | **Pattern:** Backtracking (duplicate skipping) | **Companies:** Amazon **Problem:** Given an array that may contain duplicates, return all possible unique subsets. **Brute Force:** generate all 2^n subsets, dedupe via a Set of sorted lists — wasteful. **Optimal:** sort first. At each recursion depth, when deciding whether to include `nums[i]`, skip it if `i > start && nums[i] == nums[i-1]` (only the FIRST occurrence of an equal value starts a new "branch" at a given depth; subsequent equal values would just regenerate subsets already produced by the first).

```
public List<List<Integer>> subsetsWithDup(int[] nums) {
    Arrays.sort(nums);
    List<List<Integer>> result = new ArrayList<>();
    backtrack(nums, 0, new ArrayList<>(), result);
    return result;
}
private void backtrack(int[] nums, int start, List<Integer> current,
List<List<Integer>> result) {
    result.add(new ArrayList<>(current));
    for (int i = start; i < nums.length; i++) {
        if (i > start && nums[i] == nums[i - 1]) continue;
        current.add(nums[i]);
        backtrack(nums, i + 1, current, result);
```

```
            current.remove(current.size() - 1);
        }
    }
```

Time: O(n · 2^n) worst case. Space: O(n) recursion depth. **Key insight:** contrast this `i > start` duplicate-skip (used for subsets/combinations, where each recursion "level" tries each starting value once) with Permutations II's `!used[i-1]` (used because permutations explore all positions at every depth, needing a different kind of duplicate-guard) — these are the two standard duplicate-handling idioms in backtracking, worth memorizing both distinctly.

B8. Design Problems

B8.1 Implement Queue using Stacks

Difficulty: Easy | **Pattern:** Two-Stack Simulation | **Companies:** Amazon, Microsoft **Problem:** Implement a FIFO queue using only two stacks. **Brute Force:** use one stack for input; for every `pop`/`peek`, transfer everything to a second stack, act, then transfer back — O(n) per operation even when unnecessary. **Optimal (Amortized O(1)):** maintain an `inStack` for pushes and an `outStack` for pops/peeks. Only transfer from `inStack` to `outStack` when `outStack` is empty — each element is moved at most twice in its lifetime (once into `inStack`, once into `outStack`), giving O(1) amortized per operation.

```
class MyQueue {
    private Deque<Integer> inStack = new ArrayDeque<>();
    private Deque<Integer> outStack = new ArrayDeque<>();

    public void push(int x) { inStack.push(x); }
    public int pop() { transferIfNeeded(); return outStack.pop(); }
    public int peek() { transferIfNeeded(); return outStack.peek(); }
    public boolean empty() { return inStack.isEmpty() && outStack.isEmpty(); }

    private void transferIfNeeded() {
        if (outStack.isEmpty()) {
            while (!inStack.isEmpty()) outStack.push(inStack.pop());
        }
    }
}
```

Time: O(1) amortized per operation. Space: O(n). **Key insight:** "only transfer when the output side is empty" is what converts a naive O(n)-per-op design into O(1) amortized — recognize this lazy-transfer idea as broadly applicable whenever reversing order with stacks.

B8.2 Implement Stack using Queues

Difficulty: Easy | **Pattern:** Single-Queue Rotation | **Companies:** Amazon **Problem:** Implement a LIFO stack using only queues. **Brute Force:** use two queues, always keeping the newest element accessible by shuffling — more complex than needed. **Optimal (single queue, rotate on push):** after pushing a new element, rotate the queue so that the newly pushed element moves to the front — this makes `pop`/`top` always O(1) by simply looking at the front, at the cost of O(n) per `push`.

```
class MyStack {
    private Queue<Integer> queue = new LinkedList<>();

    public void push(int x) {
        queue.offer(x);
        for (int i = 0; i < queue.size() - 1; i++) {
            queue.offer(queue.poll()); // rotate older elements behind the new one
        }
    }
    public int pop() { return queue.poll(); }
    public int top() { return queue.peek(); }
    public boolean empty() { return queue.isEmpty(); }
}
```

Time: O(n) per `push`, O(1) per `pop`/`top`. Space: O(n). **Key insight:** this is the mirror image tradeoff of Queue-via-Stacks — here we pay the cost upfront on `push` instead of lazily on `pop`, since we only have one queue to work with instead of two stacks.

B8.3 Design Twitter

Difficulty: Medium | **Pattern:** Heap + HashMap (Design) | **Companies:** Twitter, Amazon, Google **Problem:** Design a simplified Twitter: `postTweet`, `getNewsFeed` (10 most recent tweets from the user and people they follow), `follow`, `unfollow`. **Brute Force:** store all tweets globally in a list; `getNewsFeed` filters by followed-user-set and sorts by time — O(all tweets log(all tweets)) per feed request, wasteful. **Optimal:** store

each user's own tweets as a list with a global increasing timestamp. For `getNewsFeed`, use a max-heap merging the most recent tweets from the user and each followee (like Merge k Sorted Lists, A5.6) — only pull the top 10 most recent overall without scanning every tweet ever posted.

```
class Twitter {
    private int timestamp = 0;
    private Map<Integer, List<int[]>> userTweets = new HashMap<>(); // userId ->
[time, tweetId]
    private Map<Integer, Set<Integer>> following = new HashMap<>();

    public void postTweet(int userId, int tweetId) {
        userTweets.computeIfAbsent(userId, k -> new ArrayList<>()).add(new int[]
{timestamp++, tweetId});
    }

    public List<Integer> getNewsFeed(int userId) {
        PriorityQueue<int[]> maxHeap = new PriorityQueue<>((a, b) -> b[0] - a[0]);
// [time, tweetId]
        Set<Integer> people = new HashSet<>(following.getOrDefault(userId, new
HashSet<>()));
        people.add(userId);
        for (int person : people) {
            List<int[]> tweets = userTweets.getOrDefault(person, new ArrayList<>
());
            for (int i = tweets.size() - 1; i >= Math.max(0, tweets.size() - 10);
i--) {
                maxHeap.offer(tweets.get(i));
            }
        }
        List<Integer> result = new ArrayList<>();
        while (!maxHeap.isEmpty() && result.size() < 10) result.add(maxHeap.poll()
[1]);
        return result;
    }

    public void follow(int followerId, int followeeId) {
        following.computeIfAbsent(followerId, k -> new HashSet<>
()).add(followeeId);
    }
    public void unfollow(int followerId, int followeeId) {
        if (following.containsKey(followerId))
following.get(followerId).remove(followeeId);
    }
}
```

Time: `getNewsFeed` O(k log k) where k = people followed × 10 tweets considered each. Space: O(total tweets + total follow relationships). **Key insight:** only ever pulling the most recent 10 tweets per user before merging (instead of merging entire histories) bounds the heap size regardless of how prolific any single user is.

B8.4 Design Hit Counter

Difficulty: Medium | **Pattern:** Queue / Circular Buffer (sliding time window) | **Companies:** Amazon, Google **Problem:** Design a hit counter that counts hits received in the past 5 minutes (300 seconds), given timestamps arrive in non-decreasing order. **Brute Force:** store every hit's timestamp in a list; `getHits(timestamp)` scans the entire list filtering `timestamp - 300 < t <= timestamp` — O(n) per query, list grows unbounded. **Optimal (Queue as a sliding window):** store hit timestamps in a queue; on every `getHits` call, first evict timestamps older than the 300-second window from the front (they can never matter again since timestamps are non-decreasing), then the queue's size is the answer.

```
class HitCounter {
    private Queue<Integer> hits = new LinkedList<>();

    public void hit(int timestamp) {
        hits.offer(timestamp);
    }
    public int getHits(int timestamp) {
        while (!hits.isEmpty() && timestamp - hits.peek() >= 300) {
            hits.poll();
        }
        return hits.size();
    }
}
```

Time: O(1) amortized per call (each hit is evicted at most once, ever). Space: O(300) worst case in the busiest window, or O(n) if not bounded by rate. **Key insight:** "evict stale entries lazily from the front of a queue whenever you need a fresh answer" is the general sliding-time-window design pattern — also usable for rate limiters.

B8.5 Design Circular Queue

Difficulty: Medium | **Pattern:** Array-based Circular Buffer | **Companies:** Google, Amazon **Problem:** Design a fixed-capacity circular queue supporting `enQueue`, `deQueue`, `Front`, `Rear`, `isEmpty`, `isFull`. **Brute Force:** use a standard `LinkedList`/`ArrayDeque` as the backing structure — works, but doesn't demonstrate the actual circular-array indexing technique the problem is testing. **Optimal (fixed array with modular indexing):** maintain a fixed-size array, a `head` index, and a `count` of elements currently stored. Both `enQueue` and `deQueue` use modular arithmetic

((head + count) % capacity) to wrap around the array boundaries instead of shifting elements.

```
class MyCircularQueue {
    private int[] data;
    private int head = 0, count = 0, capacity;

    public MyCircularQueue(int k) { data = new int[k]; capacity = k; }

    public boolean enQueue(int value) {
        if (isFull()) return false;
        data[(head + count) % capacity] = value;
        count++;
        return true;
    }
    public boolean deQueue() {
        if (isEmpty()) return false;
        head = (head + 1) % capacity;
        count--;
        return true;
    }
    public int Front() { return isEmpty() ? -1 : data[head]; }
    public int Rear() { return isEmpty() ? -1 : data[(head + count - 1) %
capacity]; }
    public boolean isEmpty() { return count == 0; }
    public boolean isFull() { return count == capacity; }
}
```

Time: O(1) for all operations. Space: O(capacity). **Key insight:** tracking `count` explicitly (rather than trying to distinguish "empty" vs "full" purely from `head`/`tail` positions, which is ambiguous when they coincide) is the cleanest way to implement a circular buffer correctly.

B8.6 Insert Delete GetRandom O(1)

Difficulty: Medium | **Pattern:** HashMap + ArrayList (swap-to-end deletion) | **Companies:** Amazon, Facebook — clever combination of two structures **Problem:** Design a structure supporting `insert`, `remove`, and `getRandom` (uniformly random existing element), all in average O(1) time. **Brute Force:** a plain `ArrayList` gives O(1) `insert` (append) and O(1) `getRandom` (index by `random.nextInt(size)`), but `remove` is O(n) (must find the element, then shift everything after it) — fails the O(1) requirement for `remove`. **Optimal (HashMap of value→index + ArrayList, swap-with-last for deletion):** the HashMap gives O(1) lookup of any value's current index in the ArrayList. To remove in O(1), swap the target element with the **last** element in the

ArrayList (updating the swapped element's index in the map), then remove the last element (an O(1) ArrayList operation) — avoids ever shifting elements.

```
class RandomizedSet {
    private Map<Integer, Integer> valueToIndex = new HashMap<>();
    private List<Integer> values = new ArrayList<>();
    private Random random = new Random();

    public boolean insert(int val) {
        if (valueToIndex.containsKey(val)) return false;
        valueToIndex.put(val, values.size());
        values.add(val);
        return true;
    }

    public boolean remove(int val) {
        if (!valueToIndex.containsKey(val)) return false;
        int indexToRemove = valueToIndex.get(val);
        int lastValue = values.get(values.size() - 1);

        values.set(indexToRemove, lastValue);       // move last element into the
removed slot
        valueToIndex.put(lastValue, indexToRemove);  // update its index in the map

        values.remove(values.size() - 1); // O(1) removal from the end
        valueToIndex.remove(val);
        return true;
    }

    public int getRandom() {
        return values.get(random.nextInt(values.size()));
    }
}
```

Time: O(1) average for all three operations. Space: O(n). **Key insight:** "swap with the last element before removing" is the standard O(1)-deletion trick for array-backed structures where order doesn't matter — directly reusable in many "design a structure with O(1) ops" questions.

B9. More Greedy / Intervals / Math

B9.1 Meeting Rooms I

Difficulty: Easy | **Pattern:** Sorting (overlap check) | **Companies:** Facebook, Amazon
Problem: Given meeting intervals, determine if a single person could attend all of them

(i.e., no two intervals overlap). **Brute Force:** compare every pair of intervals for overlap — O(n²). **Optimal:** sort by start time; a single linear scan checking if any interval's start is before the previous interval's end is enough (no need for the full "count concurrent rooms" machinery of Meeting Rooms II since we only care about a yes/no answer).

```
public boolean canAttendMeetings(int[][] intervals) {
    Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
    for (int i = 1; i < intervals.length; i++) {
        if (intervals[i][0] < intervals[i - 1][1]) return false;
    }
    return true;
}
```

Time: O(n log n). Space: O(log n) to O(n) for sort. **Key insight:** this is the simpler "degenerate" special case of Meeting Rooms II (A8.5) — recognize when a full heap-based solution is overkill and a simple sorted scan suffices.

B9.2 Minimum Number of Arrows to Burst Balloons

Difficulty: Medium | **Pattern:** Greedy (sort by end, identical shape to Non-overlapping Intervals) | **Companies:** Amazon, Facebook **Problem:** Balloons are represented as intervals `[start, end]` on the x-axis; an arrow shot at position `x` bursts every balloon where `start <= x <= end`. Find the minimum arrows needed to burst all balloons. **Brute Force:** try every possible arrow position combination — exponential/infeasible. **Optimal:** sort by end coordinate. Greedily shoot an arrow at the end of the first (currently un-burst) balloon; this arrow also bursts every other balloon whose start is `<=` that position. Move to the next balloon not yet burst and repeat.

```
public int findMinArrowShots(int[][] points) {
    if (points.length == 0) return 0;
    Arrays.sort(points, (a, b) -> Long.compare(a[1], b[1])); // avoid overflow with
large coords
    int arrows = 1;
    long arrowPos = points[0][1];
    for (int i = 1; i < points.length; i++) {
        if (points[i][0] > arrowPos) {
            arrows++;
            arrowPos = points[i][1];
        }
    }
```

```
    return arrows;
}
```

Time: O(n log n). Space: O(log n) to O(n) for sort. **Key insight:** structurally identical to Non-overlapping Intervals (A13.5) — "sort by end, greedily keep/act on the earliest-ending option" is a recurring interval-greedy template.

B9.3 Candy

Difficulty: Hard | **Pattern:** Greedy (two-pass) | **Companies:** Google, Amazon — classic tricky greedy **Problem:** Children in a line each have a rating; every child must get at least 1 candy, and any child with a higher rating than a neighbor must get more candy than that neighbor. Find the minimum total candies needed. **Brute Force:** repeatedly scan and fix violations until no more changes occur — could take many passes to converge, inefficient. **Optimal (Two-pass greedy):** pass 1 (left to right): if `ratings[i] > ratings[i-1]`, give `candies[i] = candies[i-1] + 1` (satisfies the left-neighbor constraint). Pass 2 (right to left): if `ratings[i] > ratings[i+1]`, ensure `candies[i] = max(candies[i], candies[i+1] + 1)` (satisfies the right-neighbor constraint without breaking what pass 1 already guaranteed).

```
public int candy(int[] ratings) {
    int n = ratings.length;
    int[] candies = new int[n];
    Arrays.fill(candies, 1);

    for (int i = 1; i < n; i++) {
        if (ratings[i] > ratings[i - 1]) candies[i] = candies[i - 1] + 1;
    }
    for (int i = n - 2; i >= 0; i--) {
        if (ratings[i] > ratings[i + 1]) candies[i] = Math.max(candies[i],
candies[i + 1] + 1);
    }
    int total = 0;
    for (int c : candies) total += c;
    return total;
}
```

Time: O(n) — two linear passes. Space: O(n) for the candies array. **Key insight:** each direction's constraint (compare to left neighbor, compare to right neighbor) is handled by a separate pass in a specific order (left-to-right then right-to-left) — attempting both constraints in a single pass is a common but incorrect simplification.

B9.4 Pow(x, n)

Difficulty: Medium | **Pattern:** Fast Exponentiation (Binary Exponentiation) | **Companies:** Google, LinkedIn **Problem:** Implement `pow(x, n)` computing x raised to the power n (n can be negative). **Brute Force:** multiply `x` by itself `n` times in a loop.

```
public double myPowBrute(double x, int n) {
    double result = 1;
    long absN = Math.abs((long) n);
    for (long i = 0; i < absN; i++) result *= x;
    return n < 0 ? 1 / result : result;
}
```

Time: O(n) — far too slow for large n (e.g. n = 2^31). Space: O(1). **Optimal (Fast/Binary Exponentiation):** use the identity `x^n = (x^(n/2))^2` (adjusting for odd `n` by multiplying one extra factor of `x`) to halve the exponent at every step instead of decrementing by 1.

```
public double myPow(double x, int n) {
    long absN = Math.abs((long) n);
    double result = fastPow(x, absN);
    return n < 0 ? 1 / result : result;
}
private double fastPow(double x, long n) {
    if (n == 0) return 1;
    double half = fastPow(x, n / 2);
    return (n % 2 == 0) ? half * half : half * half * x;
}
```

Time: O(log n). Space: O(log n) recursion stack (or O(1) iteratively). **Key insight:** "halve the problem size via squaring" (binary exponentiation) is a broadly reusable O(log n) technique, also underlying fast matrix exponentiation for certain DP speed-ups.

B9.5 Excel Sheet Column Number / Title

Difficulty: Easy | **Pattern:** Math (base-26, but with a twist — no true "zero" digit) | **Companies:** Microsoft, Facebook **Problem:** Convert between a spreadsheet column title (e.g. "AB") and its column number (e.g. 28), similar to base-26 conversion but 1-

indexed (no digit represents 0 — "A"=1, ..., "Z"=26, "AA"=27). **Approach (Title to Number):** process left to right, `result = result * 26 + (char value)`.

```
public int titleToNumber(String columnTitle) {
    int result = 0;
    for (char c : columnTitle.toCharArray()) {
        result = result * 26 + (c - 'A' + 1);
    }
    return result;
}
```

Approach (Number to Title): repeatedly take `(n-1) % 26` to get the current letter (the `-1` adjustment handles the "no zero digit" quirk), then `n = (n-1) / 26`.

```
public String convertToTitle(int columnNumber) {
    StringBuilder sb = new StringBuilder();
    while (columnNumber > 0) {
        columnNumber--; // adjust for 1-indexing (no "zero" letter)
        sb.append((char) ('A' + columnNumber % 26));
        columnNumber /= 26;
    }
    return sb.reverse().toString();
}
```

Time: O(log n) for both directions. Space: O(log n) for the output string. **Key insight:** the `-1` adjustment before the modulo is the single trick that makes this "base 26 without a zero digit" system work correctly — forgetting it is the most common bug.

B9.6 Happy Number

Difficulty: Easy | **Pattern:** Floyd's Cycle Detection (applied to a numeric sequence) | **Companies:** Google, Amazon **Problem:** A number is "happy" if repeatedly replacing it with the sum of the squares of its digits eventually reaches 1. Determine if a number is happy (non-happy numbers loop endlessly in a cycle that never includes 1). **Brute Force:** use a HashSet to track seen values; if a value repeats, we've found a cycle (not happy); if we reach 1, it's happy.

```
public boolean isHappyBrute(int n) {
    Set<Integer> seen = new HashSet<>();
    while (n != 1 && seen.add(n)) {
        n = sumOfSquares(n);
    }
```

```
        return n == 1;
    }
    private int sumOfSquares(int n) {
        int sum = 0;
        while (n > 0) { int digit = n % 10; sum += digit * digit; n /= 10; }
        return sum;
    }
```

Time: O(log n) per transformation, bounded number of transformations before a cycle or 1 is reached. Space: O(cycle length) for the HashSet. **Optimal (Floyd's Cycle Detection, O(1) space):** treat the "next number" function exactly like a linked list's `next` pointer — use fast/slow pointers (same technique as Linked List Cycle, A5.5) to detect a cycle without any extra HashSet.

```
public boolean isHappy(int n) {
    int slow = n, fast = sumOfSquares(n);
    while (fast != 1 && slow != fast) {
        slow = sumOfSquares(slow);
        fast = sumOfSquares(sumOfSquares(fast));
    }
    return fast == 1;
}
private int sumOfSquares(int n) {
    int sum = 0;
    while (n > 0) { int digit = n % 10; sum += digit * digit; n /= 10; }
    return sum;
}
```

Time: O(log n) per step, bounded overall. Space: O(1) — the key improvement over the HashSet version. **Key insight:** recognizing "repeatedly applying a function until you hit a fixed point or a cycle" as structurally identical to linked-list cycle detection lets you reuse Floyd's algorithm far outside the context of literal linked lists.

B10. Advanced Strings (Pattern Matching)

B10.1 Implement strStr() (Substring Search / KMP Algorithm)

Difficulty: Easy (Medium if implementing KMP) | **Pattern:** String Matching (Naive vs. KMP) | **Companies:** Amazon, Facebook — the standard entry point into pattern-

matching algorithms **Problem:** Find the index of the first occurrence of a needle string in a haystack string, or -1 if not found. **Brute Force:** at every position in `haystack`, compare character by character against `needle`.

```
public int strStrBrute(String haystack, String needle) {
    int n = haystack.length(), m = needle.length();
    for (int i = 0; i <= n - m; i++) {
        int j = 0;
        while (j < m && haystack.charAt(i + j) == needle.charAt(j)) j++;
        if (j == m) return i;
    }
    return -1;
}
```

Time: O(n·m) worst case (e.g. haystack = "aaaa...a", needle = "aaa...ab"). Space: O(1). **Optimal (Knuth-Morris-Pratt / KMP Algorithm):** precompute a "failure function" (`lps` = longest proper prefix that is also a suffix, for every prefix of `needle`). When a mismatch occurs during matching, instead of restarting the needle comparison from scratch, use `lps` to skip directly to the next viable alignment — avoiding ever re-examining characters of `haystack` that have already been matched.

```
public int strStr(String haystack, String needle) {
    if (needle.isEmpty()) return 0;
    int[] lps = buildLPS(needle);
    int i = 0, j = 0;
    while (i < haystack.length()) {
        if (haystack.charAt(i) == needle.charAt(j)) {
            i++; j++;
            if (j == needle.length()) return i - j; // full match found
        } else if (j > 0) {
            j = lps[j - 1]; // use failure function to avoid re-comparing
        } else {
            i++;
        }
    }
    return -1;
}
private int[] buildLPS(String needle) {
    int[] lps = new int[needle.length()];
    int len = 0, i = 1;
    while (i < needle.length()) {
        if (needle.charAt(i) == needle.charAt(len)) {
            lps[i++] = ++len;
        } else if (len > 0) {
            len = lps[len - 1];
        } else {
            lps[i++] = 0;
        }
    }
```

```
        return lps;
    }
```

Time: O(n + m) — O(m) to build the LPS array, O(n) for the main scan, each haystack character examined a bounded number of times. Space: O(m) for the LPS array. **Key insight:** KMP's core idea — "precompute how much of the pattern can be reused after a partial match fails" — is the foundation for many advanced string algorithms; in most interviews, correctly explaining and using the naive O(n·m) approach is acceptable, but mentioning KMP demonstrates deeper knowledge.

B10.2 Longest Common Prefix

Difficulty: Easy | **Pattern:** String comparison (vertical/horizontal scanning) | **Companies:** Amazon, Google — common warm-up **Problem:** Find the longest common prefix string amongst an array of strings. **Brute Force (Horizontal Scanning):** start with the first string as the candidate prefix; repeatedly shrink it until it's a prefix of every other string.

```
public String longestCommonPrefixHorizontal(String[] strs) {
    if (strs.length == 0) return "";
    String prefix = strs[0];
    for (int i = 1; i < strs.length; i++) {
        while (!strs[i].startsWith(prefix)) {
            prefix = prefix.substring(0, prefix.length() - 1);
            if (prefix.isEmpty()) return "";
        }
    }
    return prefix;
}
```

Time: O(S) where S = sum of all characters across all strings (amortized, since shrinking is bounded). Space: O(1) extra (excluding output). **Alternative (Vertical Scanning) — often clearer to reason about:** compare characters column by column across all strings simultaneously; stop at the first column with a mismatch or a string that's too short.

```
public String longestCommonPrefix(String[] strs) {
    if (strs.length == 0) return "";
    for (int col = 0; col < strs[0].length(); col++) {
        char c = strs[0].charAt(col);
        for (int row = 1; row < strs.length; row++) {
            if (col == strs[row].length() || strs[row].charAt(col) != c) {
```

```
                return strs[0].substring(0, col);
            }
        }
    }
    return strs[0];
}
```

Time: O(S). Space: O(1) extra. **Key insight:** vertical scanning can short-circuit slightly earlier in some cases (stops at the first mismatched column across all strings, rather than needing full string comparisons) — either approach is acceptable in an interview, but be ready to explain both.

B10.3 Rabin-Karp Algorithm (Rolling Hash — concept + application)

Difficulty: Medium | **Pattern:** Rolling Hash | **Companies:** Google, Bloomberg — underlies "Repeated DNA Sequences" and other substring-hashing problems **Problem (general technique):** efficiently search for a pattern within text (or detect repeated substrings) using hashing instead of direct character comparison. **Brute Force:** direct substring comparison at every position — O(n·m), same as naive strStr. **Optimal (Rolling Hash):** compute a hash of the pattern and a hash of the first window of text; slide the window one character at a time, updating the hash in O(1) by removing the contribution of the character leaving the window and adding the contribution of the new character entering it (instead of recomputing the hash from scratch). Compare hashes first (cheap), and only do a full character-by-character comparison on a hash match (to rule out hash collisions).

```
public int rabinKarpSearch(String text, String pattern) {
    int n = text.length(), m = pattern.length();
    if (m > n) return -1;
    long base = 256, mod = 1_000_000_007L;
    long patternHash = 0, windowHash = 0, highOrder = 1;
    for (int i = 0; i < m - 1; i++) highOrder = (highOrder * base) % mod;

    for (int i = 0; i < m; i++) {
        patternHash = (patternHash * base + pattern.charAt(i)) % mod;
        windowHash = (windowHash * base + text.charAt(i)) % mod;
    }

    for (int i = 0; i <= n - m; i++) {
        if (windowHash == patternHash && text.substring(i, i + m).equals(pattern))
{
            return i; // verified match (guards against hash collision)
        }
```

```
            if (i < n - m) {
                windowHash = (base * (windowHash - text.charAt(i) * highOrder % mod +
mod) + text.charAt(i + m)) % mod;
            }
        }
        return -1;
    }
```

Time: O(n + m) average (amortized O(1) hash updates), O(n·m) worst case under many hash collisions (rare with a good modulus/base choice). Space: O(1) extra. **Key insight:** the "rolling hash" technique (O(1) incremental hash update as a window slides) generalizes far beyond substring search — it's also the basis for efficient duplicate-substring detection (e.g. "Repeated DNA Sequences") and some string-hashing-based LCS/similarity algorithms.

B10.4 Z-Function (concept + application to pattern matching)

Difficulty: Medium/Hard | **Pattern:** Z-Array (prefix-matching lengths) | **Companies:** Competitive-programming-adjacent, occasionally in advanced interview rounds
Problem (general technique): for a string `s`, compute `Z[i]` = the length of the longest substring starting at `i` that matches a prefix of `s`, for every position, in O(n).
Application: to search for `pattern` in `text`, form `combined = pattern + "#" + text` (the `#` separator must not appear in either string) and compute its Z-array; any position where `Z[i] == pattern.length()` marks an occurrence of `pattern` in `text`.

```java
public int[] buildZArray(String s) {
    int n = s.length();
    int[] z = new int[n];
    int left = 0, right = 0;
    for (int i = 1; i < n; i++) {
        if (i < right) {
            z[i] = Math.min(right - i, z[i - left]);
        }
        while (i + z[i] < n && s.charAt(z[i]) == s.charAt(i + z[i])) {
            z[i]++;
        }
        if (i + z[i] > right) {
            left = i;
            right = i + z[i];
        }
    }
    return z;
}
```

Time: O(n) — the `right` boundary only ever moves forward, bounding total work. Space: O(n) for the Z-array. **Key insight:** the Z-array is functionally similar in spirit to KMP's LPS array (both avoid redundant character comparisons via cleverly reused partial-match information), but is often considered more intuitive to derive and extend to related problems (e.g. counting distinct substrings, finding all border lengths of a string).

B10.5 Manacher's Algorithm (concept — O(n) Longest Palindromic Substring)

Difficulty: Hard | **Pattern:** Palindrome expansion with reused state | **Companies:** Rare in standard interviews, but an impressive "I know the optimal answer" mention for Longest Palindromic Substring (B1.2) **Problem:** Find the longest palindromic substring in O(n) time (improving on the O(n²) expand-around-center approach from B1.2). **Why the O(n²) approach isn't optimal:** each center's expansion can take O(n) in the worst case (e.g. a string of all identical characters), and there are O(n) centers, giving O(n²) total — Manacher's algorithm eliminates redundant expansion work. **Optimal (Manacher's Algorithm — conceptual overview):** transform the string by inserting a separator character (e.g. `#`) between every character (and at both ends) to uniformly handle both odd- and even-length palindromes as "odd-length" in the transformed string. Maintain the rightmost-reaching palindrome found so far (`center`, `right` boundary). For each new position, if it lies within the current rightmost palindrome, initialize its palindrome radius using the **mirror position's** already-computed radius

(reusing previous work, analogous to the Z-array and KMP's LPS array), then attempt to expand further only if needed; update `center`/`right` whenever a new palindrome extends past the previous rightmost boundary.

```
public String longestPalindromeManacher(String s) {
    StringBuilder sb = new StringBuilder("^");
    for (char c : s.toCharArray()) sb.append('#').append(c);
    sb.append("#$");
    String t = sb.toString();
    int n = t.length();
    int[] p = new int[n]; // p[i] = radius of palindrome centered at i in the
transformed string
    int center = 0, right = 0;

    for (int i = 1; i < n - 1; i++) {
        if (i < right) {
            p[i] = Math.min(right - i, p[2 * center - i]); // reuse mirror's known
radius
        }
        while (t.charAt(i + p[i] + 1) == t.charAt(i - p[i] - 1)) {
            p[i]++;
        }
        if (i + p[i] > right) {
            center = i;
            right = i + p[i];
        }
    }

    int maxLen = 0, centerIndex = 0;
    for (int i = 1; i < n - 1; i++) {
        if (p[i] > maxLen) { maxLen = p[i]; centerIndex = i; }
    }
    int start = (centerIndex - maxLen) / 2; // map back to original string
coordinates
    return s.substring(start, start + maxLen);
}
```

Time: O(n) — the `right` boundary only ever moves forward across the whole algorithm, guaranteeing linear total work despite the nested-looking expansion step. Space: O(n) for the transformed string and the `p` array. **Key insight:** Manacher's is rarely required to be derived from scratch in a standard interview, but knowing it exists (and being able to state "the O(n²) expand-around-center is what I'd implement live, but the theoretically optimal answer is Manacher's algorithm in O(n)") is a strong signal of depth if the topic comes up.

Part C: Playbook & Index

C1. Pattern Recognition Cheat Sheet

This is the single most important section to re-read the night before an interview. For each pattern: the **recognition signal** (what in the problem statement should make you think of it), a **reusable code template**, and **which problems above use it** so you can jump back for a full worked example.

C1.1 Two Pointers

Recognition signal: sorted array (or can be sorted), looking for a pair/triplet with a target property, or comparing from both ends (palindrome-style).

```
int left = 0, right = arr.length - 1;
while (left < right) {
    // evaluate arr[left], arr[right]
    if (/* need bigger sum */) left++;
    else if (/* need smaller sum */) right--;
    else { /* found */ break; }
}
```

Used in: Two Sum II (B1.1), Valid Palindrome (A2.1), 3Sum (A1.9), Container With Most Water (A1.10), Trapping Rain Water (A2.6).

C1.2 Sliding Window

Recognition signal: "contiguous subarray/substring" + a constraint on sum, distinct characters, or a target frequency map.

```
int left = 0;
for (int right = 0; right < arr.length; right++) {
    // expand window by including arr[right]
    while (/* window invalid */) {
        // shrink window by excluding arr[left]
        left++;
    }
    // window [left, right] is valid here -- update answer
}
```

Used in: Longest Substring Without Repeating Characters (A2.2), Longest Repeating Character Replacement (A2.3), Minimum Window Substring (A2.4).

C1.3 Fast & Slow Pointers (Floyd's Cycle Detection)

Recognition signal: linked list (or an implicit functional graph via array indices/digit transformations) where you need to detect a cycle or find a middle point.

```
ListNode slow = head, fast = head;
while (fast != null && fast.next != null) {
    slow = slow.next;
    fast = fast.next.next;
    if (slow == fast) { /* cycle detected */ }
}
```

Used in: Linked List Cycle (A5.5), Find the Duplicate Number (A5.10), Happy Number (B9.6), Reorder List (A5.3), Palindrome Linked List (B3.1).

C1.4 Binary Search (Classic + "On the Answer")

Recognition signal: sorted data, OR a monotonic yes/no feasibility function over a range of possible answers ("can we achieve X with capacity/speed/time = mid?").

```
int left = 0, right = /* max possible answer */;
while (left < right) {
    int mid = left + (right - left) / 2;
    if (feasible(mid)) right = mid; else left = mid + 1;
}
return left;
```

Used in: Binary Search (A4.1), Koko Eating Bananas (A4.3), Find Min in Rotated Sorted Array (A1.7), Median of Two Sorted Arrays (A4.5).

C1.5 Merge Intervals

Recognition signal: the word "intervals," or overlapping ranges that need combining or counting concurrency.

```
Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
List<int[]> merged = new ArrayList<>();
for (int[] iv : intervals) {
    if (merged.isEmpty() || merged.get(merged.size()-1)[1] < iv[0]) merged.add(iv);
    else merged.get(merged.size()-1)[1] = Math.max(merged.get(merged.size()-1)[1],
iv[1]);
}
```

Used in: Merge Intervals (A13.3), Insert Interval (A13.4), Meeting Rooms I/II (A8.5, B9.1), Non-overlapping Intervals (A13.5).

C1.6 Tree BFS (Level Order)

Recognition signal: "level by level," shortest path in an unweighted graph, or "minimum steps."

```
Queue<Node> queue = new LinkedList<>();
queue.offer(start);
while (!queue.isEmpty()) {
    int size = queue.size();
    for (int i = 0; i < size; i++) {
        Node cur = queue.poll();
        // process cur
        for (Node next : neighbors(cur)) queue.offer(next);
    }
}
```

Used in: Binary Tree Level Order Traversal (A6.6), Rotting Oranges (A10.8), Word Ladder (A10.7), Number of Islands (BFS variant, A10.1).

C1.7 Tree / Graph DFS

Recognition signal: exploring all paths, subtree computations, connected components.

```
void dfs(Node node, Set<Node> visited) {
    if (node == null || visited.contains(node)) return;
    visited.add(node);
    // process node
    for (Node next : neighbors(node)) dfs(next, visited);
}
```

Used in: Number of Islands (A10.1), Clone Graph (A10.2), Max Depth of Binary Tree (A6.2), Course Schedule (A10.3, cycle detection variant).

C1.8 Backtracking

Recognition signal: "return all possible ...", combinatorial generation (subsets, permutations, combinations, valid arrangements).

```
void backtrack(State state, List<Result> results) {
    if (isComplete(state)) { results.add(snapshot(state)); return; }
    for (Choice choice : choicesAt(state)) {
        apply(state, choice);
        backtrack(state, results);
        undo(state, choice); // the "back" in backtracking
    }
}
```

Used in: Subsets (A9.1), Combination Sum (A9.2), Permutations (A9.3), N-Queens (A9.7), Word Search (A9.4).

C1.9 Topological Sort (Kahn's Algorithm)

Recognition signal: "prerequisites," "dependencies," "must come before," directed acyclic graph ordering.

```
Queue<Node> queue = new LinkedList<>();
for (Node n : allNodes) if (inDegree[n] == 0) queue.offer(n);
List<Node> order = new ArrayList<>();
while (!queue.isEmpty()) {
    Node n = queue.poll();
    order.add(n);
    for (Node next : graph.get(n)) if (--inDegree[next] == 0) queue.offer(next);
}
// order.size() < allNodes.size() => cycle exists
```

Used in: Course Schedule I/II (A10.3), Alien Dictionary (B5.3).

C1.10 Union-Find (Disjoint Set Union)

Recognition signal: "connected components," "will adding this edge create a cycle," dynamic connectivity queries.

```
int find(int[] parent, int x) {
    if (parent[x] != x) parent[x] = find(parent, parent[x]); // path compression
    return parent[x];
}
void union(int[] parent, int a, int b) { parent[find(parent, a)] = find(parent, b);
}
```

Used in: Number of Connected Components (A10.5), Graph Valid Tree (A10.6), Redundant Connection (A10.9), Minimum Spanning Tree / Kruskal's (B5.4).

C1.11 Monotonic Stack / Deque

Recognition signal: "next greater/smaller element," histogram-style area problems, sliding window maximum.

```
Deque<Integer> stack = new ArrayDeque<>(); // store indices, values monotonic
for (int i = 0; i < arr.length; i++) {
    while (!stack.isEmpty() && arr[stack.peek()] < arr[i]) {
        int idx = stack.pop();
        // arr[i] is the "next greater element" for index idx
    }
    stack.push(i);
}
```

Used in: Daily Temperatures (A3.5), Largest Rectangle in Histogram (A3.6), Sliding Window Maximum (A2.5).

C1.12 Top-K Elements (Heap)

Recognition signal: "kth largest/smallest," "top k frequent," "k closest."

```
PriorityQueue<Integer> minHeap = new PriorityQueue<>(); // for "top k largest"
for (int num : nums) {
    minHeap.offer(num);
    if (minHeap.size() > k) minHeap.poll();
}
// minHeap now holds the k largest elements; minHeap.peek() is the kth largest
```

Used in: Kth Largest Element in an Array (A8.1), Top K Frequent Elements (A8.2), Find Median from Data Stream — two heaps (A8.3), Task Scheduler (A8.4).

C1.13 1-D Dynamic Programming

Recognition signal: "number of ways," "minimum/maximum ... ending at position i," clear recurrence depending only on a constant number of previous states.

```
int[] dp = new int[n + 1];
dp[0] = /* base case */;
for (int i = 1; i <= n; i++) {
    dp[i] = /* combine dp[i-1], dp[i-2], ... based on the problem's recurrence */;
}
return dp[n];
```

Used in: Climbing Stairs (A11.1), House Robber (A11.2), Coin Change (A11.8), Word Break (A11.5), Decode Ways (A11.7).

C1.14 2-D Dynamic Programming (Grid / Two-String)

Recognition signal: two strings being compared, or a grid where you move right/down, building up an `dp[i][j]` table.

```
int[][] dp = new int[m + 1][n + 1];
// initialize base cases (dp[0][*], dp[*][0])
for (int i = 1; i <= m; i++) {
    for (int j = 1; j <= n; j++) {
        dp[i][j] = /* combine dp[i-1][j], dp[i][j-1], dp[i-1][j-1] per the
problem's rule */;
    }
}
return dp[m][n];
```

Used in: Longest Common Subsequence (A12.2), Edit Distance (A12.4), Unique Paths (A12.1), Minimum Path Sum (A12.5).

C1.15 0/1 Knapsack (Subset-Sum Shape)

Recognition signal: "can we pick a subset of these items to exactly reach / not exceed a target," each item usable **at most once**.

```
boolean[] dp = new boolean[target + 1];
dp[0] = true;
for (int item : items) {
    for (int t = target; t >= item; t--) { // iterate HIGH to LOW -- crucial!
        dp[t] = dp[t] || dp[t - item];
    }
}
```

Used in: Partition Equal Subset Sum (A11.9), Target Sum (via subset-sum reduction).

C1.16 Unbounded Knapsack (Coin Change Shape)

Recognition signal: same as above but items are **reusable** unlimited times.

```
int[] dp = new int[target + 1];
Arrays.fill(dp, Integer.MAX_VALUE);
dp[0] = 0;
for (int t = 1; t <= target; t++) {
    for (int item : items) {
        if (item <= t && dp[t - item] != Integer.MAX_VALUE) dp[t] = Math.min(dp[t],
dp[t - item] + 1);
    }
}
```

Used in: Coin Change (A11.8), Perfect Squares (B6.5), Combination Sum IV (A11.6, counting variant with swapped loop order).

C1.17 Trie (Prefix Tree)

Recognition signal: prefix-based lookups, autocomplete, dictionary word search over a grid, wildcard pattern matching against many words.

```
class TrieNode { TrieNode[] children = new TrieNode[26]; boolean isEnd; }
// insert: walk/create children per character, mark isEnd on the last node
// search: walk children per character, check isEnd
// startsWith: walk children per character, no isEnd check needed
```

Used in: Implement Trie (A7.1), Add and Search Words (A7.2), Word Search II (A7.3).

C1.18 Greedy (Interval / Activity Selection)

Recognition signal: "minimum number of ... to remove/add," scheduling, activity selection — usually solved by sorting by a specific key (often END time) and scanning once.

```
Arrays.sort(items, (a, b) -> a[1] - b[1]); // sort by END coordinate
int lastEnd = Integer.MIN_VALUE, count = 0;
for (int[] item : items) {
    if (item[0] >= lastEnd) { count++; lastEnd = item[1]; } // keep/select this
item
}
```

Used in: Non-overlapping Intervals (A13.5), Minimum Arrows to Burst Balloons (B9.2), Jump Game (A11.10, different greedy shape — track reachable range).

C1.19 Prefix Sum

Recognition signal: repeated range-sum queries, or "subarray sum equals k"-style problems.

```
int[] prefix = new int[n + 1];
for (int i = 0; i < n; i++) prefix[i + 1] = prefix[i] + nums[i];
// sum of nums[i..j] inclusive = prefix[j+1] - prefix[i]
```

Used in: Product of Array Except Self (A1.4, prefix/suffix product variant), Path Sum III (prefix-sum + HashMap variant, mentioned in A6.14's follow-up).

C1.20 Bit Manipulation Tricks

Recognition signal: problems explicitly about bits, or "without using +/-/* " constraints.

```
n & (n - 1)   // clears the lowest set bit (Number of 1 Bits, Counting Bits)
n & (-n)      // isolates the lowest set bit
a ^ b ^ b == a // XOR self-cancellation (Single Number family, Missing Number)
```

Used in: Number of 1 Bits (A14.1), Counting Bits (A14.2), Missing Number (A14.4), Sum of Two Integers (A14.5).

C2. Master Problem Index

The following table indexes every problem covered in this guide. **P** = Part A (full treatment), **B** = Part B (condensed). Jump to any section heading above for the full breakdown.

#	Problem	Difficulty	Pattern	Section
1	Two Sum	Easy	Hashing	A1.1
2	Best Time to Buy and Sell Stock	Easy	One Pass	A1.2
3	Contains Duplicate	Easy	Hashing	A1.3
4	Product of Array Except Self	Medium	Prefix/Suffix	A1.4
5	Maximum Subarray	Medium	Kadane's DP	A1.5
6	Maximum Product Subarray	Medium	DP (max/min)	A1.6
7	Find Minimum in Rotated Sorted Array	Medium	Binary Search	A1.7
8	Search in Rotated Sorted Array	Medium	Binary Search	A1.8
9	3Sum	Medium	Sort + Two Pointers	A1.9
10	Container With Most Water	Medium	Two Pointers	A1.10
11	Valid Palindrome	Easy	Two Pointers	A2.1
12	Longest Substring Without Repeating Characters	Medium	Sliding Window	A2.2
13	Longest Repeating Character Replacement	Medium	Sliding Window	A2.3
14	Minimum Window Substring	Hard	Sliding Window	A2.4
15	Sliding Window Maximum	Hard	Monotonic Deque	A2.5
16	Trapping Rain Water	Hard	Two Pointers	A2.6
17	Group Anagrams	Medium	Hashing	A2.7

#	Problem	Difficulty	Pattern	Section
18	Valid Anagram	Easy	Hashing	A2.8
19	Valid Parentheses	Easy	Stack	A3.1
20	Min Stack	Medium	Stack	A3.2
21	Evaluate Reverse Polish Notation	Medium	Stack	A3.3
22	Generate Parentheses	Medium	Backtracking	A3.4
23	Daily Temperatures	Medium	Monotonic Stack	A3.5
24	Largest Rectangle in Histogram	Hard	Monotonic Stack	A3.6
25	Binary Search	Easy	Binary Search	A4.1
26	Search a 2D Matrix	Medium	Binary Search	A4.2
27	Koko Eating Bananas	Medium	Binary Search on Answer	A4.3
28	Find First and Last Position in Sorted Array	Medium	Binary Search	A4.4
29	Median of Two Sorted Arrays	Hard	Binary Search (Partition)	A4.5
30	Sqrt(x)	Easy	Binary Search	A4.6
31	Reverse Linked List	Easy	Pointer Reversal	A5.1
32	Merge Two Sorted Lists	Easy	Two Pointers	A5.2
33	Reorder List	Medium	Fast/Slow + Reversal	A5.3
34	Remove Nth Node From End of List	Medium	Two Pointers	A5.4
35	Linked List Cycle	Easy	Floyd's Cycle Detection	A5.5
36	Merge k Sorted Lists	Hard	Heap	A5.6
37	LRU Cache	Medium	HashMap + Doubly Linked List	A5.7

#	Problem	Difficulty	Pattern	Section
38	Copy List with Random Pointer	Medium	HashMap Mapping	A5.8
39	Add Two Numbers	Medium	Linked List Simulation	A5.9
40	Find the Duplicate Number	Medium	Floyd's Cycle Detection	A5.10
41	Invert Binary Tree	Easy	Tree DFS/BFS	A6.1
42	Maximum Depth of Binary Tree	Easy	Tree DFS	A6.2
43	Same Tree	Easy	Tree DFS	A6.3
44	Subtree of Another Tree	Easy	Tree DFS	A6.4
45	Lowest Common Ancestor of a BST	Medium	BST Property	A6.5
46	Binary Tree Level Order Traversal	Medium	BFS	A6.6
47	Validate Binary Search Tree	Medium	DFS with Bounds	A6.7
48	Kth Smallest Element in a BST	Medium	In-order Traversal	A6.8
49	Construct Binary Tree from Preorder and Inorder	Medium	Divide & Conquer	A6.9
50	Binary Tree Maximum Path Sum	Hard	Tree DFS (global state)	A6.10
51	Serialize and Deserialize Binary Tree	Hard	Tree DFS/BFS	A6.11
52	Diameter of Binary Tree	Easy	Tree DFS (global state)	A6.12
53	Balanced Binary Tree	Easy	Tree DFS	A6.13
54	Path Sum	Easy	Tree DFS	A6.14
55	Implement Trie (Prefix Tree)	Medium	Trie	A7.1

#	Problem	Difficulty	Pattern	Section
56	Design Add and Search Words Data Structure	Medium	Trie + DFS	A7.2
57	Word Search II	Hard	Trie + Grid DFS	A7.3
58	Kth Largest Element in an Array	Medium	Heap / Quickselect	A8.1
59	Top K Frequent Elements	Medium	Heap / Bucket Sort	A8.2
60	Find Median from Data Stream	Hard	Two Heaps	A8.3
61	Task Scheduler	Medium	Greedy + Heap/Math	A8.4
62	Meeting Rooms II	Medium	Heap / Sweep Line	A8.5
63	Subsets	Medium	Backtracking	A9.1
64	Combination Sum	Medium	Backtracking	A9.2
65	Permutations	Medium	Backtracking	A9.3
66	Word Search	Medium	Backtracking / Grid DFS	A9.4
67	Palindrome Partitioning	Medium	Backtracking + DP	A9.5
68	Letter Combinations of a Phone Number	Medium	Backtracking	A9.6
69	N-Queens	Hard	Backtracking	A9.7
70	Number of Islands	Medium	Grid DFS/BFS	A10.1
71	Clone Graph	Medium	Graph DFS/BFS	A10.2
72	Course Schedule	Medium	Topological Sort	A10.3
73	Pacific Atlantic Water Flow	Medium	Multi-source DFS	A10.4
74	Number of Connected Components	Medium	Union-Find	A10.5
75	Graph Valid Tree	Medium	Union-Find	A10.6
76	Word Ladder	Hard	BFS	A10.7
77	Rotting Oranges	Medium	Multi-source BFS	A10.8

#	Problem	Difficulty	Pattern	Section
78	Redundant Connection	Medium	Union-Find	A10.9
79	Climbing Stairs	Easy	DP (Fibonacci)	A11.1
80	House Robber	Medium	DP	A11.2
81	House Robber II	Medium	DP	A11.3
82	Longest Increasing Subsequence	Medium	DP + Binary Search	A11.4
83	Word Break	Medium	DP	A11.5
84	Combination Sum IV	Medium	DP	A11.6
85	Decode Ways	Medium	DP	A11.7
86	Coin Change	Medium	DP (Unbounded Knapsack)	A11.8
87	Partition Equal Subset Sum	Medium	DP (0/1 Knapsack)	A11.9
88	Jump Game	Medium	Greedy	A11.10
89	Unique Paths	Medium	2-D DP	A12.1
90	Longest Common Subsequence	Medium	2-D DP	A12.2
91	Best Time to Buy/Sell Stock with Cooldown	Medium	DP State Machine	A12.3
92	Edit Distance	Hard	2-D DP	A12.4
93	Minimum Path Sum	Medium	2-D DP	A12.5
94	Maximal Square	Medium	2-D DP	A12.6
95	Regular Expression Matching	Hard	2-D DP	A12.7
96	Jump Game II	Medium	Greedy	A13.1
97	Gas Station	Medium	Greedy	A13.2
98	Merge Intervals	Medium	Greedy + Sort	A13.3
99	Insert Interval	Medium	Greedy	A13.4
100	Non-overlapping Intervals	Medium	Greedy + Sort	A13.5

#	Problem	Difficulty	Pattern	Section
101	Number of 1 Bits	Easy	Bit Manipulation	A14.1
102	Counting Bits	Easy	DP + Bit Manipulation	A14.2
103	Reverse Bits	Easy	Bit Manipulation	A14.3
104	Missing Number	Easy	Math / XOR	A14.4
105	Sum of Two Integers	Medium	Bit Manipulation	A14.5
106	Two Sum II	Medium	Two Pointers	B1.1
107	Longest Palindromic Substring	Medium	Expand Around Center	B1.2
108	Palindromic Substrings	Medium	Expand Around Center	B1.3
109	Valid Sudoku	Medium	Hashing	B1.4
110	Rotate Array	Medium	Array Reversal	B1.5
111	Majority Element	Easy	Boyer-Moore Voting	B1.6
112	Next Permutation	Medium	Array Manipulation	B1.7
113	First Missing Positive	Hard	Cyclic Sort	B1.8
114	Rotate Image	Medium	Matrix Manipulation	B2.1
115	Spiral Matrix	Medium	Matrix Traversal	B2.2
116	Set Matrix Zeroes	Medium	Matrix Manipulation	B2.3
117	Search a 2D Matrix II	Medium	Staircase Search	B2.4
118	Game of Life	Medium	Matrix Simulation	B2.5
119	Palindrome Linked List	Easy	Fast/Slow + Reversal	B3.1
120	Intersection of Two Linked Lists	Easy	Two Pointers	B3.2
121	Odd Even Linked List	Medium	Pointer Regrouping	B3.3
122	Flatten a Multilevel Doubly Linked List	Medium	DFS/Stack	B3.4

#	Problem	Difficulty	Pattern	Section
123	Binary Tree Right Side View	Medium	BFS	B4.1
124	Count Good Nodes in Binary Tree	Medium	DFS (running max)	B4.2
125	Binary Search Tree Iterator	Medium	Controlled In-order	B4.3
126	Convert Sorted Array to BST	Easy	Divide & Conquer	B4.4
127	Delete Node in a BST	Medium	BST Manipulation	B4.5
128	Vertical Order Traversal of a Binary Tree	Hard	BFS + Bucketing	B4.6
129	Network Delay Time	Medium	Dijkstra's Algorithm	B5.1
130	Cheapest Flights Within K Stops	Medium	Bellman-Ford (bounded)	B5.2
131	Alien Dictionary	Hard	Topological Sort	B5.3
132	Connecting Cities With Minimum Cost (MST)	Medium/Hard	Kruskal's Algorithm	B5.4
133	Detect Negative Cycle	Medium/Hard	Bellman-Ford	B5.5
134	Reconstruct Itinerary	Hard	Hierholzer's Algorithm	B5.6
135	Longest Palindromic Subsequence	Medium	2-D DP (LCS shape)	B6.1
136	Interleaving String	Medium	2-D DP	B6.2
137	Distinct Subsequences	Hard	2-D DP	B6.3
138	Burst Balloons	Hard	Interval DP	B6.4
139	Perfect Squares	Medium	DP (Unbounded Knapsack)	B6.5
140	Integer Break	Medium	DP / Math	B6.6
141	Min Cost Climbing Stairs	Easy	DP	B6.7
142	Combinations	Medium	Backtracking	B7.1

#	Problem	Difficulty	Pattern	Section
143	Combination Sum II	Medium	Backtracking	B7.2
144	Permutations II	Medium	Backtracking	B7.3
145	Subsets II	Medium	Backtracking	B7.4
146	Implement Queue using Stacks	Easy	Two-Stack Simulation	B8.1
147	Implement Stack using Queues	Easy	Queue Rotation	B8.2
148	Design Twitter	Medium	Heap + HashMap	B8.3
149	Design Hit Counter	Medium	Queue / Sliding Window	B8.4
150	Design Circular Queue	Medium	Circular Buffer	B8.5
151	Insert Delete GetRandom O(1)	Medium	HashMap + ArrayList	B8.6
152	Meeting Rooms I	Easy	Sorting	B9.1
153	Minimum Number of Arrows to Burst Balloons	Medium	Greedy	B9.2
154	Candy	Hard	Greedy (Two-Pass)	B9.3
155	Pow(x, n)	Medium	Fast Exponentiation	B9.4
156	Excel Sheet Column Number/Title	Easy	Math (Base-26)	B9.5
157	Happy Number	Easy	Floyd's Cycle Detection	B9.6
158	Implement strStr() / KMP	Easy/Medium	String Matching	B10.1
159	Longest Common Prefix	Easy	String Comparison	B10.2
160	Rabin-Karp Algorithm	Medium	Rolling Hash	B10.3
161	Z-Function	Medium/Hard	Z-Array	B10.4
162	Manacher's Algorithm	Hard	Palindrome DP/Expansion	B10.5

C3. Closing Notes

C3.1 Suggested 8-Week Study Plan

This plan assumes roughly 1-2 hours per day. Adjust the pace to your timeline, but keep the **order** — patterns build on each other (e.g. two pointers before sliding window, trees before tries, DP basics before 2-D DP).

Week 1 — Foundations: Arrays, Hashing, Two Pointers

- Days 1-2: Re-read Part 0 (the universal framework) until it's automatic. Work Two Sum (A1.1), Contains Duplicate (A1.3), Valid Anagram (A2.8), Group Anagrams (A2.7) from memory, then check against the guide.
- Days 3-4: Two pointers — Valid Palindrome (A2.1), Two Sum II (B1.1), Container With Most Water (A1.10), 3Sum (A1.9).
- Days 5-7: Kadane's/prefix-sum family — Maximum Subarray (A1.5), Maximum Product Subarray (A1.6), Product of Array Except Self (A1.4). Do a timed mixed-review of all Week 1 problems.

Week 2 — Sliding Window, Stack, Binary Search

- Days 1-3: Sliding window — Longest Substring Without Repeating Characters (A2.2), Longest Repeating Character Replacement (A2.3), Minimum Window Substring (A2.4).
- Days 4-5: Stack — Valid Parentheses (A3.1), Min Stack (A3.2), Daily Temperatures (A3.5), Largest Rectangle in Histogram (A3.6).
- Days 6-7: Binary search — Binary Search (A4.1), Search in Rotated Sorted Array (A1.8), Koko Eating Bananas (A4.3, "binary search on the answer" — a very high-leverage pattern).

Week 3 — Linked Lists

- Days 1-2: Reverse Linked List (A5.1), Merge Two Sorted Lists (A5.2), Linked List Cycle (A5.5).
- Days 3-4: Remove Nth Node From End (A5.4), Reorder List (A5.3), Palindrome Linked List (B3.1).
- Days 5-7: Merge k Sorted Lists (A5.6), LRU Cache (A5.7, design — extremely common), Copy List with Random Pointer (A5.8), Add Two Numbers (A5.9).

Week 4 — Trees

- Days 1-2: Invert Binary Tree (A6.1), Max Depth (A6.2), Same Tree (A6.3), Subtree of Another Tree (A6.4).
- Days 3-4: Level Order Traversal (A6.6), Validate BST (A6.7), Kth Smallest in BST (A6.8), LCA of a BST (A6.5).
- Days 5-7: Construct Binary Tree from Preorder/Inorder (A6.9), Diameter of Binary Tree (A6.12), Binary Tree Maximum Path Sum (A6.10) — recognize the shared "combine children, update a global best" template between these last two.

Week 5 — Tries, Heaps, Backtracking

- Days 1-2: Implement Trie (A7.1), Add and Search Words (A7.2).
- Days 3-4: Kth Largest Element (A8.1, learn both heap AND quickselect), Top K Frequent Elements (A8.2), Find Median from Data Stream (A8.3).
- Days 5-7: Subsets (A9.1), Combination Sum (A9.2), Permutations (A9.3), Word Search (A9.4), N-Queens (A9.7) — internalize the "choose, explore, un-choose" skeleton so deeply you can write it without thinking.

Week 6 — Graphs

- Days 1-2: Number of Islands (A10.1), Clone Graph (A10.2).
- Days 3-4: Course Schedule (A10.3, topological sort), Number of Connected Components (A10.5), Graph Valid Tree (A10.6) — Union-Find.
- Days 5-7: Word Ladder (A10.7), Rotting Oranges (A10.8), Pacific Atlantic Water Flow (A10.4). If time allows, skim Part B5 (Dijkstra's, Bellman-Ford, MST) even if you don't fully drill them — recognizing when they're needed is often enough.

Week 7 — Dynamic Programming

- Days 1-2: Climbing Stairs (A11.1), House Robber I/II (A11.2, A11.3) — get the rolling- variable Fibonacci-shape DP into muscle memory.
- Days 3-4: Longest Increasing Subsequence (A11.4), Word Break (A11.5), Coin Change (A11.8), Partition Equal Subset Sum (A11.9).
- Days 5-7: Unique Paths (A12.1), Longest Common Subsequence (A12.2), Edit Distance (A12.4). These three cover the two most common 2-D DP shapes (grid paths, two-string comparison).

Week 8 — Greedy, Intervals, Bit Manipulation, Full Mock Interviews

- Days 1-2: Merge Intervals (A13.3), Insert Interval (A13.4), Non-overlapping Intervals (A13.5), Jump Game I/II (A11.10, A13.1).

- Days 3-4: Number of 1 Bits (A14.1), Counting Bits (A14.2), Missing Number (A14.4) — quick wins, low time investment, still commonly asked.
- Days 5-7: Full mixed mock interviews. Pick 2-3 random problems per day from across ALL sections (use the Master Index in C2), give yourself 30-45 minutes each, and explicitly practice narrating the Part 0 framework out loud, even alone.

C3.2 Frequently Asked Questions

Q: Do I really need to memorize all 162 problems? No. You need to internalize the ~20 *patterns* in the Pattern Recognition Cheat Sheet (C1). The problems are vehicles for learning the patterns — once a pattern clicks, dozens of superficially different problems become the same problem wearing a different costume.

Q: What language should I use in interviews? Whatever you're most fluent in. This guide uses Java throughout for consistency, but the same brute-force → optimal reasoning applies identically in Python, C++, JavaScript, or Go. If your target company has a strong language preference (rare, but happens), match it.

Q: How much should I talk versus code silently? Talk continuously, especially before you start typing. State the brute force and its complexity out loud within the first 2-3 minutes. Silence for more than ~60 seconds while "just thinking" reads poorly in most interview loops — narrate your thought process even when you're stuck.

Q: What if I can't find the optimal solution in the moment? Say so explicitly: "I have a working $O(n^2)$ solution; let me think about whether we can do better." Then talk through the bottleneck analysis from Part 0 (step 5) out loud. Many interviewers will nudge you toward the right idea if you show you understand *why* the brute force is slow, even if you don't independently derive the optimization.

Q: Should I use recursion or iteration for tree/backtracking problems? Default to recursion for clarity unless the interviewer specifically asks about stack overflow risk on deep/unbalanced inputs, in which case mention (or implement) the iterative version with an explicit stack — this guide shows both for several tree problems specifically so you have that alternative ready.

Q: How important is exact syntax correctness? Less than you'd think — most interviewers care far more about correct logic, complexity awareness, and edge-case handling than a missing semicolon. That said, practicing on a real keyboard (not just thinking through solutions) matters for building muscle memory under time pressure.

Q: I keep forgetting the exact duplicate-skipping conditions in backtracking (Subsets II vs Permutations II). What's the shortcut? Subsets/Combinations skip duplicates via `i > start && nums[i] == nums[i-1]` (because each recursion "level" tries each value once, moving forward). Permutations skip duplicates via `i > 0 && nums[i] == nums[i-1] && !used[i-1]` (because permutations try every unused value at every position, needing the `!used` check to enforce a consistent relative order among equal values). If you only remember one thing: subsets use a **start index** bound, permutations use a **used array** bound — that difference in mechanism is what drives the different skip condition.

Q: When should I reach for a HashMap versus a two-pointer approach? If the input is unsorted (or sortedness would be destroyed by what you need to track, e.g. original indices), use a HashMap. If the input is sorted (or you can afford to sort it and don't need original order/indices), two pointers are usually more space-efficient (O(1) vs O(n)).

Q: How do I know when a problem needs a heap versus a HashMap versus a Union-Find? Heap: whenever you need the current min/max repeatedly as data streams in or changes ("kth largest," "top k," "merge k sorted things," "median so far"). Union-Find: whenever the question is fundamentally about connectivity/grouping that changes incrementally ("will this edge create a cycle," "how many components after adding these edges"). HashMap: whenever you need O(1) existence/frequency lookups without needing an ordering guarantee.

Q: What's the single highest-leverage pattern to master first if I'm short on time? Sliding Window and Two Pointers together — they cover a disproportionate fraction of Easy/ Medium array and string questions, and the mental model transfers almost directly into several DP and greedy problems too.

Q: Are Hard-difficulty problems actually asked in real interviews? Yes, especially at Google, Meta, and quantitative finance shops (Two Sigma, Jane Street, Citadel) — but usually the "famous" hards that have well-known clean solutions (Median of Two Sorted Arrays, Trapping Rain Water, Merge k Sorted Lists, LRU Cache implementation details) rather than obscure ones. The Hard problems in Part A were deliberately chosen because they recur across many companies' loops.

Q: Should I learn Manacher's Algorithm / KMP / Z-function in depth? Only if you have spare time after mastering everything else in Part A and most of Part B. Knowing they exist and what problem class they solve (and being able to implement the $O(n^2)$/ naive version cleanly under pressure) is sufficient for the overwhelming majority of

interviews. These are included in Part B10 primarily so you recognize them by name if an interviewer asks "do you know a faster way?"

Q: How do system design and DSA interviews differ in what "good" looks like? DSA interviews reward precision (exact complexity, correct edge cases, clean code); system design interviews reward breadth and trade-off reasoning (there's rarely one "correct" answer). Don't over-engineer a DSA answer with unnecessary abstraction, and don't under- specify a system design answer by jumping straight to one solution without discussing alternatives.

Q: What if two problems in this guide seem to contradict each other's advice (e.g. sort by start time vs. sort by end time for intervals)? That's intentional and important — Merge Intervals (A13.3) sorts by **start** time (because you're combining overlapping ranges in left-to-right order), while Non-overlapping Intervals (A13.5) sorts by **end** time (because you're greedily deciding what to *keep*, and keeping the earliest-ending option leaves the most room for the future). Always ask "what am I optimizing for" before defaulting to a sort key.

C3.3 Common Mistakes and How to Avoid Them

1. **Jumping straight to the optimal solution without stating the brute force.** Interviewers often can't tell if you actually understand why an approach is slow if you skip straight to the fast one — always narrate the brute force and its complexity first, even briefly.
2. **Forgetting to ask about duplicates.** Many "Medium" problems (3Sum, Subsets II, Permutations II, Combination Sum II) become subtly wrong if you don't explicitly handle duplicate input values — always ask, and if duplicates are possible, mention how you'll handle them before coding.
3. **Off-by-one errors in binary search.** The most common bug in this entire guide's pattern family. Always be explicit about whether your search space is inclusive (`left <= right`) or exclusive (`left < right`), and be consistent about how `mid` is included or excluded from future search ranges.
4. **Mutating input during traversal without considering side effects.** Marking grid cells visited by overwriting them (Number of Islands, Word Search) is a valid and common technique, but always mention it explicitly and restore state (backtrack) if the interviewer needs the original input preserved afterward.

5. **Ignoring integer overflow.** Especially relevant in: Sum of Two Integers (bit tricks), Reverse Integer, Pow(x,n), and anywhere multiplying large counts (Distinct Subsequences). Default to `long` for accumulating products/sums when magnitudes aren't tightly bounded.
6. **Not stating space complexity.** Many candidates state time complexity reflexively but forget space — always state both, and explicitly flag when your solution uses O(1) extra space versus O(n) (interviewers often ask "can you do this with less memory?" as a deliberate follow-up).
7. **Choosing the wrong sort key for greedy interval problems.** Covered in the FAQ above — always pause and explicitly justify start-time vs. end-time sorting based on what you're trying to optimize.
8. **Treating "combination" problems as if order matters (or vice versa).** Combination Sum IV is named confusingly — it's actually counting **ordered** sequences (despite "combination" in the name), while Combination Sum I/II are genuinely order-independent. Always clarify this explicitly rather than assuming based on the problem's title.
9. **Recomputing values that were already computed in a previous recursive call.** This is the root cause of every "exponential brute force" shown throughout this guide — if you notice your recursion tree revisiting the same `(index, remaining, ...)` state repeatedly, that's your signal to add memoization or convert to bottom-up DP.
10. **Forgetting to handle empty input / single-element input / all-identical-elements edge cases before considering your solution "done."** Every full-treatment problem in Part A explicitly lists edge cases for exactly this reason — make listing edge cases a habit you do before writing a single line of code, not an afterthought after your first solution "seems to work."

C3.4 How the Patterns Combine in Real Interviews

Many real interview questions (especially at the Senior/Staff level, or in later interview rounds) don't map to exactly one pattern from the cheat sheet — they combine two or three. Recognizing this compositional structure is itself a skill worth practicing deliberately:

- **Word Search II (A7.3)** = Trie (C1.17) + Grid DFS/Backtracking (C1.8/C1.7).
- **Merge k Sorted Lists (A5.6)** = Heap (C1.12) + Linked List manipulation.

- **Course Schedule + Alien Dictionary** = deriving a graph from constraints, THEN Topological Sort (C1.9) — the graph-building step is often the actually hard part, not the topological sort itself.
- **Maximal Square / Maximal Rectangle** = 2-D DP (C1.14) or Monotonic Stack (C1.11) reused across rows — recognizing when a 1-D technique can be applied "per row" to solve a 2-D problem is a recurring escalation.
- **Reorder List / Palindrome Linked List** = Fast/Slow Pointers (C1.3) + In-place Reversal + a final merge/compare pass — three simple techniques chained together.
- **LFU Cache** (a common LRU Cache follow-up) = HashMap + Doubly Linked List (like LRU) + an additional frequency-bucketing layer (similar in spirit to C1.12's heap-based top-k idea, but implemented with ordered buckets instead).

When you get a problem that feels unfamiliar, actively ask yourself: "which 2-3 patterns from the cheat sheet, applied together, would solve this?" rather than searching your memory for an exact match to a problem you've seen before.

C3.5 Final Tips for Interview Day

- Re-read the Universal Interview Framework (Part 0) the morning of your interview — it's short, and having the 9 steps fresh keeps you from freezing at step 1 ("restate the problem") when nerves hit.
- Ask about constraints early — they tell you the expected complexity class before you've written a single line of code (e.g. `n <= 20` almost always signals exponential/bitmask is expected; `n <= 10^5` almost always rules out O(n²)).
- If you're stuck for more than a couple of minutes, explicitly say what you're considering ("I'm wondering if a monotonic stack helps here because we need 'next greater element' style information") — thinking out loud like this often gets you unstuck faster than silent thinking, and it keeps the interviewer engaged with your process.
- Test your code against the example given in the problem statement, out loud, line by line, before declaring you're done — this catches the majority of small bugs before the interviewer has to point them out.
- Don't be afraid to say "I recognize this pattern" — interviewers generally see this as a strength (efficient pattern recognition), not a weakness, as long as you can still explain *why* the pattern applies to this specific problem rather than just naming it.

C3.6 Resources for Continued Practice

- Continue solving fresh problems on LeetCode itself, filtered by company tag and by the pattern tags matching the sections in this guide, once you've internalized the core ~162 problems here.
- Revisit this guide's Pattern Recognition Cheat Sheet (C1) weekly during your job search — spaced repetition of the *patterns* (not necessarily re-solving every problem from scratch each time) is far more time-efficient than re-grinding hundreds of problems repeatedly.
- Pair this guide with mock interviews (with peers, mentors, or structured mock-interview platforms) — reading solutions and producing them live under time pressure while talking are different skills, and only the second one is what's actually being evaluated.

This guide covered 162 problems spanning every major LeetCode pattern, each solved from first principles (brute force) through to the interview-winning optimal approach, plus a compressed pattern-recognition playbook to help you generalize far beyond these specific 162 problems. The goal was never memorization — it was giving you a repeatable *process* (Part 0), a mental map of *patterns* (Part C1), and enough worked examples (Parts A and B) that any new problem you encounter gets sorted into "oh, this is basically problem X with a twist" almost immediately. That recognition speed, built through deliberate practice with this exact brute-force-then-optimal rhythm, is what actually wins interviews. Good luck.

Appendix D: Complexity Quick Reference

A last-minute reference table of the complexity classes you should be able to recite instantly for the core data structures and algorithms used throughout this guide.

Data Structure / Operation	Average Time	Worst Case Time	Space
Array access by index	O(1)	O(1)	O(1)
Array search (unsorted)	O(n)	O(n)	O(1)

Data Structure / Operation	Average Time	Worst Case Time	Space
Array search (sorted, binary search)	O(log n)	O(log n)	O(1)
ArrayList/Vector append (amortized)	O(1)	O(n) on resize	O(1) per op
HashMap get/put	O(1)	O(n) (pathological hash collisions)	O(n)
TreeMap/TreeSet get/put (balanced BST)	O(log n)	O(log n)	O(n)
Linked List access by index	O(n)	O(n)	O(1)
Linked List insert/delete at known node	O(1)	O(1)	O(1)
Stack push/pop/peek	O(1)	O(1)	O(n) total
Queue offer/poll/peek	O(1)	O(1)	O(n) total
Binary Heap insert/extract-min/max	O(log n)	O(log n)	O(n) total
Binary Heap peek	O(1)	O(1)	O(1)
Balanced BST insert/delete/search	O(log n)	O(log n)	O(n) total
Trie insert/search (length L)	O(L)	O(L)	O(total characters)
Union-Find find/union (path compression + rank)	O(α(n)) ≈ O(1)	O(log n)	O(n)
Merge Sort	O(n log n)	O(n log n)	O(n)
Quick Sort	O(n log n)	O(n²)	O(log n)
Heap Sort	O(n log n)	O(n log n)	O(1)
Counting Sort / Radix Sort	O(n + k)	O(n + k)	O(k)
BFS / DFS on a graph	O(V + E)	O(V + E)	O(V)
Dijkstra's Algorithm (with min-heap)	O(E log V)	O(E log V)	O(V + E)

Data Structure / Operation	Average Time	Worst Case Time	Space
Bellman-Ford Algorithm	O(V · E)	O(V · E)	O(V)
Kruskal's Algorithm (MST)	O(E log E)	O(E log E)	O(V)
Topological Sort (Kahn's / DFS)	O(V + E)	O(V + E)	O(V + E)
1-D DP (single loop)	O(n)	O(n)	O(n) or O(1) rolling
2-D DP (grid/two strings)	O(m·n)	O(m·n)	O(m·n) or O(min(m,n)) rolling
Backtracking (subsets)	O(2^n)	O(2^n)	O(n) recursion
Backtracking (permutations)	O(n!)	O(n!)	O(n) recursion
Backtracking (N-Queens style, pruned)	Sub-factorial in practice	O(n!) loose bound	O(n)

Appendix D.1 How to Estimate Feasible Complexity From Constraints

A fast sanity check for what time complexity is expected, based on the typical n given in constraints (assuming roughly 10^8 operations per second as a rule of thumb for a 1-2 second time limit):

Constraint on n	Expected complexity	Typical pattern
n ≤ 10-12	O(n!) or O(2^n · n)	Brute-force permutations, bitmask DP
n ≤ 20-25	O(2^n)	Subsets, bitmask DP
n ≤ 500	O(n³)	Triple nested loops, some interval DP
n ≤ 5,000	O(n²)	Nested loops, some DP
n ≤ 10^6	O(n log n)	Sorting-based, heaps, binary search combined with a scan

Constraint on n	Expected complexity	Typical pattern
n ≤ 10^8	O(n)	Single-pass, two pointers, sliding window, DP with O(1) transitions
n very large / streaming	O(log n) or O(1) per operation	Binary search, heaps, amortized structures

Always cross-check this table against the actual constraints given rather than assuming — but as a first-instinct filter within the first minute of reading a problem, it reliably narrows down which section of this guide the problem likely belongs to before you've even finished reading the full statement.

Appendix D.2 One More Reminder

Complexity analysis is a means to an end, not the end itself — the actual goal in an interview is a **correct, clearly-communicated, appropriately-optimized** solution. Use this appendix and the rest of the guide as scaffolding to get there quickly and confidently, not as a script to recite. Good luck out there.